

UNIVERSIDAD DE CASTILLA-LA MANCHA

**DEPARTAMENTO DE INGENIERÍA ELÉCTRICA,
ELECTRÓNICA, AUTOMÁTICA Y COMUNICACIONES**

**IMPACT ON SYSTEM ECONOMICS AND
SECURITY OF A HIGH PENETRATION
OF WIND POWER**

TESIS DOCTORAL

AUTOR: JUAN MIGUEL MORALES GONZÁLEZ

DIRECTORES:

ANTONIO J. CONEJO NAVARRO

JUAN PÉREZ RUIZ

Ciudad Real, Diciembre de 2010

UNIVERSIDAD DE CASTILLA-LA MANCHA

DEPARTMENT OF ELECTRICAL ENGINEERING

**IMPACT ON SYSTEM ECONOMICS AND
SECURITY OF A HIGH PENETRATION
OF WIND POWER**

PhD THESIS

AUTHOR: JUAN MIGUEL MORALES GONZÁLEZ

SUPERVISORS:

ANTONIO J. CONEJO NAVARRO

JUAN PÉREZ RUIZ

Ciudad Real, December 2010

*To Cloti, Pepe, David and Alejandro,
and to the murmurs in my memories*

Acknowledgments

In what will likely be an affront to Shakespeare's mother tongue, I would like to express my deepest gratitude to all those people and institutions that, in one way or another, have supported the realization of this work.

Heartfelt thanks go to Professor Antonio Conejo for his time and dedication, wise guidance and supervision, professionalism and sense of fairness, and above all, for providing me with a high-quality technical education.

Also, my kindest gratitude goes to Professor Juan Pérez for his permanent social and technical support, his friendship, and what's more, thanks for introducing me to the world of research.

I wish to thank the Consejería de Educación y Ciencia of the Castilla-La Mancha Government for its financial support through the grants "Ayudas para la Formación de Personal Investigador" and "Ayudas Complementarias José Castillejo". The latter has allowed me to spend part of my PhD studies at McGill University in Montreal, Canada and at Danmarks Tekniske Universitet in Copenhagen, Denmark.

Additionally, I would like to express my appreciation to the Universidad de Castilla-La Mancha for placing its facilities at my disposal and for its financial support by means of the "Ayudas a la Investigación para la realización de Tesis Doctorales."

A great and enriching experience during my PhD studies was my stay at McGill University in Montreal, Canada, in September/October 2007. There Professor Galiana, his students, and I, were often immersed in inspiring discussions within a friendly atmosphere.

I also very much appreciate the support and input I received from the research group headed by Professor Henrik Madsen at Danmarks Tekniske Universitet in Copenhagen, Denmark, from July through September 2010. No doubt I benefited from the intense and productive discussions I had with various people there.

Thanks to my family for their unconditional love. No matter where, when, how, or what for, you manage to be always there.

And thanks to all of you, to those with whom I've lived and worked side by side for these past four years in Ciudad Real. I will never forget the lab, the offices on the second floor, the illegal conversations and "los ilegales", the combos "100 montaditos" plus cinema, San Paco, the paddle matches, Chicago, the blog, the "lost" discussions, the coffee breaks, "las migas", the "hanukades" and the constant that balances my chaotic equations, my daily routine, Washington, my "paella" feasts (Ciudad Real made a good cook of me), the Paparazzi crazy parties, San Diego, the SingStar performances, our friend in the Game store, the small round table with a small heater underneath, the gym, the yoga, the bike rides, Tampa and its extras, the trips to Toledo in my tiny little Ka, Norway, my weekly visits to Eroski, the barbecues, our relaxing Friday nights, Copenhagen, the evenings of "tele y comida basura", Montreal, the tennis courses, our "carretera y manta" journeys, the arrhythmic dance classes, the Christmas meetings,...

Thanks to all of you, guys, because you are all "partyful". Dang strong men also cry!

Contents

Contents	xi
List of Figures	xvii
List of Tables	xxiii
Notation	xxv
1 Introduction	5
1.1 Background and Motivation	5
1.2 Contributions	7
1.3 Thesis Outline	8
1.4 List of Publications	9
2 The Wind Power <i>Boom</i>	11
2.1 A Look to the Near Future	11
2.2 A Stimulus for Researchers: Challenges	13
2.2.1 A Long Way Ahead	14
2.2.2 The Daily Routine	29
2.2.2.1 The Wind Producer Viewpoint	29
2.2.2.2 The MO Perspective	31
2.3 Battling with the Uncertain Character of Wind Power	36
2.4 A Preview to Whet the Reader's Appetite	38
2.4.1 Wind Producer: Offering Strategy	38
2.4.2 ISO/MO: Market Clearing and Reserve Valuation	45
2.5 Summary	48

3	Uncertainty Characterization	53
3.1	Introduction	53
3.2	Stochastic Processes	54
3.3	Scenario Generation	56
3.3.1	Scenario Tree	56
3.3.2	Scenario Generation Using ARMA models	59
3.3.3	Challenging the Stationarity Assumption	62
3.3.4	Challenging the Normality Assumption	67
3.3.5	Transforming Wind Speed into Wind Power	68
3.3.6	Modeling Statistical Dependencies	71
3.3.7	Market Price Case Study	79
3.3.8	Wind Speed Case Studies	82
3.3.8.1	Wind Speed Case Study 1	84
3.3.8.2	Wind Speed Case Studies 2, 3 and 4	91
3.4	Scenario Reduction	97
3.4.1	Motivation	97
3.4.2	Scenario Reduction Using a Probability Distance	98
3.4.3	Forward Selection	100
3.4.3.1	Variant 1: Cost function based on the norm of the difference between pairs of random vectors	103
3.4.3.2	Variant 2: Cost function based on the objective function of a single-scenario problem	103
3.4.4	Example: Forward Selection Variant 1	105
3.4.5	Example: Forward Selection Variant 2	108
3.5	Summary	113
4	Trading Wind Power in Short-Term Electricity Markets	115
4.1	Introduction	115
4.2	Decision Framework	116
4.3	Unravelling the Wind Producer Problem	118
4.3.1	Electricity Price Formation	119
4.3.2	Mechanism for Imbalance Prices	121
4.3.3	Imbalance Cost and Revenue	126

4.3.4	Certainty Gain Effect	129
4.4	Uncertainty Characterization	131
4.4.1	Day-ahead, Adjustment, and Imbalance Prices	131
4.4.2	Wind Power Production	133
4.4.3	Scenario Tree	137
4.5	Wind Producer Model	139
4.5.1	Basic Model	140
4.5.2	Offering Curves	143
4.5.3	Risk Modeling	144
4.5.4	Adjustment Market	145
4.5.5	Formulation	146
4.6	Wind Producer Example	149
4.7	Wind Producer Case Study	163
4.8	Summary	178
5	Market Clearing under Uncertainty	181
5.1	Introduction	181
5.2	Market Clearing	182
5.2.1	Overview	182
5.2.2	Procedures: Simplicity vs Optimality	183
5.2.3	Security Constraints	184
5.2.4	Approaches for Setting Reserve Requirements	185
5.2.5	Solution Algorithm	186
5.3	Stochastic Security Metrics	188
5.3.1	Security Metric Definition	188
5.3.2	Standard Metrics	188
5.3.3	Security Criteria Based on the ELNS	189
5.4	Uncertainty Characterization	190
5.4.1	Equipment Failures	191
5.4.2	Wind Power Production	192
5.4.3	Scenario Tree	193
5.5	Market-Clearing Model	194
5.5.1	Assumptions	195

5.5.2	Variables	196
5.5.3	Structure	197
5.5.4	Objective function	197
5.5.5	Electricity Market Constraints	199
5.5.6	Real-Time Operating Constraints	202
5.5.7	Linking constraints	204
5.5.8	Formulation	208
5.5.9	Wind Spillage <i>Cost</i>	212
5.6	Metrics for the Economic Appraisal of Wind Generation	212
5.6.1	Average Benefit (AB)	213
5.6.2	Average Uncertainty Cost (AUC)	213
5.6.3	Net Average Benefit (NAB)	214
5.7	Market-Clearing Example	214
5.7.1	Equipment Failures	216
5.7.2	Wind Power Production	226
5.7.3	Equipment Failures & Wind Power Production	238
5.8	Market-Clearing Case Studies	242
5.8.1	Equipment Failure Case Study	244
5.8.2	Wind Production Case Study	248
5.9	Summary	255
6	Closure	261
6.1	Summary and Conclusions	261
6.2	Recommendations for Future Work	267
A	Introduction to Stochastic Programming	271
A.1	Why stochastic programming?	271
A.2	Stochastic Programming Problems with Recourse	273
A.3	Quality Metrics	276
A.3.1	The Value of Stochastic Solution	276
A.3.2	The Expected Value of Perfect Information	278
A.4	Risk Management	279

B	24-Node System Data	285
B.1	Network data	285
B.2	Generator data	285
B.3	Demand data	291
C	Inter-Temporal Constraints of Generators	293
C.1	Ramping Limits	293
C.2	Minimum Up- and Down-Time Constraints	294
	Bibliography	297
	Index	322

List of Figures

2.1	Impact of wind integration on the load duration curve	16
2.2	Impact of wind integration on electricity prices	26
2.3	Day-ahead electricity prices registered in the Spanish market on March 1, 2010	27
2.4	ERCOT balancing market prices on March 7, 2009	28
2.5	Market structure and time framework	39
2.6	Scenario tree	43
2.7	Power sold in market A as a function of the risk aversion of the wind producer	43
2.8	Three-node system	45
2.9	Breakdown of the expected cost: a) no wind generation uncer- tainty; b) no wind generation; c) uncertain wind generation; d) network congestion; e) $\alpha = \$100/\text{MWh}$	48
3.1	Example of three-stage scenario tree	57
3.2	Example of non-stationary process	63
3.3	Logarithmic transformation of the process in Fig. 3.2	64
3.4	Differentiation of the process in Fig. 3.3	64
3.5	Example of seasonal series	65
3.6	Graphic interpretation of the normal transformation in equa- tion (3.9)	68
3.7	Wind power frequency distribution corresponding to the wind power data simulated by NREL for the the Adair site in Iowa over 2006	69
3.8	Example of a wind turbine power curve	70

3.9	Scatter plot showing the hourly wind speed and power production pairs simulated for the Adair location, Iowa, by NREL over 2006 along with the fitted and assumed power curves	71
3.10	Types of dependent stochastic processes according to their residual cross-correlogram: examples	74
3.11	Flow diagram of the algorithm to generate correlated scenarios .	77
3.12	Price series for the last three months of year 2007 in the electricity market of the Iberian Peninsula	80
3.13	Logarithmic and differentiated price series for the last three months of year 2007 in the electricity market of the Iberian Peninsula	80
3.14	96-h extract from the logarithmic and differentiated price series in Fig. 3.13	81
3.15	Set of 200 price scenarios generated for the first working day of year 2008	82
3.16	Schematic geographical distribution of wind sites and distances among them	83
3.17	Wind speed frequency distribution for site A and the pdf of the Weibull fit	85
3.18	Normality plot of the transformed data for site A	86
3.19	Cross-correlogram of the residuals obtained after the ARMA fit	87
3.20	Cross-correlograms of errors with a significance level $\alpha = 0.05$. .	87
3.21	Comparison between sample and simulated cross-correlograms .	88
3.22	A 48-hour extract from the historical and simulated wind speed series at sites A and B	90
3.23	Comparison between fitted and simulated cumulative probability functions for wind site A	90
3.24	Cross-correlograms of errors with $\alpha = 0.05$	92
3.25	Cross-correlograms of errors with $\alpha = 0.2$	95
3.26	Comparison between sample and simulated cross-correlograms .	96
4.1	Decision framework of the wind producer problem.	117

4.2	Single-period auction excluding inter-temporal and network constraints	120
4.3	Day-ahead market price formation	122
4.4	Mechanism for imbalance price formation	125
4.5	Certainty gain on wind power stochastic behavior	130
4.6	Formation of random variable r_t from the imbalance price ratios r_t^+ and r_t^-	134
4.7	Illustration of the time periods covered by indexes τ and t . . .	135
4.8	Example of wind power scenario set	136
4.9	Wind producer problem: scenario tree	137
4.10	Wind producer example: market structure and time framework .	149
4.11	Wind producer example: three-stage scenario tree	152
4.12	Wind producer example: two-stage scenario tree (excluding the adjustment market)	154
4.13	Wind producer example: imbalance cost as a function of the amount of power sold in the day-ahead market	158
4.14	Wind producer example: power sold in the day-ahead market as a function of the risk-aversion parameter β	158
4.15	Wind producer case study: market organization and time framework	164
4.16	Wind producer case study: wind speed frequency distribution of the considered location and the pdf of the Weibull fit	166
4.17	Wind producer case study: power curve of the considered turbine model	167
4.18	Wind producer case study: optimal offering curves for $\beta = 1$. .	169
4.19	Wind producer case study: expected energy traded in the day-ahead market as a function of the risk-aversion factor β	169
4.20	Wind producer case study: impact of risk aversion on expected energy deviations	170
4.21	Wind producer case study: impact of the certainty gain on expected imbalances	172
4.22	Wind producer case study: hourly reduction in expected power deviation due to the certainty gain effect	172

4.23	Wind producer case study: impact of the certainty gain effect on the energy amounts traded in both the day-ahead and adjustment markets	173
4.24	Wind producer case study: impact of the certainty gain effect on risk management (efficient frontier)	174
5.1	Scenario tree example for the stochastic security-constrained market-clearing problem	194
5.2	Three-bus system	214
5.3	Impact of the magnitude of the value of lost load on: (a) the expected social cost; (b) the expected amount of load not served (ELNS); (c) the total amount of scheduled reserve	221
5.4	Effect of the outage of line 2 during period $t = 3$ in failure scenarios $f = 5, 6$, and 7	224
5.5	Effect of the outage of unit 3 during period $t = 3$ in failure scenarios $f = 1, 2$, and 3	225
5.6	Three relevant cases: a) No wind generation uncertainty; b) no wind generation; c) uncertain wind generation	228
5.7	Base case ($P^{\text{F}, \text{max}}(2, 3) = 55$ MW). Expected cost breakdown	229
5.8	Network-constrained case ($P^{\text{F}, \text{max}}(2, 3) = 25$ MW). Expected cost breakdown	230
5.9	Network-constrained case with a cost of wind power spillage equal to \$100/MWh. Expected cost breakdown	234
5.10	Network-constrained case. Wind unit at node 2. Effect of wind spillage cost on the expected cost breakdown	235
5.11	Wind unit at node 2. Expected cost as a function of wind penetration and wind uncertainty. Base case	236
5.12	Wind unit at node 2. Expected cost as a function of wind penetration and wind uncertainty. Network-constrained case	237
5.13	a) Base case, b) $P^{\text{F}, \text{max}}(2, 3) = 25$ MW, and c) $P^{\text{F}, \text{max}}(2, 3) = 25$ MW & binding ramping constraints	238

5.14	Equipment failures and wind production. Impact of the magnitude of the value of lost load on: (a) the expected social cost, (b) the expected amount of load not served (ELNS), and (c) the total amount of scheduled reserve	241
5.15	Equipment failures and wind production. Wind unit at bus 3 and $P^{\text{F}, \max}(2, 3) = 35$ MW. Impact of the magnitude of the value of lost load on: (a) the expected social cost, (b) the expected amount of load not served (ELNS), and (c) the total amount of scheduled reserve	243
5.16	Turbine speed-power curves	250
A.1	Example of efficient frontier	281
A.2	Illustration of the concept of the Conditional Value-at-Risk . . .	283
B.1	24-node system	286

List of Tables

2.1	Generator data	47
3.1	Parameters of the SARIMA model (3.17)	82
3.2	Percentile comparison for wind site A	91
3.3	Stochastic processes for wind sites C, D, and E	93
3.4	Wind power scenarios	105
3.5	Objective function values z_ω of single-scenario problems DP_ω . .	111
4.1	Wind producer example: expected values of perfect information	156
4.2	Wind producer example: adjustment market effects on the ex- pected profit of the wind producer (increases in percentages) . .	163
4.3	Wind producer case study: stochastic processes for prices and price functions	166
4.4	Wind producer case study: optimal energy bids for $\beta = 0$	168
4.5	Actual profit per month and strategy in M€	176
5.1	Generating unit data	215
5.2	Hourly demand at bus 3	215
5.3	Scheduling program. Powers in MW	218
5.4	Involuntary load shedding (MWh)	219
5.5	Scheduling program for $V^{\text{LOL}} = \$25\,000/\text{MWh}$. Powers in MW .	219
5.6	Reserve schedule for $V^{\text{LOL}} = \$15\,000/\text{MWh}$ [$\$25\,000/\text{MWh}$]. Powers in MW	222
5.7	Involuntary load shedding (MWh) for $V^{\text{LOL}} = \$15\,000/\text{MWh}$ [$\$25\,000/\text{MWh}$]	222

5.8	Breakdown of expected social costs in \$ (%)	226
5.9	Wind power scenarios	227
5.10	Impact of network congestion on performance metrics	231
5.11	Comparison of scheduling programs. Powers in MW	232
5.12	Mean time to failure (MTTF) of generating units	245
5.13	Breakdown of the expected social cost in \$ (%) for $V^{LOL} =$ \$2000/MWh	246
5.14	Expected load not served (ELNS) and amount of scheduled re- serve for different values of lost load (V^{LOL})	247
5.15	Costs of energy and security in \$ per MWh	248
5.16	Costs of energy and security	251
5.17	Valuation of non-spinning reserves (NSR). Expected costs	252
5.18	Valuation of non-spinning reserves (NSR). CPU times	252
5.19	Expected cost increase (%) resulting from the application of several strategies to relieve the computational burden	254
5.20	Time reduction (%) achieved through the application of several strategies to relieve the computational burden	254
B.1	24-node system: reactance and capacity of transmission lines . .	287
B.2	Technical data of generating units	288
B.3	Node location of generating units	288
B.4	Size (MWh) of the energy blocks offered by generating units . .	289
B.5	Offer costs (\$/MWh) submitted by generating units	290
B.6	Start-up costs of generating units	290
B.7	Load profile	291
B.8	Node location and distribution of the total system demand . . .	292

Notation

For the sake of clarity, the symbols used to formulate the wind producer problem in Chapter 4 and the marker-clearing model in Chapter 5 are stated below separately for quick reference. Others are defined as required in the text.

List of Symbols of the Wind Producer Problem (Chapter 4)

Indices and Numbers:

t	Index of time periods spanning the market horizon, running from 1 to N_T .
τ	Auxiliary index for the time periods in between the closures of the day-ahead and adjustment markets, running from 1 to N_{T_1} .
ω	Index of scenarios in the tree (including the scenario representation of all sources of uncertainty), running from 1 to N_Ω .
N_Δ	Number of scenarios representing the difference between day-ahead and adjustment market prices.
N_D	Number of price scenarios for the day-ahead market.
N_I	Number of scenarios representing the imbalance price ratios.
N_ϖ	Number of wind power scenarios for the time interval between the closure of the day-ahead market and the end of the market horizon.

N_{ϖ_1}	Number of wind power scenarios for the time interval between the closures of the day-ahead and adjustment markets.
N_{ϖ_2}	Number of wind power scenarios for the time interval between the closure of the adjustment market and the end of the market horizon.

Real Variables:

C_t	Imbalance cost incurred by the wind producer in time period t [\$ or €].
$P_{t\omega}^A$	Power offered by the wind producer in the adjustment market for time period t and scenario ω [MW].
P_t^D	Power offered by the wind producer in the day-ahead market for time period t [MW].
$P_{t\omega}^S$	Power production scheduled for the wind producer in time period t and scenario ω [MW].
$\Delta_{t\omega}$	Total deviation of energy incurred by the wind producer with respect to the schedule in time period t and scenario ω [MWh].
$\Delta_{t\omega}^+$	Excess of energy (positive deviation) incurred by the wind producer with respect to the schedule in time period t and scenario ω [MWh].
$\Delta_{t\omega}^-$	Deficit of energy (negative deviation) incurred by the wind producer with respect to the schedule in time period t and scenario ω [MWh].
ζ	Auxiliary variable used to compute the Conditional Value at Risk, CVaR [\$ or €].
η_ω	Auxiliary variable in scenario ω used to compute the Conditional Value at Risk, CVaR [\$ or €].

Random Variables:

$\Delta\lambda_t$	Difference between adjustment and day-ahead market prices in time period t [\$/MWh or €/MWh].
λ_t^A	Adjustment market price in time period t [\$/MWh or €/MWh].
λ_t^D	Day-ahead market price in time period t [\$/MWh or €/MWh].
λ_t^{UP}	Price for upward energy resulting from the balancing market in time period t [\$/MWh or €/MWh].
λ_t^{DN}	Price for downward energy resulting from the balancing market in time period t [\$/MWh or €/MWh].
λ_t^+	Price of positive imbalances incurred in period t [\$/MWh or €/MWh].
λ_t^-	Price of negative imbalances incurred in period t [\$/MWh or €/MWh].
P_t^W	Wind power production in time period t of the market horizon [MW].
P_τ^W	Wind power production in time period τ of the time interval between the closures of the day-ahead and adjustment markets [MW].
r_t^+	Ratio between positive imbalance price and day-ahead market price in time period t .
r_t^-	Ratio between negative imbalance price and day-ahead market price in time period t .

These symbols, if augmented with subscript ω , refer to the realization of the corresponding random variable in scenario ω .

Constants:

d_t	Duration of time period t [h].
α	Per unit confidence level.
β	Risk-aversion parameter.
$\tilde{\lambda}_t^D$	Actual electricity price in the day-ahead market for time period t .
$\tilde{P}_t^D$	Actual power traded in the day-ahead market for time period t .
$P^{\text{W}, \max}$	Installed capacity of the wind farm [MW].
π_ω	Probability of occurrence of scenario ω .

Set:

Ω	Set of scenarios.
----------	-------------------

Acronyms:

CVaR	Conditional Value-at-Risk.
EVPI	Expected Value of Perfect Information.
VSS	Value of the Stochastic Solution.

Others:

$\mathcal{E}\{\cdot\}$	Expectation operator.
------------------------	-----------------------

List of Symbols of the Market-Clearing Problem (Chapter 5)

Indices and Numbers:

f	Index of failure scenarios.
i	Index of generating units, running from 1 to N_G .
j	Index of loads, running from 1 to N_L .
k	Index of contingencies.
ℓ	Index of energy bids submitted by loads, running from 1 to N_{Bjt} (number of energy bids submitted by load j in period t).
m	Index of energy blocks offered by generating units, running from 1 to N_{Oit} (number of blocks of energy offered by unit i in period t).
n, r	Indices of system buses.
q	Index of wind farms, running from 1 to N_W .
t	Index of time periods, running from 1 to N_T .
τ	Auxiliary time period index referring to the contingency occurrence intervals, running from 1 to N_T .
ω	Index of scenarios in the tree (including the scenario representation of all sources of uncertainty), running from 1 to N_Ω .
ϖ	Index of wind power scenarios, running from 1 to N_ϖ .

Real Variables:

C_{it}^{SU}	Cost due to the scheduled start-up of unit i in period t (\$). $C_{it\omega}^{\text{SU}}$ is the actual start-up cost incurred by unit i in period t and scenario ω .
----------------------	--

$P_{jt}^{L,S}$	Power scheduled for load j in period t (MW). Bounded by $P_{jt}^{L,\min}$ and $P_{jt}^{L,\max}$.
$P_{it}^{G,S}$	Power output scheduled for unit i in period t (MW). Bounded by $P_i^{G,\min}$ and $P_i^{G,\max}$.
R_{it}^U	Spinning reserve up scheduled for unit i in period t (MW). Limited to $R_{it}^{U,\max}$.
R_{it}^D	Spinning reserve down scheduled for unit i in period t (MW). Limited to $R_{it}^{D,\max}$.
R_{it}^{NS}	Non-spinning reserve scheduled for unit i in period t (MW). Limited to $R_{it}^{NS,\max}$.
R_{jt}^U	Spinning reserve up scheduled for load j in period t (MW). Limited to $R_{jt}^{U,\max}$.
R_{jt}^D	Spinning reserve down scheduled for load j in period t (MW). Limited to $R_{jt}^{D,\max}$.
$P_t^{W,S}$	Scheduled wind power in period t (MW).
$C_{it\omega}^A$	Cost due to the change in the start-up plan of unit i in period t and scenario ω (\$).
$P_{it\omega}^G$	Power output of unit i in period t and scenario ω (MW). Bounded by $P_i^{G,\min}$ and $P_i^{G,\max}$.
$p_{Git\omega}(m)$	Power output corresponding to the m -th block of energy offered by unit i in period t and scenario ω (MW). Limited to $p_{Git}^{\max}(m)$.
$p_{Ljt\omega}(\ell)$	Power consumed from the ℓ -th bid of energy submitted by load j in period t and scenario ω (MW). Limited to $p_{Ljt}^{\max}(\ell)$.
$P_{jt\omega}^L$	Power consumed by load j in period t and scenario ω (MW).
$r_{it\omega}^U$	Spinning reserve up deployed by unit i in period t and scenario ω (MW).

$r_{it\omega}^D$	Spinning reserve down deployed by unit i in period t and scenario ω (MW).
$r_{it\omega}^{NS}$	Non-spinning reserve deployed by unit i in period t and scenario ω (MW).
$r_{jt\omega}^U$	Reserve up deployed by load j in period t and scenario ω (MW).
$r_{jt\omega}^D$	Reserve down deployed by load j in period t and scenario ω (MW).
$P_{jt\omega}^{L, \text{shed}}$	Load shedding imposed on consumer j in period t and scenario ω (MW).
$S_{qt\omega}$	Wind power generation spillage from wind farm q in period t and scenario ω (MW).
$P_{t\omega}^F(n, r)$	Power flow through line (n, r) in period t and scenario ω (MW). Limited to $P^{F, \max}(n, r)$.
$\delta_{nt\omega}$	Voltage angle at node n in period t and scenario ω (rad).

Binary Variables:

u_{it}	0/1 variable equal to 1 if unit i is scheduled to be committed in period t .
$v_{it\omega}$	0/1 variable equal to 1 if unit i is online in period t and scenario ω .

Random Variables:

P_t^W	Random variable modeling the wind power generation in period t (MW). If subscript q is added to this variable, it refers to the power produced by wind farm q . Likewise, if augmented with subscript ω , it refers to the wind power realization in scenario ω .
---------	---

Random Events:

- $A(k, \tau)$ Random event “contingency k happens within the time period τ ”.
- $B(k)$ Random event “contingency k does not occur during the entire scheduling horizon”.

Constants:

- $C_{it}^{R^U}$ Offer cost of the up-spinning reserve submitted by unit i in period t (\$/MWh).
- $C_{it}^{R^D}$ Offer cost of the down-spinning reserve submitted by unit i in period t (\$/MWh).
- $C_{it}^{R^{NS}}$ Offer cost of the non-spinning reserve submitted by unit i in period t (\$/MWh).
- $C_{jt}^{R^U}$ Offer cost of the up-spinning reserve submitted by load j in period t (\$/MWh).
- $C_{jt}^{R^D}$ Offer cost of the down-spinning reserve submitted by load j in period t (\$/MWh).
- d_t Duration of time period t (h).
- λ_k Inverse of the mean time to the occurrence of contingency k (h^{-1}).
- λ_{it}^{SU} Start-up offer cost of unit i in period t (\$).
- $\lambda_{Git}(m)$ Price for the energy selling block m offered by unit i in period t (\$/MWh).
- $\lambda_{Ljt}(\ell)$ Price for the energy purchasing bid ℓ submitted by load j in period t (\$/MWh).
- $P_q^{\text{W}, \max}$ Installed capacity of wind farm q [MW].

V_{jt}^{LOL}	Value of load shed for consumer j in period t (\$/MWh).
V_{qt}^{S}	Cost of wind power spillage from wind farm q in period t (\$/MWh).
π_f	Probability of failure scenario f .
π_ω	Probability of scenario ω in the tree (including the scenario representation of all sources of uncertainty).
π_ϖ	Probability of wind power scenario ϖ .
$B(n, r)$	Absolute value of the imaginary part of the admittance of line (n, r) (per unit).

Sets:

$G(t, \omega)$	Set of available generating units in period t and scenario ω .
$\Lambda(t, \omega)$	Set of transmission lines that are functional in period t and scenario ω .
M_L	Mapping of the set of loads (consumers) into the set of buses.
$M_G(t, \omega)$	Mapping of the sets of available generating units into the set of buses in period t and scenario ω .
M_W	Mapping of the set of wind farms into the set of buses.
Ω	Set of scenarios.

Acronyms:

AB	Average benefit.
AUC	Average uncertainty cost.
EC	Expected cost.
ELNS	Expected load not served.

LOLE	Loss-of-load expectation.
LOLP	Loss-of-load probability.
MTTF	Mean time to failure.
MTTR	Mean time to repair.
NAB	Net average benefit.
NSR	Non-spinning reserve.
WP	Wind plant.

Crisis
(Camino *or a human heart*)

Mantengo.
Uno, dos, tres, cuatro.
Exhalo y alivio mis pulmones, exhaustos.
Inspiro.
Cierro los ojos, suavemente,
e imagino mis músculos, aún tensos.
Uno, dos.
Me sueño,
volando sobre vastos campos pajizos
desdibujados por el calor del infinito asfalto.
Tres, cuatro.
Me encojo,
para deshacer el miedo sobre el frío suelo.
Expiro.
Bebo de sus consuelos,
pero mi pecho en carne viva,
duele, escuece.
Morder la mano invisible
que me ahoga y me aplasta.
¿Me prometes que este es el camino?
Inspiro.
No hay respuesta.
Suspiro.
Me levanto y prosigo.

Copenhagen, 4 de julio de 2010

A stylized, handwritten signature in black ink, consisting of several loops and a long horizontal stroke at the end.

Chapter 1

Introduction

The subject matter of this thesis is inspired by the fact that today's power systems are experiencing an increasing wind power penetration and measures have to be taken to efficiently accommodate its uncertain nature. This first chapter describes the motivation for the developed methods and formally states the main contributions of this thesis. In addition, the outline of the thesis is provided and the published papers are listed.

1.1 Background and Motivation

While the power sector is far from being the only culprit of climate change, it significantly contributes to greenhouse gas emissions causing global warming. As a result, in industrialized countries, there exists strong public pressure claiming for urgent actions that drastically curb air pollution from electricity production. The options for making major reductions in the electricity sector are basically three: energy efficiency and conservation; fuel switching from coal to gas; and renewable energy.

The introduction of non-polluting technology for electricity generation is gaining much interest and relevance in the scientific community, as the threat of climate change needs to be tackled while meeting the rising demand for energy and safeguarding the security of energy supply. Furthermore, they are not few who state that the world is on the verge of an energy crisis. This technol-

ogy primarily includes power generation facilities based on renewable energy sources such as wind, solar, and biomass. Among all of them, wind power production is firmly penetrating current electric energy systems worldwide due to its technological maturity, widespread availability, speed of deployment, and the fact that there is a robust, growing industry becoming more and more global with every passing year.

The essential characteristic of the wind energy exploitation is its intrinsic dependence on weather conditions. Consequently, the facilities devoted to produce electricity from the force of the wind constitute a new source of uncertainty in the power system operation and planning. These facilities, the wind farms, cannot be scheduled or dispatched at the will of the wind producer, thus impacting basic aspects of the daily operation of today's power systems, namely:

- *System security and reliability.* The inherent randomness of wind power availability makes it necessary to have reserves that absorb unpredictable wind power fluctuations. In this thesis, a tool for *system operators* to estimate the amount and cost of these reserves within an electricity market framework is proposed.
- *Competition in electricity markets.* Nowadays, most power systems are operated in a market environment. The market organization must provide mechanisms to cover energy imbalances caused by wind generation under competition. Wind producers, in turn, should seek to reduce the costs of such imbalances through a shrewd market participation that enables them to strengthen their competitiveness. In this dissertation, a tool to devise optimal offering strategies for *wind power producers* is designed.

The two problems outlined above, the first pertaining to the power system operator and the second concerning the wind producer, can be looked at as decision-making problems under uncertainty, wherein the future wind power production constitutes input data and the main source of imperfect information at the same time. In this context, stochastic programming proves itself as a

powerful and useful mathematical framework to properly model the decision-making processes faced by these two agents. In order to make the resulting optimization problems solvable, the time evolution of wind generation needs to be represented by a finite set of plausible realizations or outcomes, referred to as *scenarios*. In this thesis, a methodology to appropriately construct scenarios reproducing the intricate nature of the wind is presented.

1.2 Contributions

The main contributions of this thesis are:

1. The review of the major economic and technical challenges originated from the large-scale integration of wind power in electric energy systems.
2. Policy recommendations for possible changes to be implemented in current procedures for the planning and operation of power systems so as to turn a renewables-based generation mix into a reality.
3. The illustration of the wind power impact on electricity market prices using both empirical and simulated results.
4. The development of an effective procedure to generate scenarios for quasi-contemporaneous stochastic processes that is especially suitable for reproducing the fundamental statistical properties of wind speed series at multiple sites, namely, marginal distributions, and temporal and spatial correlations. The scenario generation methodology is complemented with appropriate scenario reduction techniques to efficiently decrease the cardinality of the resulting scenario sets.
5. The formulation of the short-term trading problem of a wind power producer as a multi-stage stochastic programming problem with recourse, which, in addition, includes the modeling of the risk aversion of the wind producer using the Conditional Value-at-Risk.
6. A linear equivalent formulation for the multi-stage stochastic programming problem of the wind producer.

7. The design of a multi-commodity market-clearing procedure for the simultaneous scheduling of energy and reserve under wind power uncertainty and equipment failures.
8. The formulation of such a market-clearing procedure as a two-stage stochastic programming model with recourse, which translates into a mixed-integer linear optimization problem.
9. The analysis and test of the proposed mathematical models and methodologies through illustrative examples and realistic case studies.

1.3 Thesis Outline

The chapters of the thesis are outlined as follows:

Chapter 2 provides an overview of the main challenges originated from the increasing penetration of wind power in electric energy systems, and motivates the problems dealt with in this dissertation.

Chapter 3 describes procedures and tools to model the uncertainty involved in highly wind-penetrated electricity markets, namely, electricity prices and wind power production. In particular, a scenario generation methodology able to capture the intricate nature of wind speed is developed.

Chapter 4 introduces a stochastic programming model to build offering curves for wind power producers participating in an electricity pool. Diverse trading floors within the pool are considered, specifically, the day-ahead market, one adjustment market, and the balancing market. Particular emphasis is placed on this latter market, through which the energy deviations incurred by non-dispatchable energy sources in real time are covered and priced accordingly.

Chapter 5 describes and analyzes a multi-commodity market-clearing procedure based on stochastic programming to clear a day-ahead electricity market involving both energy and reserve. The reserve determination is based on an endogenous probabilistic criterion. The market-clearing procedure is especially designed to cope with wind production uncertainty and equipment failures.

Chapter 6 concludes the thesis and gives suggestions for future work.

Appendix A constitutes a brief introduction to the stochastic programming concepts used to develop and evaluate the decision-making models presented in this dissertation. Additionally, this appendix also provides basic notions for risk modeling and management via stochastic programming.

Appendix B lists the technical and economic data pertaining to the 24-node system used to test the market-clearing procedure presented in Chapter 5.

Lastly, in **Appendix C** the ramping limits and the minimum up- and down-time constraints of generating units are stated using a mixed-integer formulation.

1.4 List of Publications

The following papers have been published in the course of the work on this thesis:

1. J. M. Morales, S. Pineda, A. J. Conejo and M. Carrión. Scenario Reduction for Futures Market Trading in Electricity Markets. *IEEE Transactions on Power Systems*, 24(2):878–888, May 2009. JCR 5-year impact factor: 2.992, position 31 of 245 in Engineering, Electrical and Electronic.
2. J. M. Morales, A. J. Conejo and J. Pérez-Ruiz. Economic Valuation of Reserves in Power Systems with High Penetration of Wind Power. *IEEE Transactions on Power Systems*, 24(2):900–910, May 2009. JCR 5-year impact factor: 2.992, position 31 of 245 in Engineering, Electrical and Electronic.
3. J. M. Morales, A. J. Conejo and J. Pérez-Ruiz. Short-term Trading for a Wind Power Producer. *IEEE Transactions on Power Systems*, 25(1):554–564, Feb. 2010. JCR 5-year impact factor: 2.992, position 31 of 245 in Engineering, Electrical and Electronic.
4. A. J. Conejo, J. Nogales, M. Carrión and Juan M. Morales. Electricity pool prices: long-term uncertainty characterization for futures-market

trading and risk management. *Journal of the Operational Research Society*, 61(2):235–245, Feb. 2010. JCR 5-year impact factor: 1.565, position 26 of 73 in Operations Research and Management Science.

5. J. M. Morales, R. Mínguez and A. J. Conejo. A Methodology to Generate Statistically Dependent Wind Speed Scenarios. *Applied Energy*, 87(3):843–855, Mar. 2010. JCR 5-year impact factor: 2.193, position 24 of 70 in Energy and Fuels.
6. J. M. Morales, L. Baringo, A. J. Conejo and R. Mínguez. Probabilistic Power Flow with Correlated Wind Sources. *IET Generation, Transmission & Distribution*, 4(5):641–651, May 2010. JCR 5-year impact factor: 0.856, position 135 of 245 in Engineering, Electrical and Electronic.
7. J. M. Morales, A. J. Conejo and J. Pérez-Ruiz. Simulating the Impact of Wind Production on Locational Marginal Prices. *IEEE Transactions on Power Systems*, in press, 2010. JCR 5-year impact factor: 2.992, position 31 of 245 in Engineering, Electrical and Electronic.

Additionally, part of the work developed in this dissertation has been included in the book entitled “Decision Making Under Uncertainty in Electricity Markets” by A. J. Conejo, M. Carrión, and J. M. Morales, which has been published in 2010 by Springer, New York, US, within the *International Series in Operations Research and Management Science*.

Chapter 2

The Wind Power *Boom*

2.1 A Look to the Near Future

The issue of climate change has sparked discussion on the benefits of limiting industrial emissions of greenhouse gases in comparison with the costs that such alterations would entail. Reaching an agreement on this controversial subject seems to be difficult due to the time scales and uncertainties involved. For the time being, climate change can only be strictly appraised in terms of the projected impacts for the coming decades, centuries, or even millennia of increasing temperatures, rising sea levels, heat waves, droughts, reduced crop yields and species extinction. However, beyond this controversy, one thing seems clear: if we desire to avoid the worst and irreversible damages of a plausible climate change, then global greenhouse gas emissions must begin to decline in the near future.

Governments in industrialized nations are currently subject to public opinion demanding actions to head off the worst impacts of climate change. These actions require addressing fundamental, long term structural adjustments to the global economy. A significant part of these structural changes is to be carried out in the electricity generation sector, where curbing emissions of greenhouse gases causing global warming is nowadays one of the most pressing issues. In this sector, the measures undertaken to meet emission reduction targets have basically consisted in increasing the level of penetration of renewable

and low carbon electricity generation resources, with wind power generation being the resource par excellence. In fact, at this time, wind energy is the only power generation technology that can enable the cuts in CO₂ emissions from the power sector necessary for fulfilling the emission reduction targets agreed in some industrialized countries [69, 92].

According to the Global Wind Energy Council (GWEC) in [61], the world's total installed wind capacity amounted to 158.5 GW at the end of 2009, more than 38 GW of which came online in 2009 alone, representing a 41.5% growth rate with respect to the wind capacity that was installed just in 2008 and a year-on-year increase of 31.7%. These 158.5 GW of global wind capacity are expected to save around 200 million tons of CO₂ every year. Furthermore, wind energy is firmly penetrating the world's energy markets, with a market for turbine installations valued at about \$63 billion during 2009. This emerging economic activity is responsible for the creation of many new jobs. The GWEC estimates that over half a million people were working in the wind industry in 2009, a number that is expected to increase up to millions in the near future. The GWEC also forecasts that in 2014 the global wind generating capacity will reach 409 GW, 62.5 GW of which will be added precisely during 2014, compared to 38.3 GW of wind capacity introduced in 2009. The annual growth rates during the period 2010-2014 will average 20.9% in terms of total installed capacity.

The continued expansion of global wind generation capacity is driven by three regions: North America, Europe, and Asia. In the United States, a package of measures have been approved at the state level, from renewable portfolio standards (RPSs) and renewable energy certificates (RECs) [218], to regional greenhouse gas emission control schemes [187]. In the same vein, the US Federal government has put forward new energy and environmental policies and targets, which are likely to substantially increase the use of renewable energy. Indeed, the US Department of Energy (DOE) describes in [163] a model-based scenario in which wind energy provides the 20% of the US electricity demand in 2030, and discusses a collection of technical and economic challenges that need to be overcome for this scenario to come true.

The European Union is currently pursuing the implementation of its ambi-

tious 20/20/20 goals, which intend by 2020 to reduce greenhouse gas emissions by 20%, as compared to 1990, increase the amount of renewable energy to 20% of the energy supply, and reduce the total energy consumption by 20% through energy efficiency [79]. Several European countries, such as Germany, Spain, Portugal, and Denmark, have already attained a high penetration of wind power in the range of 7 to 20% of electricity consumption. The fast growth in installed wind power capacity is expected to continue in both the United States and Europe [80, 81, 156, 163].

Asia is expected to be the region in the world with the highest growth of wind power capacity in the near future. This growth will be mainly led by China, which has been doubling its installed capacity every year for the past few years, and is set to continue the rapid increase of its wind capacity to become the world's largest annual market. In fact, annual additions are expected to reach more than 20 GW in China by 2014 [61]. This spectacular development is strongly spurred on by government policies intended to promote the diversification of electricity supply, encourage the development of the domestic industry, and undertake the transmission network expansion and reinforcement required to get the electricity to the market.

2.2 A Stimulus for Researchers: Challenges

Wind generation is free of emissions and promotes sustainable development. The energy source, the force of the wind, is indigenous and, as such, geopolitically generous, encouraging the self-sufficiency in energy of nations. Moreover, it uses no water. Technically, wind power is fast to deploy, economical and competitive, and generates employment. As wind farms do not consume fuel, they can reduce fuel costs and offer a hedge against fuel price volatility. Wind power plants have low forced outage rates and can contribute to lessen the need for polluting generation sources.

However, wind production cannot be dispatched in a traditional sense due to the inherent randomness of the natural phenomenon involved: the wind. As a result, wind generation is variable and uncertain, and its large-scale integration into a power system is posing and is expected to pose unique challenges

for system operators and planners in almost all the fields of the electric energy systems [17, 18, 66, 67]. But here are the good news: researchers love being challenged.

Actually, the management of uncertainties in a power system is not new for practitioners. The energy demand is also variable and uncertain, just like wind generation, and system agents have been coping with the natural variability and stochasticity of demand since the dawn of the power industry. Then, what is the problem? Where is the challenge?

The answers to these questions are manifold and certainly, not trivial. In the following, we strive to touch on the multiple facets of the large-scale integration of wind power into current power systems.

2.2.1 A Long Way Ahead

The introduction of wind generation into a power system entails the consideration of additional amounts of *uncertainty* and *variability* in the operation of the system. This basic fact becomes of the utmost importance if taking into account the exigent wind penetration marks that many countries worldwide have set out to achieve.

The variability of wind power production requires operating the power system with a higher degree of *flexibility* in order to coordinate the following of the fluctuating net load, which is defined at each time instant as the difference between the total energy consumption and the total wind energy production. This flexibility determines the ability of a power system to integrate wind.

The need for enhancing the flexibility to accommodate wind variability is contingent on several features of a power system, namely:

Wind and demand correlation. If the daily pattern of wind power production coincides with that of demand, wind variability can be *passively* absorbed by consumers, with the consequent reduction in net load fluctuations. Unfortunately, in most cases, demand and wind availability are not correlated. For instance, in Northern Europe, even if the historical records reveal that wind power production is greater in winter than in summer, there is not, however, a significant coincidence of winter periods

with high load and high wind generation [107]. Similarly, in the state of Texas, the wind power production from the West zone is anti-correlated with the energy consumption in the power system operated by the Electric Reliability Council of Texas (ERCOT). In this region, wind tends to blow more in winter, spring, and autumn than in summer, when the energy demand for air-conditioning reaches its maximum; and more during off-peak hours than on-peak.

Wind power dispersion. Correlation between wind speeds at two different locations usually decreases with their distance [151]. Moreover, power output fluctuations from wind farms fed with uncorrelated winds cancel out and as a result, the overall wind generation variability diminishes. Therefore, the geographical spreading of wind farms has a beneficial smoothing effect on wind power variations. Nevertheless, decisions on the siting of wind plants are generally constrained to the availability of wind resources that can be profitably utilized. In the ERCOT system, for example, most of the exploitable wind resources are concentrated in the west zone, far away from the big load pockets.

Transmission network. Wind farms are typically placed in areas where wind resources are plentiful and can meet certain requirements. In addition, these areas are well distant from the large demand centers (e.g., big cities) [162]. Therefore, a network topology enabling wind farms to evacuate their energy production towards the consumption nodes is a determining factor in the integration of wind. Furthermore, owing to the vast spread of intermittent energy sources throughout the transmission and distribution grids, the range of possible power-flow configurations is expected to be enormous [98, 215].

Ancillary services. Power systems rely on a number of services necessary to support the transmission of electric power from producers to consumers by guaranteeing the stability [125, 165], security, and reliability [33] of the electrical infrastructure. Broadly speaking, these services are reserves, regulation and load following, reactive power and voltage control, loss

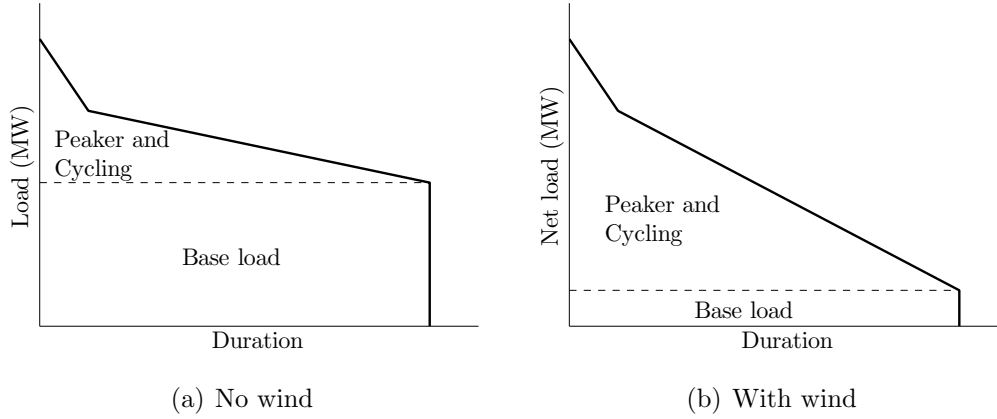


Figure 2.1: Impact of wind integration on the load duration curve

compensation, and system protection [122,205]. The role played by these services in the large-scale integration of wind power should be carefully examined [168], while appraising the impact of the massive use of wind on the stability, security, and reliability of power systems.

Generation mix. Undoubtedly, the ability of a power system to integrate wind generation is strongly dependent on its generation mix. Production technologies able to efficiently alter their power output as required by the net load variations are of necessity to sustain an electricity supply paradigm primarily based on renewable, but intermittent, energy sources. In fact, given the usually unfavorable correlation between wind and demand, not only the need for peak and cycling units is expected to increase substantially, as graphically illustrated in Fig. 2.1, but also the operating regime of these units is likely to change significantly. In this respect, flexibility on the generation-side of a power system usually leads to operate conventional units at productions levels higher or lower than their optimum in an attempt to accommodate the inherent variability of wind generation by ramping up or down. Further, these ramping excursions may often end up with the start-up or shut-down of conventional units.

According to the data provided by Gas Natural Fenosa, one of the most important generation companies in the Electricity Market of the Iberian

Peninsula [3], the number of start-ups of its combined-cycle gas turbines during the first quarter of 2010 increased 145% with respect to the same time period in 2009. However, the daily energy production per unit experienced a 25% decrease. These numbers are the result of a temporary situation characterized by the combination of the remarkable wind power penetration in Spain and the large amounts of power produced by hydro plants due to the abnormally high water inflow recorded in that period. Notwithstanding the circumstantial character of this situation, it could be seen as a preview of the things to come if the committed renewable targets are eventually fulfilled. Consequently, if large variations of wind generation are to be accommodated by the conventional generation, this may result in conventional power plants operating in a less efficient way, thus reducing the quantity of pollutant emissions that are abated by the contribution of wind farms to the electricity supply [69].

The abatement of pollution achieved by integrating wind is also dependent on the type of production and fuel that is being replaced at the moment that wind farms are generating. In theory, this should be the production form in use with the highest marginal cost, which may correspond to old coal-fired plants, which emit about 800-900 gCO₂/kWh. However, during some hours of the year, wind power generation could displace other production forms such as gas-fired plants, with CO₂ emissions around 400-600 gCO₂/kWh, or even CO₂-free generation, e.g. hydro or nuclear power. In [109], Holttinen and Tuhkanen estimate a CO₂ reduction of 700 gCO₂/kWh thanks to wind power production in the Nordic system, which is a mixed system of thermal and hydro production.

There is no doubt then that wind power integration has a major impact on the generation mix of a power system. The proper evaluation of this impact requires accounting for the *capacity credit* of wind generation [18]. The capacity credit is a measure of the amount of conventional generation that can be replaced by the intermittent production without making the system any less reliable. Hence, capacity credits of wind plants do not approach nameplate ratings due to the non-controllability of the

underlying energy source. Nonetheless, it is also true that, since no generating plant is 100% reliable, there is always a finite risk of not having enough capacity available to satisfy the demand, and wind generation may be available at the critical moment when demand is high and many other units may fail. In practice, this means that 1 MW of installed wind capacity is not sufficient to cover 1 MW of demand in a reliable and secure manner, but it is enough to adequately satisfy a certain percentage of it.

If the power system is not flexible enough to integrate wind, this will have to be irremediably curtailed [161]. The occurrence of wind spillage events should be understood as a clear signal that *active* actions on the features above need to be implemented in order to properly accommodate larger amounts of wind power. We provide below a brief description of some of these actions:

Demand-side management. If wind production and demand are not *naturally* correlated and we, human beings, cannot alter weather conditions at our own will yet, how can we encourage loads to move the bulk of their electricity consumption to periods of high wind power production? One way to introduce *demand responsiveness* is to use the real-time pricing (RTP) of electricity [36, 37, 94, 106], according to which retail electric prices change frequently to reflect variations in the energy supply and the cost of serving load. Thus, consumer demand could be made to follow the supply of wind generation by using a price signal. If wind generation is high/low, electricity demand will increase/decrease as a result of low/high electricity prices [201, 202].

The implementation and success of demand-side management programs will be greatly facilitated by the establishment of *smart grids* [10], which, among other things, are conceived to allow consumers to play an important role in optimizing the operation of power systems through intelligent monitoring, control, and communication.

New loads. New *controllable* loads can be added to the power system to make use of the otherwise wasted wind power production. Examples of these loads include space and process heating in the form of heat pumps, electric heating, and electric boilers, among others; fuel production such as

hydrogen via electrolysis or shale oil heating; and the electrification of transportation using electric vehicles or plug-in hybrid vehicles, which is becoming a promising potential large-scale application [42, 123, 135, 136]. In this line, the concept of *vehicle-to-grid* (V2G) refers to a system in which electric or plug-in hybrid vehicles communicate with the power grid to provide demand response services by either injecting or retrieving electricity into or from the network [68, 121, 136]. Battery charging can be timed to periods of greatest generation from renewable intermittent sources, while charging rates can be controlled to provide reserves.

Electricity storage. The scientific community agrees that the large-scale integration of intermittent energy sources could be greatly facilitated by developing large-scale storage capabilities. Even though nowadays there exists a number of energy storage technologies that can contribute to this end (e.g., batteries, flywheels, electric vehicles, superconducting magnetic energy storage systems, compressed air, molten-salt or pumped hydro storage systems) [68, 114, 207, 215], it seems that new advances and more incentives in this field are still needed for the massive electricity storage to become a reality in the required terms [76, 176].

Flexible generation. As previously mentioned, the ability of a power system to integrate wind generation is largely contingent on the capability of its conventional generating units to follow wind power fluctuations. Therefore, both the building of new and more flexible conventional generators and the increase in flexibility of existing ones constitute effective measures to accommodate larger amounts of wind.

Wind technology upgrading. Part of the electricity supply needs to be covered by generators able to provide the ancillary services required to guarantee the secure operation of the power system. If wind farms are not equipped with the appropriate technology to contribute to these services (e.g., voltage control devices), wind energy curtailments become inevitable in situations where the sum of the wind power production and the conventional generation necessary for security reasons exceeds energy

consumption. In some countries, grid companies are obliging producers to install dedicated equipment for voltage and reactive power control regardless of the applied technology [18]. This means that no exception is made for wind power, as has been often the case so far. Moreover, the large-scale penetration of wind generation will necessitate that wind farms remain connected to the transmission network in the aftermath of a fault. In Spain, for instance, *Red Eléctrica de España* (REE) [7], the Independent System Operator of the Spanish electric energy system, is preparing a standard operating procedure to force new installed wind farms to support voltage and frequency control [64]. This regulation comes after the successful application of a previous operating procedure according to which wind farms are obliged to stand voltage dips without disconnecting [63].

Optimal use and expansion of the transmission network. Network bottlenecks may become a major obstacle in the large-scale integration of wind power. Unfortunately, the transmission system expansion required to allow for wind connection at sites far from the traditional load and generation centers usually involves significant lead-time and investment, aside from being exposed to the public reluctance to extensive new-build infrastructure. Consequently, procedures for the optimal allocation of wind capacity are needed to make the most of existing transmission resources, especially, in the short run [45]. Notwithstanding this, transmission expansion projects aimed to reinforce system interconnections can considerably increase the value of wind generation [70]. For example, wind power could be much more easily integrated if spread all over a hypothetical well-interconnected Europe.

Tools for power system operation. At the beginning of this section, we stressed that wind production involves additional amounts of *uncertainty* and *variability* in the operation of a power system. We intentionally made this distinction to motivate the notion of *reserve* and underline its cardinal importance in wind integration. Certainly, wind power is highly variable. Part of this variability can be predicted some hours or days

ahead, and as such can be absorbed by an optimal *energy* dispatch of system agents in accordance with their flexibility. The unpredictable part of this variability is managed, however, with reserves. Basically, a generating unit or load that provides reserve is willing to alter its production as required in a short period of time. Reserves required to accommodate unforeseen wind power fluctuations are fundamentally slow, i.e., with activation times around or greater than 15 minutes [67, 72]. Even so, the deployment of reserves translates into relatively fast energy re-dispatches, which significantly increases the necessity for flexibility of power system components. As a result, wind power uncertainty entails additional operating costs [66, 67, 82, 148], which can be minimized by appropriate *reserve* allocation and deployment. In short, tools for the optimal energy and reserve scheduling are crucial to make best use of wind, avoid unnecessary wind energy curtailments, and reduce the costs derived from its variability, either unpredictable or not.

We have until now omitted the fact that, in most countries, the exponential growth in wind power is taking place within a restructured electricity sector governed by market laws. Electricity transactions are thus agreed in a marketplace to which producers and consumers submit their energy offers and bids in a *competitive fight* for maximizing their benefits or minimizing their procurement costs. A third entity, usually referred to as *market operator* (MO), is in charge of determining the accepted offers and bids, as well as the electricity price. To this end, the MO uses a market-clearing tool, whose aim is to maximize the net social welfare.

The competitive trade of the commodity *electricity* is challenging by itself due to the physical limitations imposed by the power systems and their agents, and the uncertainties affecting them. Market outcomes should be aimed at increasing the economic performance of power systems while guaranteeing a secure operation of the electrical infrastructure. Roughly speaking, when attempting to reconcile the economic efficiency and technical feasibility of electricity transactions, two basic aspects of electricity markets need to be examined, namely:

Market design. Each electricity market in the world is organized differently.

Typically, electricity markets include several trading arenas to facilitate energy commerce between producers and consumers in the long and short run. Trading floors for short-term energy transactions may comprise a day-ahead market, several adjustment markets, and balancing markets. The set of these sequentially arranged short-term trading floors is usually referred to as the *pool*.

In the day-ahead market, as its name suggests, energy transactions to be materialized the next day are negotiated. The adjustment markets enable market agents to modify the energy sold or purchased in the day-ahead market in response to changes in their expected productions or consumptions. Lastly, the balancing markets, also called *real-time markets*, allow *last minute* energy transactions to cope with actual conditions and ensure the real-time balance between generation and demand.

This way, the Electricity Market of the Iberian Peninsula, for instance, includes one day-ahead market, six adjustment markets sequentially arranged over the physical delivery horizon, and hourly balancing markets [3]. In contrast, the Electric Reliability Council of Texas (ERCOT) market does not currently have day-ahead or adjustment markets, but does count on a real-time energy market [13].

By purchasing a certain amount of electricity, the consumer acquires the right of using the same amount of *energy* under *secure* conditions. Therefore, electricity involves a physical product, *energy*, and an intangible asset, *security*. Security is possible thanks to the ancillary services, which, like energy production and consumption, can also be traded and priced in markets [120, 122, 220, 223]. As an example, we underline the markets for *reserve*, which is the ancillary service most affected by wind power uncertainty, and as such plays a key role in its integration.

Market-clearing tools. The outcomes of an electricity market are conditional on the algorithm used by the market operator to clear it. Market-clearing procedures currently in practice are very diverse, varying from single auction processes to complex mechanisms for the simultaneous procurement

of energy and reserve using a formulation based on an optimal power flow. The complexity of a market-clearing tool is, in general, related to the power system characteristics that are accounted for in the algorithm. Thus, procedures for market clearing may take into account, e.g., intertemporal constraints such as the ramping capabilities or the minimum up and down times of generating units. Logically, as the accuracy of the physical system modeling is increased, the economic performance of the electricity market is expected to improve at the cost of losing simplicity and transparency.

The mechanism for pricing electricity is also contingent on the market-clearing tool. Thus, the modeling of the transmission network in the procedure to clear the electricity market naturally leads to the notion of *locational marginal prices*, according to which the electricity price at each node of the system is different as a result of active power losses or network congestion [115, 195, 205]. In contrast, if network constraints are disregarded in the algorithm, a single electricity price is obtained.

The large-scale integration of wind power can be seen as an opportunity to reexamine the design of electricity markets and the tools used to clear them, since large amounts of wind generation are likely to alter their economic performance for the following two reasons at least:

Increased uncertainty. Wind power uncertainty is passed on to both the variables determining the operational functioning of power systems (voltages, line flows, active power losses, etc) [147, 213] and the electricity market outcomes (e.g. electricity prices) [52, 77, 116, 145, 150, 153, 203]. From a technical point of view, this means that power systems are expected to operate over a much wider range of working conditions as the wind power penetration increases. Power system operating point will fluctuate over time as a result of real-time energy re-dispatches in the form of reserve deployment aimed to accommodate the actual realizations of wind generation. Consequently, the growth of wind power is expected to significantly increase the need for reserve and its associated cost [62, 70, 82, 105, 108, 133, 172]. From an economic point of view, elec-

tricity markets should provide mechanisms to allocate and minimize the costs of such energy re-dispatches by competition. This poses the following important questions:

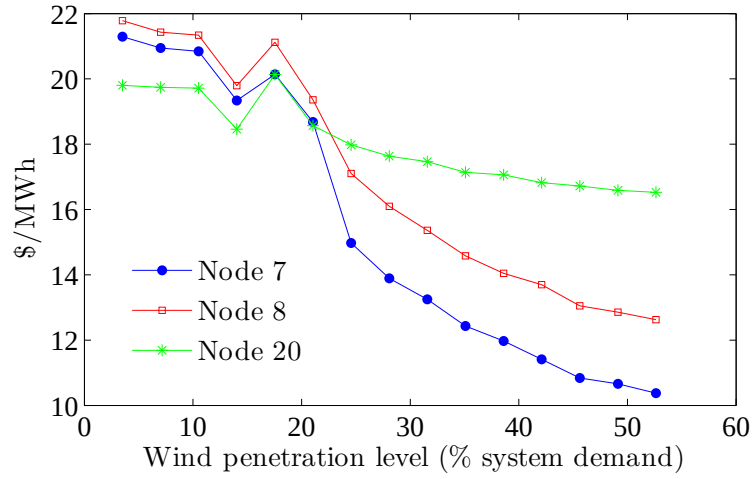
1. On whom the costs of the additional reserves required to integrate wind power should be fallen?
2. How the reserve level and cost increase as the wind power penetration grows?
3. Given that adjustment markets are cleared closer to the actual energy delivery, wind producers can participate in these markets with a reduced level of uncertainty about its actual production (as predictions improve as the forecasting time span decreases). How this type of trading floors can help to minimize the cost of wind uncertainty?
4. Does a day-ahead market, based on anticipation, make sense in highly wind-penetrated electricity markets?
5. Is a real-time market the most suitable settlement mechanism to cope with wind power uncertainty?
6. How market-clearing algorithms can be improved to produce the best energy and reserve scheduling in highly wind-penetrated markets, where the reserve service is not an “ancillary” service anymore, but a primary one?

Zero generation cost (plus subsidies?). Wind power production, while uncertain, is characterized by a zero generation cost, if we neglect operating expenses. Consequently, as wind generation penetrates electricity markets, the electricity price, while becoming more volatile, is expected to decrease in average terms. Fig. 2.2 shows the evolution of the mean and standard deviation of electricity prices at nodes 7, 8 and 20 as the penetration level of wind generation in the 24-bus system described in Appendix C increases [150]. The wind power penetration level is expressed as a percentage of the total system demand. The wind power

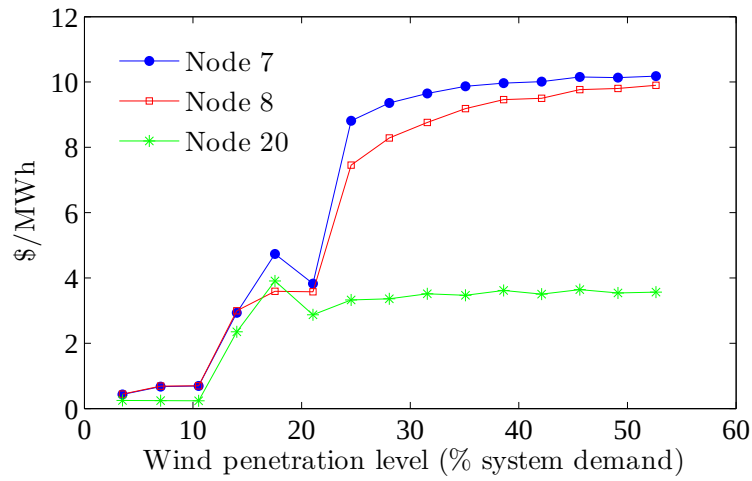
production comes from nodes 7 and 8, where two wind farms are located. Node 20 can be seen as representative of those buses electrically far from the nodes where the wind generation is injected. This evolution has been obtained by simulating an electricity market in which wind producers offer their production at zero prices. The market is cleared using a single-period market-clearing procedure which accounts for network constraints and active power losses, but disregards inter-temporal constraints.

The simulation verifies that, in general, the impact of a growing wind generation on electricity prices translates into a decrease of their means, but an increase of their standard deviations. The sudden changes observed in the evolution of the electricity prices for penetration levels between 10 and 20% are caused by a change in the production technology that is more likely to be replaced by wind generation. The technology that is displaced is not always the one with the highest production cost in the system, since other aspects such as network capacity and technical limits of generating units can decisively influence the energy dispatch and consequently, the locational marginal prices.

For low wind penetration levels ($< 21\%$), the wind energy injected at nodes 7 and 8 is able to *reach* every bus in the system and therefore, the means and standard deviations of electricity prices are all very similar. The existing differences stem from losses. However, from a wind power penetration level higher than 21%, the curves represented in Fig. 2.2 start to diverge in a significant manner. The different behaviors exhibited by the means and standard deviations of locational prices from this point on are mainly due to the fact that situations in which the network becomes congested begin to happen, or statistically speaking, begin to be probable. In such situations, the network caps the amount of wind energy that can be transferred from nodes 7 and 8 to the rest of buses in the system and as a result, the effect of wind generation on electricity prices becomes *locally* accentuated. This effect is twofold: on the one hand, the probability of loads at nodes 7 and 8 being supplied by free



(a) Mean



(b) Standard deviation

Figure 2.2: Impact of wind integration on electricity prices

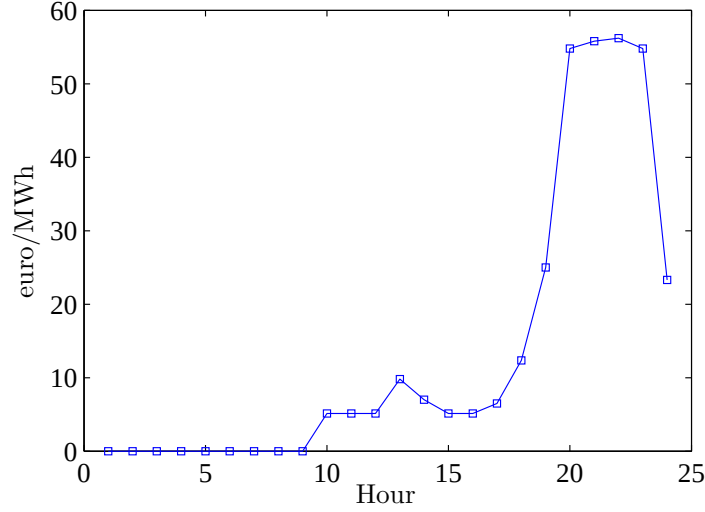


Figure 2.3: Day-ahead electricity prices registered in the Spanish market on March 1, 2010

wind energy increases, which gives rise to zero price situations, and hence the sharp decrease in the means of the corresponding locational prices; on the other hand, the energy supply from wind sources is inherently uncertain and such an uncertainty is transmitted to electricity prices in the form of a considerable increase of their standard deviations.

The results obtained by simulation here are in accordance with the historical series of prices recorded in highly wind-penetrated markets worldwide. As an example, Fig. 2.3 illustrates the 24 hourly day-ahead prices registered in the Electricity Market of the Iberian Peninsula [3, 6] on March 1, 2010. This day, zero prices occurred for hours in which the renewable and hydro production exceeded energy demand.

Furthermore, if wind producers receive an extra payment for each generated MWh as a subsidy outside the market, they may be willing to drop its offer price below zero and consequently, the resulting electricity price may be negative under certain conditions. This is what happened on March 7, 2009 at the West zone of the ERCOT market [13] (see Fig. 2.4), where most of the wind farms in the state of Texas are con-

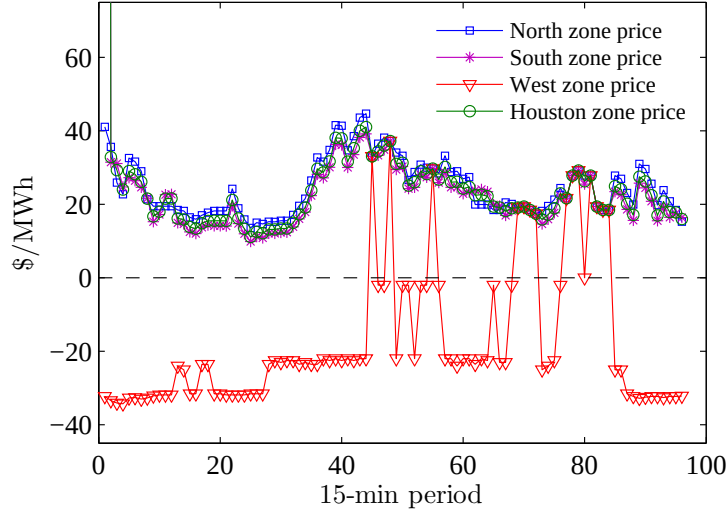


Figure 2.4: ERCOT balancing market prices on March 7, 2009

centrated. This day, the concurrence of offers by wind producers below \$0/MWh and network bottlenecks caused negative electricity prices in that zone.

If situations like the ones represented in Figs. 2.3 and 2.4 start to occur frequently as the level of wind production increases, we should wonder if highly wind-penetrated electricity markets will provide by themselves the adequate economic incentives for electricity producers to invest in conventional generation, especially, in the flexible production technologies required to continue integrating larger amounts of wind power. Perhaps, such incentives can be obtained from the markets for reserve, which probably will become a primary commodity in renewables-based electricity sectors if appropriate mechanisms for reserve cost allocation and pricing are designed [24, 87]. Or perhaps, subsidies are to be considered to encourage flexibility in generation technologies.

Anyhow, there is just one thing we can be sure about: researchers have a long way ahead to turn sustainable development in power systems from words into reality.

2.2.2 The Daily Routine

In this dissertation, from the wide range of challenges and issues arising from the large-scale integration of wind power, we focus on two basic problems that need to be daily faced by the market operator and the wind producers, respectively. Specifically, these short-term problems are the following.

1. A wind producer needs to establish an effective strategy to offer its production in the different short-term trading floors available in an electricity market.
2. The market operator (MO) should use an adequate market-clearing procedure able to efficiently cope with the uncertain nature of wind generation.

Next, we elaborate on these problems and provide some insight into the mathematical tools that we use to solve them in the subsequent chapters of this thesis.

2.2.2.1 The Wind Producer Viewpoint

When it comes to the energy-selling business, three basic aspects differentiate the conventional producer (nuclear and thermal producer) and the wind power producer, namely:

1. Wind power constitutes a non-polluting energy source, and as such, its development is spurred on by governments willing to face the issue of global warming. Economically speaking, this usually translates into subsidies that increase the value of the MWh produced by a wind farm.
2. Wind energy production does not require fuel consumption, and as a result, its associated fuel cost is zero.
3. Wind power is non-dispatchable and plagued by the major uncertainty that constitutes wind availability. Decisions on the amount of wind energy to be traded are therefore risky.

While the features indicated in points 1 and 2 above clearly boost the integration of wind generation in power systems, the characteristic expressed in point 3, the wind availability uncertainty, constitutes the major obstacle to the natural incorporation of wind producers into a market framework in both technical and economic terms. How can a wind power producer *stay afloat* by itself in a competitive environment so far dominated by conventional producers?

System energy imbalances caused by non-dispatchable agents, such as wind producers, are generally fixed by flexible generating units using a real-time or balancing market. The energy required to restore the system balance is extracted from reserves. In this thesis, we consider that the cost of this energy is assumed by the market agents incurring energy deviations, that is, wind producers among others. Since flexible generation is normally expensive, the competitiveness of wind producers that often need to acquire balancing energy in the real-time market may considerably diminish. In plain words, the *invisible hand* governing market laws might bring about the downfall of the less sharp wind producers.

Roughly speaking, in response to this situation, three different attitudes can be adopted by regulators:

1. Wind generation is managed through the electricity market (as a negative demand), but wind power producers are paid a regulated tariff for their actual energy production. This way, wind producers are released from the risk stemming from wind availability at the expense of losing the opportunity of making a higher profit through a shrewd market participation.
2. Wind producers compete in the different trading floors available in the electricity market, including the balancing market, in which they must face the cost of their energy deviations. For each sold MWh, they are paid the price resulting from the market-clearing process plus a subsidy intended to strengthen the competitiveness of wind producers in the electricity market.
3. According to a pure competitive rationale, wind producers must bear the

burden of the market as any other market participant. Consequently, this situation is achieved by eliminating the subsidies in the previous point.

Within a decision-making framework, situation 3 is the one of higher interest, and as such, its analysis and modeling from the wind producer's viewpoint constitutes the *leitmotiv* of Chapter 4 in this thesis. In order for a wind power producer to *survive* in such a strictly competitive panorama, three main lines of action have been proposed in the technical literature. One of them is based on the combined and coordinated use of wind power and energy storage technologies (pumped-storage facilities, compressed air facilities, etc), [20,21,30,32,50,51,60,89,124]. However, plants including storage facilities have no clear incentive to associate with wind farms to build a joint offering strategy instead of providing balancing energy in real-time markets. Another relevant area of possibilities is the use of financial options as a tool for wind producers to hedge against wind generation uncertainty, [99]. Nevertheless, no clear financial instruments are available to hedge against the risk of production unavailability. The third line of action focuses on designing stochastic models intended to produce optimal offering strategies for a wind producer participating in an electricity market on its own, [31,141,178].

Chapter 4 of this thesis explores this last area of research and, in particular, introduces advances in the line of investigation initiated in [141], where stochastic programming is used to generate optimal offers for trading wind generation in short-term energy markets. These advances have to do with improvements in the mathematical formulation of the problem, the consideration of risk management, and the analysis of the benefits of including an adjustment market within the market organization from a wind producer perspective.

2.2.2.2 The MO Perspective

In order to accommodate the unpredictable nature of wind power, the energy dispatch settled in the electricity market needs to be modified during the real-time operation of the power system. The reserve is the service traded in the market to materialize physically the required adjustments.

Therefore, the efficient integration of wind power into power systems calls

for markets decisions relying on economic criteria while maintaining or even improving the reliability of the electrical infrastructure by means of a proper reserve management program. In this respect, there are several issues related to reserves that need to be assessed [17]:

1. Scheduling and allocation of reserves: how much capacity has to be allocated for each hour. The system operator must ensure that an adequate amount of reserve is kept in power plants in order to face the unpredictable variability of wind generation.
2. Deployment or use of reserves: how the scheduled reserves are utilized in real time as the wind uncertainty is revealed.

At this point, we find ourselves having to make it clear that we consider the *market operator* as a non-profit entity that is responsible for the correct functioning and the clearing of electricity markets, including the markets for reserves. In some market structures, like those in the US, these tasks are carried out by the so-called *independent system operator*.

Decisions on the production and consumption levels of generating units and loads, and on the amount and allocation of reserves required to deal with power system uncertainties can be made either jointly in a co-optimized energy-reserve market [19, 87, 185, 205], or independently in separate and sequential markets, one for energy and one or several for reserves [198, 200]. However, in power systems affected by a high penetration of wind power, energy and reserve dispatches becomes strongly interrelated. Consequently, the sequential scheduling, although based on simpler market-clearing algorithms, can lead to important inefficiencies.

The technical literature offers a number of methodologies to set the amount of operating reserves required in a power system to cope with wind power uncertainty. Specifically, in [72, 108, 166, 167, 204] different procedures are provided to compute off-line the reserve needs. In the pioneering work by Söder [204], the margins of instantaneous, fast, and slow reserves are estimated for each hour according to possible forced outages of thermal units and uncertain load and wind speed forecasts. As the paper was published many

years ago (1993), the most recent advances in probabilistic methods and market development are not reflected. In the paper by Doherty and O'Malley [72], generator outages rates and load and wind power forecasts are also accounted for when quantifying the amount of reserve needed to attain a predefined reliability target. However, the proper specification of this target constitutes a problem by itself, and may lead to economically suboptimal solutions considering the cost and benefit associated with reserves. Holttinen [108] presents an approach for setting reserve requirements of a power system that takes wind power forecasts into account as well, and applies it to appraise the impact of wind production variations on the system operation in the Nordic countries. The result is that the hourly load-following reserve requirements grow as the wind power penetration increases. Ortega-Vazquez and Kirschen [167] propose a procedure based on Monte Carlo simulation to compute the optimal amount of spinning reserve requirements using the cost-benefit analysis method that they describe in a previous paper [166]. The Monte Carlo technique is used to simulate wind and load forecasting errors, and the reserve requirements are set by solving an optimization problem whose objective function is to minimize the expected total cost of reserves.

The works in [118, 142, 170] propose on-line decision-making tools for the determination of optimal reserves. Karapidakis and Hatziaargyriou [118] describe a methodology for reserve determination that is based on decision trees built for a set of selected disturbances (including those pertaining to wind power uncertainty). Reference [170] constitutes an improved version of [118] where the costs of energy and primary spinning reserves are simultaneously minimized. Finally, Matos and Bessa [142] design a tool to support the system operator in the on-line definition of operating reserve needs for the daily and adjustment markets, and discuss decision strategies such as guaranteeing an acceptable risk level or resolving a trade-off between economic issues and the risk of loss of load.

However, none of the previous references provide insight into how wind power integration affects the unit commitment problem, which constitutes the centerpiece of market clearing in many electricity markets, particularly in the US. Further, numerous studies reveal that high penetration of wind power has

a remarkable impact on unit commitment, which, in turn, motivates the co-optimized trading of energy and reserve. The work by Schlueter et al. [194] is one of the first papers that draws attention to the costs and benefits of the changes caused by wind generation intermittency in unit commitment. Tuohy et al. [211] conclude that accounting for the stochastic nature of wind and load in the unit commitment and dispatch of power systems with high levels of wind power can considerably reduce the cost of wind integration and produce better-performing schedules than a traditional deterministic approach. The analysis is conducted using the WILMAR software [16, 28], which is basically a planning model and consequently, neglects real-time operational issues such as network bottlenecks. References [212], [65], and [137] make use of unit commitment models to examine, respectively, the benefits of a large share of combined heat and power units, improved wind power forecasts and increased ramping capabilities to mitigate the costs associated with wind integration. All these works, either explicit or implicitly, show that the uncertainty and variability of wind power require the revision of current security-constrained unit commitment algorithms to perform secure-economic short-term market-based scheduling of generation, load, and reserves.

In this line, Wang et al. [216] propose a stochastic formulation to manage the intermittency and variability of wind power generation in the unit commitment problem. The uncertainty in wind power output is modeled via scenarios. The resulting large-scale mixed-integer problem is tackled by Bender's decomposition technique. Thus, the unit commitment is solved for the predicted wind power generation in the master problem. Then, a collection of scenarios representing wind power variability is generated. In each sub-problem, generation re-dispatch is performed to accommodate the simulated wind power realization. If the re-dispatch leads to constraint violations, then Benders cuts are added to the master problem and the commitment solution is revised accordingly. The iterative process ends when the simulated wind power scenarios can be *absorbed* by re-dispatching. The algorithm does not allow for load curtailment or wind spillage, and does not consider the possibility of scheduling reserve, in addition to that required for contingencies, to specifically cope with wind generation volatility. Bouffard and Galiana [38] develop

a stochastic security-constrained market-clearing procedure that accounts for load shedding and wind spillage actions and in which wind generation uncertainty is managed by the optimal allocation and deployment of reserves. The procedure is based on the concept of the *error-free scenario*. This scenario, built by analogy with the pre-contingency state in contingency analysis, is defined as the realization which perfectly follows the forecast net load (the difference between the predicted demand and wind generation). Thus, as in [216], the outputs resulting from the market-clearing procedure are the decision variables corresponding to this particular scenario, which does not give up being, all in all, a specific realization of the uncertain variables (load and wind production). For this reason, we argue that the explicit tie between this error-free scenario and the market results constitutes a deterministic trace which limits an optimal management of the uncertainty associated with wind. As the deviations of wind power around the forecast values may be high in power systems with a high penetration of wind power, the rationale behind the use of a specific realization of wind production when clearing the market, or making unit commitment decisions, undermines the value of the stochastic scheduling.

Chapter 5 of this thesis provides a stochastic security-constrained market-clearing procedure in which reserve requirements are driven by the tradeoff between the value of lost load and the energy and reserve offers submitted by market participants. This procedure averts the need for binding the market outputs to a specific realization of wind power production, embeds the advantages of the co-optimized energy and reserve markets and makes use of the modeling capability of the stochastic programming [35]. The procedure is then formulated as a two-stage stochastic programming problem, where the first stage represents the electricity market, its constraints and rules, and the second stage represents the power system, its operation and physical limitations. Thus, the variables pertaining to the first stage correspond to the market decisions, which consider the realizations of uncertain parameters (wind production and equipment failures) without being explicitly linked to one specific realization, but to all of them.

Lastly, we should point out that schemes for pricing energy and reserve in the context of security-constrained electricity markets are described in [24, 87,

219].

2.3 Battling with the Uncertain Character of Wind Power: Stochastic Programming

Wind farms cannot be dispatched using traditional methods because of the volatility related to their final wind power production. As a result, the uncertainty caused by the connection of multiple wind farms to the grid can decrease the efficiency of the system operation. As formerly discussed, wind power uncertainty poses unique challenges for system operators and planners, which should take preventive and corrective actions (e.g., scheduling and deploying additional reserves) in order to maintain system security and reliability, potentially threatened by the unpredictable and variable nature of wind resources. Likewise, in a liberalized electricity sector, wind producers should gather as much accurate information as possible on their future wind power production to make informed decisions concerning electricity trading and risk management, thus enhancing the market position of wind energy in comparison with other forms of energy.

Broadly speaking, there exist two different courses of action to mitigate the costs of wind uncertainty, namely:

Improving methods for wind power forecasting. There are two different schools of thought with regard to short-term wind power prediction: the physical and the statistical approach. Some models are built on a combination of both in order to join the advantages of both approaches, thus improving the forecasts. The physical methods focus on the description of the wind flow around and inside the wind farm to subsequently provide an estimation of the wind power output by using the manufacturer's power curve. The statistical methods seek to exploit the relationship between meteorological predictions, historical measurements, and generation output using statistical models whose parameters are estimated from data without taking into account any physical phenomenon. The state of the art of wind power forecasting models can be found in numerous publica-

tions, such as [59, 91, 128, 129, 146, 221]. Even if future improvements on current techniques for wind power forecasting can lead to significant reductions in wind integration costs, decisions under wind uncertainty need to be made by system operators and wind producers in the meantime. Further, wind will be always uncertain by nature.

Optimal decision making under uncertainty. No perfect forecasting methodologies exist. That's life. However, we can inspect and analyze the statistical properties of the forecast errors [91, 146] and use this information to make *better* decisions. Here it is where we can take full advantage of *stochastic programming*, which constitutes an appropriate modeling framework in which problems of decision making under uncertainty are adequately formulated, [35, 117]. Stochastic programming relies on the knowledge of the probability distributions of the uncertain parameters involved in the underlying optimization problem, e.g., wind power production or electricity prices. For tractability, each uncertain parameter is usually modeled as a set of finite outcomes or scenarios, where each outcome represents a plausible realization of the uncertain parameter with an associated weight or probability of occurrence. *Good* sets of scenarios should reflect the statistical properties of the corresponding uncertain parameters. For instance, scenario sets for wind generation should retain the spatial correlations existing among the power outputs of different wind farms, as these correlations may have a significant impact on multiple facets of wind integration [18, 147, 150, 214, 217]. To this end, several scenario generation methodologies and forecasting models for wind speed and power have been proposed in the technical literature [144, 151, 171, 209, 222]. Specifically, Miranda and Dunn [144] introduce a spatially correlated wind speed model using a multivariate time-series approach, which models the correlation between various sites in the UK. Xie and Billinton [222] proposes a model for power system reliability assessment that preserves some of the statistical characteristics of the wind speed distributions at two different locations, namely, their means and standard deviations, and allows simulating a given contem-

poraneous wind speed cross-correlation between these locations through a time-shifting technique. The optimal shifting time at the two sites is determined using a linear interpolation algorithm. In [171], stochastic dependence is modeled by means of special functions, named *copulae*, once random variables have been transformed to the rank/uniform domain. Tastu et al. [209] propose nonlinear models to capture the interdependence structure of wind power forecast errors and demonstrate its existence considering the case study of western Denmark. Morales et al. [151] design a scenario generation technique able to preserve the main statistical properties (marginal distributions, autocorrelations and cross-correlations) of wind speeds at different sites.

The number of scenarios required to properly represent a given uncertain parameter is usually very large, which may render the underlying optimization problem intractable. To regain tractability, scenario-reduction techniques [74, 100, 152] are used to decrease the cardinality of the scenario sets while retaining as much as possible the statistical properties of the uncertain parameters.

Chapter 3 of this dissertation provides further insight into scenario generation and reduction techniques, with particular emphasis on wind power and electricity price modeling.

2.4 A Preview to Whet the Reader's Appetite

In this section, we give the highlights of what the reader will find in the following chapters of this dissertation through two motivating and illustrative examples.

2.4.1 Wind Producer: Offering Strategy

Let us consider a wind power producer willing to participate in a pool-based electricity market with the structure and time framework represented in Fig. 2.5. Market A is cleared one hour prior to the energy delivery period, while Market B closes right before it starts. In the balancing market, the energy

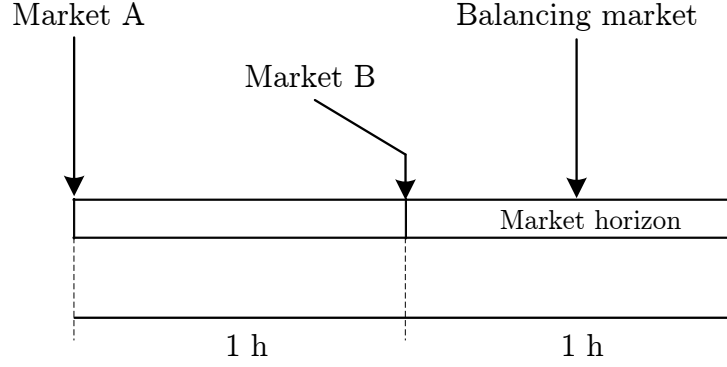


Figure 2.5: Market structure and time framework

deviations incurred by market agents during the actual delivery period are computed, with respect to the dispatch program agreed in markets A and B, and priced accordingly.

The sequence of stages and decisions that the wind producer has to face is as follows:

1. Decide the amount of energy to be sold in market A with inaccurate information about prices in markets A and B, prices in the balancing market, and wind power production.
2. Decide the amount of energy to be sold/bought in/from market B. In this stage, the price in market A is known as well as the wind power value materialized in between the closures of markets A and B. This value constitutes information of utmost importance for the wind producer to improve and update the wind production forecasts spanning the market horizon.
3. Finally, once the actual wind power production is known, the wind producer must cover the amount of energy deviated from that scheduled in markets A and B by selling or purchasing its generation surplus or shortage, respectively, in the balancing market.

Therefore, the offering strategy problem of the wind power producer is subject to four sources of uncertainty, namely, the price in market A, the price

in market B, the price in the balancing market, and the wind availability. Examples of scenario generation techniques for electricity prices, balancing market prices, and wind speed are provided in [46, 56, 180], [164] and [151], respectively.

As the wind power producer needs to rely more and more on the balancing market to fix its energy production deviations, its profit drops rapidly. This is so because generation shortages are often covered in the balancing market by expensive, but easily dispatchable, energy production technologies. Likewise, selling prices for generation surpluses in the balancing market are normally lower than prices in previous markets (e.g., markets A and B in Fig. 2.5), because the leftover energy needs to be repurchased by producers owning flexible units and willing to reduce quickly the production levels that were settled in preceding markets. Consequently, the aim of the wind producer should be to maximize its expected profit from trading in markets A and B, while minimizing as much as possible the need for resorting to the balancing market to amend its energy deviations.

In qualitative terms, the offering strategy problem of a wind power producer can be stated as follows:

Maximize Expected profit + β Risk measure
subject to

- Capacity limit of the wind farm.
- Non-anticipativity constraints imposed on:
 - Decisions made in market A.
 - Decisions made in market B.

The risk measure allows controlling profit variability through the weighting factor β . As this scalar increases ($\beta > 0$), the wind producer becomes more risk averse prevailing the minimization of profit volatility over the maximization of the expected profit. Several risk measures have been proposed in the technical literature, e.g., the standard deviation, the Value-at-Risk (VaR), and the Conditional Value-at-Risk (CVaR), [25, 75, 174, 175, 188–190, 210].

In stochastic programming, non-anticipativity constraints guarantee that decisions in each stage are made according to the information on stochastic processes available until that stage. In this specific problem, these constraints ensure that the decision on the amount of energy to be sold in market A is made with uncertain market prices and wind power production, whereas the decision on the amount of energy to be traded in market B is made with known price in market A, known wind power produced in between markets, but uncertain prices in market B and in the balancing market, and uncertain wind power production during the market horizon.

Basically, the key factors impacting in a particular manner the short-term trading of wind energy are:

1. *Uncertain production.* Wind generation in a future time is indeterminate and decisions on the amount of wind energy to be traded are risky in consequence.
2. *Mechanism for imbalance price formation*, i.e., the mechanism through which energy deviations incurred by market agents are priced in the balancing market. For simplicity, in this illustrative example, we assume that the price of the energy required to cover generation shortages is 20% higher than the price in market A. Similarly, generation surpluses are sold in the balancing market at a price 50% smaller than the price in market A. More complex mechanisms for imbalance price formation, implemented in different European electricity markets, are described in [141, 149, 164] and in Chapter 4 of this thesis.
3. *Certainty gain effect*, defined as the beneficial impact of having market B following the clearing of market A, and thus, closer to the real-time operation. This impact is related to the value of updating wind forecasts [29, 154].

Let us suppose that:

1. The price in market A (λ^A) is equal to \$50/MWh with a probability of 0.40 and equal to \$20/MWh with a probability of 0.60.

2. The price in market B (λ^B) is identical to the day-ahead market price.
3. The quantity (P_1^W) of power produced by a certain 100-MW wind farm during the hour between the closures of markets A and B can be *high* (60 MW) with a probability of 0.4 or *low* (30 MW) with a probability of 0.6.
4. If the wind power production P_1^W is *high* (60 MW), the amount of power produced during the market horizon (P_2^W) can be *extremely high* (100 MW) with a probability of 0.75 or *relatively high* (50 MW) with a probability of 0.25.
5. If the wind power production P_1^W is *low* (30 MW), the amount of power produced during the market horizon (P_2^W) can be *extremely low* (0 MW) or *relatively low* (40 MW), both with a probability of 0.5.

The probabilistic information provided above can be arranged in a tree, as depicted in Fig. 2.6, where π_ω , $\omega = 1, \dots, 8$, is the probability of occurrence of scenario $\omega = (\lambda_\omega^A, P_{1\omega}^W, \lambda_\omega^B, P_{2\omega}^W)$. The scenario tree is made up of three sets of nodes corresponding to the three stages of the wind producer decision-making process.

First, let us suppose that the wind producer ignores the existence of market B. In this case, the wind producer offers 100 MW in market A, which yields an expected profit of \$1164.8. However, according to the scenario tree represented in Fig. 2.6, the maximum expected profit that the wind producer could make if it were able to *anticipate* its future production would be \$1504. Therefore, the uncertain nature of the wind power production entails an opportunity cost of \$339.2.

Fig. 2.7 illustrates the evolution of the power offer submitted to market A as the wind producer becomes more risk-averse, i.e., as the parameter β increases. Note that the amount of power sold in this market abruptly decreases for certain values of β for which the wind producer becomes more sensitive to risk. The reason for this is that selling in the market A is riskier than selling in the balancing market. As an example, if the wind producer sells 100 MW in market A at \$50/MWh, but it eventually produces nothing (scenario $\omega = 4$

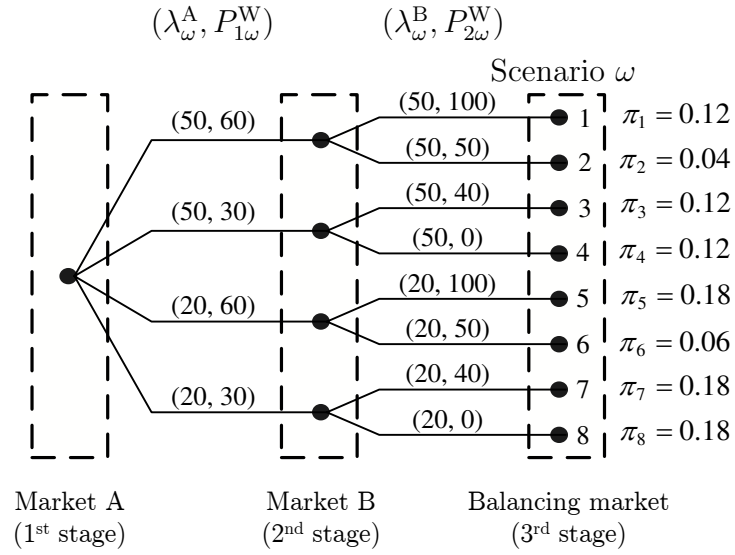


Figure 2.6: Scenario tree

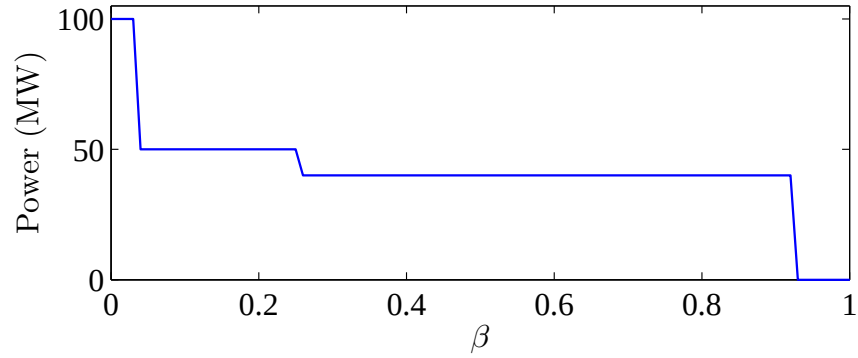


Figure 2.7: Power sold in market A as a function of the risk aversion of the wind producer

in the tree), its profit will be $-\$1000$, because it has to cover its generation shortage in the balancing market at a price 20% higher ($\$60/\text{MWh}$). However, if the wind producer decides to wait and sell its actual production in the balancing market, its profit will be $\$0$ in the same situation. In contrast, selling in market A is potentially more profitable than selling in the balancing market. For instance, if the resulting price in market A is $\$50/\text{MWh}$ and the eventual wind production is 100 MW (scenario $\omega = 1$), the profit obtained by the wind producer will be $\$5000$ or $\$2500$ depending on whether those 100 MW are sold in market A (at a price equal to $\$50/\text{MWh}$) or in the balancing market (at a price 50% lower), respectively.

To conclude this example, we assume next that the wind producer decides to make use of market B. In this case, the best strategy that the wind producer can adopt is to transfer its whole trading activity from market A to market B, because both markets offer the same energy prices, but the future wind power production is known better in market B. In this respect, note that, from market A, the wind producer foresees a future power production ranging from 0 to 100 MW. In contrast, from market B, it forecasts a power production varying either from 50 to 100 MW or from 0 to 40 MW depending on whether the wind power produced in between the closures of markets A and B, $P_{1\omega}^W$, is *high* (60 MW) or *low* (30 MW), respectively. This way, if $P_{1\omega}^W$ is *high*, the wind power producer sells 100 MW in market B, while if it is *low*, its energy offer in this market is reduced to 40 MW. The expected profit associated with this strategy offer is $\$1395.2$, which means an increase of 19.78% with respect to the expected profit obtained if market B is disregarded.

In reality, energy prices in markets A and B are expected to be different. Indeed, think of markets A and B as the day-ahead and adjustment markets, respectively. Given that the trading volume in adjustment markets is usually lower than in the day-ahead market, adjustment market prices are generally more volatile. As a result, the wind producer must face a trade-off between selling in the day-ahead market at less volatile prices and selling in the adjustment market with a reduced level of uncertainty about its future wind power production. This trade-off can be resolved by means of the stochastic-approach-based tool illustrated in this example and comprehensively examined

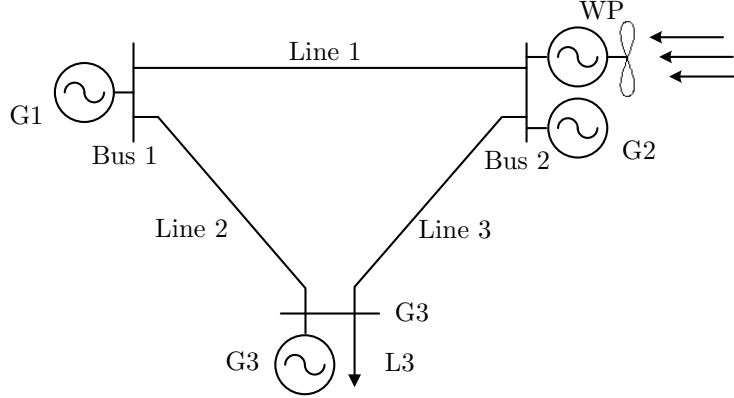


Figure 2.8: Three-node system

in [149] and in Chapter 4 of this thesis.

2.4.2 ISO/MO: Market Clearing and Reserve Valuation

Consider the three-node system shown in Fig. 2.8. This small system includes three thermal units, one inelastic load at node 3, and a 50-MW wind farm placed at node 2. For a given time period in the future, the system operator must determine *here and now* both the energy and reserve needs. Naturally, reserves are required to cope with the uncertain wind power production, which is represented via three scenarios: *high* (50 MW), *average* (25 MW), and *low* (0 MW), with probabilities 0.2, 0.6, and 0.2, respectively.

Specifically, the sequence of stages and decisions that the system operator has to face is as follows:

1. Determine the production levels of thermal units, the quantity and allocation of reserves required to deal with the uncertain wind power production, and the prices at which both services, energy and reserve, need to be paid. In other words, the system operator must clear the market(s) organized for energy and reserve transactions.
2. Deploy reserves in the real-time operation of the power system to accommodate the actual value of the wind power production.

If pursuing a co-optimized energy-reserve dispatch, the market operator needs to solve a market-clearing problem that can be schematically formulated as follows:

Minimize Expected cost + α spilled wind

subject to

- Electricity-market constraints:
 - Market equilibrium.
 - Scheduled reserve determination constraints.
 - Offer and bid limits.
- Real-time operating constraints:
 - Power balances.
 - Capacity limits of transmission lines.
 - Technical limits of production units.
- Linking constraints:
 - Deployed reserve determination constraints.

The objective function includes two terms. The first one, referred to as *expected cost*, comprises both the costs pertaining to the functioning of the electricity market (energy and reserve scheduling) and the costs incurred during the real-time operation of the power system (reserve deployment and involuntary load shedding). These costs are computed from the energy and reserve offers submitted by market agents to the electricity market. In this illustrative example, we assume that each thermal unit offers a single block of energy and reserve at prices λ_E and λ_R , respectively. The value of these prices are shown in Table 2.1 together with the minimum ($P^{G,\min}$) and maximum ($P^{G,\max}$) power outputs of the generating units. The second term in the objective function accounts for the subjective value of the wind production that is curtailed, and it allows the system operator to promote wind energy usage through the weighting factor α , which can be understood as the *cost* of wind spillage.

Table 2.1: Generator data

Unit	1	2	3
$P^{\text{G}, \min}$ (MW)	10	10	10
$P^{\text{G}, \max}$ (MW)	100	100	50
λ_{E} (\$/MWh)	30	40	20
λ_{R} (\$/MWh)	9	12	15

As for the constraints, three different groups can be distinguished: equations modeling the market rules (electricity-market constraints), equations modeling the real-time operation of the power system (real-time operating constraints), and finally, the linking constraints, which couple market outcomes with the real-time system operation through the deployment of reserves.

Let us consider a level of demand equal to 120 MW. Fig. 2.9 shows the breakdown of the expected cost (excluding the wind spillage term of the objective function) for five different cases, namely:

- a) The future wind power production is perfectly known and it coincides with the *average* scenario (25 MW).
- b) All the wind power produced is *spilled* at no cost, and consequently, load at node 3 is exclusively supplied by thermal units.
- c) Wind generation is uncertain in keeping with the three-scenario representation indicated above.
- d) Wind generation is uncertain as in c) and line 2-3 is congested.
- e) Wind generation is uncertain as in c) and the wind spillage cost (α) is equal to \$100/MWh.

By comparing case a) or c) with case b), it becomes clear that wind generation leads to a significant reduction in the costs of energy production, since an important portion of the demand, which otherwise would be covered by expensive thermal generation, is satisfied with renewable and free energy instead. However, the expected cost in case a) is slightly smaller than in case c). The difference corresponds mainly to the cost of reserves required to cope with

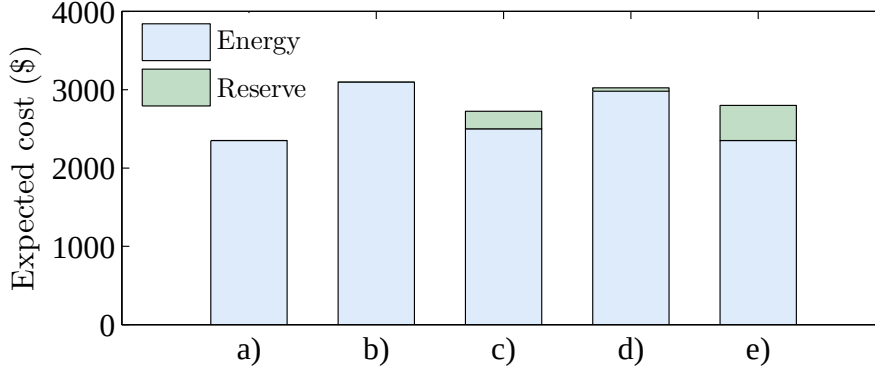


Figure 2.9: Breakdown of the expected cost: a) no wind generation uncertainty; b) no wind generation; c) uncertain wind generation; d) network congestion; e) $\alpha = \$100/\text{MWh}$

the wind production uncertainty. Case d) reveals the key role played by the network in the large-scale integration of wind generation into power systems. If, for instance, the capacity of line 2-3 is not high enough to completely evacuate the likely wind power produced at node 2, a part of this production is expected to be wasted and the economic performance of the system worsens. Finally, case e) illustrates how a positive value of parameter α (wind spillage cost) can be used to increase the wind energy usage by enhancing the flexibility of the system through the scheduling of additional amounts of reserve. As a result, costs pertaining to the energy production diminishes at the expense of increasing the cost of reserves.

The tool to clear highly wind-penetrated electricity markets that has been outlined in this section is comprehensively analyzed in [148] and in Chapter 5 of this thesis.

2.5 Summary

The progress of wind energy around the world in recent years has been impressive, and the policies intended to curb emissions of greenhouse gases causing global warming are expected to guarantee the steady growth of wind power generation in the electricity sector, or even speed it up.

In this chapter, we review some of the most important challenges derived from the large-scale integration of wind power into power systems and put forward the need to replace old-fashioned conceptions and methods to succeed in the attempt.

We also introduce the wind-related issues that are dealt with in this dissertation, namely, the short-term trading problem of a wind power producer and the clearing of highly wind-penetrated electricity markets.

Personajes
(Verdad *or* just thanks)

Como mosaico de colores,
tejen ellos mis versos,
los narran a fuego sobre mi piel,
y avivan con su aliento
la razón primera de mi ser.

Los llevo todos:
Tonos vivos, tonos oscuros,
alegres y divertidos,
también los más tristes y amargos.

Aparecen como única verdad en mis caminos.
Son carcajadas, risas y llantos,
son cobijo, refugio y amparo.
Son amigos, compañeros y hermanos.
También rostros desconocidos
y palabras de desencanto.

Todos anudan mis trozos,
colorean mis sentidos,
llenan de brío mis pasos,
y por ello, por cuanto vivido
he aquí en mi soberana simpleza
que os brindo un guiño agradecido.

Copenhague, 5 de julio de 2010

A stylized, handwritten signature in black ink, featuring a large, looping 'Z' or 'S' shape with a horizontal line extending to the right.

Chapter 3

Uncertainty Characterization in Electricity Markets with a High Wind Power Penetration

3.1 Introduction

Numerous problems in engineering and science are exposed to uncertainty due to the inherent randomness of the natural phenomena involved and/or to the erratic evolution of the variables determining the functional state of human-created structures. Within such a hazardous world, computational methods capable of dealing with imperfect information enable engineers and scientists to propose solutions less sensitive to environmental conditions and uncontrollable parameters, while simultaneously attaining cost reduction, profit gains, and reliability improvement.

Decision-making problems related to electricity markets are no exception, but quite the opposite. These problems are plagued with unknown data in practically all their facets, if not in all. In fact, the own mechanisms governing the functioning of these markets can be considered responsible for the existence of uncertainties influencing market agents' behavior. As an example, electricity prices are unknown when producers and consumers have to submit their selling offers and purchasing bids, respectively, to the pool. Consequently, decisions

on the amount and price of the energy to be sold or purchased are irremediably made with inaccurate knowledge of the final market outcome. Likewise, the time gap existing between agreements on energy transactions and their physical implementation causes that a producer must face the trading process with a certain degree of uncertainty as for the availability of its power sources. This is an issue of particular significance for wind power producers, whose production is conditioned to the uncertain nature of wind.

Therefore, most decisions need to be made with incomplete information in practice. Fortunately, over the past few decades, outstanding endeavors in the field of operations research and management science have brought forth the rapid development of mathematical tools able to efficiently tackle decision-making problems under uncertainty. This is the case of *stochastic programming*, [35, 117, 181, 192]. According to *The Stochastic Programming Community* [12], stochastic programming can be defined as “a framework for modeling optimization problems that involve uncertainty”, i.e., as the set of modeling tools aimed to find optimal decisions in problems including uncertain data. Appendix A provides a brief introduction of those stochastic programming concepts on which the mathematical models presented throughout this dissertation are built.

Prior to undertaking the solution of a stochastic programming problem, uncertainties on the problem parameters need to be first characterized. Roughly speaking, the characterization of an uncertain parameter consists in identifying and isolating the part of its nature that can be explained in a deterministic sense from its purely random component. In this chapter, we provide algorithms especially tailored to characterize the uncertain data involved in the optimization models proposed in this thesis, namely, market prices and wind power production. These algorithms make use of well-known statistical analysis tools developed from the notion of *stochastic process*.

3.2 Stochastic Processes

Within a modeling framework based on stochastic programming, input data affected by uncertainty are conceptually described as stochastic processes, [41,

43, 54, 173]. A stochastic process λ is defined as a collection of dependent random variables $\lambda = \{\lambda_t, t \in T\}$. That is, for each t in the index set T , λ_t is a random variable. We often interpret t as time and call λ_t the state of the process at time t . Therefore, a stochastic process can be intuitively seen as a random variable whose value evolves over time. Instances of stochastic processes in the field of electric energy systems are countless. For example, the monthly energy demand consumed at a given node of the transmission grid over a year can be mathematically studied as a stochastic process. Likewise, an instance of a stochastic process closely related to electricity markets is the set of the 24 hourly electricity prices of a day.

Additionally, a stochastic process λ is said to be *continuous* or *discrete* depending on whether its component random variables $\{\lambda_t, t \in T\}$ are continuous or discrete, respectively. If we denote the wind speed in hour t at a given site A by v_t^A , then the set of random variables $V^A = \{v_t^A, t = 1, 2, \dots, 24\}$, describing wind speed at this site throughout a day, is an example of a continuous stochastic process. In contrast, if binary variable u_{it} is used to indicate the availability of a generating unit i in hour t (being equal to 1 if it is functional in that hour and 0 otherwise), the collection of random variables $U_i = \{u_{it}, t = 1, 2, \dots, 168\}$, representing the availability of unit i throughout a week, is a case of a discrete stochastic process.

A discrete stochastic process can be represented by a finite set of real vectors, referred to as *scenarios*, resulting from the combinations of all the discrete values that its component random variables can adopt. In mathematical terms, if λ is a discrete stochastic process, it can be expressed as $\lambda = \{\lambda_\omega, \omega = 1, 2, \dots, N_\Omega\}$, where ω is the scenario index and N_Ω is the number of possible scenarios. In order for the discrete stochastic process to be perfectly determined, a probability of occurrence π_ω needs to be associated with each realization λ_ω such that $\pi_\omega = P(\omega | \lambda = \lambda_\omega)$ with $\sum_{\omega=1}^{N_\Omega} \pi_\omega = 1$.

Solving optimization problems where uncertainty on input data is modeled by continuous stochastic processes is very difficult, or even impossible in most cases. On the contrary, discrete stochastic processes can be easily embedded into an optimization problem using the so-called *deterministic equivalent problem* (see Appendix A). For this reason, previous to finding the solu-

tion of a stochastic programming problem, every continuous stochastic process is to be replaced by an approximate discrete one. For example, a continuous stochastic process γ can be approximated by a discrete one λ such that $\gamma \approx \lambda = \{\lambda_\omega, \omega = 1, 2, \dots, N_\Omega\}$, with each scenario λ_ω having an associated probability of occurrence equal to π_ω . Intuitively, a scenario so obtained can be understood as a plausible realization of the stochastic process that it represents. For simplicity, it is customary to simply state that the continuous stochastic process γ is approximated or represented through a set of scenarios $\{\gamma_\omega, \omega = 1, 2, \dots, N_\Omega\}$, where, in fact, $\gamma_\omega = \lambda_\omega$ for all ω .

The following sections of this chapter focus on the theoretical fundamentals required to design efficient and effective algorithms for the scenario representation of the typical stochastic processes involved in electricity market problems with a high presence of wind power, in particular, market prices and wind production.

3.3 Scenario Generation

3.3.1 Scenario Tree

The set of realizations or scenarios used to model a stochastic process are normally arranged in a *scenario tree*, which graphically expresses the sequential order in which decisions are made and uncertainty is revealed in a stochastic programming problem. Figure 3.1 illustrates an example of a scenario tree for a three-stage decision-making process in which uncertainties are represented by four scenarios.

Every scenario tree is structurally composed of a set of nodes and branches [35, 117]. The nodes are grouped by stages, which represent points in time where decisions are made. Each node has a single predecessor and may have several successors. The transition between two stages is graphically indicated by the branches, which are also referred to as *arcs*. Each branch or arc corresponds to a particular realization of one or some of the stochastic processes involved in the decision-making problem. Such a realization constitutes relevant information conditioning future decisions in subsequent stages. Thus,

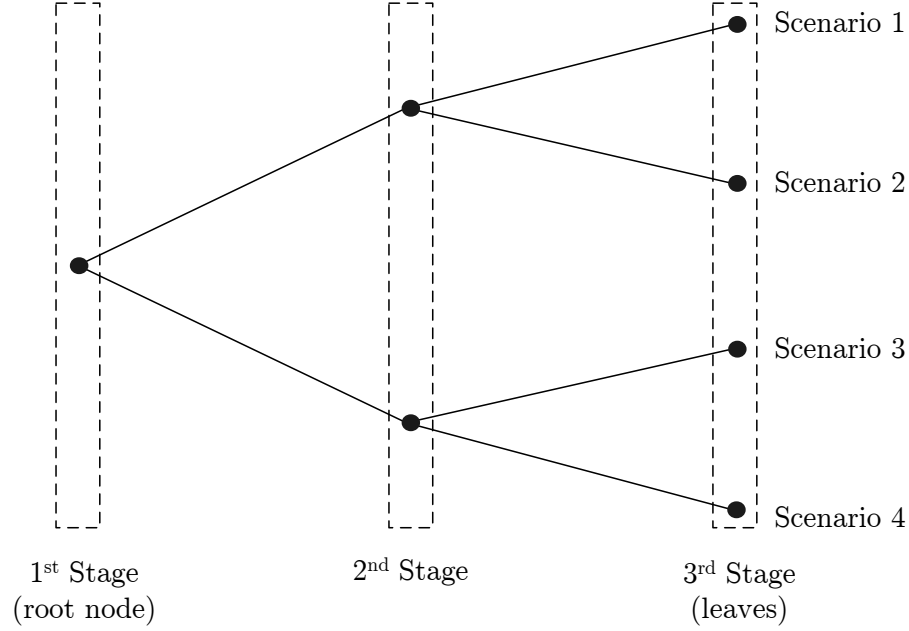


Figure 3.1: Example of three-stage scenario tree

each node in the tree represents a state of the “world” at a certain instant of the decision-making process, which is characterized by the set of input parameters that are still uncertain and the values adopted by those already known to the decision maker. The first node in the tree is named the root node and it corresponds to the beginning of the planning horizon. Accordingly, in this node, all the input parameters that are modeled as uncertain by the decision maker are unknown. The decisions made in these circumstances are called *first-stage decisions* and their optimization constitutes the ultimate purpose of the stochastic programming problem. The nodes connected to the root node are the second-stage nodes and represent the points where the second-stage decisions are made. The number of nodes in the last stage equals the number of scenarios. These nodes are referred to as *leaves*. Thus, in the leaf nodes, all uncertain parameters are unveiled.

When it comes to make decisions under uncertainty using stochastic programming, the building of scenario trees that properly represent the uncertain input parameters constitutes a preliminary task of utmost importance [119]. In

this sense, the optimal decisions derived from stochastic programming models may be indeed remarkably sensitive to the scenario characterization of uncertain data. In addition, the appropriate representation of stochastic processes using a finite set of realizations or scenarios can be computationally more costly than formulating and solving the resulting stochastic programming problem. For these reasons, a considerable effort has been made by the research community to design effective and efficient scenario generation methods. We provide below a brief description of the most relevant ones:

Moment matching techniques [112, 113, 143], which approximate the continuous probability distributions of uncertain parameters by discrete distributions such that a prefixed set of the original statistical properties (e.g., moments, correlation matrix, percentiles, etc.) are preserved in the approximation.

Internal sampling [103, 104], which consists in a continuous sampling process from the distribution functions characterizing the uncertain parameters during the solution procedure.

Generation of data trajectories or path-based methods [73], which produce complete paths (scenarios) using econometric and time series models. The set of scenarios so obtained is named *fan*. Once the fan is generated, the scenarios are clustered to form the scenario tree.

Scenario reduction techniques [74, 95, 100, 101, 152], whose aim is to fabricate a scenario set Ω_S of prescribed cardinality by selecting or eliminating scenarios from an initial set Ω such that Ω and Ω_S are as close as possible according to a *probability distance*.

The scenario generation procedures described in this dissertation result from combining path-based methods with scenario reduction techniques. Specifically, in the following section, we elaborate on how to produce scenario sets for a continuous stochastic process using the time series theory based on ARMA models. The accuracy of the resulting algorithm is contingent on the number N_Ω of the generated scenarios. In fact, from a theoretical viewpoint, as this

number approaches infinity, the scenario set used as a discrete approximation *converges* to the continuous form of the stochastic process under consideration. As a result, obtaining good scenario trees often requires the generation of a large number of scenarios, which may render the underlying optimization problem intractable. To overcome this computational hurdle, two efficient scenario reduction techniques are provided in Section 3.4 of this chapter to trim down the number of scenarios in the tree, thus regaining tractability while deteriorating, at least as possible, the accuracy of the approximation.

3.3.2 Scenario Generation Using ARMA models

The probabilistic structure of a stochastic process Y is determined by identifying the joint distribution of its random variables $\{y_t, t \in T\}$. Roughly speaking, this distribution explains both the probabilistic behavior of each random variable y_t on its own (marginal distributions), and the interrelations existing among all of them (statistical dependencies).

In practice, the determination of the joint distribution is usually a complex and cumbersome endeavor. However, this estimation becomes much easier under the following two assumptions:

1. The joint distribution is a multivariate Gaussian distribution and therefore, is determined by just specifying the mean vector and the variance-covariance matrix of the random variables constituting the stochastic process.
2. The stochastic process under analysis is *stationary*, which means that neither the mean vector nor the variance-covariance matrix depend on time t .

The two assumptions above imply that the marginal distributions of the random variables involved are all identical univariate Gaussian distributions. Being so and for simplicity, hereafter we will just refer to the marginal distribution associated with the considered stochastic process.

With reference to assumption 1, the non-diagonal elements of a variance-covariance matrix are generally referred to as *autocovariances*. Through a

simple standardization process consisting in just dividing the autocovariances by the corresponding product of standard deviations, the *autocorrelations* are obtained. These are also known as *temporal correlations* and quantify the statistical interdependencies among the random variables.

The time series theory based on *autoregressive moving average* (ARMA) models relies on these two premises [41]. An ARMA(p, q) process Y is mathematically expressed as

$$y_t = \sum_{i=1}^p \phi_i y_{t-i} + \varepsilon_t - \sum_{j=1}^q \theta_j \varepsilon_{t-j} \quad (3.1)$$

with p autoregressive parameters $\phi_1, \phi_2, \dots, \phi_p$, and q moving average parameters $\theta_1, \theta_2, \dots, \theta_q$. The term ε_t in equation (3.1) stands for an uncorrelated normal stochastic process with mean zero and variance σ_ε^2 , and is also uncorrelated with $y_{t-1}, y_{t-2}, \dots, y_{t-p}$. Stochastic process ε_t is also referred to as *white noise*, *innovation term*, or *error term*.

Time series analysis based on ARMA models provides a comprehensive mathematical framework to represent an important group of stationary processes that result from imposing a linear dependence among the process variables and a series of white noises. In particular, the autoregressive part of the ARMA(p, q) model in (3.1), $\text{AR}(p)=y_t - \sum_{i=1}^p \phi_i y_{t-i}$, establishes that the present realization y_t of the stochastic process Y directly depends on its last p realizations. The inclusion of the moving-average component, $\text{MA}(q)=\varepsilon_t - \sum_{j=1}^q \theta_j \varepsilon_{t-j}$, in combination with the autoregressive one, allows obtaining simple and compact ARMA(p, q) models, i.e., with p and q sufficiently small [173].

Equation (3.1) can be alternatively written as

$$\phi(B)y_t = \theta(B)\varepsilon_t, \quad (3.2)$$

where B is the delay operator, $B^j(y_t) = y_{t-j}$, and polynomials $\phi(B)$ and $\theta(B)$

are defined as

$$\phi(B) = 1 - \sum_{i=1}^p \phi_i B^i \quad (3.3)$$

$$\theta(B) = 1 - \sum_{j=1}^q \theta_j B^j. \quad (3.4)$$

The estimation and adjustment of ARMA models basically comprises the following phases:

Phase 1: Formulation of the model assuming the normality and stationarity hypotheses for the underlying stochastic process.

Phase 2: Identification of the model terms.

Phase 3: Estimation of the model parameters.

Phase 4: Model validation, which essentially boils down to checking the normality and stationarity assumptions.

We address the interested reader to [41] for specific information on the fitting of time series models.

Once an appropriate ARMA model has been fitted to the stochastic process Y under consideration, the procedure to generate a set of scenarios basically comes down to the sampling of the innovation term from its distribution $\varepsilon_t \sim N(0, \sigma_\varepsilon)$. The proposed scenario-generation algorithm using an ARMA model is outlined below.

Algorithm

Step 1: Initialize the scenario counter: $\omega \leftarrow 0$.

Step 2: Update the scenario counter and initialize the time period counter:
 $\omega \leftarrow \omega + 1, t \leftarrow 0$.

Step 3: Update the time period counter: $t \leftarrow t + 1$.

Step 4: Randomly generate $\varepsilon_t \sim N(0, \sigma_\varepsilon)$.

Step 5: Evaluate expression (3.1) to obtain $y_{t\omega}$.

Step 6: If $t < N_T$ go to Step 3), else go to Step 7).

Step 7: If $\omega < N_\Omega$ go to Step 2), else the scenario-generation process concludes.

3.3.3 Challenging the Stationarity Assumption: Scenarios for Market Prices

Many stochastic processes involved in problems related to electric energy systems and electricity markets are non-stationary in terms of mean and variance. Intuitively, the stationarity property requires that the overall behavior of the time series under study remains the same over time. If, on the contrary, systematic changes in mean or in variance are observed, the series is said to be non-stationary.

Most of the probability theory of time series is built on the stationarity hypothesis, and for this reason time-series analysis often requires one to transform a non-stationary series into a stationary one so as to make use of this theory. In the context of this dissertation, such a transformation is necessary for using the ARMA-based scenario generation technique described in Section 3.3.2.

There are several types of systematic changes that can be identified in non-stationary processes, namely:

Variance changes. Consider the time series $\{y_t\}$ represented in Fig. 3.2. It is clear that as the mean level of the process decreases, its variance, called *volatility* in the finance literature, diminishes.

In order to attain stationarity for the variance, the so-called *Box-Cox transformations* are commonly used. These transformations basically boil down to applying either the logarithm or the square root functions to the process with non-constant variance. In particular, the logarithmic transformation is indicated if the standard deviation of the time series appears to be directly proportional to the mean, as it occurs for the series depicted in Fig. 3.2. Fig. 3.3 shows the time series $\{y_t\}$ after applying the logarithmic conversion. Note that the series so obtained does not exhibit variance changes over time.

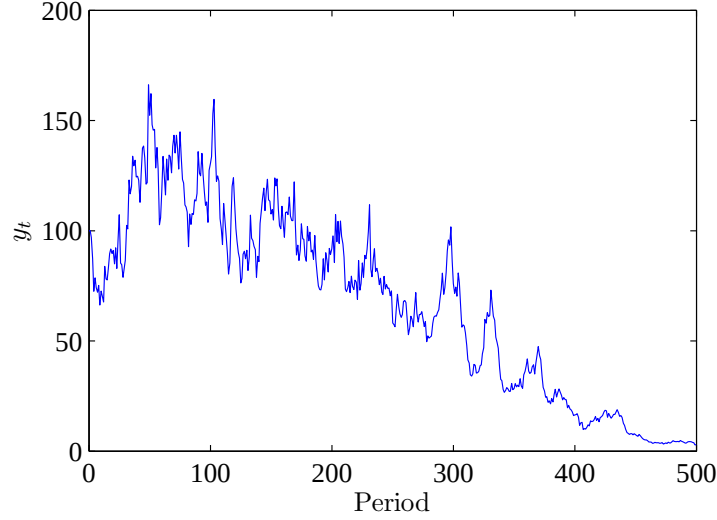


Figure 3.2: Example of non-stationary process

Trend. Broadly speaking, a trend may be defined as a long-term change in the mean level, where the notion of *long term* is relative to the time horizon under consideration. Observe the apparent trend exhibited by the stochastic process represented in Fig. 3.3.

A well-known procedure to stabilize the mean of a time series is to differentiate it. In general, differencing a process $\{y_t\}$ with order d , denoted by $\nabla^d y_t$, is written as

$$\nabla^d y_t = (1 - B)^d y_t = y_t - y_{t-d}, \quad (3.5)$$

where B is the backshift operator, i.e., $B^j y_t = y_{t-j}$.

Usually, a difference order equal to 1 is enough to obtain a stable mean. Fig. 3.4 shows the process $\{\log(y_t)\}$ in Fig. 3.3 after differencing. Note that the resulting time series is stationary in terms of both variance and mean.

The combination of the differencing technique and the ARMA models gives rise to the concept of ARIMA processes for modeling non-stationary time series. ARIMA stands for *autoregressive integrated moving average*,

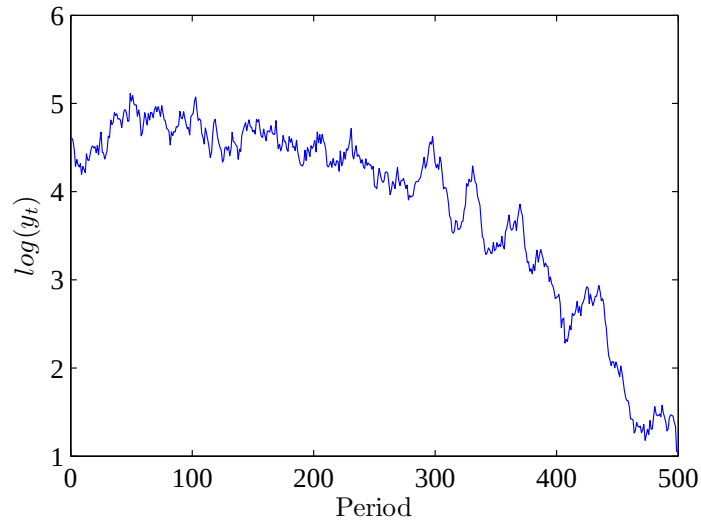


Figure 3.3: Logarithmic transformation of the process in Fig. 3.2

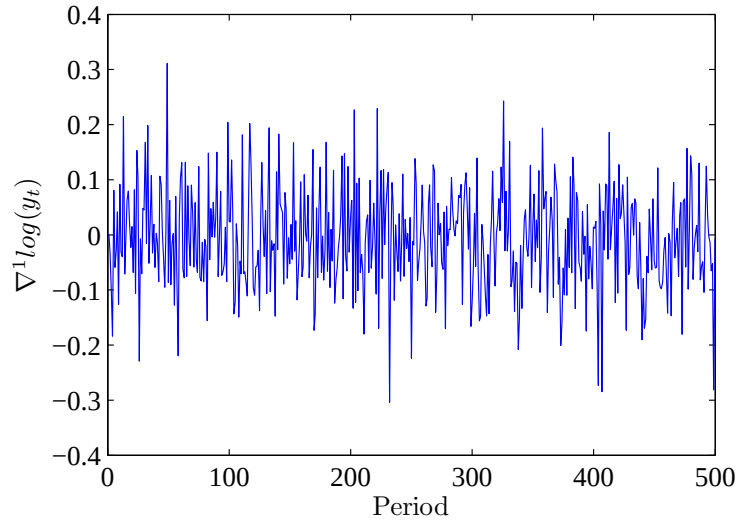


Figure 3.4: Differentiation of the process in Fig. 3.3

where *integrated* indicates that the original non-stationary process is obtained by integrating the stationary process resulting after differencing. In this way, ARIMA models are defined by three parameters (p, d, q) corresponding to the number of autoregressive terms, the differencing order,

and the number of moving-average terms, respectively. The general expression of an ARIMA model with parameters (p, d, q) is

$$\left(1 - \sum_{j=1}^p \phi_j B^j\right) (1 - B)^d y_t = \left(1 - \sum_{j=1}^q \theta_j B^j\right) \varepsilon_t. \quad (3.6)$$

Seasonal or cyclic changes. Time series data often exhibit a strong cyclical or seasonal behavior. For example, the demand for electricity in Nordic countries is known to be systematically higher during the days of Winter than during the rest of the year as a result of the energy consumption increase for heating in that season. Fig. 3.5 shows an example of a stochastic process that presents a periodic pattern.

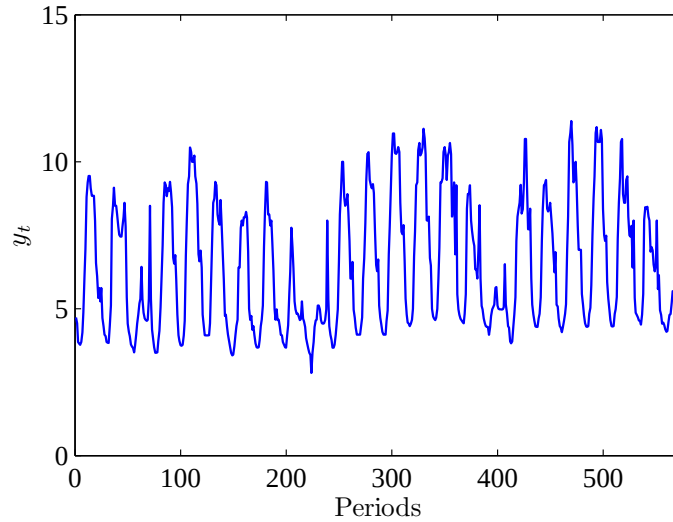


Figure 3.5: Example of seasonal series

To deal with cyclical variations, seasonal ARIMA models (SARIMA) are used. Let us consider a stochastic process with a seasonality of order S . The general expression of a seasonal ARIMA model with parameters

$(p, d, q) \times (P, D, Q)_S$ is

$$\begin{aligned} \left(1 - \sum_{j=1}^p \phi_j B^j\right) \left(1 - \sum_{j=1}^P \Phi_j B^{jS}\right) (1 - B)^d (1 - B^S)^D y_t = \\ \left(1 - \sum_{j=1}^q \theta_j B^j\right) \left(1 - \sum_{j=1}^Q \Theta_j B^{jS}\right) \varepsilon_t \end{aligned} \quad (3.7)$$

with a seasonal component of P autoregressive parameters $\Phi_1, \Phi_2, \dots, \Phi_P$, Q moving average parameters $\Theta_1, \Theta_2, \dots, \Theta_Q$, and a differentiation order D .

Note that, in the SARIMA expression (3.7), the technique of differencing has been adapted to deal with a possible trend in the seasonal component of period S by introducing the lag- S difference operator ∇_S defined by

$$\nabla_S y_t = y_t - y_{t-S} = (1 - B^S) y_t. \quad (3.8)$$

Trend and seasonality are usually detected by inspecting the graph of the (possibly transformed) series. However, they are also characterized by sample autocorrelation functions (autocorrelograms) which are slowly decaying and nearly periodic, respectively [41, 173].

The series of electricity market prices is strongly non-stationary. Indeed, it exhibits a complex dynamic structure that other commodities such as oil and gas do not present in general. The complexity of the electricity price behavior is principally caused by the non-storability character of electricity and the imperfect nature of electricity markets, which are incomplete and not sufficiently liquid. This translates into properties such as high frequency and volatility, non-constant mean and variance, multiple seasonality (daily and weekly periodicity), calendar effect (holidays), and a high percentage of unusual prices (outliers) [56, 58, 159]. In Section 3.3.7, we succinctly highlight some of these features from a series of electricity prices recorded in the Electricity Market of the Iberian Peninsula [3].

3.3.4 Challenging the Normality Assumption: Scenarios for Wind Speed

Observe in the ARMA general form (3.1) that y_t boils down to a linear combination of white noises, and as such, the marginal distribution associated with the stochastic process Y is necessarily normal, which is in accordance with the normality assumption on which the ARMA models are based. However, for a number of stochastic processes, such an assumption does not hold. This is the case of the speed of local winds, which is not normally distributed, but follows a Weibull distribution. In fact, the technical literature is rich in references on this issue [83, 88, 102, 127, 157], including relevant works dealing with the problem of how to estimate the parameters of the Weibull distribution that best describes a given wind speed frequency distribution [53, 134, 183, 196]. In [26], Weibull distributions for different wind sites all over the world are provided.

In order to preserve the original marginal distribution associated with the stochastic process Y while making use of the modeling capability of the ARMA models, a new stochastic process, Z , with a standard normal marginal distribution is defined through the transformation

$$Z = \Phi^{-1} [F_Y(Y)], \quad (3.9)$$

where F_Y is the cumulative distribution function (cdf) of the marginal distribution associated with the original stochastic process Y and $\Phi(\cdot)$ is the cumulative distribution function of the standard normal random variable.

Transformation (3.9) comes from [132] and is graphically illustrated in Fig. 3.6, where the path in direction d (direct) represents the referred transformation.

The ARMA model (3.1) is adjusted to the transformed stochastic process Z . From the fitted model, a *transformed* set of scenarios Ω_Z is first generated, and then *untransformed* in order to produce a set of scenarios, Ω_Y , characterizing the original stochastic process, Y . The inverse transformation can be written as

$$Y = F_Y^{-1} [\Phi(Z)], \quad (3.10)$$

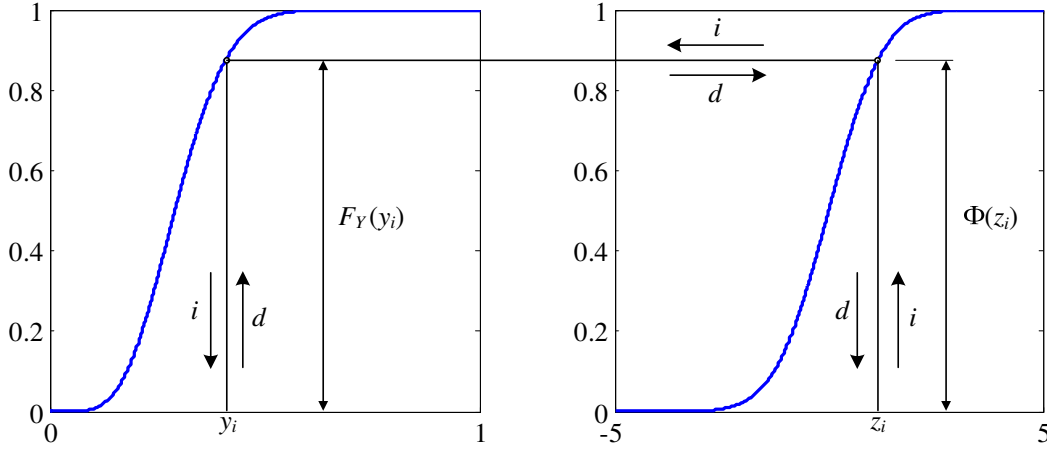


Figure 3.6: Graphic interpretation of the normal transformation in equation (3.9)

and is graphically represented by the path in direction i (inverse) of Fig. 3.6.

3.3.5 Transforming Wind Speed into Wind Power

The ARMA-based technique introduced in Section 3.3.2 can be equally used to generate both wind speed and wind power scenarios. However, unlike the speed of local winds, the power produced by a wind farm cannot be generally fitted to a standard probability distribution [91, 179]. As a result, a non-parametric fit, such as quantile regression [179], has to be considered instead when applying the marginal transformation (3.10). Fig. 3.7 shows the frequency distribution for the power produced during the year 2006 by a simulated 380.5-MW wind farm located at the Adair site in Iowa, US. Specifically, the site coordinates are $41^{\circ} 27'$ North and $94^{\circ} 34' 12''$ West. The wind power data set used to build Fig. 3.7 has been simulated by the *National Renewable Energy Laboratory* (NREL) of the US Department of Energy in the context of the *Eastern Wind Integration and Transmission Study* (EWITS). These data can be freely downloaded from [126]. Further information on the software employed to simulate these wind power data can be found in [44].

Needless to say, the wind power frequency distribution depicted in Fig. 3.7 does not look like any standard probability distribution. This is mainly due to the highly non-linear shape of a wind turbine power curve, which describes

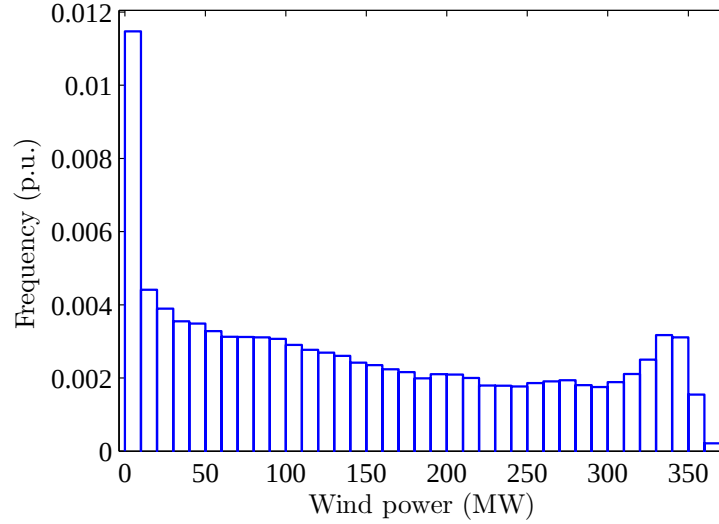


Figure 3.7: Wind power frequency distribution corresponding to the wind power data simulated by NREL for the the Adair site in Iowa over 2006

the power output of the turbine as a function of the wind speed. Fig. 3.8 illustrates a typical wind turbine power curve. Three wind speed values in this curve are of special interest, namely:

Cut-in speed. This is the minimum wind speed at which the wind turbine starts to generate usable power.

Rated speed. This is the minimum wind speed at which the wind turbine produces its designated rated output power.

Cut-out speed. This is the wind speed at which the turbine is shut-down as a safety measure to protect it from damage.

In addition to the non-standard nature of the wind power probability distribution, power production data from wind farms are usually proprietary and fiercely guarded by their owners, the wind producers. Thus, these data are not publicly available in general. For these reasons, in this dissertation, we decided to produce wind power scenarios by transforming wind speed scenarios that have been previously generated using the ARMA-based scenario generation algorithm described in Section 3.3.2. The transformation process is as simple as converting wind speed values into wind power values through the power

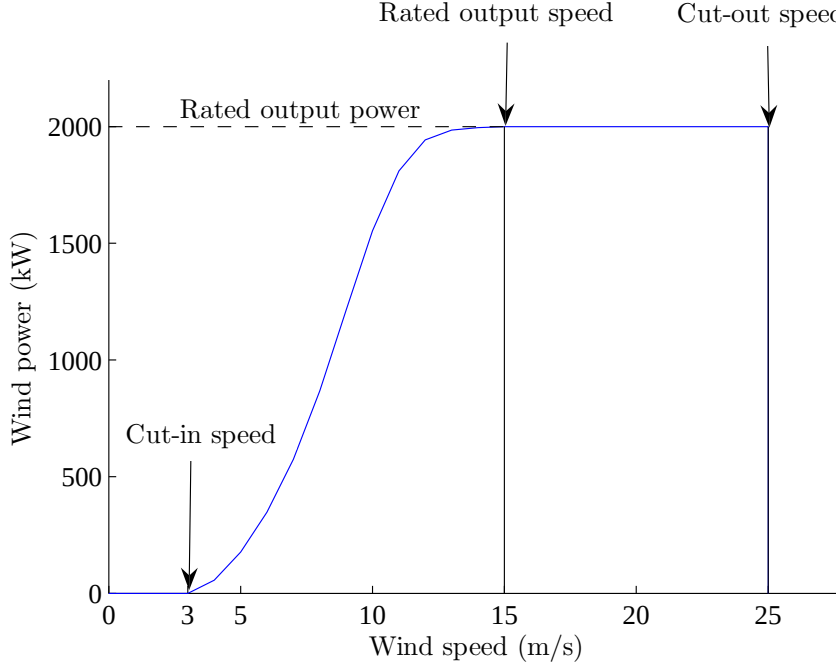


Figure 3.8: Example of a wind turbine power curve

curve of the turbine model installed in the wind farm under consideration. The wind farm power output is then obtained by just aggregating the production of every turbine in the plant. Logically, this is an approximation based on the assumptions that the characteristics of the wind are the same all over the plant at each instant and all the turbines therein are 100% reliable, i.e., they do not fail unexpectedly. Therefore, we disregard the power losses stemming from unit availability, wake effect, soiling, etc., some of which are of a stochastic nature. If these losses are accounted for, the wind farm power curve proves to be a swarm of (wind speed, wind production) points as shown in Fig. 3.9 for the Adair location in Iowa. In this figure, the red line represents a possible piece-wise polynomial fit of this curve, while the black line is the power curve that results from the approximation made in this thesis. As expected, observe that the approximated curve overestimates the power produced by the wind farm. Nevertheless, this is irrelevant to the purposes of this thesis and to the conclusions drawn from the mathematical models proposed in it.

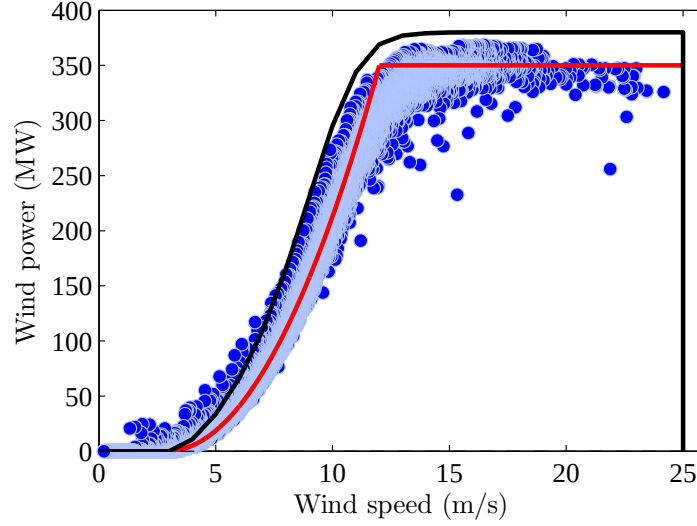


Figure 3.9: Scatter plot showing the hourly wind speed and power production pairs simulated for the Adair location, Iowa, by NREL over 2006 along with the fitted and assumed power curves

3.3.6 Modeling Statistical Dependencies: Scenarios for Multiple Wind Sites

The winds at nearby locations usually exhibit similar behaviors over time. As a result, the power productions from spatially close wind farms frequently follow related patterns. In statistical terms, we say that the wind speed at a particular site can be described by a stochastic process that is spatially correlated with the stochastic processes describing wind speeds at other sites. Recognizing and accounting for this fact can be very valuable both for producers owning wind farms at several locations to make informed decisions on electricity trading and for system operators, which are in charge of maintaining system security and reliability, potentially threatened by the unpredictable and variable nature of wind resources.

Therefore, procedures capable of generating scenario sets incorporating dependencies among wind speeds at multiple sites are of interest. With this in mind, we develop below a procedure of this type.

Let us consider first two stochastic processes, Y^a and Y^b , that can be

separately described by means of the following ARMA(p, q) models:

$$y_t^a = \sum_{i=1}^{p^a} \phi_i^a y_{t-i}^a + \varepsilon_t^a - \sum_{j=1}^{q^a} \theta_j^a \varepsilon_{t-j}^a \quad (3.11)$$

$$y_t^b = \sum_{i=1}^{p^b} \phi_i^b y_{t-i}^b + \varepsilon_t^b - \sum_{j=1}^{q^b} \theta_j^b \varepsilon_{t-j}^b. \quad (3.12)$$

As stated in Section 3.3.2, the series of errors ε^a and ε^b are not autocorrelated, i.e., $\mathcal{E}\{\varepsilon_t^a \varepsilon_{t-j}^a\} = 0$ and $\mathcal{E}\{\varepsilon_t^b \varepsilon_{t-j}^b\} = 0, \forall j \neq 0$, where $\mathcal{E}\{\cdot\}$ is the expectation operator. However, if stochastic processes Y^a and Y^b are statistically dependent, such a dependency is *transferred* to the series of residuals ε^a and ε^b through equations (3.11) and (3.12). Consequently, ε^a and ε^b should be indeed *cross-correlated*, i.e., $\mathcal{E}\{\varepsilon_t^a \varepsilon_{t-j}^b\} \neq 0$ and/or $\mathcal{E}\{\varepsilon_t^b \varepsilon_{t-j}^a\} \neq 0$ for some j . The dependency structure of the residual series ε^a and ε^b can be analyzed by means of their *cross-correlogram*, in which the cross-correlation coefficients for different time lags are represented [41]. Specifically, the positive and negative lag- k cross-correlation coefficients, ρ_k and ρ_{-k} , of ε^a and ε^b are defined, respectively, as $\mathcal{E}\{\varepsilon_t^a \varepsilon_{t-k}^b\} = \rho_k$ and $\mathcal{E}\{\varepsilon_{t-k}^a \varepsilon_t^b\} = \rho_{-k}$, where the equality holds for any t by appealing to the stationarity assumption. The cross-correlation coefficient corresponding to the lag zero, ρ_0 , is usually referred to as *contemporaneous correlation*.

According to the main features characterizing the shape of the residual cross-correlogram, three different types of dependent stochastic processes can be distinguished:

Contemporaneous stochastic processes. The residual cross-correlogram of two contemporaneous stochastic processes is characterized by a single significant correlation coefficient placed at lag zero, that is, by a significant contemporaneous correlation. The rest of coefficients can be considered statistically zero. Contemporaneous stochastic processes are usually *driven* by a common physical cause. Fig. 3.10(a) illustrates a residual cross-correlogram typical of this type of dependent processes.

Quasi-contemporaneous stochastic processes. The residual cross-correlogram of

two quasi-contemporaneous stochastic processes presents a triangular structure in which the most significant correlation coefficients form a group around the lag zero, with the lag-0 correlation coefficient being precisely the largest one. This structure is typical of stochastic processes with a common, but delayed, physical cause. The triangle base length gives an idea of the delay magnitude. The narrower and the more pointed this triangle is, the more contemporaneous the stochastic processes are. Fig. 3.10(b) shows an example of the residual cross-correlogram of two quasi-contemporaneous stochastic processes.

Non-contemporaneous stochastic processes. With this name, we refer to any other pair of stochastic processes that does not fit into any of the two previous categories. The modeling of the dependencies existing between two non-contemporaneous stochastic processes is generally more complex and cumbersome. Fig. 3.10(c) constitutes an example of a possible residual cross-correlogram of this kind of processes.

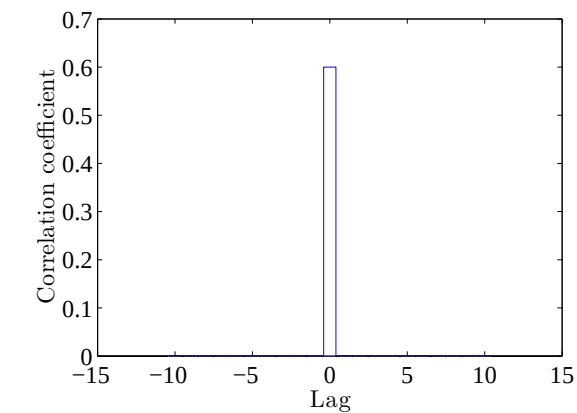
As shown below through various case studies (see Section 3.3.8), wind speeds at relatively close sites constitute an example of quasi-contemporaneous stochastic processes.

Next, we propose a methodology to generate statistically correlated scenarios for this kind of processes. The proposed methodology seeks to produce correlated scenarios by simulating a series of errors, ε^a and ε^b , that reproduces the residual cross-correlogram of the stochastic processes, Y^a and Y^b , under consideration.

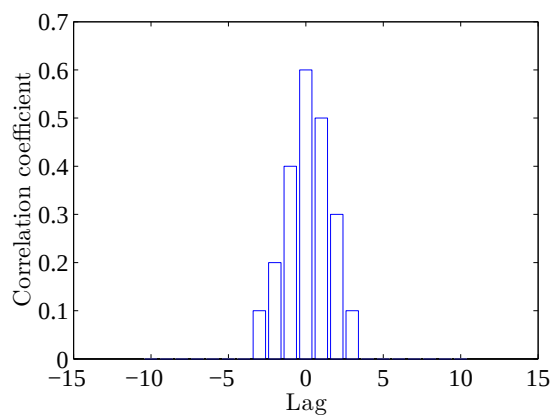
Mathematically, the cross-correlations between the series of errors ε^a and ε^b can be described by means of a variance-covariance matrix G . This matrix is symmetric by definition, and therefore, can be always diagonalized. In other words, an orthogonal transformation can be used to model such series of errors as

$$\varepsilon = \begin{pmatrix} \varepsilon^a \\ \varepsilon^b \end{pmatrix} = B\xi = B \begin{pmatrix} \xi^a \\ \xi^b \end{pmatrix}, \quad (3.13)$$

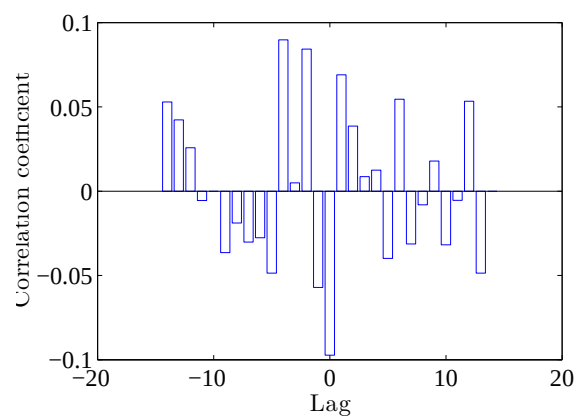
in which B is the orthogonal matrix of the transformation, and ξ is normal and



(a) Contemporaneous



(b) Quasi-contemporaneous



(c) Non-contemporaneous

Figure 3.10: Types of dependent stochastic processes according to their residual cross-correlogram: examples

satisfies that $\mathcal{E}\{\xi\xi^T\} = I$, being I the identity matrix. That is, ξ is neither autocorrelated, nor cross-correlated.

This way, white noises (independent standard normal errors) can be generated, and then, cross-correlated according to the variance-covariance matrix G by using the orthogonal transformation as expressed in (3.13).

In most engineering applications, matrix G , besides being symmetric, is also positive definite, and as such, can be decomposed through the computationally advantageous Cholesky decomposition, which avoids the calculation of eigenvalues and eigenvectors. The Cholesky decomposition can be stated as

$$G = LL^T, \quad (3.14)$$

where L is an inferior triangular matrix that turns out to be the orthogonal matrix required for transformation (3.13), i.e., $B = L$, as shown below.

The variance-covariance matrix G can be computed as

$$\begin{aligned} G &= \text{cov}(\varepsilon, \varepsilon^T) = \text{cov}(B\xi, \xi^T B^T) = B \text{cov}(\xi, \xi^T) B^T \\ &= B \mathcal{E}\{\xi\xi^T\} B^T = B I B^T = B B^T, \end{aligned} \quad (3.15)$$

where $\text{cov}(\cdot, \cdot)$ stands for the covariance operator.

By using the Cholesky decomposition (3.14), equation (3.15) yields

$$G = LL^T = BB^T \Rightarrow (B^{-1}L)(L^T(B^T)^{-1}) = I \Rightarrow B = L.$$

The simulation of the cross-correlated error ε becomes more costly as the number of significative cross-correlation coefficients increases. In this sense, the proposed methodology is efficacious provided that the statistical dependencies among residuals can be explained by a finite set of cross-correlation coefficients, and its efficaciousness is dependent on the cardinality of this set. In other words, this methodology is suitable for contemporaneous and quasi-contemporaneous stochastic processes. In fact, the former should be considered as a simplified case of the latter.

In the following, the procedure to generate a scenario set for a number N_S of quasi-contemporaneous stochastic processes is described step by step. The

algorithm is also illustrated through the flow chart depicted in Fig. 3.11.

Algorithm

Step 1: For each stochastic process, fit an ARMA model to the corresponding series of historical values. This fitting process yields non-autocorrelated normal residuals (historical errors).

Step 2: Although the historical error series obtained in the preceding step are not *autocorrelated*, they should be *cross-correlated* if the stochastic processes under analysis are indeed interrelated. In this step, the cross-correlations are computed and, consequently, the variance-covariance matrix G is built. Considering for the sake of simplicity just two stochastic processes, Y^a and Y^b , the structure of matrix G is as

$$G = \left(\begin{array}{c|c} \varepsilon^a & \varepsilon^b \\ \hline G_{11} & G_{12} \\ G_{21} & G_{22} \end{array} \right) \begin{array}{c} \varepsilon^a \\ \varepsilon^b \end{array} \quad (3.16)$$

where $G_{11} = \sigma_{\varepsilon^a}^2 I_{K+N_T}$, $G_{22} = \sigma_{\varepsilon^b}^2 I_{K+N_T}$, and

A diagram of a 2D lattice with diagonal dashed lines. The lattice is enclosed in a rounded rectangle. The top-right corner is labeled '0' and the bottom-left corner is labeled '0'. The diagonal lines are labeled with ρ_{-K} , ρ_{-k} , ρ_0 , ρ_k , and ρ_K from bottom-left to top-right.

ρ_k is the positive lag- k cross-correlation coefficient, K is the maximum lag (positive or negative) with a significant cross-correlation coefficient, and N_T is the number of time periods to be covered in the decision-making process.

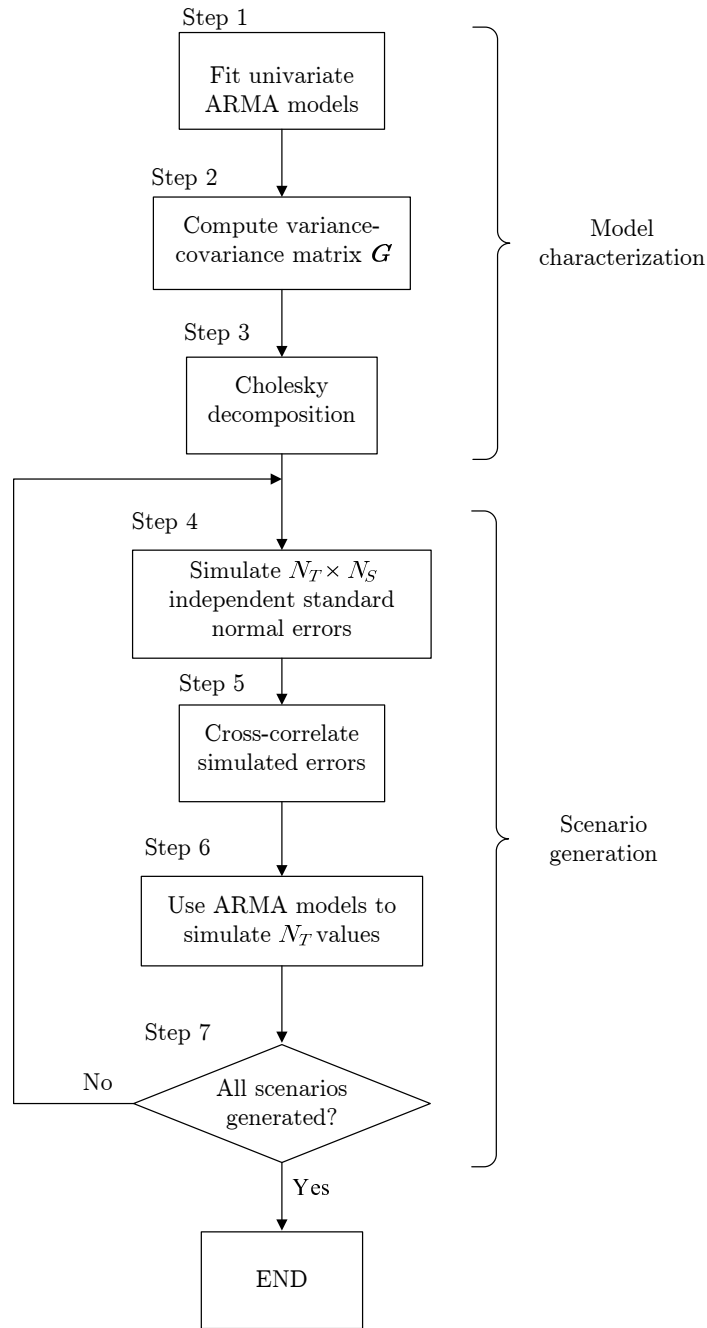


Figure 3.11: Flow diagram of the algorithm to generate correlated scenarios

The size of the square and symmetric matrix G is $(K + N_T) \times N_S$, where N_S is the number of stochastic processes under consideration. If $K \ll K + N_T$, then it is computationally advantageous to treat G as a sparse matrix.

Step 3: Apply Cholesky decomposition to G (i.e., compute L such that $G = LL^T$) so as to obtain the matrix B of the orthogonal transformation (3.13).

Step 4: Simulate $N_T \times N_S$ independent standard normal errors, ξ .

Step 5: Cross-correlate the $N_T \times N_S$ standard normal errors generated in the previous step by using the orthogonal transformation (3.13). Specifically, for two stochastic processes, Y^a and Y^b , this cross-correlation process is performed as

$$\begin{pmatrix} \hat{\varepsilon}_{n-K+1}^a \\ \vdots \\ \hat{\varepsilon}_n^a \\ \hline \varepsilon_{n+1}^a \\ \vdots \\ \varepsilon_{n+N_T}^a \\ \hline \hat{\varepsilon}_{n-K+1}^b \\ \vdots \\ \hat{\varepsilon}_n^b \\ \hline \varepsilon_{n+1}^b \\ \vdots \\ \varepsilon_{n+N_T}^b \end{pmatrix} = L \begin{pmatrix} \varepsilon_{n-K+1}^a \\ \vdots \\ \varepsilon_n^a \\ \hline \xi_{n+1}^a \\ \vdots \\ \xi_{n+N_T}^a \\ \hline \varepsilon_{n-K+1}^b \\ \vdots \\ \varepsilon_n^b \\ \hline \xi_{n+1}^b \\ \vdots \\ \xi_{n+N_T}^b \end{pmatrix}$$

where n is the sample size of the available historical data set, $\varepsilon_{n-K+1}^a, \dots, \varepsilon_n^a, \varepsilon_{n-K+1}^b, \dots, \varepsilon_n^b$ are the last K error terms of the historical series, $\hat{\varepsilon}_{n-K+1}^a, \dots, \hat{\varepsilon}_n^a, \hat{\varepsilon}_{n-K+1}^b, \dots, \hat{\varepsilon}_n^b$ are their corresponding transformed values, and $\xi_{n+1}^a, \dots, \xi_{n+N_T}^a, \xi_{n+1}^b, \dots, \xi_{n+N_T}^b$ are the $N_T \times N_S$ ($N_S = 2$) independent standard normal errors simulated in Step 4 above.

Finally, the error series $\varepsilon_{n+1}^a, \dots, \varepsilon_{n+N_T}^a$ and $\varepsilon_{n+1}^b, \dots, \varepsilon_{n+N_T}^b$ are those to be introduced in the ARMA models fitted in Step 1.

Step 6: For each stochastic process, use, as explained in Section 3.3.2, the corresponding N_T cross-correlated normal errors to simulate N_T values by means of the corresponding univariate ARMA model fitted in Step 1.

Step 7: Repeat Steps 4-6 until the desired number (N_Ω) of scenarios is obtained.

3.3.7 Market Price Case Study

The technical literature is rich in references on the modeling, forecasting, and scenario generation of electricity market prices, e.g., [48, 56–58, 130, 138, 159, 164, 180, 208], among others. For this reason, in this case study, we limit ourselves to highlight just the basic features that characterize the dynamic behavior of electricity prices. To this end, we consider the historical series of day-ahead prices depicted in Fig. 3.12, which belongs to the electricity market of the Iberian Peninsula [3]. Specifically, these price data correspond to the working days of the last three months of year 2007 (1488 hourly periods), and are publicly available in [6]. We focus our study on working days to sidestep the analysis of the weekly seasonality that electricity prices usually exhibit. This cyclical component is mostly caused by the differences between weekdays and weekends in terms of price behavior and electricity consumption pattern.

From the graph of the price series data in Fig. 3.12, it can be inferred that neither its mean nor its variance are constant. The series is therefore non-stationary. We can observe that the price volatility becomes higher as the mean level of the price increases, which justifies the use of the logarithmic transformation to stabilize the variance. It is also worth pointing out the price spike of almost 130 €/MWh that takes place in December, since this type of discordant observations or outliers are usual in electricity price series.

Fig. 3.13 shows the price series after differencing, once the logarithmic conversion has been performed.

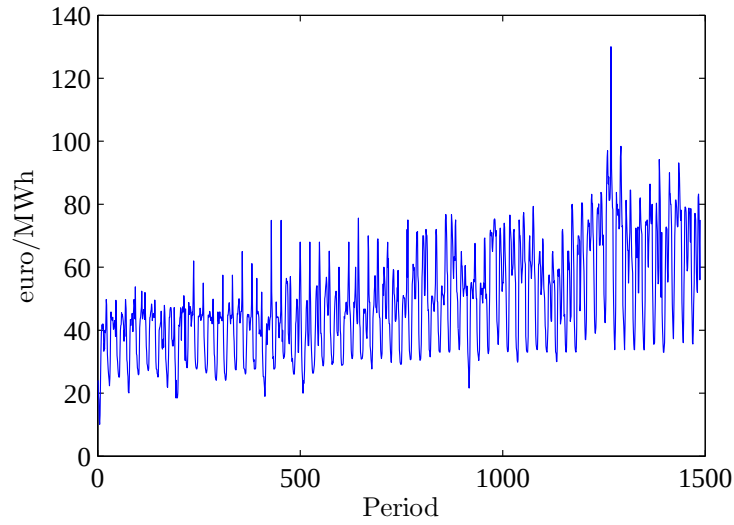


Figure 3.12: Price series for the last three months of year 2007 in the electricity market of the Iberian Peninsula

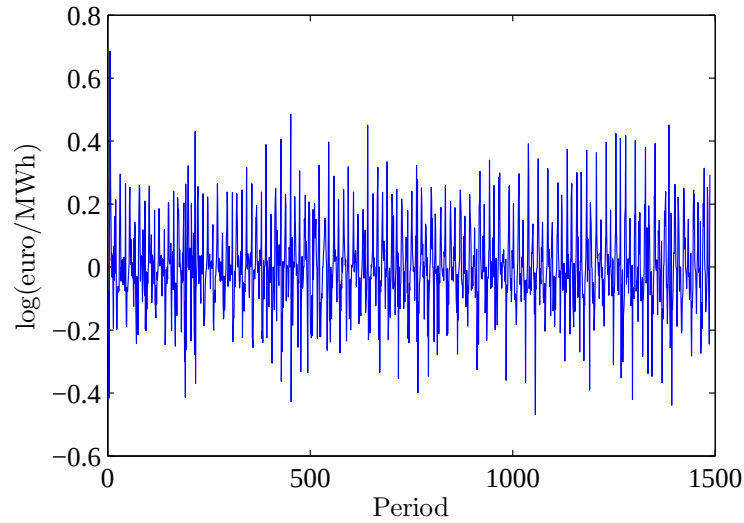


Figure 3.13: Logarithmic and differentiated price series for the last three months of year 2007 in the electricity market of the Iberian Peninsula

The transformed series is still non-stationary because of its marked daily seasonality, which can be easily detected in Fig. 3.14, where the first 96 price

data are represented.

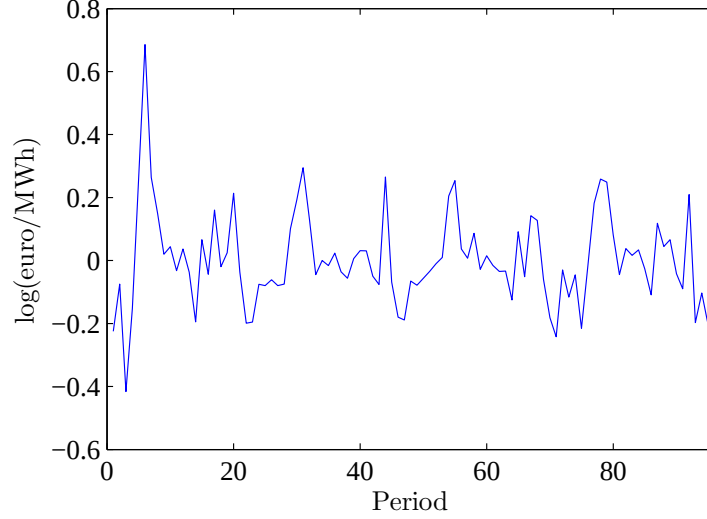


Figure 3.14: 96-h extract from the logarithmic and differentiated price series in Fig. 3.13

In order to account for both the regular and seasonal components of the price data series, we make use of the following SARIMA model,

$$\begin{aligned} (1 - \phi_1 B^1)(1 - B^1)(1 - B^{24}) \ln(\lambda_t^D) = \\ (1 - \theta_1 B^1)(1 - \Theta_1 B^{24} - \Theta_2 B^{48}) \varepsilon_t, \end{aligned} \quad (3.17)$$

where λ_t^D represents the day-ahead market price in time period t .

The standard deviation of the error term ε_t is equal to $0.07613 \ln(\text{€/MWh})$. The values of the parameters in (3.17), which are provided in Table 3.1, as well as the standard deviation of ε_t can be estimated using commercially available software for time series analysis, e.g., SCA [9, 131], SAS [8], or SPSS [11].

Finally, the SARIMA model (3.17) can be used to generate price scenarios by randomly simulating error terms, ε_t , as explained in Section 3.3.2. Fig. 3.15 illustrates a set of 200 price scenarios for the 24 hours after the time interval spanned by the historical data. The black line represents the forecast values.

Table 3.1: Parameters of the SARIMA model (3.17)

ϕ_1	=	0.650	θ_1	=	0.920
			Θ_1	=	0.669
			Θ_2	=	0.078

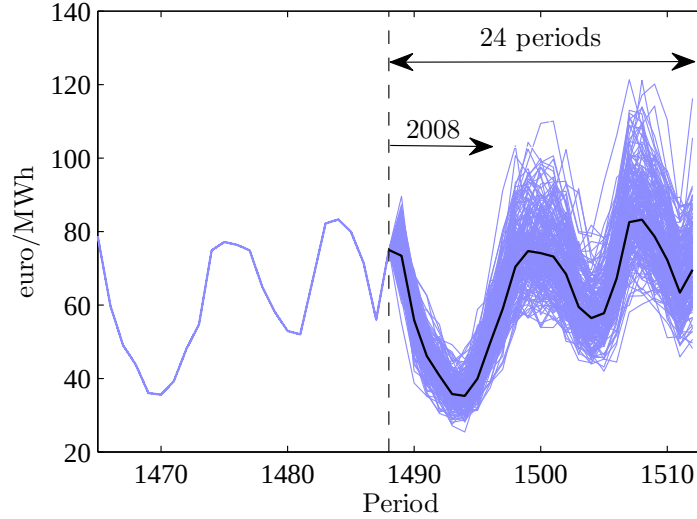


Figure 3.15: Set of 200 price scenarios generated for the first working day of year 2008

3.3.8 Wind Speed Case Studies

Wind speeds at two different sites that are geographically close constitute an example of quasi-contemporaneous stochastic processes. Hence, the statistical dependence existing between both sites can be embedded into a scenario set by means of the methodology introduced in Section 3.3.6. The proximity between sites is required for their winds to be spatially correlated.

In this section, the aforesaid methodology is tested on four different case studies based on the wind data collected at five sites in Massachusetts, US. All the historical wind speed series employed are publicly available in [186]. The data sets comprise 10-min averaged speed measurements that have been aggregated into hourly time intervals. An approximate geographical distribution of the sites as well as the distances among them are shown in Fig. 3.16.

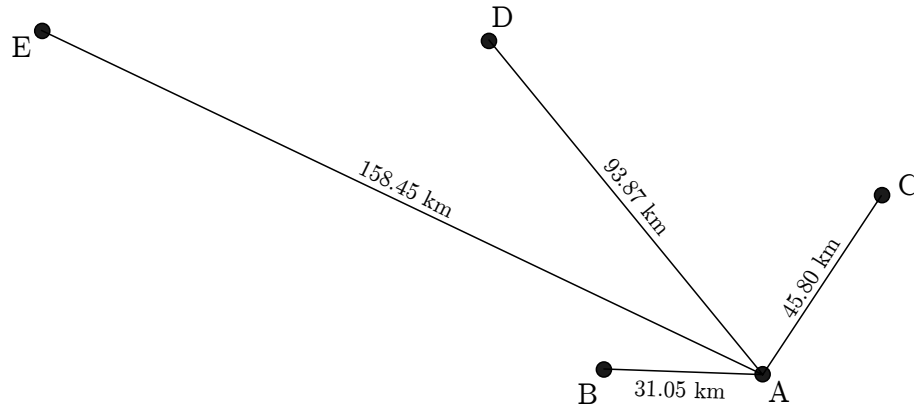


Figure 3.16: Schematic geographical distribution of wind sites and distances among them

Next, a brief description of each wind site is provided:

Site A: The wind monitoring station is located at Bishop and Clerks, on the top of a lighthouse placed within the three-mile state limit of Massachusetts's waters, at $41^{\circ} 34' 27.6''$ North and $70^{\circ} 14' 59.5''$ West. The anemometer is mounted at a height of 15 m relative to the mean low water level.

Site B: The measurement station is located at the town water treatment plant in Falmouth. The location of the tower base is at $41^{\circ} 36' 21.6''$ North, $70^{\circ} 37' 15.60''$ West. The wind monitoring equipment includes anemometers at three different heights on the tower: 10 m, 30 m and 39 m. Redundant anemometers exist at 30 m and 39 m. The wind data series used in this paper from this site corresponds to a height of 39 m.

Site C: The site is located at a parking lot very near the beach on Ocean View Drive in Wellfleet. The location of the tower base is at $41^{\circ} 56' 2.4''$ North, $69^{\circ} 58' 48''$ West. Wind monitoring equipment comprises anemometers situated at three heights on the tower: 50 m, 38 m, and 20 m. Redundant anemometers are positioned at 50 m and 38 m. The wind speed series considered here is the one obtained at a height of 38 m.

Site D: The monitoring station is located at the WBZ radio tower, near the salt marsh on the west coast of the Hull isthmus, on the southern portion of the Boston harbor. The site coordinates are $42^\circ 16' 44.11''$ North by $70^\circ 52' 34.39''$ West, and the anemometer from which the wind data employed comes from is placed at a height of 61 m.

Site E: The monitoring station is located at the Yankee Network Tower on Mount Asnebumskit, southeast of the town of Paxton. Site coordinates are $42^\circ 18' 11.6''$ North by $71^\circ 53' 50.9''$ West. The wind data series collected from this site corresponds to a height of 78 m.

3.3.8.1 Wind Speed Case Study 1

In this case study, the referred methodology is applied to model wind speed stochastic processes, Y^A and Y^B , at sites A and B. Moreover, this first case study also serves us to illustrate the algorithm in Fig. 3.11 in more detail.

Prior to using this algorithm, we should deal with the fact that wind speed is not usually normally distributed, but follows a Weibull distribution. As an example, Fig. 3.17 plots both the wind speed frequency distribution of site A and the corresponding probability density function (pdf) of the fitted Weibull.

Consequently, in order to reconcile the normality assumption on which ARMA models are based with the non-Gaussian nature of local winds, we make use of the marginal transformation (3.9) to *normalize* the wind speed historical data, i.e.,

$$\begin{aligned} Z^A &= \Phi^{-1} [F_{Y^A}(Y^A)] , \\ Z^B &= \Phi^{-1} [F_{Y^B}(Y^B)] , \end{aligned}$$

where F_{Y^A} and F_{Y^B} are the cumulative distribution functions (cdf) of the Weibull distributions fitted for wind speeds at sites A and B, respectively, and Φ is the cdf of the standard normal random variable (with zero mean and unit standard deviation).

The procedure described in Fig. 3.11 is then applied to the resulting *trans-*

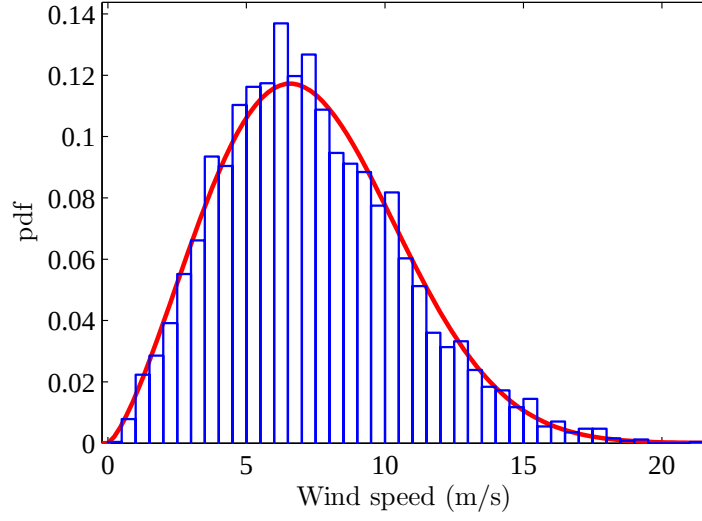


Figure 3.17: Wind speed frequency distribution for site A and the pdf of the Weibull fit

formed series.

Fig. 3.18 is the normality plot of the transformed data corresponding to site A. Needless to say, the normal transformation is not perfect because neither is the Weibull fit. However, note that it constitutes a good approximation.

The next step of the algorithm consists in modeling the autocorrelations inferred from the historical data series by adjusting univariate ARMA(p, q) models. This fitting process provides the following time series models:

Site A: ARMA(1,2)

$$z_t^A = 0.925z_{t-1}^A + \varepsilon_t^A + 0.183\varepsilon_{t-1}^A - 0.025\varepsilon_{t-2}^A.$$

Site B: ARMA(1,2)

$$z_t^B = 0.915z_{t-1}^B + \varepsilon_t^B + 0.092\varepsilon_{t-1}^B - 0.040\varepsilon_{t-2}^B.$$

The residual errors derived from the univariate time series analysis should not be autocorrelated, but they are cross-correlated if there exists a spatial

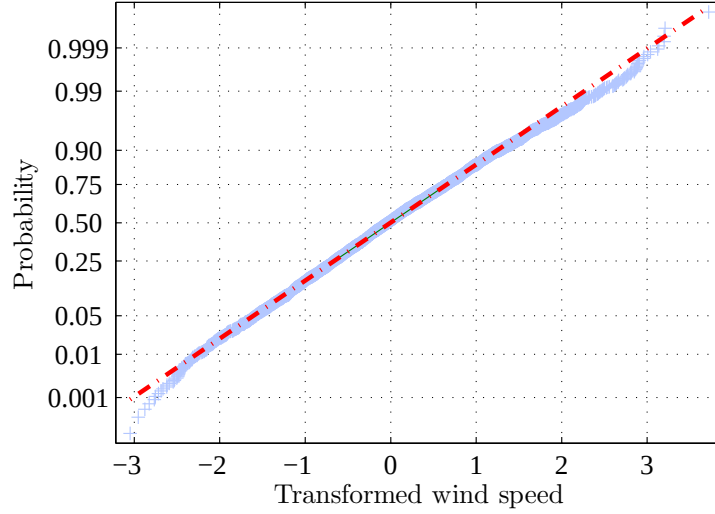


Figure 3.18: Normality plot of the transformed data for site A

interaction between wind sites A and B. It is important to remark that the cross-correlation among wind speeds at multiple sites stems from their geographical closeness. This is the reason why this type of cross-correlation is also referred to as *spatial correlation*. In Fig. 3.19, the cross-correlogram of the residuals (which we often call *residual cross-correlogram*) is depicted. Observe the triangular structure, typical of quasi-contemporaneous stochastic processes, with the most significant correlation coefficients around the lag zero.

From this point, the algorithm focuses on simulating errors that reproduce the cross-correlogram of Fig. 3.19. Nevertheless, not all the cross-correlation coefficients should be reproduced, but just those considered significant enough from a statistical point of view. The significance of a given cross-correlation coefficient is appraised by means of a *p-value* testing the hypothesis of no correlation [193]. This p-value is the probability of obtaining a correlation as large as the observed one by random chance, when the true correlation is zero. In Fig. 3.20(a), only those coefficients with a p-value smaller than a *significance level* $\alpha = 0.05$ are represented.

The simulation process starts generating $2 \times N_T$ independent standard normal errors, which are subsequently cross-correlated according to the variance-

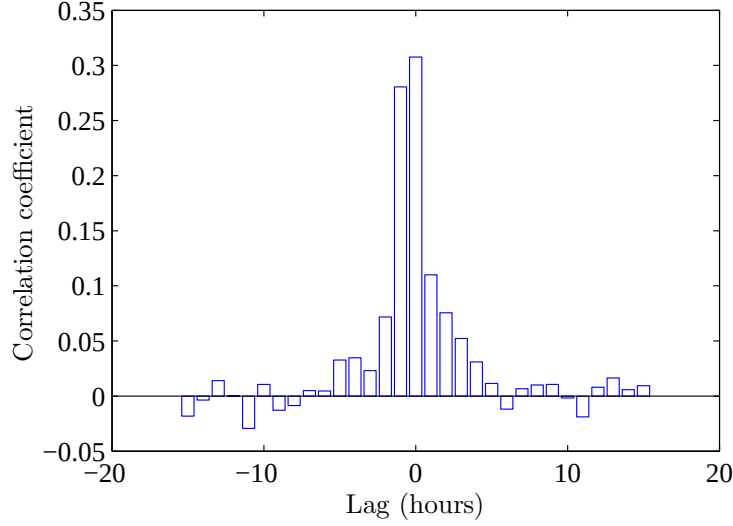
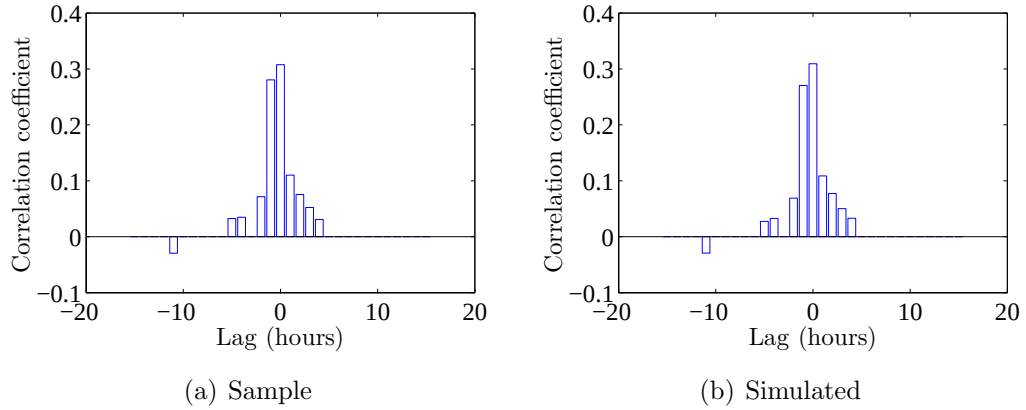


Figure 3.19: Cross-correlogram of the residuals obtained after the ARMA fit

Figure 3.20: Cross-correlograms of errors with a significance level $\alpha = 0.05$

covariance matrix G defined by the cross-correlations coefficients of Fig. 3.20(a) and built as explained in Step 2 of the algorithm described in Section 3.3.6. For comparative purposes, the cross-correlogram of the simulated errors for $N_T = 50\,000$ is depicted in Fig. 3.20(b). The time span to be covered is selected long enough to properly observe and assess the stationary properties of the simulated processes. The fidelity of the reproduction is apparent, which

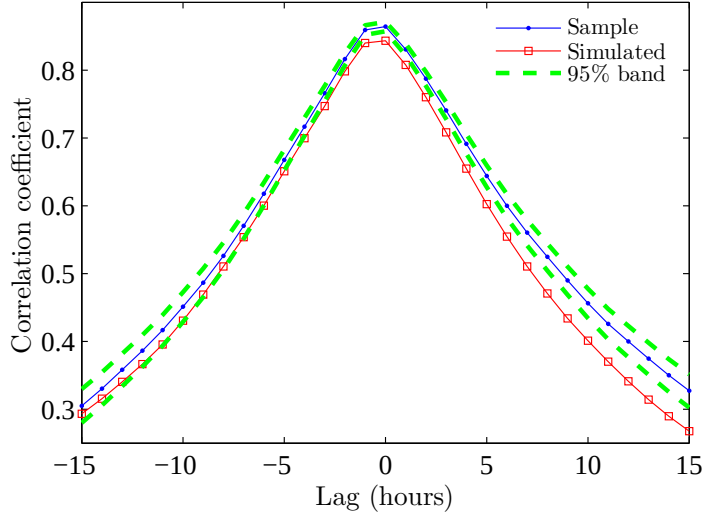


Figure 3.21: Comparison between sample and simulated cross-correlograms

highlights the effectiveness of the cross-correlation procedure.

Once the cross-correlated errors have been generated, they are introduced into the previously fitted ARMA models in order to obtain the simulated series of *normalized* wind speeds for sites A and B. Finally, we apply the inverse marginal transformation (3.10) to these normalized series in order to produce wind speed values distributed according to the Weibull fits, i.e.,

$$\begin{aligned} Y^A &= F_{Y^A}^{-1} [\Phi(Z^A)] , \\ Y^B &= F_{Y^B}^{-1} [\Phi(Z^B)] . \end{aligned}$$

In Fig. 3.21, the cross-correlogram of the simulated wind speed series, obtained using the above methodology, is compared with that estimated from the historical data. The 95% confidence limits are also shown. The lower and upper confidence bounds of the population lag- k cross-correlation coefficient, r_k^{lo} and r_k^{up} , respectively, are approximately computed by applying the Fisher

Transformation [111] to the sample cross-correlation coefficient ρ_k , that is,

$$r_k^{\text{lo}} = \tanh \left[\frac{1}{2} \log \left(\frac{1 + \rho_k}{1 - \rho_k} \right) + \frac{\text{erfinv}(\beta - 1) \sqrt{2}}{\sqrt{n - k - 3}} \right] \quad (3.18)$$

$$r_k^{\text{up}} = \tanh \left[\frac{1}{2} \log \left(\frac{1 + \rho_k}{1 - \rho_k} \right) - \frac{\text{erfinv}(\beta - 1) \sqrt{2}}{\sqrt{n - k - 3}} \right] \quad (3.19)$$

where $\text{erfinv}(\cdot)$ is the value of the inverse error function, n is the sample size of the available historical data set, and β is the confidence level (equal to 0.05 in this case).

Note that the methodology designed to deal with quasi-contemporaneous stochastic processes succeeds in capturing the shape of the sample cross-correlogram with a remarkable degree of precision, albeit some cross-correlation coefficients fall out of the confidence limits. This is mostly due to the irremediable inaccuracies associated with the fitting processes (both ARMA and Weibull fits) and with the cross-correlogram reproduction task. It is also important to underline the significant spatial correlation existing between wind speeds at sites A and B, with a high contemporaneous correlation coefficient equal to 0.8643, which is due to the geographical proximity of both sites.

This strong correlation can be graphically appreciated in Fig. 3.22(a), where a 48-h sample from the historical time series is depicted. Fig. 3.22(b) illustrates a 48-h extract from the simulated wind speed series. By simple inspection, one can intuitively realize that the simulation replicates the cross-correlated behavior.

Finally, in order to illustrate the efficaciousness of transformation (3.9)–(3.10) to preserve the original marginal distribution of the wind stochastic processes, Fig. 3.23 shows, for wind site A, a comparison between the empirical cdf of the simulated series and the cdf of the Weibull distribution estimated from the historical data. Their similarity is clear from this figure. So as to complement this graphical contrast by giving simulation error values, Table 3.2 offers a numerical comparison between the 5, 25, 50, 75, and 95% percentiles computed from both the simulated and the historical wind series. The last

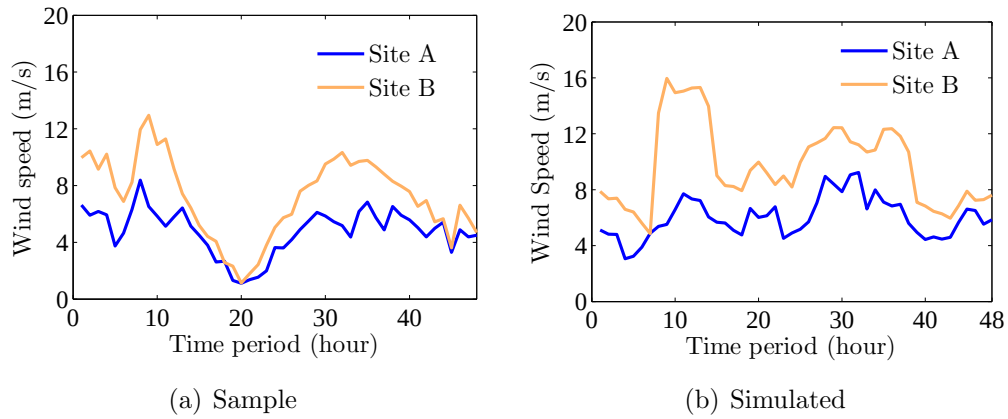


Figure 3.22: A 48-hour extract from the historical and simulated wind speed series at sites A and B

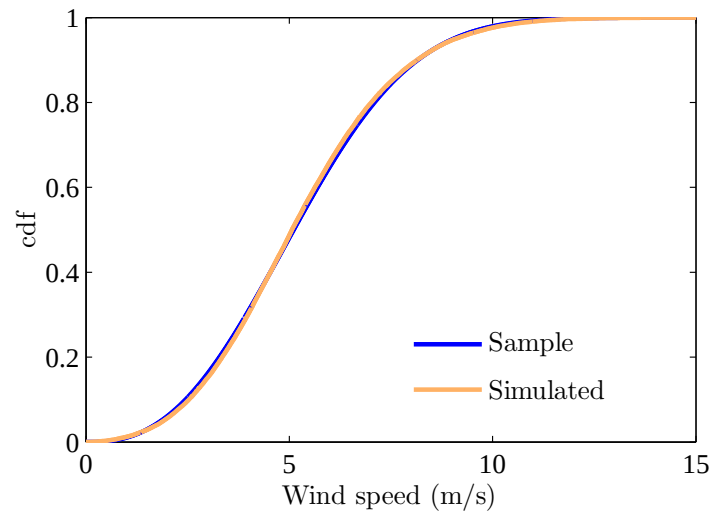


Figure 3.23: Comparison between fitted and simulated cumulative probability functions for wind site A

row of this table provides the error of the simulation relative to the sample percentile. Note that this error keeps reasonably low.

Table 3.2: Percentile comparison for wind site A

	Percentile				
	5	25	50	75	95
Sample	1.85	3.62	5.11	6.69	9.04
Simulated	1.91	3.68	5.06	6.61	9.09
Error (%)	3.31	1.63	0.85	1.22	0.56

3.3.8.2 Wind Speed Case Studies 2, 3 and 4

In the following, the results obtained for three additional case studies are presented, discussed and compared. These case studies are described below:

Case study 2 tackles the modeling of wind speeds at sites A and C (see Fig. 3.16). The historical data used for this purpose are comprised between June 1 and December 31, 2007.

Case study 3 analyzes wind speeds at sites A and D (see Fig. 3.16). The data set employed in this analysis includes wind speed measurements from June 1 to December 31, 2007.

Case study 4 focuses on modeling wind speeds at sites A and E (see Fig. 3.16). The wind data series used in this study were collected in the time span between June 1 and December 31, 2004.

We should point out that, due to data limitations, the time interval covered by the historical wind speed series in case studies 2 and 3 is different from that of case study 4.

The type of ARMA model fitted for wind sites C, D, and E is indicated in Table 3.3.

In Fig. 3.24, the cross-correlograms of both the residuals derived from the ARMA fits and the errors simulated by the proposed methodology are plotted for the three case studies. Only those correlation coefficients with a p-value smaller than a significance level $\alpha = 0.05$ are depicted. The rest of them are considered to be nil. In view of these cross-correlograms, the following observations are in order:

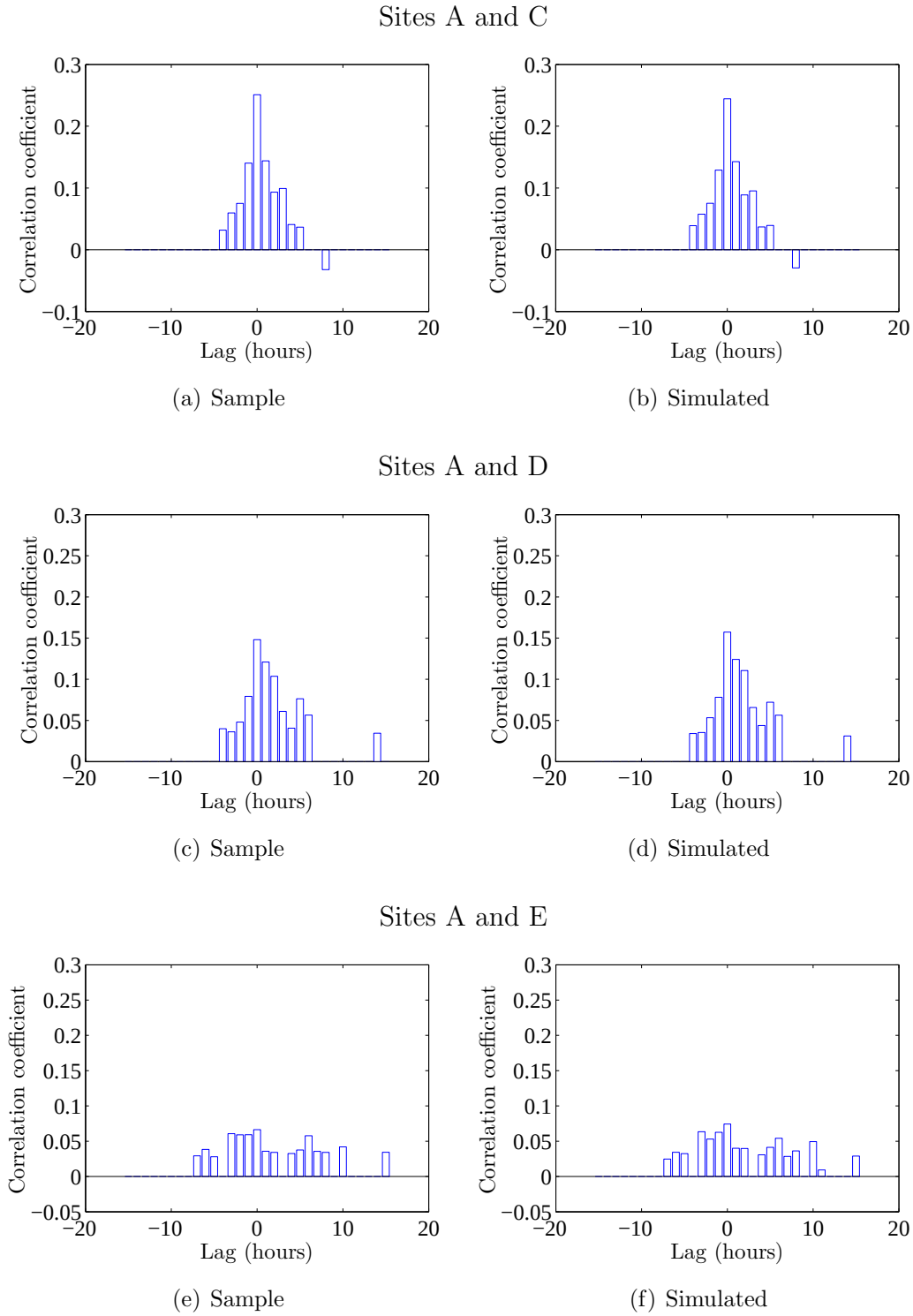
Figure 3.24: Cross-correlograms of errors with $\alpha = 0.05$

Table 3.3: Stochastic processes for wind sites C, D, and E

Site	Time series model
C	ARMA(1,3)
D	ARMA(1,2)
E	ARMA(2,2)

1. In terms of spatial correlation, case studies 1 and 2 are very similar. They are both characterized by cross-correlograms with a well-defined triangular shape in which the contemporaneous correlation coefficient stands out among the rest. This similarity is mainly due to the fact that distances A-B (31.05 km) and A-C (45.80 km) are of the same order of magnitude. However, the contemporaneous correlation between wind sites A and B is a slightly greater than that between sites A and C. In this sense, note that A is closer to B than to C.
2. As the distance between wind sites increases, the triangular structure characterizing the cross-correlograms flattens. In fact, the cross-correlogram of sites A and E is not a triangle, rather it can be described as a rectangle made up of a finite number of correlation coefficients with none particularly standing out from the others.
3. The suggested technique succeeds in reproducing by simulation the sample cross-correlograms. Moreover, in light of the results obtained from these case studies, its accuracy (in absolute terms) is observed to be independent of the distance between wind sites.

Fig. 3.25 is analogous to Fig. 3.24, but considering a significance level α equal to 0.2. Consequently, a greater number of correlation coefficients are taken into account in the analysis. It should be noted that this increase in the significance level introduces important modifications into the cross-correlogram of wind sites A and E, to which a notable number of correlations coefficients, previously disregarded, are now added. Furthermore, these new correlations

are not as *negligible* as they are in the other case studies if compared with the rest of coefficients composing such cross-correlogram.

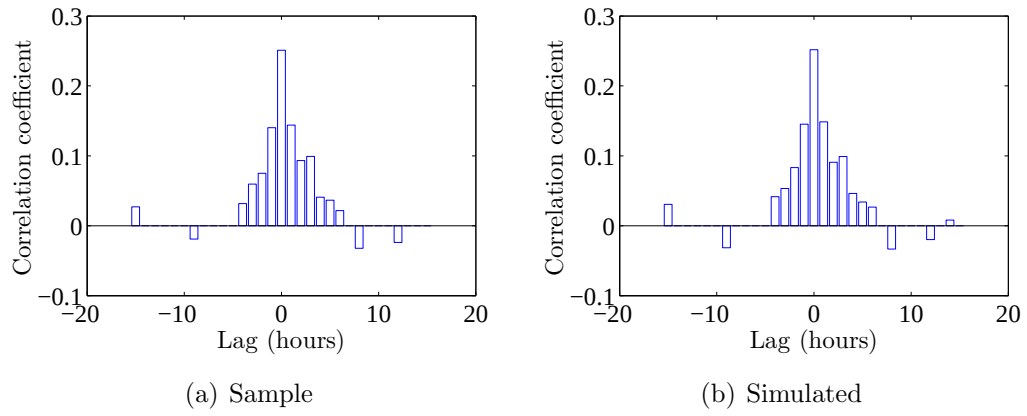
In Fig. 3.26, the cross-correlograms of the sample and simulated wind speed series are compared for each case study. The 95% confidence limits obtained from the historical series are also drawn. The difference between the graphics labelled as “ $\alpha = 0.05$ ” and those tagged as “ $\alpha = 0.2$ ”, stems from the significance level imposed on the residual cross-correlations coefficients to be reproduced in the simulation process. Note that the consideration of a higher significance level, or equivalently, a greater number of residual cross-correlations coefficients, does not have a significant effect on the simulated cross-correlograms in case studies 2 and 3 (A-C and A-D), but just slight improvements are observed, especially for high lags. The reason for this is that the residual cross-correlation coefficients making up the characteristic triangular structure referred to above explain almost completely the spatial interrelations between the wind sites defining these two case studies. In contrast, the impact of the considered significance level on the simulated cross-correlogram of case study 3 (A-E) is much more remarkable. In this respect, it should be noted that, for $\alpha = 0.05$, the simulated cross-correlogram is far from falling within the 95% confidence limits, as opposed to what happens if $\alpha = 0.2$. In this case, the residual cross correlations that are disregarded if the significance level is reduced from 0.2 to 0.05 are important in relative terms, that is, in comparison with the magnitude of the rest of coefficients.

Concerning computational issues, all the simulations have been carried out using Matlab 7.0 on an Intel Core Duo, 1.83 GHz and 1 GB of RAM. CPU time required to generate 1000 wind speed scenarios covering 1 day (24 hourly periods) is around 3 minutes for all the considered case studies and their variants ($\alpha = 0.05$ or $\alpha = 0.2$). If the number of hourly periods is increased up to 2200 (i.e., wind speed scenarios spanning three months approximately), the CPU time keeps below 7 minutes in all cases.

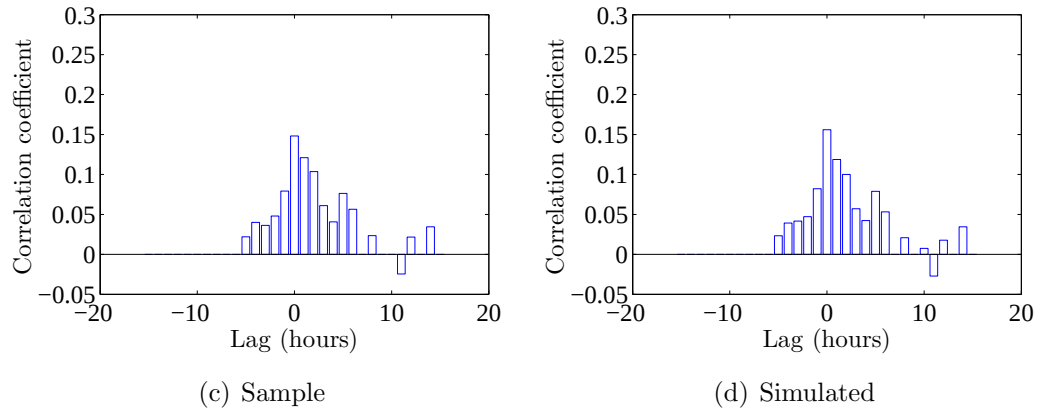
In short, the results presented in these four case studies allow us to conclude that the proposed wind scenario generation procedure performs satisfactorily in:

1. Capturing the dynamic relationships in time series data by means of the

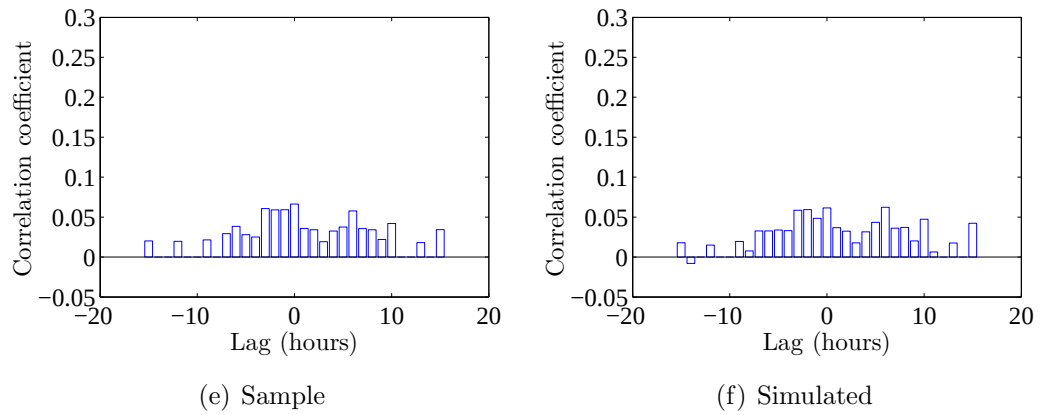
Sites A and C



Sites A and D



Sites A and E

Figure 3.25: Cross-correlograms of errors with $\alpha = 0.2$

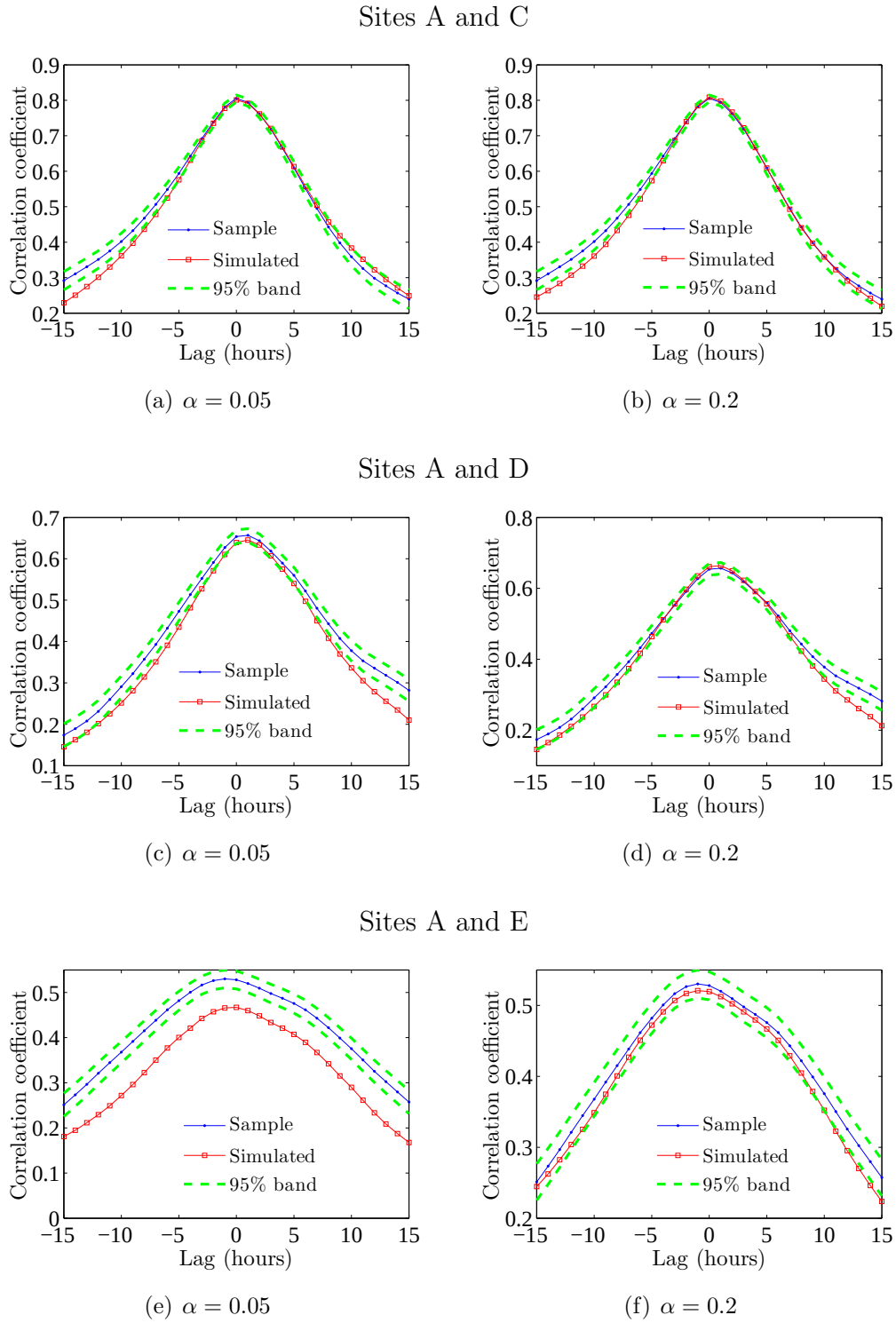


Figure 3.26: Comparison between sample and simulated cross-correlograms

appropriate fit of ARMA models (Section 3.3.2).

2. Reproducing the marginal distribution of the stochastic process modeling wind speed at each site (Section 3.3.4).
3. Preserving the main cross-correlations among the stochastic processes representing wind speed at different locations (Section 3.3.6).

3.4 Scenario Reduction

3.4.1 Motivation

The scenario generation techniques described in Section 3.3 are based on a sampling approach. That is, after identifying the time series model that best represents the continuous stochastic process under study, a repeated random generation of white noises is performed to produce a discrete approximation in the form of a scenario set. Consequently, in order for this approximation to be accurate, a high number of scenarios is usually required. Given that the computational burden of a stochastic programming model rapidly increases with the number of scenarios, a mathematical tool aimed to efficiently reduce such a number becomes a need. In other words, the necessity of reconciling scenario generation and computational tractability is what justifies the existence of *scenario reduction techniques*.

In more detail, a scenario reduction methodology seeks to downsize a scenario set while still keeping as intact as possible the stochastic information embedded in it. Logically, the quality of the reduction process is measured in terms of the solution to the underlying optimization problem. In this line, a *good* reduction process results in a reduced scenario set that yields an optimal solution close in value to the solution obtained from the initial one.

Next, we explain in detail two scenario reduction procedures, both relying on the concept of *probability distance*. Roughly speaking, a probability distance allows us to quantify the *closeness* of two different scenario sets that represents the same stochastic process.

3.4.2 Scenario Reduction Using a Probability Distance

In two-stage stochastic linear programming problems, it is possible to reduce a large scenario set to a simpler one that is close to the original one if measured by a so-called *probability distance*. Under mild conditions on the problem data, it can be shown that the optimal value of the simpler problem (considering the reduced scenario set) is close to the value of the solution to the original problem (considering the original scenario set) if the scenario sets are sufficiently close in terms of such a probability distance. Reference [182] constitutes a comprehensive review of the theoretical bases behind the notion of probability distance. Likewise, its application to scenario reduction is analyzed in detail in [74].

Consider a probability distribution Q defined over the scenario set Ω . The problem of how to reduce optimally the set Ω can be rigorously stated as follows: determine a scenario subset $\Omega_S \subset \Omega$ of prescribed cardinality N_{Ω_S} or accuracy and assign new probabilities to the preserved scenarios such that the corresponding reduced probability distribution Q' defined over the subset Ω_S is the closest to the original distribution Q in terms of a probability distance.

The most common probability distance used in stochastic programming is the *Kantorovich distance*, $D_K(\cdot)$, defined between two probability distributions Q and Q' by the problem

$$D_K(Q, Q') = \inf_{\eta} \left\{ \int_{\Omega \times \Omega} \nu(\omega, \omega') \eta(d\omega, d\omega') : \right. \\ \left. \int_{\Omega} \eta(\cdot, d\omega') = Q, \quad \int_{\Omega} \eta(d\omega, \cdot) = Q' \right\}. \quad (3.20)$$

Problem (3.20) is known as *Monge-Kantorovich mass transportation problem*. Further details on this problem are provided in [182]. In (3.20), $\nu(\omega, \omega')$ is a nonnegative, continuous, symmetric function, often referred to as *cost function*, and the minimum is taken over all joint probability distributions defined on $\Omega \times \Omega$. In addition, $D_K(\cdot)$ can only be properly called Kantorovich distance if function $\nu(\cdot)$ is given by a norm.

In the case that the probability distributions Q and Q' are discrete distribu-

tions defined over the set of scenarios Ω and Ω_S , respectively, the Kantorovich distance (3.20) boils down to

$$D_K(Q, Q') = \min_{\eta} \left\{ \sum_{\substack{\omega \in \Omega \\ \omega' \in \Omega_S}} \nu(\omega, \omega') \eta(\omega, \omega') : \right. \\ \eta(\omega, \omega') \geq 0, \forall \omega \in \Omega, \forall \omega' \in \Omega_S, \\ \sum_{\omega' \in \Omega_S} \eta(\omega, \omega') = \pi_{\omega}, \forall \omega \in \Omega, \\ \left. \sum_{\omega \in \Omega} \eta(\omega, \omega') = \tau_{\omega'}, \forall \omega' \in \Omega_S \right\}, \quad (3.21)$$

where π_{ω} and $\tau_{\omega'}$ represent the probabilities of scenarios ω and ω' in sets Ω and Ω_S according to probability distributions Q and Q' , respectively. Observe that the Kantorovich distance $D_K(Q, Q')$ in (3.21) represents the optimal value of a linear transportation problem.

Given that the decision-making problems addressed in this thesis solely present stochasticity in the objective function and right-hand sides, the Kantorovich distance (3.21) can be equivalently determined as

$$D_K(Q, Q') = \sum_{\omega \in \Omega \setminus \Omega_S} \pi_{\omega} \min_{\omega' \in \Omega_S} \nu(\omega, \omega'). \quad (3.22)$$

Further information on this equivalency can be found, e.g., in [74].

Expression (3.22) can be used to derive several heuristics for generating reduced scenario sets that are close to an original set. Specifically, two different heuristic algorithms can be developed, namely, the *backward reduction* or the *forward selection*, depending on whether the subset $\Omega_S \subset \Omega$ is built by *eliminating* or *selecting* scenarios from the initial set Ω .

The decision-making problems dealt with in this dissertation are computationally intensive in general, and as such, the scenario sets representing the stochastic processes involved (market prices and wind power production) often require a preliminary drastic reduction so as to enable tractability and/or enhance efficiency. In these cases, the forward selection algorithm exhibits a

better performance. For this reason, we focus our study on this algorithm hereinafter.

In more detail, the forward selection algorithm is an iterative greedy process starting with an empty set ($\Omega_S = \emptyset$). In each iteration, from the set of non-selected scenarios ($\Omega \setminus \Omega_S$), the scenario which minimizes the Kantorovich distance between the reduced and original sets is selected. Then, this scenario is included in the reduced set Ω_S . The algorithm stops if either a specified number of scenarios or a certain Kantorovich distance is attained.

Note that this scenario reduction technique is a heuristic, with no known performance guarantee. The reduced scenario set generated by the forward selection algorithm is not necessarily the closest in terms of the Kantorovich distance to the original set (over all reduced sets of the same cardinality). Moreover, we have no guarantee that the reduced set gives a good approximation to the optimal value of the original problem. Nevertheless, empirical results reported in the literature (e.g., in [74, 100, 152]) indicate that the reduced sets defined by the forward selection algorithm perform well in practice.

We describe below the forward selection algorithm step by step.

3.4.3 Forward Selection

Let us denote the original scenario set as Ω . The forward selection algorithm produces a succession of reduced sets $\Omega_S^{[0]}, \Omega_S^{[1]}, \dots, \Omega_S^{[i]}, \dots, \Omega_S^*$, of increasing cardinality, where the set Ω_S^* is the target of the search. To this end, the forward selection algorithm works as indicated in the following.

Step 0: The initial step consists in computing the function $\nu(\cdot)$ defining the Kantorovich distance (3.22) for each pair of scenarios. Two different variants of this algorithm are defined depending on how the cost function $\nu(\cdot)$ is calculated. These variants are explained below in detail.

Step 1: The iterative process begins with the choice of a starting scenario, that is, with the scenario from which the reduced scenario set is built.

Mathematically, the starting scenario (ω_1) is obtained from

$$\omega_1 = \arg\left\{\min_{\omega' \in \Omega} \sum_{\omega \in \Omega} \pi_{\omega} \nu(\omega, \omega')\right\}. \quad (3.23)$$

The starting scenario can be interpreted as the most equidistant one from the rest. In other words, this first scenario can be seen as the *average* scenario.

Step i : Starting from the scenario selected in step 1, in each iteration a new scenario is added to the reduced scenario set until this is considered to be close enough to the original one. For this purpose, this selection is carried out using

$$\omega_i = \arg\left\{\min_{\omega' \in \Omega_J^{[i-1]}} \sum_{\omega \in \Omega_J^{[i-1]} \setminus \{\omega'\}} \pi_{\omega} \min_{\omega'' \in \Omega_S^{[i-1]} \cup \{\omega\}} \nu(\omega, \omega'')\right\}, \quad (3.24)$$

where $\Omega_J^{[i]}$ represents the set composed of those scenarios which have not been selected in the first i steps of the algorithm and $\Omega_S^{[i]}$ represents the set comprising the selected scenarios until step i . Note that $\Omega_J^{[i]} \cup \Omega_S^{[i]} = \Omega$, $\Omega_J^{[0]} = \Omega$, $\Omega_S^{[0]} = \emptyset$, $\Omega_J^{[i]} = \Omega_J^{[i-1]} \setminus \{\omega_i\}$ and $\Omega_S^{[i]} = \Omega_S^{[i-1]} \cup \{\omega_i\}$.

This step is repeated $N_{\Omega_S^*} - 1$ times, where $N_{\Omega_S^*}$ is the number of scenarios comprising the reduced set Ω_S^* .

Step $N_{\Omega_S^*} + 1$: In this step, an optimal redistribution of probabilities is carried out. It consists in adding the probabilities of those scenarios which have not been finally selected ($\omega \in \Omega_J^*$, with $\Omega_J^* \cup \Omega_S^* = \Omega$) to the probabilities of those comprising the reduced set ($\omega \in \Omega_S^*$).

Mathematically, this redistribution of probabilities can be accomplished as follows,

$$\pi_{\omega}^* = \pi_{\omega} + \sum_{\omega' \in J(\omega)} \pi_{\omega'}, \quad (3.25)$$

where $J(\omega)$ is defined as the set of scenarios $\omega' \in \Omega_J^*$ such that $\omega =$

$$\arg \min_{\omega'' \in \Omega_S^*} \nu(\omega'', \omega').$$

In other words, the probability of each non-selected scenario is aggregated to the probability of the closest selected scenario according to the function $\nu(\cdot)$. Thus, the reduced scenario set is made up of the scenarios $w \in \Omega_S^*$ with associated probability π_w^* .

A particularly efficient implementation of the algorithm described above is presented next and is referred to as *fast forward selection* algorithm, [100, 101]. Specifically, the efficiency of the algorithm is enhanced by updating cost function $\nu(\cdot)$ each time a new scenario is selected, thus substantially reducing the computational burden associated with the operation (3.24).

Algorithm

Step 0: Compute cost function $\nu(\omega, \omega')$ for each pair of scenarios ω and ω' in Ω .

Step 1: Compute $d_\omega = \sum_{\substack{\omega'=1 \\ \omega' \neq \omega}}^{N_\Omega} \pi_{\omega'} \nu(\omega, \omega')$, $\forall \omega \in \Omega$.
 Select $\omega_1 \in \arg \min_{\omega \in \Omega} d_\omega$.
 Set $\Omega_J^{[1]} = \{1, \dots, N_\Omega\} \setminus \omega_1$.

Step i : Compute

$$\begin{aligned} \nu^{[i]}(\omega, \omega') &= \min\{\nu^{[i-1]}(\omega, \omega'), \nu^{[i-1]}(\omega, \omega_{i-1})\}, \forall \omega, \omega' \in \Omega_J^{[i-1]} \\ d_\omega^{[i]} &= \sum_{\omega' \in \Omega_J^{[i-1]} \setminus \{\omega\}} \pi_{\omega'} \nu^{[i]}(\omega', \omega), \quad \forall \omega \in \Omega_J^{[i-1]} \\ \text{Select } \omega_i &\in \arg \min_{\omega \in \Omega_J^{[i-1]}} d_\omega^{[i]}. \\ \text{Set } \Omega_J^{[i]} &= \Omega_J^{[i-1]} \setminus \omega_i. \end{aligned}$$

Step $N_{\Omega_S^*} + 1$: It follows that

$\Omega_J^* = \Omega_J^{[N_{\Omega_S^*}]}$ is the final set of indices of discarded scenarios.

$\Omega_S^* = \Omega \setminus \Omega_J^*$ is the set of indices of selected scenarios.

The probabilities of selected scenarios $\omega \in \Omega_S^*$ are computed as

$$\begin{aligned} \pi_\omega^* &\leftarrow \pi_\omega + \sum_{\omega' \in J(\omega)} \pi_{\omega'}, \quad \forall \omega \in \Omega_S^*, \text{ where} \\ J(\omega) &= \{\omega' \in \Omega_J^* | \omega = j(\omega')\}, \quad j(\omega') \in \arg \min_{\omega'' \in \Omega_S^*} \nu(\omega'', \omega'). \end{aligned}$$

Next, we present two variants of this scenario reduction algorithm based on two different manners of computing the cost function $\nu(\cdot)$ in Step 0 above.

3.4.3.1 Variant 1: Cost function based on the norm of the difference between pairs of random vectors

Consider a stochastic process λ characterized by an initial set of scenarios λ_ω , for all $\omega \in \Omega$, i.e., $\lambda = \{\lambda_\omega, \omega = 1, 2, \dots, N_\Omega\}$.

Cost function $\nu(\cdot)$ can be computed as the norm of the difference between pairs of scenarios, i.e.,

$$\nu(\omega, \omega') = \|\lambda_\omega - \lambda_{\omega'}\|, \quad (3.26)$$

where $\|\cdot\|$ stands for the norm operator.

Note that, in order for the cost function $\nu(\cdot)$ to be computed according to (3.26), just the original scenario discretization $\{\lambda_\omega, \forall \omega \in \Omega\}$ of the stochastic process λ is required and therefore, this first variant of the forward selection algorithm yields the same reduced set Ω_S irrespective of the stochastic programming problem to be solved.

Further details on this variant can be found in [74, 100].

3.4.3.2 Variant 2: Cost function based on the objective function of a single-scenario problem

Consider the deterministic equivalent formulation (3.27)–(3.30) of a two-stage stochastic programming problem (see Appendix A), namely

$$\begin{aligned} & \text{Minimize}_{x, y_\omega} \\ z = & c^\top x + \sum_{\omega \in \Omega} \pi_\omega q_\omega^\top y_\omega \end{aligned} \quad (3.27)$$

subject to

$$Ax = b \quad (3.28)$$

$$W_\omega y_\omega = h_\omega - T_\omega x, \forall \omega \in \Omega \quad (3.29)$$

$$x \geq 0, y_\omega \geq 0; \forall \omega \in \Omega, \quad (3.30)$$

where, for simplicity, we just write q_ω , y_ω , T_ω , W_ω , and h_ω , instead of $q(\lambda_\omega)$, $y(\lambda_\omega)$, $T(\lambda_\omega)$, $W(\lambda_\omega)$, and $h(\lambda_\omega)$. For convenience, this problem is denoted by SP henceforth.

We also define $DP_\mathcal{E}$ as the deterministic expected value problem associated with SP, i.e., $DP_\mathcal{E}$ is equal to problem SP in which the random vector λ is replaced by its expected value, $\hat{\lambda} = \mathcal{E}\{\lambda\}$. Mathematically, this problem is formulated as

$$\text{Minimize}_{x,y(\hat{\lambda})} \quad (3.31)$$

$$z_\mathcal{E} = c^\top x + q(\hat{\lambda})^\top y(\hat{\lambda}) \quad (3.32)$$

subject to

$$Ax = b \quad (3.33)$$

$$W(\hat{\lambda})y(\hat{\lambda}) = h(\hat{\lambda}) - T(\hat{\lambda})x \quad (3.34)$$

$$x \geq 0, y(\hat{\lambda}) \geq 0. \quad (3.35)$$

Let us denote the optimal value of the first-stage variables in problem $DP_\mathcal{E}$ as $\hat{x}$.

Next, we build the single-scenario problem DP_ω as the one resulting from problem SP if the stochastic process λ is replaced by its realization in scenario ω , $\lambda(\omega)$, and its first-stage variables are fixed to $\hat{x}$, i.e.,

$$\text{Minimize}_{y_\omega} \quad (3.36)$$

$$z_\omega = c^\top \hat{x} + q_\omega^\top y_\omega \quad (3.37)$$

subject to

$$W_\omega y_\omega = h_\omega - T_\omega \hat{x} \quad (3.38)$$

$$y_\omega \geq 0. \quad (3.39)$$

Finally, the cost function $\nu(\cdot)$ is defined as

$$\nu(\omega, \omega') = |z_\omega - z_{\omega'}|, \quad (3.40)$$

where z_ω is the optimal value of the objective function (3.37) of problem DP_ω .

From expression (3.40), it can be inferred that, unlike in variant 1, the reduction process in this second variant of the forward selection algorithm is dependent on the stochastic programming problem SP to be solved to obtain z_ω . In other words, this variant incorporates information on the structure of the problem into the cost function $\nu(\cdot)$, which, in practice, results in reduced sets substantially *closer* to the original one. However, even though each single-scenario problem DP_ω is much easier to solve than the target problem SP, it is, by far, more involved than the calculation of a norm. Consequently, this second variant is computationally more demanding than the first one.

The following two examples are aimed to illustrate the two described variants of the *fast* forward selection algorithm.

3.4.4 Example: Forward Selection Variant 1

Suppose that the power production P^W of a 100-MW wind farm in a given time period can be statistically represented by the four wind power scenarios $P_\omega^W, \omega = 1, \dots, 4$, listed in Table 3.4, with associated probabilities $\pi_\omega, \omega = 1, \dots, 4$. It follows that $N_\Omega = 4$. Likewise, note that, since just one period is considered, the wind power stochastic process boils down to a random variable.

Table 3.4: Wind power scenarios

Scenario ω	1	2	3	4
P_ω^W (MW)	5	40	60	100
π_ω	0.25	0.20	0.20	0.35

Suppose also that we need to reduce the number of wind power scenarios to two, i.e., $N_{\Omega_S^*} = 2$. For this purpose, we decide to use variant 1 of the fast

forward selection algorithm, which works as follows:

Step 0: Before starting the iterative process, we compute cost function $\nu(\omega, \omega')$
 $= \|P_\omega^W - P_{\omega'}^W\|, \forall \omega, \omega' \in \Omega$, where $\Omega = \{1, 2, 3, 4\}$. The values of function ν can be conveniently arranged into a symmetric matrix with zero diagonal elements,

$$\nu = \begin{pmatrix} 0 & 35 & 55 & 95 \\ 35 & 0 & 20 & 60 \\ 55 & 20 & 0 & 40 \\ 95 & 60 & 40 & 0 \end{pmatrix} \text{ MW.}$$

Step 1: For each scenario ω candidate to be selected from Ω , we choose the one that minimizes the resulting Kantorovich distance between the reduced and original sets, Ω_S and Ω , respectively,

$$\begin{aligned} d_1 &= \pi_2 \nu(1, 2) + \pi_3 \nu(1, 3) + \pi_4 \nu(1, 4) = 51.25 \text{ MW} \\ d_2 &= \pi_1 \nu(2, 1) + \pi_3 \nu(2, 3) + \pi_4 \nu(2, 4) = 33.75 \text{ MW} \\ d_3 &= \pi_1 \nu(3, 1) + \pi_2 \nu(3, 2) + \pi_4 \nu(3, 4) = 31.75 \text{ MW} \\ d_4 &= \pi_1 \nu(4, 1) + \pi_2 \nu(4, 2) + \pi_3 \nu(4, 3) = 43.75 \text{ MW.} \end{aligned}$$

Therefore,

$$\begin{aligned} \Omega_S^{[1]} &= \{3\}, \\ \Omega_J^{[1]} &= \{1, 2, 4\}. \end{aligned}$$

Step 2: Next, we update the cost matrix as follows:

$$\begin{aligned} \nu^{[2]}(1, 2) &= \min\{\nu(1, 2), \nu(1, 3)\} = 35 \text{ MW} \\ \nu^{[2]}(1, 4) &= \min\{\nu(1, 4), \nu(1, 3)\} = 55 \text{ MW} \\ \nu^{[2]}(2, 1) &= \min\{\nu(2, 1), \nu(2, 3)\} = 20 \text{ MW} \\ \nu^{[2]}(2, 4) &= \min\{\nu(2, 4), \nu(2, 3)\} = 20 \text{ MW} \\ \nu^{[2]}(4, 1) &= \min\{\nu(4, 1), \nu(4, 3)\} = 40 \text{ MW} \end{aligned}$$

$$\nu^{[2]}(4, 2) = \min\{\nu(4, 2), \nu(4, 3)\} = 40 \text{ MW.}$$

Consequently,

$$\nu^{[2]} = \begin{pmatrix} 0 & 35 & 55 & 55 \\ 20 & 0 & 20 & 20 \\ 55 & 20 & 0 & 40 \\ 40 & 40 & 40 & 0 \end{pmatrix} \text{ MW.}$$

Considering the new cost matrix $\nu^{[2]}$, we select from $\Omega_J^{[1]}$ the scenario ω that, if introduced into the subsequent reduced set $\Omega_S^{[2]}$, minimizes the Kantorovich distance between this set and the original one, Ω ,

$$d_1^{[2]} = \pi_2 \nu^{[2]}(2, 1) + \pi_4 \nu^{[2]}(4, 1) = 18.00 \text{ MW}$$

$$d_2^{[2]} = \pi_1 \nu^{[2]}(1, 2) + \pi_4 \nu^{[2]}(4, 2) = 22.75 \text{ MW}$$

$$d_4^{[2]} = \pi_1 \nu^{[2]}(1, 4) + \pi_2 \nu^{[2]}(2, 4) = 17.75 \text{ MW.}$$

Hence,

$$\Omega_S^{[2]} = \Omega_S^* = \{3, 4\}$$

$$\Omega_J^{[2]} = \Omega_J^* = \{1, 2\}.$$

Step 3: The scenario reduction algorithm ends with the optimal *transfer* of probabilities from the set of non-selected scenarios, Ω_J^* , to the set of selected ones, Ω_S^* .

Given that

1. Scenario 3 in Ω_S^* is the *closest* one to scenario 1 in Ω_J^* ($\nu(1, 3) = 55$, while $\nu(1, 4) = 95$), and
2. scenario 3 in Ω_S^* is the *closest* one to scenario 2 in Ω_J^* ($\nu(2, 3) = 20$, while $\nu(2, 4) = 60$),

it follows,

$$\pi_3^* = \pi_3 + \pi_2 + \pi_1 = 0.65$$

$$\pi_4^* = \pi_4 = 0.35.$$

In short, the variant 1 of the forward selection algorithm provides a reduced scenario set $\Omega_S^* = \{3, 4\}$ with associated probabilities $\pi_3^* = 0.65$ and $\pi_4^* = 0.35$.

3.4.5 Example: Forward Selection Variant 2

In this example, we aim to reduce the wind power scenario set provided in Table 3.4 to two scenarios by using the variant 2 of the fast forward selection algorithm. Previous to tackling the reduction process itself, this variant requires to define the problem to be solved considering this scenario set (problem SP).

Let us assume that the producer owning the wind farm is interested in selling its wind power production to the day-ahead electricity market. Since the final power produced by its wind farm is uncertain, the producer has no option other than to submit an energy offer (P^D) to the day-ahead market, subsequently *fixing* its likely excess (Δ^+) or deficit of generation (Δ^-) through a real-time balancing process. Resorting to this balancing mechanism entails a cost for the wind producer, usually referred to as *imbalance cost*.

The decision on the amount of energy to be sold in the day-ahead market with a view to minimize the imbalance cost can be properly made by means of the following optimization problem,

$$\begin{aligned} & \text{Maximize}_{P^D; \Delta_\omega^+, \forall \omega; \Delta_\omega^-, \forall \omega} \\ z = & \sum_{\omega=1}^{N_\Omega} \pi_\omega (\lambda^D P_\omega^W - \lambda^D (1 - r^+) \Delta_\omega^+ - \lambda^D (r^- - 1) \Delta_\omega^-) \end{aligned} \quad (3.41)$$

subject to

$$0 \leq P^D \leq P^{W, \max} \quad (3.42)$$

$$\Delta_\omega^+ - \Delta_\omega^- = P_\omega^W - P^D, \forall \omega \quad (3.43)$$

$$0 \leq \Delta_\omega^+ \leq P_\omega^W, \forall \omega \quad (3.44)$$

$$0 \leq \Delta_\omega^- \leq P^{W, \max}, \forall \omega, \quad (3.45)$$

where λ^D and $P^{W, \max}$ are, respectively, the day-ahead market price and the

installed capacity of the wind farm (100 MW). Parameters r^+ and r^- are the ratios between the positive and negative imbalance prices and the day-ahead market price. As the wind power production P^W , constants λ^D , r^+ and r^- are indeed stochastic processes. Nevertheless, for simplicity, we consider them perfectly known and equal to \$20/MWh, 0.95, and 1.4, respectively.

The problem formulated above constitutes a simplified version of the short-term trading problem of a wind power producer, comprehensively analyzed in Chapter 4 of this thesis.

Once the target problem SP is rightly defined, we can proceed with the scenario reduction process:

Step 0: Calculate cost function $\nu(\omega, \omega') = |z_\omega - z_{\omega'}|, \forall \omega, \omega' \in \Omega$.

In order to determine the objective function values $z_\omega, \forall \omega \in \Omega$, we need to solve first the following deterministic expected-value problem $DP_\mathcal{E}$,

$$\begin{aligned} & \text{Maximize}_{\hat{P}^D, \hat{\Delta}^+, \hat{\Delta}^-} \\ z_\mathcal{E} &= (1125 - \hat{\Delta}^+ - 8\hat{\Delta}^-) \end{aligned}$$

subject to

$$\begin{aligned} 0 &\leq \hat{P}^D \leq 100 \\ \hat{\Delta}^+ - \hat{\Delta}^- &= 56.25 - \hat{P}^D \\ 0 &\leq \hat{\Delta}^+ \leq 56.25 \\ 0 &\leq \hat{\Delta}^- \leq 100, \end{aligned}$$

where the wind power random variable P^W has been replaced by its expected value: $\mathcal{E}\{P^W\} = 5 \times 0.25 + 40 \times 0.20 + 60 \times 0.20 + 100 \times 0.35 = 56.25$ MW. Likewise, parameters $P^{W, \max}$, λ^D , r^+ , and r^- have been replaced by their numerical values.

The solution to this problem is trivial: $\hat{P}^D = 56.25$ MW, $\hat{\Delta}^+ = \hat{\Delta}^- = 0$, and $z_\mathcal{E} = \$1125$.

Finally, the objective function values $z_\omega, \forall \omega \in \Omega$, are obtained from the family of single-scenario problems $DP_{\omega, \omega} = 1, \dots, 4$, which result

from problem SP if the wind power stochastic process is replaced by its realization in scenario ω , P_ω^W , and its first-stage variable P^D is fixed to $\hat{P}^D = 56.25$ MW. Specifically, each problem DP_ω is formulated as follows:

$$\begin{aligned} & \text{Maximize}_{\Delta_\omega^+, \Delta_\omega^-} \\ z_\omega &= (20P_\omega^W - \Delta_\omega^+ - 8\Delta_\omega^-) \end{aligned}$$

subject to

$$\begin{aligned} \Delta_\omega^+ - \Delta_\omega^- &= P_\omega^W - 56.25, \forall \omega \\ 0 &\leq \Delta_\omega^+ \leq P_\omega^W \\ 0 &\leq \Delta_\omega^- \leq 100. \end{aligned}$$

The solution to each problem DP_ω is also straightforward and is given by

$$\Delta_\omega^+ \text{ (MW)} = \begin{cases} 0, & \text{if } \omega = 1, 2 \\ P_\omega^W - 56.25, & \text{if } \omega = 3, 4. \end{cases}$$

$$\Delta_\omega^- \text{ (MW)} = \begin{cases} 56.25 - P_\omega^W, & \text{if } \omega = 1, 2 \\ 0, & \text{if } \omega = 3, 4. \end{cases}$$

$$z_\omega \text{ (\$)} = \begin{cases} 28P_\omega^W - 450, & \text{if } \omega = 1, 2 \\ 19P_\omega^W + 56.25, & \text{if } \omega = 3, 4. \end{cases}$$

Table 3.5 lists the numerical value of each objective function variable $z_\omega, \omega \in \Omega$, and its associated probability π_ω , i.e., the probability pertaining to the wind power realization P_ω^W from which the single-scenario problem DP_ω is built.

Therefore, we already have the required information to compute cost matrix ν , which results in

Table 3.5: Objective function values z_ω of single-scenario problems DP_ω

Scenario ω	1	2	3	4
z_ω (\$)	-310	670	1196.25	1956.25
π_ω	0.25	0.20	0.20	0.35

$$\nu = \$ \begin{pmatrix} 0 & 980 & 1506.25 & 2266.25 \\ 980 & 0 & 526.25 & 1286.25 \\ 1506.25 & 526.25 & 0 & 760 \\ 2266.25 & 1286.25 & 760 & 0 \end{pmatrix}.$$

From this point on, this second variant of the fast forward selection algorithm proceeds similarly to variant 1.

Step 1: Determine the scenario that, if selected, minimizes the resulting probability distance between the reduced and original sets, Ω_S and Ω , respectively,

$$d_1 = \pi_2 \nu(1, 2) + \pi_3 \nu(1, 3) + \pi_4 \nu(1, 4) = \$1290.4375$$

$$d_2 = \pi_1 \nu(2, 1) + \pi_3 \nu(2, 3) + \pi_4 \nu(2, 4) = \$800.4375$$

$$d_3 = \pi_1 \nu(3, 1) + \pi_2 \nu(3, 2) + \pi_4 \nu(3, 4) = \$747.8125$$

$$d_4 = \pi_1 \nu(4, 1) + \pi_2 \nu(4, 2) + \pi_3 \nu(4, 3) = \$975.8125.$$

Consequently,

$$\Omega_S^{[1]} = \{3\},$$

$$\Omega_J^{[1]} = \{1, 2, 4\}.$$

Step 2: Update the cost matrix and select the new scenario to be introduced into the reduced set,

$$\nu^{[2]}(1, 2) = \min\{\nu(1, 2), \nu(1, 3)\} = \$980$$

$$\nu^{[2]}(1, 4) = \min\{\nu(1, 4), \nu(1, 3)\} = \$1506.25$$

$$\nu^{[2]}(2, 1) = \min\{\nu(2, 1), \nu(2, 3)\} = \$526.25$$

$$\nu^{[2]}(2, 4) = \min\{\nu(2, 4), \nu(2, 3)\} = \$526.25$$

$$\nu^{[2]}(4, 1) = \min\{\nu(4, 1), \nu(4, 3)\} = \$760$$

$$\nu^{[2]}(4, 2) = \min\{\nu(4, 2), \nu(4, 3)\} = \$760.$$

Therefore,

$$\nu^{[2]} = \$ \begin{pmatrix} 0 & 980 & 1506.25 & 1506.25 \\ 526.25 & 0 & 526.25 & 526.25 \\ 1506.25 & 526.25 & 0 & 760 \\ 760 & 760 & 760 & 0 \end{pmatrix},$$

$$d_1^{[2]} = \pi_2 \nu^{[2]}(2, 1) + \pi_4 \nu^{[2]}(4, 1) = \$371.25$$

$$d_2^{[2]} = \pi_1 \nu^{[2]}(1, 2) + \pi_4 \nu^{[2]}(4, 2) = \$511$$

$$d_4^{[2]} = \pi_1 \nu^{[2]}(1, 4) + \pi_2 \nu^{[2]}(2, 4) = \$481.8125.$$

Hence,

$$\Omega_S^{[2]} = \Omega_S^* = \{1, 3\}$$

$$\Omega_J^{[2]} = \Omega_J^* = \{2, 4\}.$$

Step 3: Optimal redistribution of probabilities.

Considering that:

1. Scenario 3 in Ω_S^* is the *closest* one to scenario 2 in Ω_J^* ($\nu(2, 3) = 526.25$, while $\nu(2, 1) = 980$), and
2. scenario 3 in Ω_S^* is the *closest* one to scenario 4 in Ω_J^* ($\nu(4, 3) = 760$, while $\nu(4, 1) = 2266.25$),

it follows,

$$\pi_1^* = \pi_1 = 0.25$$

$$\pi_3^* = \pi_3 + \pi_2 + \pi_4 = 0.75.$$

In summary, the variant 2 of the forward selection algorithm provides a reduced scenario set $\Omega_S^* = \{1, 3\}$ with associated probabilities $\pi_1^* = 0.25$ and $\pi_3^* = 0.75$.

The optimal solution to the wind producer problem (3.41)–(3.45) for the original set of wind power scenarios (Table 3.4) is $P^{D*} = 5$ MW with an associated expected profit $z^* = \$1073.8$. If this problem is solved for the two-scenario sets provided by the variants 1 and 2 of the fast forward selection algorithm, the optimal amount of wind power to be traded in the day-ahead market results in 60 and 5 MW, respectively, with objective function values of \$1466 and \$883.75, in that order. Therefore, the variant 2 of the algorithm provides a reduced scenario set that yields a solution closer to the actual one in terms of both optimal decision variables and objective function at the expense of a higher computational cost. Notwithstanding the illustrative character of this example, these results are in accordance with those reported in the technical literature [152, 177].

3.5 Summary

Prior to solving a decision-making problem, uncertainties on input data need to be identified and modeled. Within a stochastic programming framework, such a modeling endeavor is based on the notion of *stochastic process* and usually requires generating a set of scenarios representing plausible realizations of the random parameters throughout the decision-making horizon.

This chapter introduces several strategies and techniques to produce scenario sets using time series analysis. These techniques are especially tailored to the modeling of electricity market prices and wind power production. Thus, strategies are provided to deal with the non-stationary character of electricity prices and the non-Gaussian nature of the speed of local winds. Likewise, a dedicated algorithm is developed to account for the statistical dependencies among wind speeds at different sites.

The number of scenarios required to appropriately represent stochastic processes in optimization problems is often large, which may lead to computational intractability. For this reason, this chapter concludes describing two efficient

scenario reduction procedures to trim down considerably the number of scenarios while preserving, as much as possible, the statistical information contained in them.

Chapter 4

Trading Wind Power in Short-Term Electricity Markets

4.1 Introduction

The amount of wind generation is rapidly increasing in many electric energy systems across the world. The large-scale electricity production based on the exploitation of wind energy resources is currently feasible and is gradually becoming a profitable business attracting the attention of investors. Three main driving forces are responsible for the growing relevance of wind generation: the increasing trend of fossil fuel prices (natural gas, oil, and coal) as less and less reserves are available, the public's reluctance to the installation of new nuclear power plants, and the need to curb carbon emissions to preserve a functioning atmosphere.

Given the vertiginous spread of wind power production, the transition of the wind generation activity towards its deregulation is a natural step in line with the current existence of markets for the competitive trade of electricity. However, wind power is non-dispatchable and uncertain. The unpredictable character of wind availability constitutes indeed the major obstacle to the otherwise straightforward incorporation of wind producers into a market framework in both technical and economic terms. The question here is “how can a wind power producer *survive* by itself in a competitive environment especially

designed for conventional (hydro-thermal) producers?”

From a technical point of view, the pernicious effect of wind generation uncertainty is eliminated by a balancing mechanism that corrects the lack or excess of production/consumption in real time. The balancing mechanism is provided by energy sources that are generally expensive but dispatchable, such as combined-cycle gas turbines. The balancing energy is made available in a succession of real-time markets and also in the form of reserves, which are dealt with in the next chapter.

From an economic perspective, in this thesis we consider that the cost of the energy required to maintain the system balance in real time is assumed by the market agents incurring energy deviations, that is, wind producers among others. Therefore, as a wind producer needs to rely more and more on the balancing mechanism to fix its energy production imbalances, its profitability and competitiveness deteriorates quickly. Thus, a key concern of a wind producer should be to reduce its need for balancing energy.

In this chapter, we introduce a stochastic programming model to derive hourly offering strategies for a wind producer operating within an electricity market that involves various short-term trading floors and a balancing mechanism. This model enables the wind producer to limit the risk of its profit variability at the cost of a small reduction in its expected profit.

4.2 Decision Framework

Without loss of generality, let us consider a wind power producer participating in a pool-based electricity market that includes three representative, independent, and successive short-term trading floors: the day-ahead market, the adjustment market, and the balancing market. We also assume that the wind producer has no market-power capability in either of the aforementioned markets, that is, it has no influence on the electricity prices settled in these markets. Its objective is to maximize its expected profits from trading energy in the day-ahead and adjustment markets, while reducing as much as possible the cost incurred in the balancing market due to energy deviations.

Each of these three markets is cleared through a single auction process.

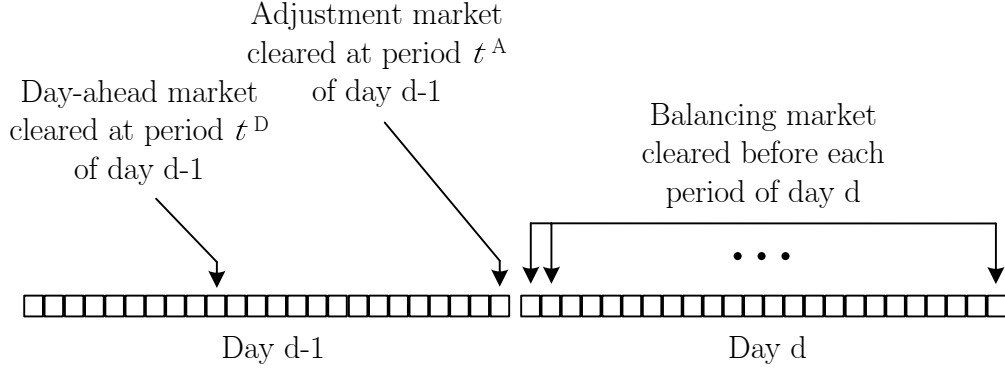


Figure 4.1: Decision framework of the wind producer problem.

More details about market-clearing algorithms are provided in Sections 4.3.1 and 4.3.2, as well as in Chapter 5. The time framework for market clearing is as represented in Fig. 4.1. The day-ahead market, which involves energy transactions coming into effect during the whole day d , is cleared at a given time period t^D of day $d - 1$. Due to the usual delay between the closure of this market and the beginning of the energy delivery horizon (day d), the electricity pool includes an adjustment market, where it is possible to take corrective actions. These actions are intended to modify the outcomes of the day-ahead market in order to make them closer to the expected production for day d . This adjustment market is therefore of especial usefulness for wind producers as it allows them to incorporate into their offers the certainty gained on production availability during the time interval between the closure of both markets by updating their generation forecasts and thus, reducing the associated uncertainty. This phenomenon is formally described in Section 4.3.4. For convenience, we consider that the adjustment market is cleared just before the delivery horizon (day d), as indicated in Fig. 4.1. Consequently, energy transactions in this market cover the whole day d . Nevertheless, additional adjustments markets can be sequentially arranged throughout the time span comprising the closure of the day-ahead market and the end of day d , and as a result, energy transactions resolved in them may cover the whole day d or just a part of it. Finally, the balancing market ensures the real-time balance

between generation and demand by offsetting the differences between the real-time operation and the last energy program settled in the previous markets. For that reason, this market is cleared just before each time period of day d . Therefore, through this last market, energy imbalances are corrected and priced according to the settlement mechanism for imbalance prices explained in Section 4.3.2 below.

In short, a wind power producer competing within a market structure as the one described above and graphically represented in Fig. 4.1 should: i) decide the offering curve to be submitted to the day-ahead market for each time period of day d ; ii) modify in the adjustment market the energy offers submitted to the day-ahead market according to the wind power forecast updates; and iii) amend its energy deviations in the balancing market for each time period of day d .

The short-term decision-making process of a wind power producer is analyzed in Section 4.4.3 from a stochastic programming viewpoint. Hereinafter, the time horizon spanning this process is assumed to be comprised of hourly periods.

4.3 Unravelling the Wind Producer Problem

In this section, we analyze in detail the key factors having a special bearing on the decision-making process of a wind power producer that seeks to maximize its profit in an electricity market framework, namely, the *imbalance cost* and the *certainty gain effect*.

The imbalance cost stems from the economic losses suffered by a wind producer that incurs energy deviations in real time, while the certainty gain effect is related to the beneficial impact that an adjustment market following the clearing of the day-ahead market has on the wind producer gains.

To fully comprehend the origin of the imbalance cost, we first review the basics of how energy prices are generally formed in an electricity market. This brief revision allows us to subsequently set out the mechanism considered in this dissertation to price the real-time energy deviations incurred by market agents.

4.3.1 Electricity Price Formation

The price of electricity is generally obtained as a result of a *market-clearing procedure*. For a given set of energy offers by producers and energy bids by consumers, a market-clearing procedure is an algorithm used by the market operator to determine i) accepted offers, ii) accepted bids, and iii) clearing prices at which electricity transactions are valued. Further information on market-clearing procedures is provided in the next chapter.

Offers by producers are usually forced to be monotonically increasing, while bids by consumers, monotonically decreasing.

The simplest market-clearing procedure consists in a single-period auction based on a single-node model of the power system. In a single-period auction, time periods covering the scheduling horizon are considered independently, thus disregarding the inter-temporal coupling derived from operating constraints such as ramping limits and minimum up and down times of generating units. In turn, in a single-node model of the power system, the physical limitations imposed by the transmission grid are simply neglected and consequently, the procedure fails to rule out energy transactions that are infeasible in practice due to eventual network bottlenecks.

The market operator collects generating offers (increasing in price) by producers and load bids (decreasing in price) by consumers and clears the market by maximizing the social welfare [140], or equivalently, minimizing the social cost. We often talk about the *declared social welfare* or the *declared social cost* since producers/consumers may not submit actual costs/utilities. Thus, the market-clearing price is the intercept between the aggregated power supply and demand curves, as exemplified in Fig. 4.2.

Mathematically, the single-period auction can be formulated as

$$\underset{p_{Gi}(m), \forall i, m; p_{Lj}(\ell), \forall j, \ell}{\text{Minimize}} \quad \text{SC} = \sum_{i=1}^{N_G} \sum_{m=1}^{N_{O_i}} \lambda_{Gi}(m) p_{Gi}(m) - \sum_{j=1}^{N_L} \sum_{\ell=1}^{N_{B_j}} \lambda_{Lj}(\ell) p_{Lj}(\ell) \quad (4.1)$$

subject to

$$0 \leq p_{Gi}(m) \leq p_{Gi}^{\max}(m) \quad \forall i, \forall m \quad (4.2)$$

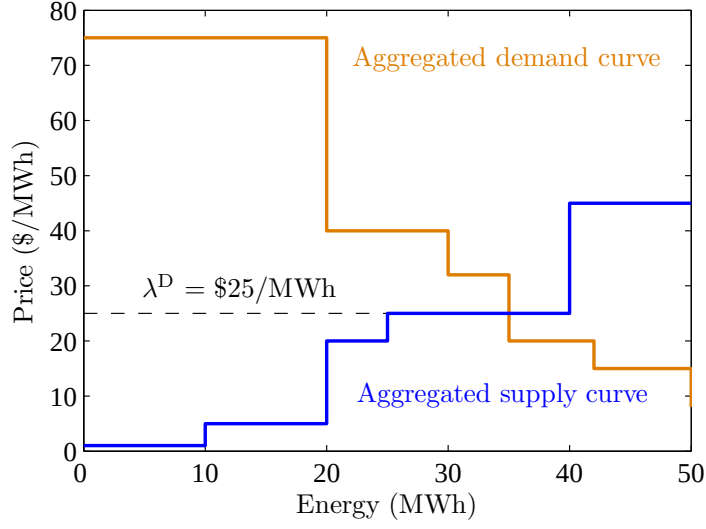


Figure 4.2: Single-period auction excluding inter-temporal and network constraints

$$0 \leq p_{Lj}(\ell) \leq p_{Lj}^{\max}(\ell) \quad \forall j, \forall \ell \quad (4.3)$$

$$\sum_{j=1}^{N_L} \sum_{\ell=1}^{N_{Bj}} p_{Lj}(\ell) = \sum_{i=1}^{N_G} \sum_{m=1}^{N_{O_i}} p_{Gi}(m), \quad (4.4)$$

where SC is the single-period social cost (objective function). As previously stated, to this auction, every producer i submits a series of N_{O_i} selling offers comprising each an energy block $p_{Gi}^{\max}(m)$ and a price $\lambda_{Gi}(m)$. Analogously, every consumer j submits a set of N_{B_j} purchasing bids including each a block of energy $p_{Lj}^{\max}(\ell)$ and a price $\lambda_{Lj}(\ell)$.

Constraints (4.2) establish bounds and declare nonnegativity of the power block offers by generating units. Likewise, constraints (4.3) establish bounds and declare the nonnegativity of the demand power block bids. Finally, constraint (4.4) enforces the market equilibrium, i.e., that the total accepted demand equals the total accepted generation.

The market-clearing price is then obtained as the dual variable associated with constraint (4.4), which coincides either with the price of the most expensive generation block (see Fig. 4.2) or the price of the cheapest demand block

that is accepted in the clearing procedure.

A model for the transmission network can be embedded into formulation (4.1)–(4.4). In such a case, constraint (4.4) should be replaced by the set of equations representing the power balance at every node of the power system and transmission capacity limits should be added. The consideration of the network to clear the electricity market naturally leads to the notion of *locational marginal pricing* of electricity, according to which the electricity price at each node of the system may be different. The differences stem from the active power losses and network bottlenecks, [115, 205].

For simplicity, in this chapter, we assume that both the day-ahead and adjustment market prices are obtained from a market-clearing algorithm similar to the auction (4.1)–(4.4). In turn, imbalance prices, which represent the value of the energy required to maintain the real-time balance between generation and demand under unpredictable disturbances, are determined by a somewhat more complex mechanism that takes the electricity price in the day-ahead market as a reference. This mechanism is described in detail below.

4.3.2 Mechanism for Imbalance Prices

In this dissertation, we call *balancing market* the trading floor available in an electricity market to guarantee the balance between the energy supplied by producers and the energy demanded by consumers on a real-time basis. The development of this function requires the balancing market to be cleared as close as possible to the period of physical energy delivery so that the available production means and the actual consumption needs are known with sufficient accuracy. This allows us to assume that every wind power producer attends the balancing market with perfect information on its wind power production.

Any market agent expecting a final production below or above the last energy schedule resulting from trading in the day-ahead and adjustment markets is *commanded* to amend its energy deviations in the balancing market. These imbalances can be positive or negative. We consider a positive energy deviation as a higher production or lower consumption than scheduled. Analogously, a negative energy deviation is defined as a lower production or higher

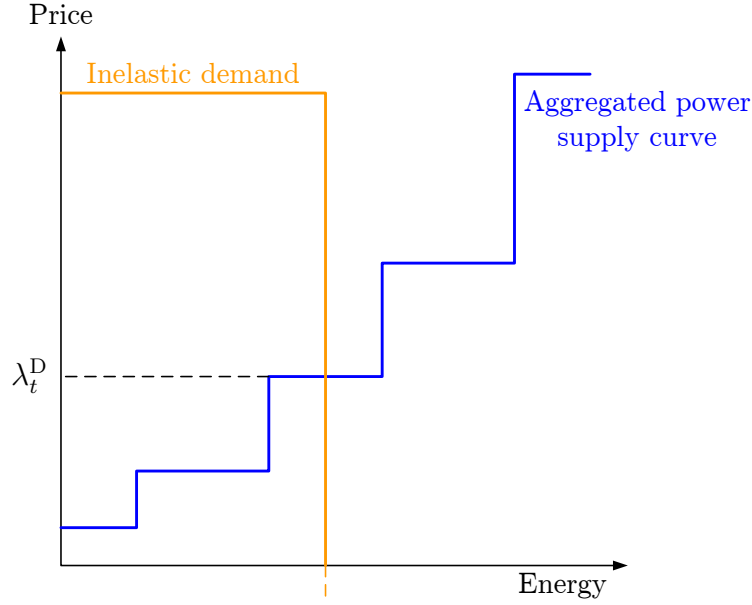


Figure 4.3: Day-ahead market price formation

consumption than scheduled. In practice, energy imbalances in a power system are caused by consumers, wind producers, and conventional producers suffering from an unexpected unit failure.

Thus, every *deviated* producer must sell its generation surplus or buy its generation deficit in the balancing market at a price referred to as *imbalance price*. The selling and purchasing prices are obtained from a single auction process. Without loss of generality, we describe next the general rationale behind such a process, a rationale fundamentally shared by many electricity markets worldwide, but with slight variants.

Let us start considering a day-ahead market price λ_t^D for a given time period t of the market horizon. As stated in Section 4.3.1, this price is obtained as the intercept between the aggregated power supply and demand curves as shown in Fig. 4.3, where for simplicity the behavior of consumers is supposed to be completely inelastic in this explanation. The aggregated supply function is built from the energy offers submitted by producers to the day-ahead market. If the functioning of this market follows the rules of perfect competition, which is undoubtedly an utopian paradigm in practice, energy offers by producers

equal their marginal production costs, and so, the aggregated supply curve depicted in Fig. 4.3 constitutes the minimum cost function at which any level of demand can be satisfied. Likewise, if we extend the assumption of perfect competition to the balancing market, the cost of energy deviations around the intercept defining the day-ahead price in Fig. 4.3 should be given by the same aggregated supply curve as the one of the day-ahead market for the referred time period t . This line of argument constitutes the logic supporting the mechanism for imbalance price formation described below. However, electricity markets are far from being perfectly competitive and thus, the aggregated supply curves resulting in both markets can be very different. Further, some generating units may not be able to alter their productions levels in the time frame required to provide balancing energy. As a result, some conventional producers may not participate in the balancing market.

In the balancing market, a price λ_t^+ for positive energy deviations (higher productions or lower consumptions than scheduled) and a price λ_t^- for negative energy deviations (lower productions or higher consumptions than scheduled) are settled for each time period. The rationale is that these prices represent the cost of the energy required to counteract the unplanned total energy imbalance of the power system and consequently, they depend on the sign, positive or negative, of the imbalance as a whole (system imbalance). Specifically:

1) If the system imbalance Θ_t is negative (deficit of generation in the system), $\Theta_t \leq 0$, then

$$\lambda_t^+ = \lambda_t^D \quad (4.5)$$

$$\lambda_t^- = \max(\lambda_t^D, \lambda_t^{UP}). \quad (4.6)$$

where λ_t^{UP} is the price of the energy (upward energy) that needs to be added to the system. This situation is graphically represented in Fig. 4.4(a). Note that a generation shortage can be equivalently interpreted as an identical increase in the level of demand to be supplied.

In this case, the operator organizes a balancing market to which producers submit additional selling offers. Again assuming a rational behavior, producers are willing to produce the amount of energy required to cover the deficit of

generation at a price (λ_t^{UP}) higher than the day-ahead market-clearing price (λ_t^{D}). The cost of this transaction falls on those market participants responsible for the system imbalance. In this sense, note that, as the price of this additional energy is higher than the actual price of the energy traded in the day-ahead market, the balancing process implies a profit loss for these market participants with respect to the profit they would have obtained by selling only their actual productions in the day-ahead market. On the other hand, those market participants producing more than agreed on the day-ahead market and, consequently, contributing to mitigate the negative system imbalance are paid at λ_t^{D} for their overproduction, and hence they do not incur imbalance cost.

Note that taking the maximum in equation (4.6) above guarantees that the price of the energy bought in this market to cover generation deficits is higher than or equal to the day-ahead market price. This needs to be enforced owing to the fact that electricity markets are not perfectly competitive environments.

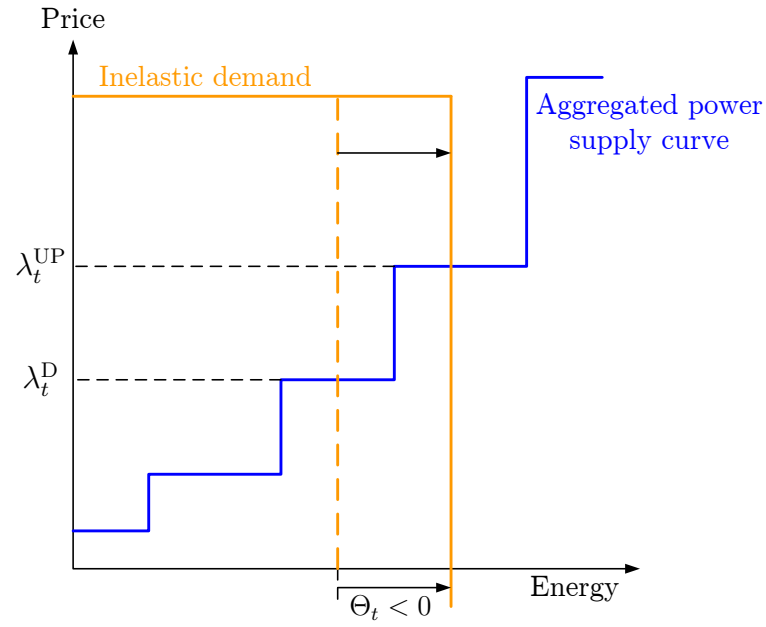
2) If the system imbalance Θ_t is positive (excess of generation in the power system), $\Theta_t \geq 0$, then

$$\lambda_t^+ = \min(\lambda_t^{\text{D}}, \lambda_t^{\text{DN}}), \quad (4.7)$$

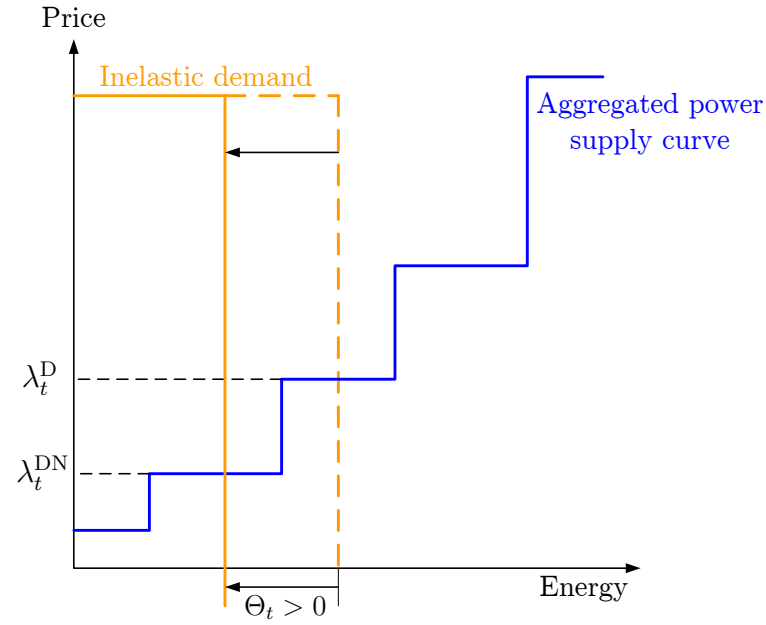
$$\lambda_t^- = \lambda_t^{\text{D}}, \quad (4.8)$$

where λ_t^{DN} is the price of the energy (downward energy) that needs to be removed from the system. This situation is illustrated in Fig. 4.4(b). Note that a generation surplus can be equivalently understood as an identical decrease in the level of demand to be met.

In order to palliate the excess of generation in the system, the operator uses the balancing market where producers submit offers to repurchase some of the energy previously sold in the day-ahead market. Assuming a rational behavior, producers are only willing to repurchase the excess of energy at a price (λ_t^{DN}) smaller than the day-ahead market-clearing price (λ_t^{D}). Thus, those market participants contributing to the positive system imbalance are paid for their overproduction at λ_t^{DN} . As a result, they obtain a profit smaller than the profit they would have achieved if they had sold their overproduction in the day-ahead market. On the contrary, those market participants producing



(a) Negative system imbalance



(b) Positive system imbalance

Figure 4.4: Mechanism for imbalance price formation

less energy than that resulting from the day-ahead market and, consequently, helping to alleviate the positive system imbalance are just paid at λ_t^D for their actual productions and so, they do not incur opportunity cost.

Note that taking the minimum in equation (4.7) above guarantees that the generation surplus is sold at a price smaller than or equal to the day-ahead market price. Again, this needs to be imposed due to the fact that, in general, the rules of perfect competition do not hold in electricity markets.

From the discussion above, it follows that $\lambda_t^+ \leq \lambda_t^D$ and $\lambda_t^- \geq \lambda_t^D$.

It is important to emphasize that, according to these criteria, those participants who offset the system imbalance, i.e., those participants who contribute to restore the system balance, are not *punished* by the market, which is a sound, albeit debatable, principle.

Once we have carefully set out the mechanism through which imbalance prices are formed in the balancing market, we are in a position to understand the notion of *imbalance cost* and to derive the consequent expression for the wind producer revenue.

4.3.3 Imbalance Cost and Revenue

Consider a wind power producer that offers and gets accepted a level of energy E_t^D into the day-ahead market for time period t , but actually generates E_t^W . For the sake of simplicity, no adjustment market is considered in this exposition. The revenue R_t of this wind power producer for period t is

$$R_t = \lambda_t^D E_t^D + I_t \quad (4.9)$$

where I_t is the imbalance income resulting from the balancing process and may be negative, i.e., it may represent a cost. The total deviation Δ_t incurred by the wind producer is defined as

$$\Delta_t = E_t^W - E_t^D = d_t(P_t^W - P_t^D), \quad (4.10)$$

where d_t , P_t^W , and P_t^D are, respectively, the time span between two consecutive time periods (1 hour), the actual power produced by the wind power producer

in period t , and the level of power it sells in the day-ahead market for such a period.

Consequently, I_t is given by

$$I_t = \begin{cases} \lambda_t^+ \Delta_t, & \Delta_t \geq 0 \\ \lambda_t^- \Delta_t, & \Delta_t < 0. \end{cases} \quad (4.11)$$

According to expression (4.10), a positive deviation means production surplus and a negative deviation represents a deficit of production. Therefore, λ_t^+ (λ_t^-) is the price at which the wind producer will be paid (charged) for its excess (deficit) of generation. Lastly, if we define

$$r_t^+ = \frac{\lambda_t^+}{\lambda_t^D}, \quad r_t^+ \leq 1, \quad (4.12)$$

$$r_t^- = \frac{\lambda_t^-}{\lambda_t^D}, \quad r_t^- \geq 1 \quad (4.13)$$

then

$$I_t = \begin{cases} \lambda_t^D r_t^+ \Delta_t, & \Delta_t \geq 0 \\ \lambda_t^D r_t^- \Delta_t, & \Delta_t < 0. \end{cases} \quad (4.14)$$

Note that definitions (4.12) and (4.13) are consistent under the assumption that the hourly electricity prices in the day-ahead market are necessarily positive, i.e., $\lambda_t^D > 0$. This corresponds to the general behavior of energy prices in most electricity markets. If not, ratios r_t^+ and r_t^- need to be redefined in a different manner, or simply, the imbalance income function I_t should be expressed in the form of (4.11).

A wind power producer that needs to correct its energy deviations in the balancing market incurs an opportunity cost as it loses the chance of trading the deviated energy through the day-ahead market at a more competitive price. We can easily reformulate the revenue function (4.9) so that it explicitly reflects such an opportunity cost. We distinguish two cases:

- 1) The energy deviation incurred by the wind power producer is positive,

i.e., $\Delta_t > 0$. According to (4.9) and (4.14), we can write

$$R_t = \lambda_t^D E_t^D + \lambda_t^D r_t^+ \Delta_t. \quad (4.15)$$

By using the definition (4.10) of the total deviation Δ_t , we can replace E_t^D in (4.15) by $E_t^W - \Delta_t$. Thus, we obtain

$$R_t = \lambda_t^D (E_t^W - \Delta_t) + \lambda_t^D r_t^+ \Delta_t = \lambda_t^D E_t^W - \lambda_t^D (1 - r_t^+) \Delta_t, \quad (4.16)$$

with $\Delta_t \geq 0$.

2) The energy deviation incurred by the wind power producer is negative, i.e., $\Delta_t < 0$. Following a reasoning similar to that of the previous case, we arrive at the revenue expression below,

$$R_t = \lambda_t^D (E_t^W - \Delta_t) + \lambda_t^D r_t^- \Delta_t = \lambda_t^D E_t^W + \lambda_t^D (r_t^- - 1) \Delta_t, \quad (4.17)$$

with $\Delta_t < 0$.

Equations (4.16) and (4.17) can be expressed in the following general form

$$R_t = \lambda_t^D E_t^W - C_t, \quad (4.18)$$

where

$$C_t = \begin{cases} \lambda_t^D (1 - r_t^+) \Delta_t, & \Delta_t \geq 0 \\ -\lambda_t^D (r_t^- - 1) \Delta_t, & \Delta_t < 0. \end{cases} \quad (4.19)$$

The term $\lambda_t^D E_t^W$ in (4.18) constitutes the maximum level of revenue that the wind power producer could collect from trading its energy production in a situation free of wind uncertainty, that is, in a position with perfect information on its future wind generation. The term C_t represents the aforementioned opportunity cost, which results from trading the energy deviations in the balancing market at a less attractive price. This opportunity cost is usually referred to as *imbalance cost* and can be also interpreted as the cost of imperfect wind power forecasts. In the specific case for which a market participant is not charged for its deviation because this deviation helps to restore the sys-

tem balance, it holds that C_t is zero. In addition, note that, since the term $\lambda_t^D E_t^W$ is an uncontrollable component within the revenue expression (4.18), maximizing R_t boils down to minimizing the imbalance cost C_t .

The definition of imbalance cost becomes trickier if an adjustment market is taken into account in the previous analysis due to the price differences in the day-ahead and adjustment markets. Notwithstanding this complication, it is clear that, even if an adjustment market is available, every market agent forced to correct its energy deviations in the balancing market incurs an opportunity cost.

We conclude this section pointing out that, as the generation cost of a wind power producer is zero, revenue equals profit.

Next, we formally introduce one of the most relevant aspects characterizing the decision-making process of a wind power producer: a phenomenon that we refer to as *certainty gain effect*.

4.3.4 Certainty Gain Effect

Adjustments markets are short-term trading floors intended to allow market agents to modify their production or consumption schedule in response to previously obtained market results that are unfavorable or unforeseen events, e.g., unexpected equipment failures or sudden changes in loads' behavior. For this reason, these markets are organized after the clearing of the day-ahead market and may cover the whole energy delivery horizon or just a part of it. Apart from this service, adjustment markets provide market participants with additional trading opportunities to make more profit by taking advantage of the energy price differences existing between this market and the day-ahead market.

As indicated in Section 4.2, for simplicity, we solely work with one adjustment market cleared just before the beginning of the energy delivery horizon and covering all of it. Fig. 4.5 illustrates the temporal position of this trading floor within the considered market structure.

Adjustment markets are especially useful for wind power producers, because these markets are cleared closer to the delivery time and therefore, en-

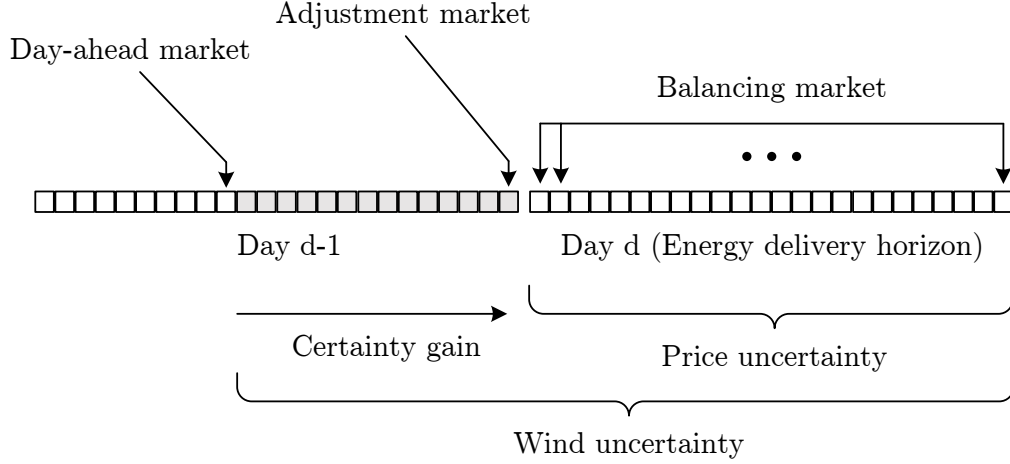


Figure 4.5: Certainty gain on wind power stochastic behavior

ergy trading in them can be carried out with a diminished level of uncertainty as for the availability of wind energy. Wind production volatility is strongly dependent on the magnitude of the time interval up to its realization. For example, the forecasts of wind one hour ahead are much more accurate than the forecasts available forty hours ahead. In the particular case represented in Fig. 4.5, this means that the likely realization of wind during the hours comprising the energy delivery horizon is better known in the adjustment market than in the day-ahead market. We say that some certainty on wind power stochastic behavior is gained during the time periods in between the closures of these two markets. This certainty gain is a phenomenon inherent to the physics of the short-term trading problem of a wind power producer and as such, it exists irrespective of the approach used to tackle its solution.

When it comes to trade in the day-ahead market, the wind power producer should account for this phenomenon to build its strategy offer, because its consideration is expected to have an impact on the decisions to be made in this market. This impact is what we call *certainty gain effect*.

Formally, the certainty gain effect can be defined as the economic surplus that the wind power producer obtains if it builds its offering strategy for the day-ahead market considering that the subsequent energy trading in the ad-

justment market can be done with a greater level of certainty on its future wind energy production.

From a mathematical point of view, the consideration of such a wind certainty gain affects the modeling of the wind producer problem in two aspects: i) the generation of wind power scenarios, which must reflect a level of wind volatility in the adjustment market lower than that in the day-ahead market (see Section 4.4.2), and ii) the constraints pertaining to the decision making in the adjustment market (see Section 4.5.4).

The physical phenomenon described above can be also extended to the stochastic processes describing energy prices in the balancing market, although such an extension is not made in this chapter for simplicity. Notwithstanding this, we should point out that the modeling of the certainty gain effect related to both wind production and imbalance prices are mathematically analogous.

4.4 Uncertainty Characterization

The wind producer problem described above is subject to four sources of uncertainty, namely, the day-ahead market price, the adjustment market price, the prices for imbalances and the wind power generation. Among these sources of uncertainty, the one associated with the wind power production constitutes the keystone of the decision-making process. Given that the wind production is characterized by a zero generation cost, its unpredictable nature is directly responsible for the likely loss of gains incurred by the wind power producer through the imbalance cost.

In order to account for the impact of these sources of uncertainty on the wind producer problem, we propose to characterize them as stochastic processes, which are conveniently discretized in the form of scenarios as stated in Chapter 3.

4.4.1 Day-ahead, Adjustment, and Imbalance Prices

The day-ahead market price is mathematically treated as a stochastic process λ^D spanning the N_T periods of the market horizon, i.e., $\lambda^D = \{\lambda_t^D, t =$

$1, 2, \dots, N_T\}$. This stochastic process is represented through a finite set of N_D scenarios.

Analogously, the adjustment market price can be modeled as a stochastic process $\boldsymbol{\lambda}^A = \{\lambda_t^A, t = 1, 2, \dots, N_T\}$, which is discretized in N_A scenarios.

In practice, prices $\boldsymbol{\lambda}^D$ and $\boldsymbol{\lambda}^A$ in the day-ahead and adjustment markets, respectively, are unquestionably correlated. The existing relationship is direct (or positive), that is, periods of high day-ahead price realizations often coincide with periods of high adjustment price values. This strong correlation may have a significant impact on the decisions to be made by market participants concerning energy trading and risk management and therefore, should not be hastily ignored.

To deal with the statistical correlation between $\boldsymbol{\lambda}^D$ and $\boldsymbol{\lambda}^A$, we can make use of multivariate analysis techniques such as those based on transfer function and dynamic regression models [41, 56, 97, 158, 159], and artificial neural networks [110, 169, 184, 191, 208, 225], among others. Nevertheless, a simple alternative to these more sophisticated methods consists in working on the price difference $\Delta\boldsymbol{\lambda}$ between markets, i.e.,

$$\boldsymbol{\lambda}^A = \boldsymbol{\lambda}^D + \Delta\boldsymbol{\lambda}, \quad (4.20)$$

where the price difference $\Delta\boldsymbol{\lambda}$ is a stochastic process $\Delta\boldsymbol{\lambda} = \{\Delta\lambda_t = \lambda_t^A - \lambda_t^D, t = 1, 2, \dots, N_T\}$ discretized in N_Δ scenarios.

Thus, the N_A adjustment market price scenarios are obtained from the scenarios corresponding to the day-ahead price and the price difference as expressed in (4.20). It holds that $N_A = N_D \times N_\Delta$.

Additionally, for each period t of the market horizon, we have two imbalance prices, namely, λ_t^+ and λ_t^- , for positive and negative energy deviations incurred in that period, respectively. The two series of values describing the realizations of these prices throughout the market horizon constitute each a stochastic process. However, these series are deterministically interrelated due to the rules governing the mechanism for imbalance price formation explained in Section 4.3.2. These rules can be easily summarized by considering the imbalance price ratios r_t^+ and r_t^- instead of the actual imbalance prices λ_t^+ and

λ_t^- as follows:

1. If $r_t^+ < 1$, then the net system balance is positive and as a result, it holds that $r_t^- = 1$.
2. If $r_t^- > 1$, then the net system imbalance is negative and consequently, $r_t^+ = 1$.
3. If the net system imbalance is zero, then it holds that $r_t^- = r_t^+ = 1$.

Thus, we can create a new random variable r_t as

$$r_t = \begin{cases} r_t^+, & \text{if } r_t^+ \neq 1, \\ r_t^-, & \text{if } r_t^+ = 1, \end{cases} \quad (4.21)$$

which naturally incorporates the three previous rules in its definition. In addition, it turns out that random variable r_t can be expressed linearly as $r_t = r_t^+ + r_t^- - 1$. Such a deduction is straightforward and is graphically illustrated in Fig. 4.6. Moreover, note that random variable r_t does not share the discrete character of the imbalance prices ratios r_t^+ and r_t^- . Hence, the stochastic behavior of r_t is easier to model by employing the time-series-based techniques presented in Chapter 3.

The stochastic process $\mathbf{r}$ defined as the collection of random variables r_t throughout the market horizon, i.e., $\mathbf{r} = \{r_t, t = 1, 2, \dots, N_T\}$, is represented by a set of N_I scenarios. From each one of these scenarios, a series of pairs (r_t^+, r_t^-) , $\forall t = 1, 2, \dots, N_T$, is obtained by applying definition (4.21).

4.4.2 Wind Power Production

Similarly to market prices, wind power production is modeled as a stochastic process $\mathbf{P}^W = \{P_t^W, t = 1, 2, \dots, N_T\}$ represented by a finite set of N_ϖ scenarios.

An appropriate modeling of the uncertainty characterizing the wind producer problem should include the knowledge of the wind generation stochastic process gained during the time interval between the closures of the day-ahead and adjustment markets. As explained in Section 4.3.4, this gain of certainty

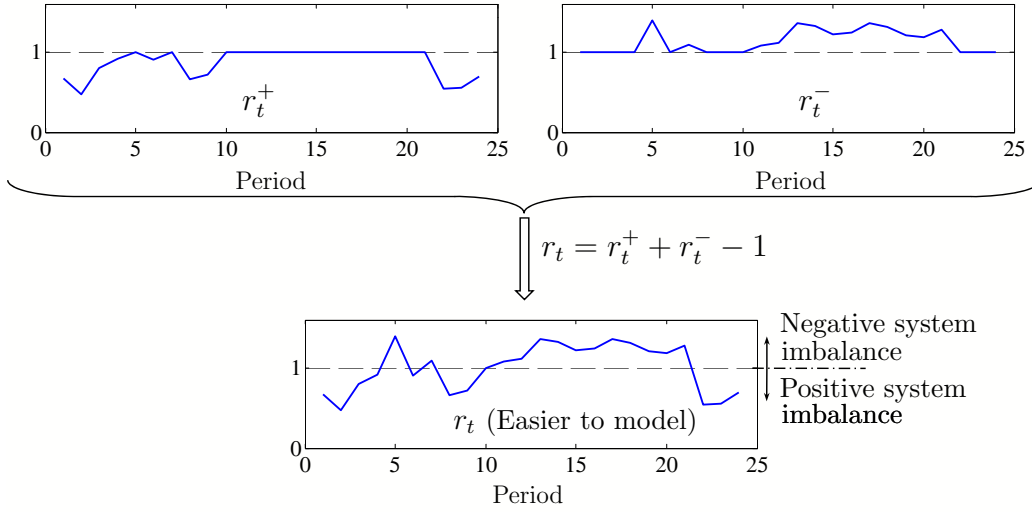


Figure 4.6: Formation of random variable r_t from the imbalance price ratios r_t^+ and r_t^-

is due to the fact that the energy selling/buying offer to submit in the adjustment market can be decided taking into account the realized values of the wind generation stochastic process for the time periods after the day-ahead market clearing. In order to reflect this circumstance within the scenario representation, we propose to generate N_ϖ wind power scenarios as follows:

1. Simulate N_{ϖ_1} wind power scenarios for the periods making up the time interval between the closures of the day-ahead and adjustment markets. In other words, if such time periods are indexed by τ , our aim in this step is then to create N_{ϖ_1} series of wind power values in the form $\{P_\tau^W, \tau = 1, 2, \dots, N_{T_1}\}$, where N_{T_1} is the number of periods comprising the time lapse between the day-ahead and adjustment market-clearing processes.
2. For each one of the wind power scenarios generated in the previous step, simulate N_{ϖ_2} wind power scenarios for the time interval spanning from the closure of the adjustment market and the end of the market horizon. Therefore, this second step yields $N_\varpi = N_{\varpi_1} \times N_{\varpi_2}$ series of wind power values in the form $\{P_t^W, t = 1, 2, \dots, N_T\}$, where t is the index of time periods covering the market horizon.

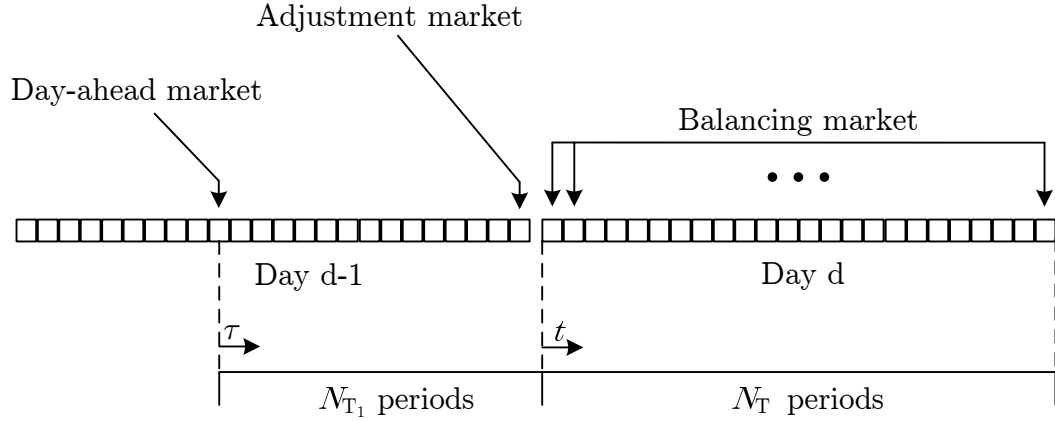


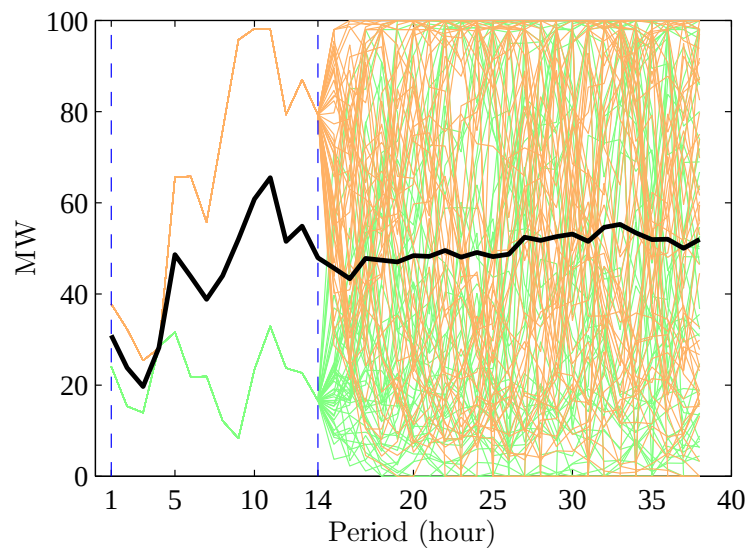
Figure 4.7: Illustration of the time periods covered by indexes τ and t

The roles played by indexes τ and t in the above process are graphically clarified in Fig. 4.7.

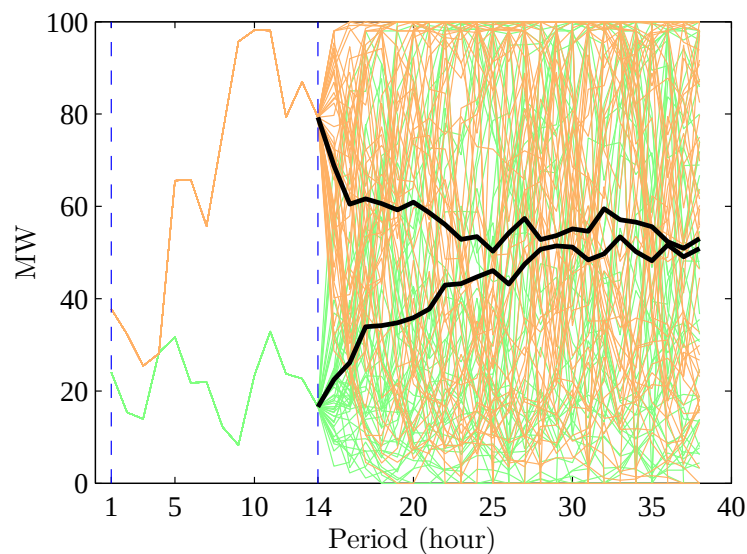
It is important to stress that, in order for the certainty gain effect to be properly embedded into the scenario representation of the wind power production, the scenario generation process carried out in step 2 above must be *fed* with the wind power scenarios previously built in step 1.

Fig. 4.8 shows a possible wind power scenario set for a market structure including one day-ahead and one adjustment markets separated by a time interval of 14 hours, both covering a trading horizon of 24 hours. This set is made up of $N_{\varpi} = N_{\varpi_1} \times N_{\varpi_2} = 2 \times 50 = 100$ wind power scenarios generated using the time-series-based approach described in Chapter 3. Note that from each one of the 2 scenarios modeling the wind power production during the 14 hours in between the closures of the day-ahead and adjustment markets, a new bunch of 50 scenarios representing plausible wind power realizations throughout the market horizon springs up.

The bold lines in Figs. 4.8(a) and 4.8(b) represent the series of wind power expected values observed from the day-ahead and adjustment markets, respectively. Note that the view of the wind producer about its future production throughout the market horizon is more accurate in the adjustment market than in the day-ahead market, especially, for the first few hours. This is so because



(a) Wind power average values observed from the day-ahead market



(b) Wind power average values observed from the adjustment market

Figure 4.8: Example of wind power scenario set

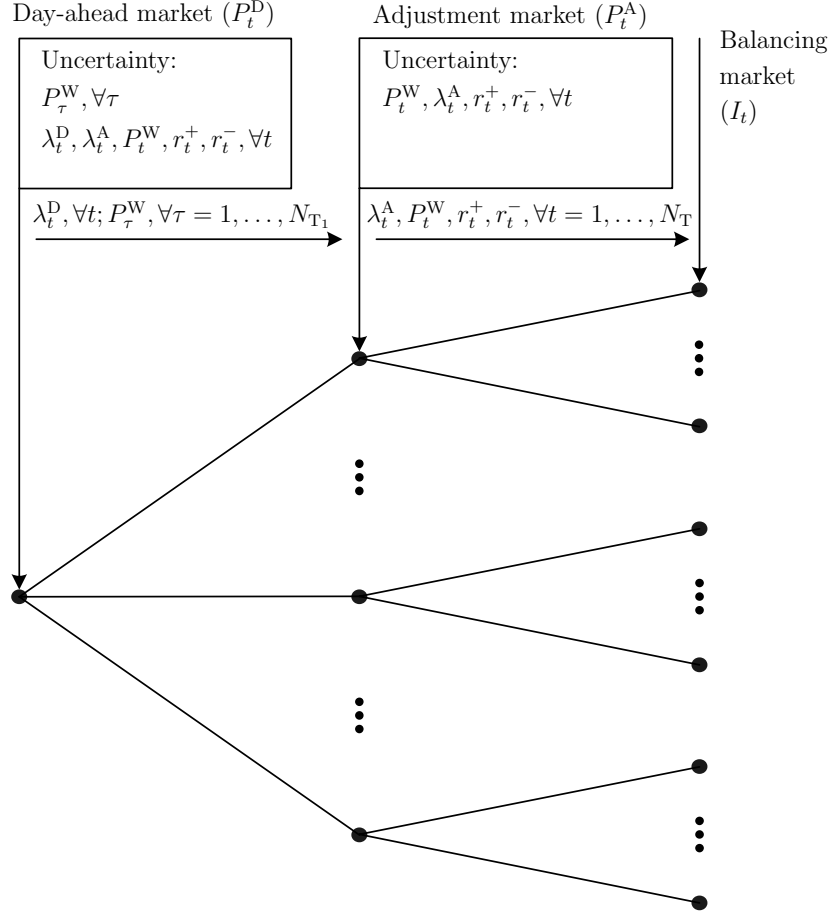


Figure 4.9: Wind producer problem: scenario tree

the wind power scenarios split up into two subsets at the clearing time of the adjustment market in order to incorporate the information on the wind power values realized during the periods in between both markets.

4.4.3 Scenario Tree

The sets of scenarios characterizing the uncertainty associated with market prices and wind generation in the wind producer problem can be arranged in a symmetric scenario tree with the form depicted in Fig. 4.9.

Specifically, this tree is built as follows:

1. Generate N_D price scenarios for the day-ahead market.
2. For each realization of the day-ahead market prices, simulate N_Δ scenarios modeling the difference between the day-ahead and adjustment market prices.
3. For each scenario of the adjustment market prices, generate $N_\varpi = N_{\varpi_1} \times N_{\varpi_2}$ wind power realizations.
4. Finally, for each wind power realization, simulate N_I imbalance price ratio scenarios.

Each scenario ω in the tree so built is made up of a set of vectors representing plausible realizations of the stochastic processes involved in the wind producer problem, namely, market prices and wind generation. Mathematically, this can be expressed as:

$$\text{Scenario } \omega = \{\lambda_{t\omega}^D, P_{\tau\omega}^W, \lambda_{t\omega}^A, P_{t\omega}^W, r_{t\omega}^+, r_{t\omega}^-\},$$

$$\forall t = 1, \dots, N_T, \forall \tau = 1, \dots, N_{T_1}, \quad (4.22)$$

where $\lambda_{t\omega}^A = \lambda_{t\omega}^D + \Delta\lambda_{t\omega}$.

Likewise, each scenario ω in the tree has a probability of occurrence π_ω computed as the product of the probabilities associated with vectors or pairs of vectors $\lambda_{t\omega}^D$, $\Delta\lambda_{t\omega}$, $(P_{\tau\omega}^W, P_{t\omega}^W)$, and $(r_{t\omega}^+, r_{t\omega}^-)$. Needless to say, the summation of all these probabilities π_ω over the whole set of scenarios Ω must be equal to 1, i.e., $\sum_{\omega \in \Omega} \pi_\omega = 1$. The total number of scenarios composing the tree is $N_\Omega = N_D N_\Delta N_\varpi N_I$.

It is important to point out that the procedure described above to build the scenario tree implicitly assumes that market prices and wind production are independent stochastic processes. If, on the contrary, they are correlated, the scenario tree construction calls for a more complex joint scenario generation process for market prices and wind production based on multivariate analysis. However, this is outside the scope of this dissertation.

Lastly, the scenario tree is fed into a three-stage stochastic programming model wherein each stage represents a market. The sequence of stages and

decisions is as follows:

1. Design the offer strategy for the day-ahead market and submit the resulting energy selling offers to this market for each period of the market horizon. In this stage, decisions are made based on plausible realizations of the stochastic processes involved, namely, market prices (day-ahead, adjustment, and balancing) and wind power production.
2. Once the day-ahead market price is known for each time period, decide the amount of energy to be sold/bought in/from the adjustment market. Specifically, at the moment that this decision is made, the 24 hourly day-ahead market prices and the wind power generated in the time periods between the closures of the day-ahead and adjustment markets are known. On the contrary, the adjustment and balancing market prices, and the wind power production for the rest of the market horizon are still uncertain. Therefore, in this second stage, and for every day-ahead market price realization, decisions are made based on plausible scenarios of adjustment and balancing prices and of wind power production. Note that the wind production scenarios, spanning the whole market horizon and available at this second stage, are generated knowing the actual wind production in the periods between the closures of the day-ahead and adjustment markets.
3. The last stage of the stochastic programming approach is determined by the materialization of the adjustment market prices, the imbalance prices and the wind power generated in the time periods spanning the whole market horizon. Thus, the deviation incurred by the wind power producer in each one of these periods is known and the consequent cost for imbalance can be computed.

4.5 Wind Producer Model

In this section, we describe the mathematical modeling of the wind power producer problem in detail, starting up with the basics and finishing up with

its complete formulation.

4.5.1 Basic Model

Firstly, no adjustment market is considered. Then, the optimization problem aimed to maximize the expected profit of a wind power producer trading its production in the day-ahead and balancing markets can be formulated as follows:

$$\begin{aligned} & \text{Maximize}_{P_t^D, \forall t; \Delta_{t\omega}, \forall t, \forall \omega} \\ \mathcal{E}\{R\} = & \sum_{\omega=1}^{N_\Omega} \sum_{t=1}^{N_T} \pi_\omega (\lambda_{t\omega}^D P_t^D d_t + I_{t\omega}) \end{aligned} \quad (4.23)$$

subject to

$$I_{t\omega} = \begin{cases} \lambda_{t\omega}^D r_{t\omega}^+ \Delta_{t\omega}, & \Delta_{t\omega} \geq 0 \\ \lambda_{t\omega}^D r_{t\omega}^- \Delta_{t\omega}, & \Delta_{t\omega} < 0 \end{cases}, \quad \forall t, \forall \omega \quad (4.24)$$

$$0 \leq P_t^D \leq P^{\text{W}, \max}, \quad \forall t \quad (4.25)$$

$$\Delta_{t\omega} = d_t(P_{t\omega}^{\text{W}} - P_t^D), \quad \forall t, \forall \omega. \quad (4.26)$$

The objective function (4.23) represents the expected profit obtained by the wind power producer from trading its production in the day-ahead and balancing markets. As explained in Section 4.3.3, the profit equals the revenue, and consequently, maximizing the expected profit boils down to maximizing the expected revenue. Equation (4.24) constitutes the definition, per period and scenario, of the imbalance income function. Constraints (4.25) limit the amount of power that can be sold in the day-ahead market to the installed capacity of the wind farm, and equation (4.26) defines the total deviation incurred by the wind producer in each period and scenario.

Problem (4.23)–(4.26) cannot be solved through optimization techniques as it stands, since the imbalance income $I_{t\omega}$ is a piecewise function. A simple way to circumvent this hurdle is to define a binary variable $y_{t\omega}$, per period and

scenario, indicating the direction, positive or negative, of the energy deviation. Thus, problem (4.23)–(4.26) can be recast as

$$\begin{aligned} & \text{Maximize}_{P_t^D, \forall t; \Delta_{t\omega}, \forall t, \forall \omega} \\ \mathcal{E}\{R\} = & \sum_{\omega=1}^{N_\Omega} \sum_{t=1}^{N_T} \pi_\omega (\lambda_{t\omega}^D P_t^D d_t + \lambda_{t\omega}^D r_{t\omega}^+ \Delta_{t\omega} (1 - y_{t\omega}) + \lambda_{t\omega}^D r_{t\omega}^- \Delta_{t\omega} y_{t\omega}) \end{aligned} \quad (4.27)$$

subject to

$$0 \leq P_t^D \leq P^{W, \max}, \quad \forall t \quad (4.28)$$

$$\Delta_{t\omega} = d_t (P_{t\omega}^W - P_t^D), \quad \forall t, \forall \omega \quad (4.29)$$

$$\Delta_{t\omega} \leq M(1 - y_{t\omega}), \quad \forall t, \forall \omega \quad (4.30)$$

$$-\Delta_{t\omega} \leq M y_{t\omega}, \quad \forall t, \forall \omega \quad (4.31)$$

$$y_{t\omega} \in \{0, 1\}, \quad \forall t, \forall \omega, \quad (4.32)$$

where M is a large enough positive constant exceeding any maximum feasible value of $|\Delta_{t\omega}|$. Note that $y_{t\omega}$ is equal to 1 if the energy deviation incurred by the wind power producer in period t under scenario ω is negative (generation shortage), and 0 otherwise.

Problem (4.27)–(4.32) is both integer, because of the use of binary variables $y_{t\omega}$, and non-linear, due to the product of variables $\Delta_{t\omega}(1 - y_{t\omega})$ and $\Delta_{t\omega}y_{t\omega}$ appearing in the objective function. Finding the optimum of a mixed-integer non-linear programming problem is, in general, complex owing to the lack of theoretical results guaranteeing the optimum existence and uniqueness. Fortunately for us, problem (4.27)–(4.32) can be easily transformed into a mixed-integer linear one. To this end, we decompose the total energy imbalance $\Delta_{t\omega}$ into the sum of a positive and a negative imbalances, $\Delta_{t\omega}^+$ and $\Delta_{t\omega}^-$, respectively. Thanks to this simple decomposition, the previous products of variables are not needed anymore, as illustrated in the formulation below,

$$\text{Maximize}_{P_t^D, \forall t; \Delta_{t\omega}^+, \forall t, \forall \omega; \Delta_{t\omega}^-, \forall t, \forall \omega}$$

$$\mathcal{E}\{R\} = \sum_{\omega=1}^{N_{\Omega}} \sum_{t=1}^{N_T} \pi_{\omega} (\lambda_{t\omega}^D P_t^D d_t + \lambda_{t\omega}^D r_{t\omega}^+ \Delta_{t\omega}^+ - \lambda_{t\omega}^D r_{t\omega}^- \Delta_{t\omega}^-) \quad (4.33)$$

subject to

$$0 \leq P_t^D \leq P^{W, \max}, \quad \forall t \quad (4.34)$$

$$\Delta_{t\omega} = d_t(P_{t\omega}^W - P_t^D), \quad \forall t, \forall \omega \quad (4.35)$$

$$\Delta_{t\omega} = \Delta_{t\omega}^+ - \Delta_{t\omega}^-, \quad \forall t, \forall \omega \quad (4.36)$$

$$0 \leq \Delta_{t\omega}^+ \leq M_1(1 - y_{t\omega}), \quad \forall t, \forall \omega \quad (4.37)$$

$$0 \leq \Delta_{t\omega}^- \leq M_2 y_{t\omega}, \quad \forall t, \forall \omega \quad (4.38)$$

$$y_{t\omega} \in \{0, 1\}, \quad \forall t, \forall \omega, \quad (4.39)$$

where M_1 and M_2 are large enough positive constants that exceed any maximum feasible value of $\Delta_{t\omega}^+$ and $\Delta_{t\omega}^-$, respectively. Through simple reasoning, we can find appropriate values for constants M_1 and M_2 : maximum positive energy deviations occur in scenarios where the wind power producer does not sell any amount of energy in the day-ahead market, $P_t^D = 0$, for a given time period t , but it eventually produces $P_{t\omega}^W$ MW of wind power during that period. Therefore, we can fix M_1 to $P_{t\omega}^W$. Similarly, maximum negative energy deviations happen in scenarios where the wind power producer sells its full capacity in the day-ahead market, $P_t^D = P^{W, \max}$, for a period t , but its final production in that period is $P_{t\omega}^W$. Hence, we can make M_2 equal to $P^{W, \max} - P_{t\omega}^W$.

Optimization problem (4.33)–(4.39) naturally tends to minimize the imbalance cost as underlined in Section 4.3.3. As a result, given the total energy deviation $\Delta_{t\omega} = \Delta_{t\omega}^+ - \Delta_{t\omega}^-$ incurred by the wind power producer in period t and scenario ω , the solution to problem (4.33)–(4.39) is guaranteed to be achieved always with one of the variables $\Delta_{t\omega}^+$ or $\Delta_{t\omega}^-$ equal to zero due to the fact that $r_{t\omega}^+ \leq 1$ and $r_{t\omega}^- \geq 1$. That is, any other feasible solution for which $\Delta_{t\omega}^+$ and $\Delta_{t\omega}^-$ are simultaneously different from zero in period t and scenario ω is economically less profitable. Further, if $r_{t\omega}^+ = r_{t\omega}^- = 1$, the term $\lambda_{t\omega}^D r_{t\omega}^+ \Delta_{t\omega}^+ - \lambda_{t\omega}^D r_{t\omega}^- \Delta_{t\omega}^-$ in the objective function (4.33) boils down to $\lambda_{t\omega}^D \Delta_{t\omega}$ and as a result, the solution is also unique in such a case. Consequently, binary variables $y_{t\omega}$ are in fact not necessary and thus, the mixed-integer programming problem (4.33)–

(4.39) can be equivalently transformed into a linear one with the consequent gain in robustness, simplicity, and computational efficiency. The basic linear formulation of the wind producer problem is presented below,

$$\begin{aligned} & \text{Maximize}_{P_t^D, \forall t; \Delta_{t\omega}^+, \forall t, \forall \omega; \Delta_{t\omega}^-, \forall t, \forall \omega} \\ \mathcal{E}\{R\} = & \sum_{\omega=1}^{N_\Omega} \sum_{t=1}^{N_T} \pi_\omega (\lambda_{t\omega}^D P_t^D d_t + \lambda_{t\omega}^D r_{t\omega}^+ \Delta_{t\omega}^+ - \lambda_{t\omega}^D r_{t\omega}^- \Delta_{t\omega}^-) \end{aligned} \quad (4.40)$$

subject to

$$0 \leq P_t^D \leq P^{W, \max}, \quad \forall t \quad (4.41)$$

$$\Delta_{t\omega} = d_t (P_{t\omega}^W - P_t^D), \quad \forall t, \forall \omega \quad (4.42)$$

$$\Delta_{t\omega} = \Delta_{t\omega}^+ - \Delta_{t\omega}^-, \quad \forall t, \forall \omega \quad (4.43)$$

$$0 \leq \Delta_{t\omega}^+ \leq P_{t\omega}^W d_t, \quad \forall t, \forall \omega \quad (4.44)$$

$$0 \leq \Delta_{t\omega}^- \leq (P^{W, \max} - P_{t\omega}^W) d_t, \quad \forall t, \forall \omega. \quad (4.45)$$

Model (4.40)-(4.45) can be solved by decomposition into N_T optimization problems, one for each time period t , because it does not contain any constraint or any decision variable coupling different time periods, i.e., it does not include any *inter-temporal* constraint or variable.

4.5.2 Offering Curves

The mathematical model presented above is intended to obtain the optimal quantities (in a stochastic programming sense) to be sold in the day-ahead market (e.g., 20 MWh in hour 1, 65 MWh in hour 2, etc). However, it may be convenient to derive optimal offering curves for every hour of the day-ahead market. For this purpose, variables P_t^D representing the power traded in this market for each time period t are made dependent on scenarios ($P_t^D \rightarrow P_{t\omega}^D$), as in [27, 180], and the following constraints are added to model (4.40)–(4.45),

$$P_{t\omega}^D - P_{t\omega'}^D \leq 0, \quad \forall t, \forall \omega, \omega' : O(\lambda_{t\omega}^D) + 1 = O(\lambda_{t\omega'}^D) \quad (4.46)$$

$$P_{t\omega}^D = P_{t\omega'}^D, \quad \forall t, \forall \omega, \omega' : \lambda_{t\omega}^D = \lambda_{t\omega'}^D. \quad (4.47)$$

Constraints (4.46) make offering curves non-decreasing, which is a requirement in most markets. Equations (4.47) are non-anticipativity constraints, which model the fact that only one offering curve can be submitted to the day-ahead market for each hour irrespective of the wind power and imbalance price realizations. That is, this equation enforces non-anticipativity for the wind power and imbalance price realizations. Note that this model is less constrained than the basic one. In fact, the basic model can be obtained from this one by rewriting the non-anticipativity constraints (4.47) as $P_{t\omega}^D = P_{t\omega'}^D, \forall t, \forall \omega, \omega'$.

Despite including the above constraints, the offering strategy model continues being decomposable by time period (hour).

4.5.3 Risk Modeling

A simple manner to control the risk of profit variability in the wind producer problem is to introduce the Conditional Value-at-Risk at the α confidence level (α -CVaR) in the problem formulation. As explained in Appendix A, this metric can be expressed linearly within an optimization problem and exhibits good mathematical properties. If maximizing a discrete profit distribution, α -CVaR can be defined approximately as the expected profit of the $(1-\alpha)100\%$ scenarios with lowest profit.

The risk-constrained formulation of the problem faced by the wind producer is provided below,

$$\begin{aligned} & \text{Maximize}_{P_{t\omega}^D, \forall t, \forall \omega; \Delta_{t\omega}^+, \forall t, \forall \omega; \Delta_{t\omega}^-, \forall t, \forall \omega; \eta_\omega, \forall \omega; \zeta} \\ & \mathcal{E}\{R\} + \beta \left(\zeta - \frac{1}{1-\alpha} \sum_{\omega=1}^{N_\Omega} \pi_\omega \eta_\omega \right), \end{aligned} \quad (4.48)$$

subject to constraints (4.41)–(4.47) plus the following ones required for the linear formulation of the CVaR,

$$- \sum_{t=1}^{N_T} (\lambda_{t\omega}^D P_{t\omega}^D d_t + \lambda_{t\omega}^D r_{t\omega}^+ \Delta_{t\omega}^+ - \lambda_{t\omega}^D r_{t\omega}^- \Delta_{t\omega}^-) + \zeta - \eta_\omega \leq 0, \forall \omega \quad (4.49)$$

$$\eta_\omega \geq 0, \quad \forall \omega. \quad (4.50)$$

The objective function (4.48) to be maximized includes the expected profit of the wind producer ($\mathcal{E}\{R\}$ in (4.40)) and the CVaR of the profit multiplied by the weighting parameter $\beta \in [0, \infty)$. This weighting factor enforces the tradeoff between expected profit and risk in such a way that the higher the value of β , the more risk averse the wind producer is. Thus, if risk is not considered (risk-neutral case), the value of β is set to 0.

Since risk is considered throughout the entire market horizon, model (4.41)–(4.50) cannot be decomposed by time period. This is so because variables ζ and η_ω , used to compute the CVaR over the total revenue distribution, couple the decisions pertaining to different hours. That is, equations (4.49) are inter-period constraints.

At this point, we should clarify that risk management makes sense in a short-term problem like the one considered in this chapter because, as seen later on, profit variability can be significantly reduced with a slight decrease in the expected profit.

4.5.4 Adjustment Market

The consideration of an adjustment market in the wind producer problem entails three changes or additions to the mathematical model presented in Section 4.5.1:

1. First, a term accounting for the revenue obtained from trading energy in the adjustment market needs to be included in objective function (4.40), i.e.,

$$\mathcal{E}\{R\} = \sum_{\omega=1}^{N_\Omega} \sum_{t=1}^{N_T} \pi_\omega (\lambda_{t\omega}^D P_{t\omega}^D d_t + \lambda_{t\omega}^A P_{t\omega}^A d_t + \lambda_{t\omega}^D r_{t\omega}^+ \Delta_{t\omega}^+ - \lambda_{t\omega}^D r_{t\omega}^- \Delta_{t\omega}^-), \quad (4.51)$$

where $P_{t\omega}^A$ is the wind power traded (sold or purchased) in the adjustment market. Note that term $\lambda_{t\omega}^A P_{t\omega}^A d_t$ in (4.51) can be negative if the wind power producer decides to purchase energy from this market ($P_{t\omega}^A < 0$).

Likewise, the risk-related constraint (4.49) needs to be modified accordingly.

2. Second, the total energy deviation Δ_{tw} is computed with respect to the final energy schedule $(P_{tw}^S d_t)$, that is, the summation of the amounts of energy traded in the day-ahead and adjustment markets,

$$P_{tw}^S = P_{tw}^D + P_{tw}^A, \forall t, \forall \omega \quad (4.52)$$

$$0 \leq P_{tw}^S \leq P^{W, \max}, \forall t, \forall \omega \quad (4.53)$$

$$\Delta_{tw} = d_t(P_{tw}^W - P_{tw}^S), \forall t, \forall \omega. \quad (4.54)$$

3. Lastly, non-anticipativity constraints for the decisions made in the adjustment market have to be added,

$$P_{tw}^A = P_{tw'}^A, \forall t, \forall \omega, \omega' : (\lambda_{tw'}^D = \lambda_{tw}^D, \forall t) \\ \text{and } (P_{\tau\omega'}^W = P_{\tau\omega}^W, \forall \tau = 1, 2, \dots, N_{T_1}), \quad (4.55)$$

where N_{T_1} is the number of time periods between the closures of the day-ahead and adjustment markets.

Constraints (4.55) express that the amount of energy traded in the adjustment market may be different depending on the day-ahead market price realization and the wind power generated after and before the closure of the day-ahead and adjustment markets, respectively. Specifically, the second term of the “and” condition in (4.55) models the certainty gain effect explained in Section 4.3.4. Furthermore, these constraints also ensure non-anticipativity with respect to the adjustment market and imbalance price realizations, and the wind power realization after the closure of the adjustment market.

4.5.5 Formulation

The short-term trading problem of a wind power producer participating in an electricity market as the one represented in Fig. 4.1 (with one day-ahead

market, one adjustment market and a balancing market) can be formulated as follows:

$$\begin{aligned}
 & \text{Maximize}_{P_{t\omega}^D, \forall t, \forall \omega; P_{t\omega}^A, \forall t, \forall \omega; \Delta_{t\omega}^+, \forall t, \forall \omega; \Delta_{t\omega}^-, \forall t, \forall \omega; \eta_\omega, \forall \omega; \zeta} \\
 & \sum_{\omega=1}^{N_\Omega} \sum_{t=1}^{N_T} \pi_\omega [\lambda_{t\omega}^D P_{t\omega}^D d_t + \lambda_{t\omega}^A P_{t\omega}^A d_t + \lambda_{t\omega}^D r_{t\omega}^+ \Delta_{t\omega}^+ - \lambda_{t\omega}^D r_{t\omega}^- \Delta_{t\omega}^-] \\
 & + \beta \left(\zeta - \frac{1}{1-\alpha} \sum_{\omega=1}^{N_\Omega} \pi_\omega \eta_\omega \right) \tag{4.56}
 \end{aligned}$$

subject to

$$0 \leq P_{t\omega}^D \leq P^{\text{W}, \max}, \forall t, \forall \omega \tag{4.57}$$

$$P_{t\omega}^S = P_{t\omega}^D + P_{t\omega}^A, \forall t, \forall \omega \tag{4.58}$$

$$0 \leq P_{t\omega}^S \leq P^{\text{W}, \max}, \forall t, \forall \omega \tag{4.59}$$

$$\Delta_{t\omega} = d_t (P_{t\omega}^W - P_{t\omega}^S), \forall t, \forall \omega \tag{4.60}$$

$$\Delta_{t\omega} = \Delta_{t\omega}^+ - \Delta_{t\omega}^-, \forall t, \forall \omega \tag{4.61}$$

$$0 \leq \Delta_{t\omega}^+ \leq P_{t\omega}^W d_t, \forall t, \forall \omega \tag{4.62}$$

$$0 \leq \Delta_{t\omega}^- \leq (P^{\text{W}, \max} - P_{t\omega}^W) d_t, \forall t, \forall \omega \tag{4.63}$$

$$P_{t\omega}^D - P_{t\omega'}^D \leq 0, \forall t, \forall \omega, \omega' : O(\lambda_{t\omega}^D) + 1 = O(\lambda_{t\omega'}^D) \tag{4.64}$$

$$P_{t\omega}^D = P_{t\omega'}^D, \forall t, \forall \omega, \omega' : \lambda_{t\omega'}^D = \lambda_{t\omega}^D \tag{4.65}$$

$$P_{t\omega}^A = P_{t\omega'}^A, \forall t, \forall \omega, \omega' : (\lambda_{t\omega'}^D = \lambda_{t\omega}^D \forall t)$$

$$\text{and } (P_{\tau\omega'}^W = P_{\tau\omega}^W, \forall \tau = 1, 2, \dots, N_{T1}) \tag{4.66}$$

$$- \sum_{t=1}^{N_T} [\lambda_{t\omega}^D P_{t\omega}^D d_t + \lambda_{t\omega}^A P_{t\omega}^A d_t + \lambda_{t\omega}^D (r_{t\omega}^+ \Delta_{t\omega}^+ - r_{t\omega}^- \Delta_{t\omega}^-)] + \zeta - \eta_\omega \leq 0, \forall \omega \tag{4.67}$$

$$\eta_\omega \geq 0, \quad \forall \omega. \tag{4.68}$$

The objective function (4.56) comprises two terms: i) the expected profit

and ii) the CVaR multiplied by the weighting factor β , which allows controlling the risk-aversion degree of the wind producer.

Constraints (4.57)–(4.59) limit the amount of energy that can be traded in both the day-ahead and adjustment markets to the installed capacity of the wind farm. The set of equations (4.60)–(4.63) determines the total, positive and negative energy deviations incurred by the wind producer per period and scenario. Constraints (4.64) enforce the non-decreasing condition of the offering curves in the day-ahead market. Constraints (4.65) and (4.66) constitute the non-anticipativity conditions related to the decisions made in the day-ahead and adjustment markets, respectively. Finally, equations (4.67)–(4.68) are required to compute the CVaR.

It is important to note that the three-stage model (4.56)–(4.68) above is intended to produce optimal offering strategies for the day-ahead market. Once the position in this market is fixed, a detailed two-stage stochastic programming formulation analogous to (4.40)–(4.45) can be built to generate optimal offers for the adjustment market. Specifically, such a model can be formulated as follows:

$$\begin{aligned} & \text{Maximize}_{P_t^A, \forall t; \Delta_{t\omega}^+, \forall t, \forall \omega; \Delta_{t\omega}^-, \forall t, \forall \omega} \\ & \sum_{\omega=1}^{N_\Omega} \sum_{t=1}^{N_T} \pi_\omega (\lambda_{t\omega}^A P_t^A d_t + \tilde{\lambda}_t^D r_{t\omega}^+ \Delta_{t\omega}^+ - \tilde{\lambda}_t^D r_{t\omega}^- \Delta_{t\omega}^-) \end{aligned} \quad (4.69)$$

subject to

$$0 \leq \tilde{P}_t^D + P_t^A \leq P^{\text{W}, \max}, \quad \forall t \quad (4.70)$$

$$\Delta_{t\omega} = d_t (P_{t\omega}^{\text{W}} - \tilde{P}_t^D - P_t^A), \quad \forall t, \forall \omega \quad (4.71)$$

$$\Delta_{t\omega} = \Delta_{t\omega}^+ - \Delta_{t\omega}^-, \quad \forall t, \forall \omega \quad (4.72)$$

$$0 \leq \Delta_{t\omega}^+ \leq P_{t\omega}^{\text{W}} d_t, \quad \forall t, \forall \omega \quad (4.73)$$

$$0 \leq \Delta_{t\omega}^- \leq (P^{\text{W}, \max} - P_{t\omega}^{\text{W}}) d_t, \quad \forall t, \forall \omega, \quad (4.74)$$

where $\tilde{\lambda}_t^D$ and $\tilde{P}_t^D$ are, respectively, the actual day-ahead market price and the actual power traded in this market in time period t .

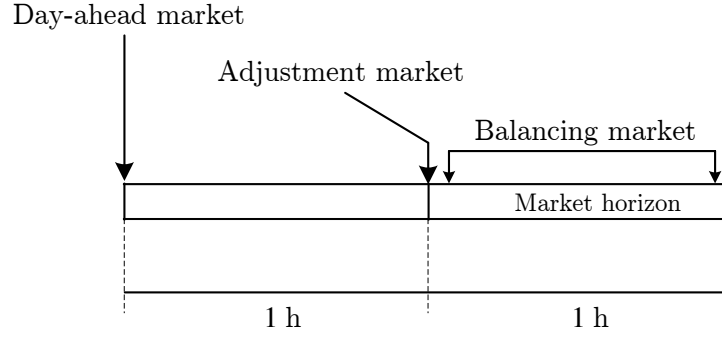


Figure 4.10: Wind producer example: market structure and time framework

4.6 Wind Producer Example

In this example, the offering strategy model formulated in Section 4.5.5 is illustrated considering an electricity market with the short-term trading structure and time framework represented in Fig. 4.10.

The market horizon only spans one hour. Consequently, subscript t can be omitted. The day-ahead market is cleared one hour in advance, while the adjustment market closes just before the single delivery period. Note that the name *day-ahead* market is somewhat misleading here. Rather, it should be called *1-hour ahead* market. Finally, in the balancing market, the energy deviations incurred by market participants during the 1-hour market horizon are covered and priced accordingly.

Let us assume that:

1. The day-ahead market price λ^D is equal to \$50/MWh with a probability of 0.40 and equal to \$20/MWh with a probability of 0.60.
2. The difference $\Delta\lambda$ between day-ahead and adjustment market prices is -10\$/MWh with a probability of 0.35 and \$3/MWh with a probability of 0.65.

Therefore, the adjustment market price λ^A can adopt the following four possible values:

1. $\lambda^A = \$50/\text{MWh} - 10\$/\text{MWh} = \$40/\text{MWh}$ with a probability $0.40 \times 0.35 = 0.14$.
2. $\lambda^A = \$50/\text{MWh} + 3\$/\text{MWh} = \$53/\text{MWh}$ with a probability $0.40 \times 0.65 = 0.26$.
3. $\lambda^A = \$20/\text{MWh} - 10\$/\text{MWh} = \$10/\text{MWh}$ with a probability $0.60 \times 0.35 = 0.21$.
4. $\lambda^A = \$20/\text{MWh} + 3\$/\text{MWh} = \$23/\text{MWh}$ with a probability $0.60 \times 0.65 = 0.39$.

Suppose also that random variable r is conveniently discretized in the following two scenarios:

1. Variable r is equal to 1.5 with a probability of 0.7.
2. Variable r is equal to 0.9 with a probability of 0.3.

According to definition (4.21), the above description is equivalent to the next one:

1. Pair (r^+, r^-) is equal to $(1, 1.5)$ with a probability of 0.7.
2. Pair (r^+, r^-) is equal to $(0.9, 1)$ with a probability of 0.3.

Lastly, consider that the amount of power produced by a certain 100-MW wind farm during the hour between the closures of the day-ahead and adjustment markets ($\tau = 1$) can be described by means of two scenarios ($N_{\omega_1} = 2$), namely, scenarios *high* and *low* corresponding to wind power values of 60 MW and 30 MW, respectively.

On the other hand, the amount of power that the wind farm is expected to produce during the hour comprising the market horizon ($t = 1$) is contingent on the scenario, *high* or *low*, realized in the previous hour. Such a dependency can be expressed as follows:

1. If the wind power production in $\tau = 1$ is *high* (60 MW), then the amount of power produced in $t = 1$ (market horizon) can be *extremely high* (100 MW) or *relatively high* (50 MW), with probabilities 0.1 and 0.3, respectively.

2. If the wind power production in $\tau = 1$ is *low* (30 MW), then the amount of power produced in $t = 1$ (market horizon) can be *extremely low* (0 MW) or *relatively low* (40 MW), with probabilities 0.35 and 0.25, respectively.

Therefore, the wind power production during the 1-h market horizon can be statistically represented by means of four scenarios, namely, scenario *extremely high* (100 MW), scenario *relatively high* (50 MW), scenario *relatively low* (30 MW), and scenario *extremely low* (0 MW) with probabilities 0.1, 0.3, 0.25, and 0.35 in that order.

The scenario tree that results from the scenario representation of the day-ahead market price, the price difference between day-ahead and adjustment markets, the imbalance price ratios and the wind power production provided above is shown in Fig. 4.11. Prices are in \$/MWh, powers in MW and ratios are non-dimensional numbers.

Some of the probabilities π_ω are computed below for illustrative purposes. The rest of them are shown in the aforementioned figure. The computed ones are

$$\pi_1 = 0.40 \times 0.35 \times 0.10 \times 0.70 = 0.0098$$

$$\pi_2 = 0.40 \times 0.35 \times 0.10 \times 0.30 = 0.0042$$

$$\vdots$$

$$\pi_{15} = 0.40 \times 0.65 \times 0.25 \times 0.70 = 0.0455$$

$$\pi_{16} = 0.40 \times 0.65 \times 0.25 \times 0.30 = 0.0195$$

$$\vdots$$

$$\pi_{31} = 0.60 \times 0.65 \times 0.25 \times 0.70 = 0.06825$$

$$\pi_{32} = 0.60 \times 0.65 \times 0.25 \times 0.30 = 0.02925.$$

Note that this example is intended to be a more realistic version of that described in Section 2.4.1.

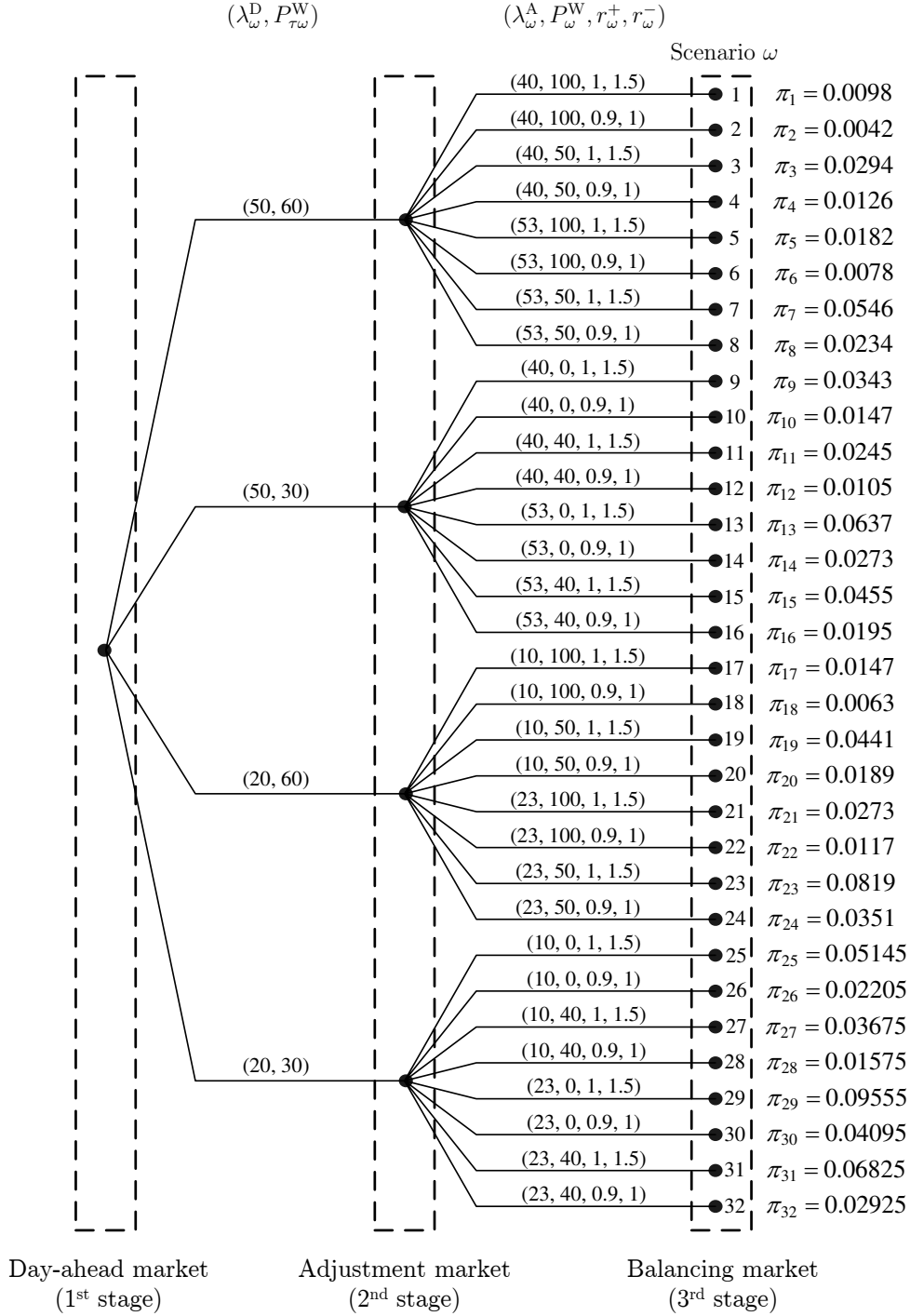


Figure 4.11: Wind producer example: three-stage scenario tree

To begin with the fundamentals, we first consider the basic model introduced in Section 4.5.1, which produces optimal energy quantities (not offering curves) to be offered in the day-ahead market at price zero by a risk-neutral wind power producer that ignores the presence of the adjustment market. Being so, in this case, the scenario tree in Fig. 4.11 can be reduced to half of its former size, i.e., it can be shrunk from 32 to 16 scenarios. Every pair of scenarios in the tree sharing the same realizations for the day-ahead market prices, imbalance price ratios and wind power production can be merged into a single one by adding their probabilities. That is, scenarios $\omega = i + j$ and $\omega = i + j + 4$, with $i = 1, 2, 3, 4$ and $j = 0, 8, 16, 24$, are equivalent. The tree so condensed is shown in Fig. 4.12. As an example, probability π_1 in this tree is obtained as follows,

$$\pi_1 = 0.0098 + 0.0182 = 0.028,$$

where 0.0098 and 0.0182 are, respectively, the probabilities associated with scenarios $\omega = 1$ and $\omega = 5$ in the original tree.

When it comes to deal with uncertainties in an optimization problem, it is a good practice to ask oneself first about the convenience of using a stochastic solution approach instead of a simpler deterministic one. In order to solve this dilemma, we calculate the Value of the Stochastic Solution (VSS), which allows precisely assessing the effectiveness of a stochastic programming approach. The concept of the VSS is explained in Appendix A. In a few words, it is defined as the difference between the expected profit of the stochastic model and the expected profit obtained from the problem where decisions variables are fixed to those resulting from the associated deterministic problem, i.e., from the problem in which stochastic processes are replaced by their respective expected values. In the considered case, the deterministic approach to tackle the short-term trading problem of a wind power producer is naïve and boils down to submitting the forecast wind power production to the day-ahead market. According to the scenario tree in Fig. 4.12, the predicted wind power produced during the single period comprising the market horizon, $\hat{P}^W$, can be calculated

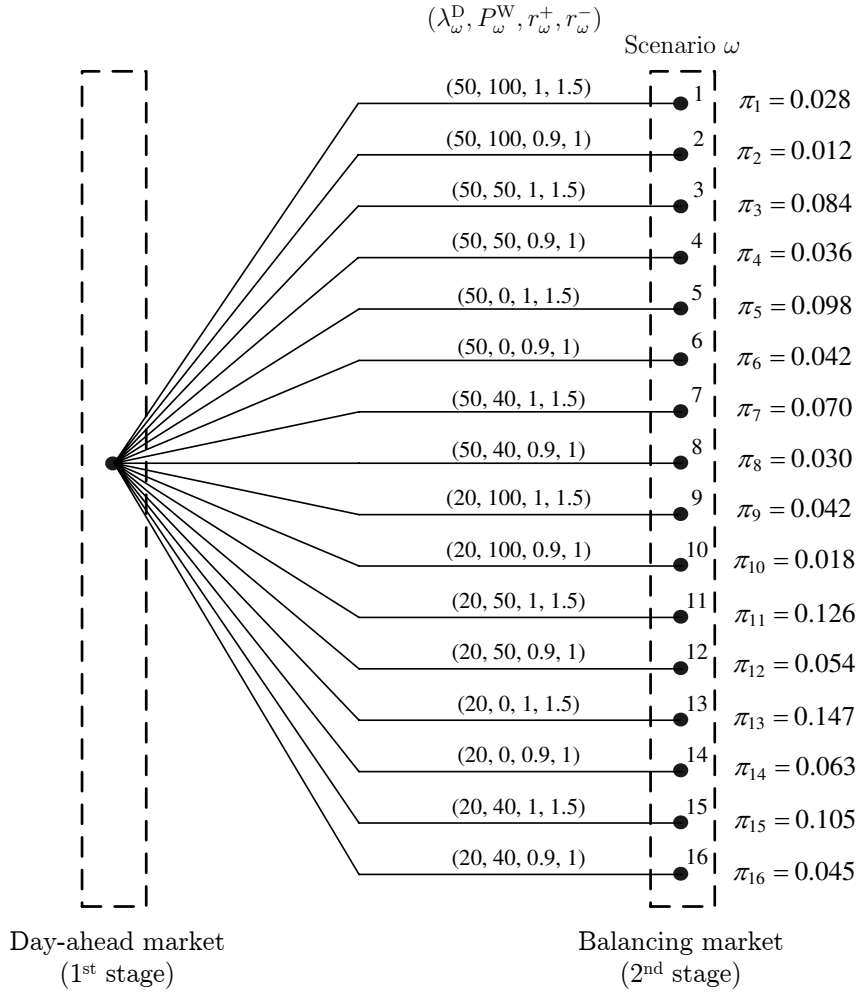


Figure 4.12: Wind producer example: two-stage scenario tree (excluding the adjustment market)

as

$$\begin{aligned}\hat{P}^W = \mathcal{E}\{P^W\} &= \sum_{\omega=1}^{16} \pi_{\omega} P_{\omega}^W = (0.028 + 0.012 + 0.042 + 0.018) \times 100 \\ &+ (0.084 + 0.036 + 0.126 + 0.054) \times 50 \\ &+ (0.098 + 0.042 + 0.147 + 0.063) \times 0 \\ &+ (0.070 + 0.030 + 0.105 + 0.045) \times 40 = 35\text{MW}.\end{aligned}$$

The expected profit resulting from model (4.40)-(4.45) if the first-stage variable P^D is fixed to 35 MW is equal to \$971.04. For convenience, we denote the expected profit so obtained as z^{d*} , with superscript d standing for *deterministic*. On the other hand, the stochastic solution of this model yields an expected profit z^{s*} equal to \$1086.4. Note that superscript s indicates *stochastic*. Consequently, the Value of Stochastic Solution is given by

$$\text{VSS} = z^{s*} - z^{d*} = 1086.4 - 971.04 = \$115.36,$$

which approximately represents a percentage of 10.6% with respect to z^{s*} . This significant value justifies the use of a stochastic approach to tackle the wind producer problem instead of a prediction-based one, and reveals that the short-term decision-making process of a wind power producer is highly influenced by the uncertainties involved, namely, day-ahead market prices, imbalance price ratios, and wind power production.

In this respect, it is relevant to quantify the relative economic impact of each source of uncertainty on the decision-making undertaken by a wind producer. To this end, we can make use of the Expected Value of Perfect Information (EVPI), which measures how much the wind producer would be willing to pay to know the future realizations of the stochastic processes under consideration (market prices and wind availability). It is computed as the difference between the expected profit resulting from the problem where the decisions to be made *anticipate* the subsequent outcomes of the random variables (wait-and-see problem) and the expected profit of the stochastic model. In the wait-and-see problem, therefore, the optimal amount of power to be traded in

the day-ahead market is determined for each possible scenario. Further details on the EVPI can be found in Appendix A. In the example under study, we determine the EVPI by assuming perfect information on each stochastic process separately in order to isolate the economic effect of each uncertainty source from the rest.

Table 4.1: Wind producer example: expected values of perfect information

Perfect information on	λ^D	P	(r^+, r^-)	all
EVPI (%)	0	3.09	3.09	3.09

Table 4.1 shows the expected values of perfect information as improvements in percentage with respect to the expected profit obtained from model (4.40)–(4.45). In light of the results presented in this table, the following two comments are in order. First, the economic improvement produced by a perfect knowledge of day-ahead market prices is smaller than the other two, which evidences that the uncertainty associated with these prices plays a secondary role in the wind producer offering problem. Moreover, this economic improvement is nil. This is due to the fact that, as highlighted in Section 4.3.3, the maximization of the expected profit (4.40) boils down to the minimization of the imbalance cost. This cost depends on r^+ and r^- , but is independent of λ^D . Second, the economic improvements owing to perfect information on wind production and imbalance price ratios are equal. The reason for this is that knowing the values of r^+ and r^- in advance is equivalent to knowing perfectly the future wind production availability from a strategic point of view. As a result, the total EVPI that is obtained assuming perfect information on all three stochastic processes, namely, day-ahead market prices, imbalance price ratios and wind power production, is also 3.09%.

According to model (4.40)–(4.45), the optimal offering strategy of a wind power producer facing the two-stage scenario tree depicted in Fig. 4.12 comes down to not participating in the day-ahead market, and thus, trading all its actual production in the balancing market. That is, $P^D = 0$. The ultimate purpose behind this strategy is to avoid producing less than settled in the day-ahead market because:

1. The probability that the net system imbalance is negative (deficit of generation) is high (0.7), while the probability of being positive (excess of generation) is comparatively low (0.3).
2. If the net system imbalance is indeed negative, the price at which the wind producer has to buy energy in the balancing market to cover a possible generation shortage is 50% greater than the day-ahead market price ($r^- = 1.5$). However, it can sell its possible overproduction at a price equal to the day-ahead market price ($r^+ = 1$).
3. In contrast, if the net system imbalance is positive, the price at which the wind power producer can sell its generation surplus is just 10% lower than the day-ahead market price ($r^+ = 0.9$).

Therefore, the wind producer has little to lose by trading all its actual production in the balancing market. This attitude is even easier to justify by appealing to the fact that the maximization of the expected profit (4.40) is equivalent to the minimization of the imbalance cost defined in (4.19).

In Fig. 4.13(a), the imbalance cost incurred by the wind producer is plotted as a function of the amount of power sold in the day-ahead market. Observe that the minimum imbalance cost is achieved for $P^D = 0$. The shape of the imbalance cost function can be easily altered by changing the possible values of the imbalance price ratios r^+ and r^- as illustrated in Figs. 4.13(b), 4.13(c), and 4.13(d). Note that these figures represent situations in which the best energy offer to the day-ahead market turns out to be 40, 50, and 100 MW, respectively.

Another effective manner to alter the optimal offering strategy of a wind power producer consists in introducing a measure allowing to control the risk of profit variability into the offering model. The mathematical adjustments required to this end are explained in Section 4.5.3, where the referred measure is the Conditional Value-at-Risk (CVaR).

Let us consider the case represented in Fig. 4.13(d), in which the pair (r^+, r^-) is equal to $(1, 1.02)$ with a probability of 0.7 and equal to $(0.1, 1)$ with a probability of 0.3. As indicated above, the best offer to the day-ahead market in

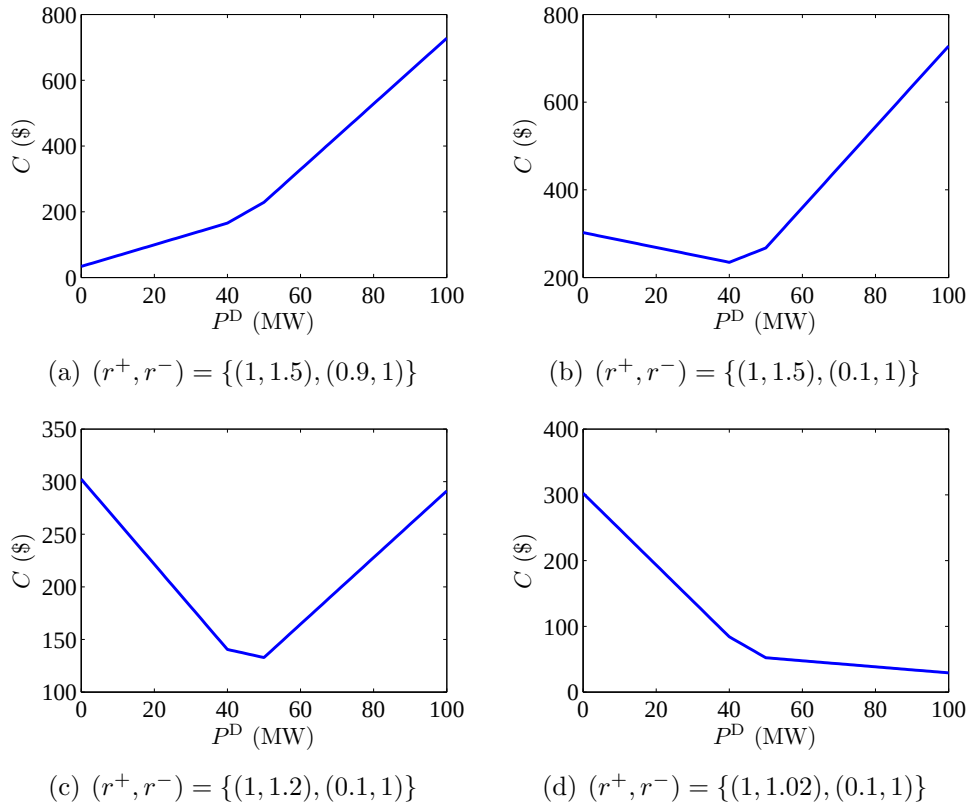


Figure 4.13: Wind producer example: imbalance cost as a function of the amount of power sold in the day-ahead market

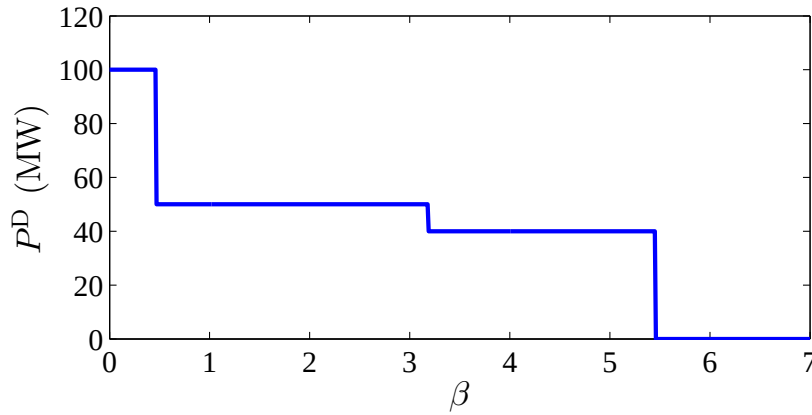


Figure 4.14: Wind producer example: power sold in the day-ahead market as a function of the risk-aversion parameter β

this case is 100 MW. Actually, it is 100 MW as long as the wind power producer adopts a risk-neutral attitude ($\beta = 0$). Fig. 4.14 illustrates the evolution of the best offer as the wind producer becomes more and more risk averse, i.e., as the parameter β increases. Observe that the amount of power sold in the day-ahead market abruptly diminishes for certain values of β determining a change in the risk perception of the wind producer. In fact, this evolution reflects the basic strategy that a wind power producer can adopt to hedge against profit volatility in the absence of an adjustment market: trading less energy in the day-ahead market in the hope that its overproduction can be sold in the balancing market at a still competitive price. In this sense, it is important to note that, according to the mechanism for imbalance price formation described in Section 4.3.2, where negative electricity prices are *prohibited*, negative energy imbalances can entail economic losses (profit below zero) for the wind producer, while positive energy imbalances can lead to nil profit at the most. Therefore, in general terms, suffering negative deviations is riskier than incurring positive deviations.

So far we have ignored the possibility of submitting an offering curve to the day-ahead market instead of just a single energy quantity. The mathematical adjustments and considerations required to design such a curve are provided in Section 4.5.2 of this chapter. It is based on adding constraints (4.46) and (4.47) to the offering strategy model of the wind power producer. As explained in the referred section, on the one hand, constraints (4.46) enforce the non-decreasing nature of the offering curve, required in most electricity markets while, on the other hand, constraints (4.47) constitute a *loosened* implementation of the non-anticipatory character of information on day-ahead market prices, which precisely allows us to obtain the optimal energy-price pairs making up the offering curve. In particular, for the scenario tree represented in Fig. 4.12, these constraints boil down to

$$\begin{aligned} P_9^D - P_1^D &\leq 0, \\ P_1^D &= P_2^D = P_3^D = P_4^D = P_5^D = P_6^D = P_7^D = P_8^D, \\ P_9^D &= P_{10}^D = P_{11}^D = P_{12}^D = P_{13}^D = P_{14}^D = P_{15}^D = P_{16}^D. \end{aligned}$$

Therefore, the offering curve resulting from the wind producer model is simply composed of two different energy-price pairs ($1 \text{ h} \times P_1^D, \lambda_1^D$) and ($1 \text{ h} \times P_9^D, \lambda_9^D$) corresponding to the two possible values of the day-ahead market price, \$50/MWh and \$20/MWh, respectively. More realistic offering curves are obtained in the case study presented in Section 4.7.

Due to the fact that the generation cost of a wind farm is zero, offering curves are, a priori, less useful for a wind producer than they are for a thermal one. Actually, offering curves enable thermal producers to account somehow for the shape of their quadratic generation cost function in the trading process. Needless to say, this is not an issue for a wind producer. As an example of this, let us consider the four cases represented in Fig. 4.13. In all these instances, the optimal energy-price curve resulting from the offering strategy model including constraints (4.46) and (4.47) is such that $P_1^D = P_9^D$. In plain words, there is no curve, but just a point, a single optimal energy quantity to be offered at price zero.

Notwithstanding the remark above, there are some special circumstances in which submitting offering curves to the day-ahead market can be potentially advantageous for a wind producer. These circumstances occur in electricity markets where the value of the day-ahead market price is indicative of the direction, positive or negative, and/or the magnitude of the real-time net system imbalance. Statistically speaking, we refer to those situations in which the day-ahead market price and the imbalance price ratios are correlated. As an example, let us consider the following two opposite hypotheses:

Hypothesis A: if the day-ahead market price λ^D is high (\$50/MWh), the net system imbalance is negative and the pair of imbalance price ratios (r^+, r^-) resulting from the balancing market is equal to (1, 1.5) with a probability of 0.7 and equal to (1, 1.3) with a probability of 0.3. Conversely, if the day-ahead market price is low (\$20/MWh), the net system imbalance is positive and the imbalance price ratios (r^+, r^-) are (0.6, 1) and (0.9, 1) with probabilities 0.7 and 0.3, respectively.

Submitting an offering curve to the day-ahead market is useless under this hypothesis due to the enforced non-decreasing character of the curve.

Actually, the optimal energy-price pairs making up the curve in this case are (0 MWh, \$20/MWh) and (0 MWh, \$50/MWh). Thus, all the wind power production is traded in the balancing market.

Hypothesis B: if the day-ahead market price λ^D is high (\$50/MWh), the net system imbalance is positive and the pair of imbalance price ratios (r^+, r^-) is equal to (0.6, 1) with a probability of 0.7 and equal to (0.9, 1) with a probability of 0.3. On the contrary, if the day-ahead market price λ^D is low (\$20/MWh), the net system imbalance is negative and the pair of imbalance price ratios (r^+, r^-) is equal to (1, 1.5) with a probability of 0.7 and equal to (1, 1.3) with a probability of 0.3. Note that hypothesis B is right the opposite of hypothesis A.

Offering a curve results advantageous in this situation, where the optimal energy-price pairs turn out to be (0 MWh, \$20/MWh) and (100 MWh, \$50/MWh). In this second case, the wind producer can allow itself to trade a greater amount of energy (100 MWh) in the day-ahead market for a high price realization (\$50/MWh) without having to worry about the imbalance cost entailed by a possible generation shortage. Specifically, if the wind producer submits the previous optimal energy-price curve to the day-ahead market in this second instance, the achieved economic improvement in terms of expected profit amounts to over 13.1% with respect to the expected profit obtained if the opportunity of submitting a curve is wasted.

To conclude this example, we assess how much the expected profit of the wind power producer increases if it decides to make use of the adjustment market available within the short-term trading structure illustrated in Fig. 4.10. Then, the scenario tree to be considered is the one depicted in Fig. 4.11. As pointed out in Section 4.3.4, such an increase is caused by two different effects, namely:

- i) As a trading floor, the adjustment market provides the wind producer with an additional opportunity to buy or sell energy at a price different than the day-ahead market price. For convenience, hereinafter, we refer to this effect as *additional trading effect*.

- ii) The clearing time of the adjustment market is closer to the energy delivery horizon and consequently, trading in this market can be performed with a reduced level of uncertainty about the future wind power production. This is what we call *certainty gain effect*.

In order to isolate both effects, we manipulate the non-anticipativity constraints (4.55) related to the decisions made in the adjustment market. As indicated in Section 4.5.4, the certainty gain effect is mathematically implemented by means of the second term of the “and” condition appearing in these constraints. Therefore, if we ignore this term, the certainty gain effect is not accounted for in the offering strategy model. In particular, for the scenario tree depicted in Fig. 4.11, non-anticipativity constraints (4.55) are

$$\begin{aligned}
P_1^A &= P_2^A = P_3^A = P_4^A = P_5^A = P_6^A = P_7^A = P_8^A, \\
P_9^A &= P_{10}^A = P_{11}^A = P_{12}^A = P_{13}^A = P_{14}^A = P_{15}^A = P_{16}^A, \\
P_{17}^A &= P_{18}^A = P_{19}^A = P_{20}^A = P_{21}^A = P_{22}^A = P_{23}^A = P_{24}^A, \\
P_{25}^A &= P_{26}^A = P_{27}^A = P_{28}^A = P_{29}^A = P_{30}^A = P_{31}^A = P_{32}^A.
\end{aligned}$$

However, if the certainty gain effect is disregarded, such non-anticipativity constraints become

$$\begin{aligned}
P_1^A &= P_2^A = P_3^A = P_4^A = P_5^A = P_6^A = P_7^A = P_8^A = P_9^A = P_{10}^A = \\
&= P_{11}^A = P_{12}^A = P_{13}^A = P_{14}^A = P_{15}^A = P_{16}^A, \\
P_{17}^A &= P_{18}^A = P_{19}^A = P_{20}^A = P_{21}^A = P_{22}^A = P_{23}^A = P_{24}^A = P_{25}^A = P_{26}^A = \\
&= P_{27}^A = P_{28}^A = P_{29}^A = P_{30}^A = P_{31}^A = P_{32}^A.
\end{aligned}$$

Observe that this last version of the non-anticipativity constraints is *tighter* than the previous one and therefore, the consideration of the certainty gain effect is expected to improve the solution to the wind producer problem in terms of profit.

Let us consider again the four scenario representations (a), (b), (c), and (d) of the imbalance price ratios (r^+ , r^-) used as examples to build the imbalance cost functions in Figs. 4.13(a), 4.13(b), 4.13(c), and 4.13(d), respectively.

Table 4.2: Wind producer example: adjustment market effects on the expected profit of the wind producer (increases in percentages)

Effect	Case			
	(a)	(b)	(c)	(d)
Additional trading	14.27	11.83	8.64	4.99
Certainty gain	0	14.03	4.24	2.05
Total	14.27	25.86	12.88	7.04

Table 4.2 shows, for all these cases, the economic improvements due to the two effects indicated above in percentages with respect to the expected profit obtained if the adjustment market is not taken into account.

It is relevant to underline that, according to the results provided in Table 4.2, the economic impact of the certainty gain effect may be as significant as the additional trading effect and therefore, overlooking it in the wind producer problem should be seen as a mistake. In contrast, the certainty gain effect is nonexistent in the short-term trading problem of a conventional (thermal) producer.

4.7 Wind Producer Case Study

In this section, the features of the proposed stochastic offering model are further illustrated using a realistic case study based on an electricity market that includes one day-ahead market and one adjustment market arranged throughout the trading horizon as shown in Fig. 4.15.

Additionally, a balancing market is organized before each trading hour of the market horizon (day d) to guarantee the real-time energy balance between generation and demand. For this purpose, this market remains open until 10 minutes before the delivery hour under consideration. For instance, the balancing market intended to ensure the system energy balance during the hour 8:00-9:00 pm of day d is closed at 7:50 pm of this day. We assume therefore that 10 minutes prior to the energy delivery hour, energy deviations incurred by market participants in that hour are perfectly known. In this

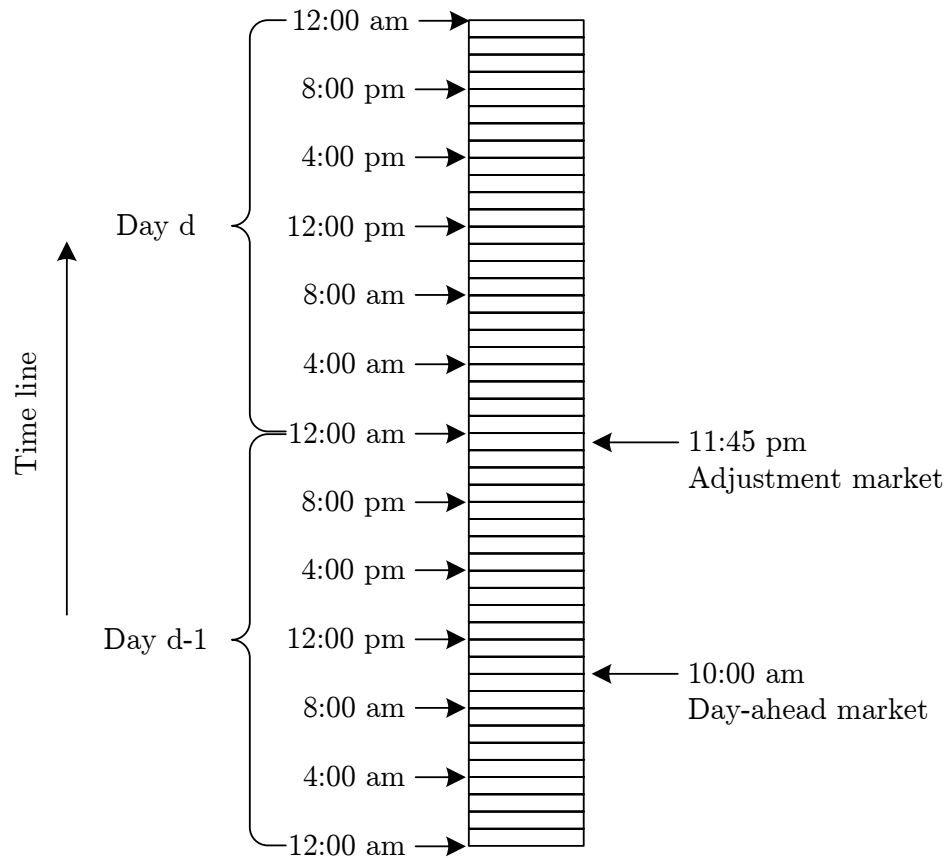


Figure 4.15: Wind producer case study: market organization and time framework

respect, we emphasize that, from a theoretical point of view, the balancing market should be understood just as a mechanism through which real-time energy deviations are covered and priced under competition.

The day-ahead market is closed at 10:00 am of day $d - 1$, i.e., 14 hours before the beginning of the energy delivery period (day d). The adjustment market, which is cleared at 11:45 pm of day $d - 1$, can be used by market agents (wind producers, among others) to alter their scheduled productions (or consumptions in the case of loads) for the trading periods after its closure, that is, for the 24 hours of day d .

We consider a wind farm having 40 2.5-MW commercial wind generators. The total installed wind power capacity of the plant is therefore 100 MW. All the turbines making up the wind farm are assumed to be operating for the considered trading day.

Stochastic processes describing wind speed and market prices are modeled via the time-series-based approach described in Chapter 3. Specifically, to characterize wind speed uncertainty, the following second-order autoregressive model is used,

$$Z_t = \phi_1 Z_{t-1} + \phi_2 Z_{t-2} + a_t, \quad (4.75)$$

where Z_t represents the *normalized* wind speed in time period t , a_t is a white noise term, and ϕ_1 and ϕ_2 , with values equal to 1.175 and -0.244, respectively, are parameters estimated from the historical wind speed data collected by Dr. G. L. Johnson from a location in the state of Kansas (US). This set of data is publicly available in [1]. The historical records used comprise average hourly wind speed measurements from January 25 to March 31, 1996. As explained in Section 3.3.4, a normalization of the wind speed series is performed in order to preserve its marginal distribution, usually a Weibull. Fig. 4.16 represents both the wind speed frequency distribution of the considered site and the probability density function (pdf) of the fitted Weibull.

As stated in Section 3.3.5, to convert wind speed scenarios into wind power scenarios, the power curve of the turbine model installed in the wind farm (model Nordex N80/250 with a hub height of 100 m) is used. Such a power curve is depicted in Fig. 4.17, can be found in the manufacturer database, and

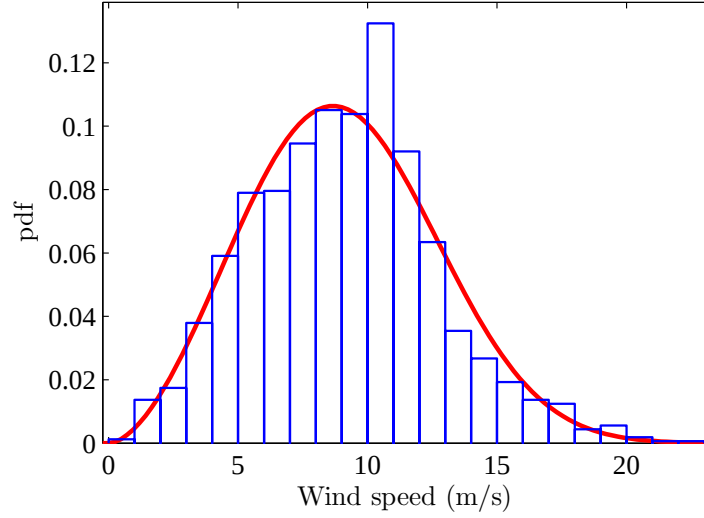


Figure 4.16: Wind producer case study: wind speed frequency distribution of the considered location and the pdf of the Weibull fit

Table 4.3: Wind producer case study: stochastic processes for prices and price functions

Variable	Time series model
$\log(\lambda_t^D)$	ARIMA(2, 0, 1)(1, 1, 1)_{24}
$\Delta\lambda_t = \lambda_t^A - \lambda_t^D$	ARIMA(1, 0, 11)(1, 0, 1)_{24}
$\sqrt{r_t} = \sqrt{r_t^+ + r_t^- - 1}$	ARIMA(2, 0, 1)(1, 0, 1)_{24}

is provided in [26]. Then, on the hypothesis that the characteristics of the wind are the same all over the plant at each instant, scenarios representing the power output of the wind farm can be obtained by just aggregating the production of each available turbine in the plant.

For simplicity, the stochastic processes describing price behavior in the day-ahead, adjustment, and balancing markets are modeled using simplified seasonal ARIMA models. The ARIMA models fitted for the day considered in this case study (April 1, 2008) are provided in Table 4.3. The historical data used for the fitting process belong to the electricity market of the Iberian Peninsula [3] and are publicly available in [6]. These data correspond to the

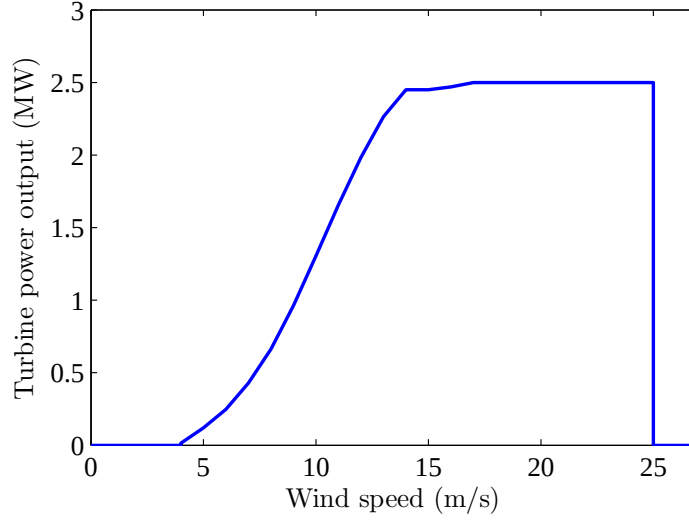


Figure 4.17: Wind producer case study: power curve of the considered turbine model

working days between October 2007 and March 2008. Only working days are considered to simplify the ARIMA models by eliminating the weekly seasonality.

The number of generated scenarios for the day-ahead, adjustment, and balancing markets are, respectively, 1000, 1000, and 1000. The number of scenarios simulated to characterize the wind generation uncertainty is also 1000, resulting from the combination of 20 (N_{ϖ_1}) scenarios representing the wind power realizations in time periods between the closures of the day-ahead and adjustment markets, and 50 (N_{ϖ_2}) scenarios to embrace the possible wind power realizations after the clearing of the adjustment market.

The total number of scenarios comprising the original scenario tree is therefore 10^9 , which clearly yields an optimization problem that is intractable. Hence, to achieve tractability, the variant 1 of the scenario reduction technique described in Section 3.4.3.1 is applied to each stochastic process, resulting in 10, 5, and 10 scenarios for the day-ahead, adjustment and balancing markets, respectively, and 20 scenarios for the wind farm production ($N_{\varpi_1} \times N_{\varpi_2} = 5 \times 4$). Thus, the reduced scenario tree includes 10 000 scenarios.

Table 4.4: Wind producer case study: optimal energy bids for $\beta = 0$

Period t	7	15	20	21
Offer (MWh)	56.68	69.56	91.16	81.18

A confidence level $\alpha = 0.95$ is used to compute the Conditional Value-at-Risk (CVaR) in all instances.

Firstly, no adjustment market is considered. Being so, the Value of the Stochastic Solution (VSS) is 1.87%, expressed as a percentage of the expected profit obtained from model (4.40)–(4.47). This value justifies the interest in using a stochastic approach for solving the wind producer problem instead of a naïve deterministic one. If the wind producer decides to adopt a risk-neutral attitude ($\beta = 0$), offers to the day-ahead market just consist of an optimal energy quantity submitted at zero price for each time period. However, these single quantities may turn into curves if the wind producer becomes risk-averse ($\beta \neq 0$). Table 4.4 shows the energy quantities offered to the day-ahead market for selected hours in the risk-neutral situation and Fig. 4.18 depicts the resulting offer curves in the same hours if $\beta = 1$. As an example, the offer curve for period 21 should be interpreted as follows: the wind producer offers 40 MWh at zero price. It is willing to sell 14 additional MWh of energy approx. (i.e., up to 54 MWh in total) if the resulting price in the day-ahead market price is above €70/MWh. For prices in between 67.5 and €70/MWh, the amount of energy that the wind producer offers in the day-ahead market is given by the linear function linking points (40 MWh, €67.5/MWh) and (54 MWh, €70/MWh) in the curve. It is important to note that for a market requiring step-wise offer stacks, the linear blocks of the curves in Fig. 4.18 should be approximated by step-wise functions.

The changes undergone by the offering curves for increasing β values are intended to reduce the expected energy traded in the day-ahead market as the wind producer becomes more risk-averse. This trend is illustrated in Fig. 4.19, where the expected amount of energy sold in this market is represented as a function of the risk-aversion factor β . As pointed out in the wind producer

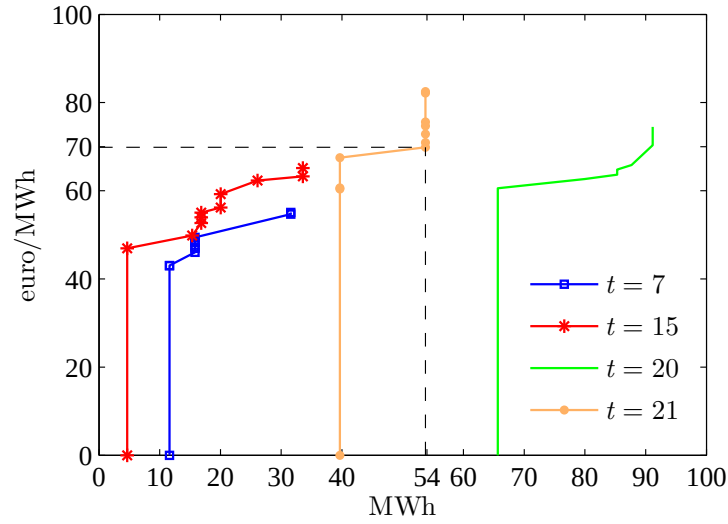


Figure 4.18: Wind producer case study: optimal offering curves for $\beta = 1$

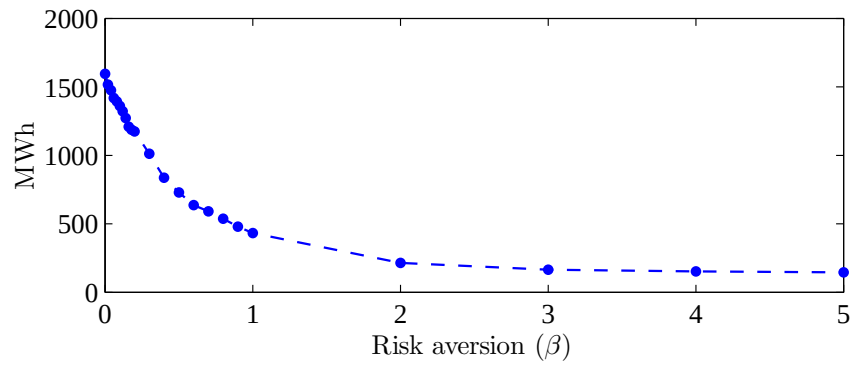


Figure 4.19: Wind producer case study: expected energy traded in the day-ahead market as a function of the risk-aversion factor β

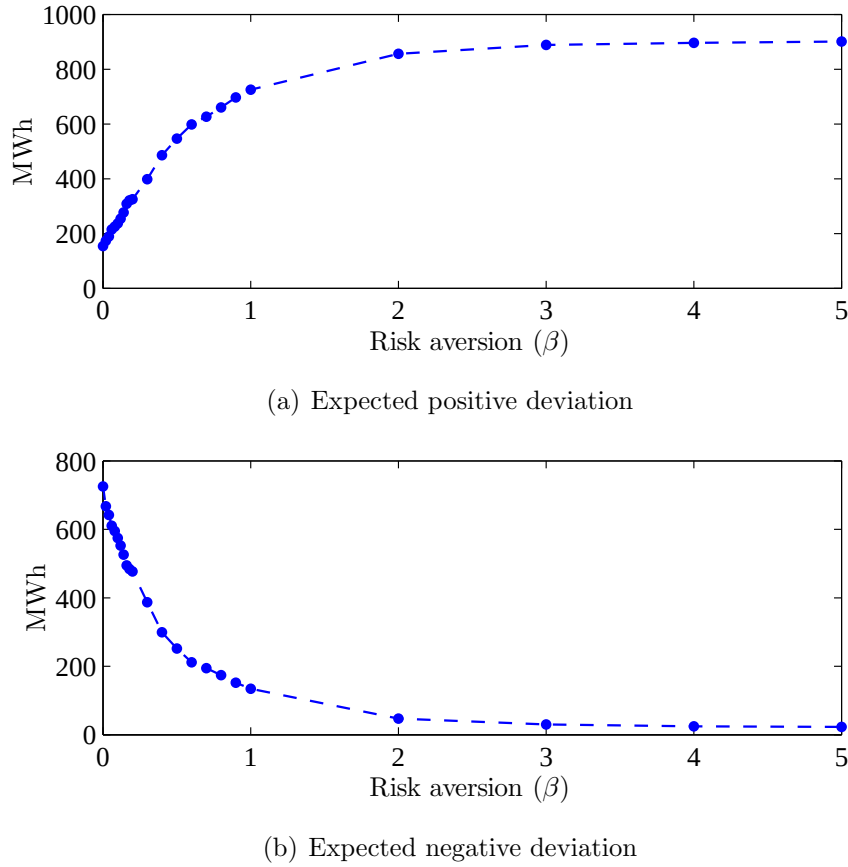


Figure 4.20: Wind producer case study: impact of risk aversion on expected energy deviations

example of Section 4.6, the only measure that a wind power producer can take to hedge against profit variability in the absence of an adjustment market is to trade less energy in the day-ahead market in the hope of selling its generation surplus in the balancing market at a good price. Consequently, as the risk aversion increases, the expected negative deviation incurred by the wind producer all over the market horizon decreases at the expense of rising the expected positive deviation. This is shown in Fig. 4.20. Nevertheless, unlike submitting just a single energy value for each time period, offering curves allows reducing the expected energy placed in the day-ahead market while still keeping the possibility of trading greater amounts of energy for high price realizations.

Next, we analyze the enhancement in competitiveness that the wind producer achieves if it takes full advantage of the certainty gain associated with the availability of an adjustment market within the market structure. For this purpose, we perform a comparison between the results provided by model (4.56)–(4.68) depending on whether the certainty gain introduced by the adjustment market is considered or not, i.e., depending on whether the second term of the “and” condition in constraints (4.66) is included or not.

To start with, if an adjustment market is taken into account in the offering strategy model, the Value of the Stochastic Solution for the risk-neutral case increases up to 3.61%, i.e., 1.74% more than the value obtained if this market is ignored. However, the VSS is reduced to 2.58% (1.03% lower!) if the associated certainty gain effect is not accounted for. This result reveals that one of the most important advantages of using a stochastic approach to tackle the wind producer problem, instead of just a deterministic one, is indeed to consider this effect.

In Fig. 4.21 the expected total deviations incurred by the wind producer all over the market horizon are compared for different values of the weighting factor β . The reduction in the expected total amount of imbalances due to the certainty gain associated with trading in the adjustment market is apparent. Fig. 4.22 shows the value of this reduction per time periods and $\beta = 0$.

Observe that the decrease in expected deviation is considerably more significant for the first few periods, in which the volatility pertaining to wind power generation is comparatively smaller. Reduction levels in energy deviations similar to those obtained in these first periods could be achieved for the rest of hours by including additional adjustment trading floors sequentially arranged all over the market horizon.

The certainty gain effect involves noticeable changes in the offering strategy of the wind producer as can be inferred from Fig. 4.23.

To grasp the reason for such changes, we must bear in mind that the certainty gain effect translates into a better knowledge on how to offer in the adjustment market in order to decrease the expected level of imbalances. Thus, for high enough levels of risk aversion ($\beta > 0.2$), there is a transfer of trading from the day-ahead market to the adjustment market, where the information

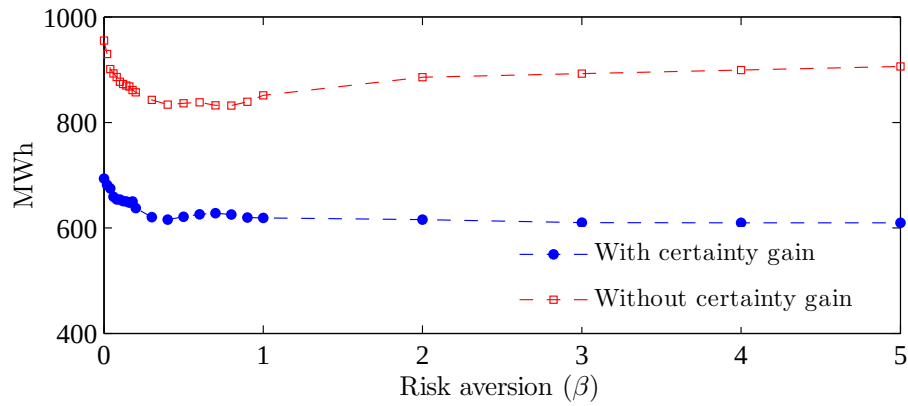


Figure 4.21: Wind producer case study: impact of the certainty gain on expected imbalances

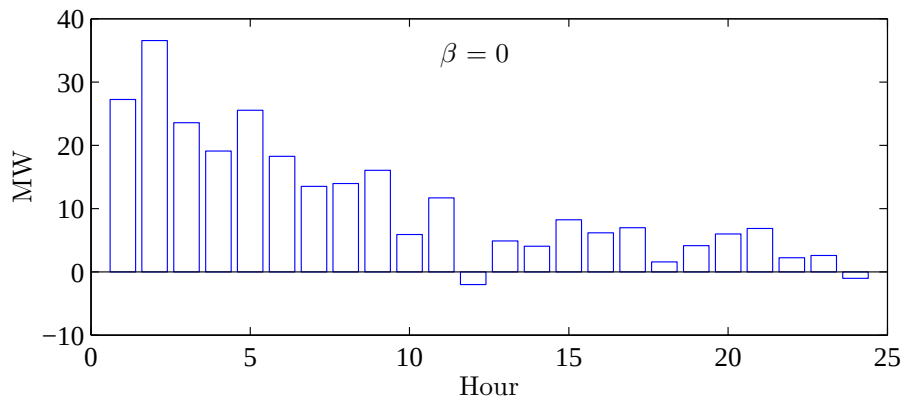
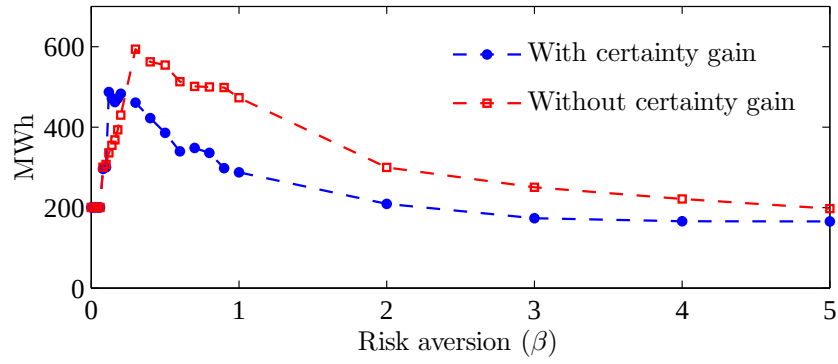
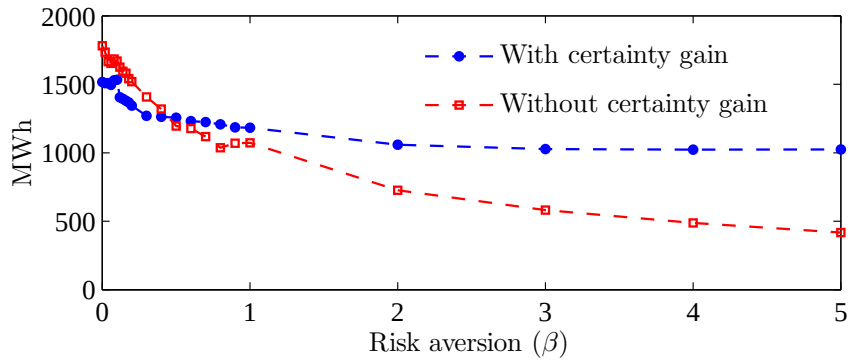


Figure 4.22: Wind producer case study: hourly reduction in expected power deviation due to the certainty gain effect



(a) Energy traded in the day-ahead market



(b) Energy traded in the adjustment market

Figure 4.23: Wind producer case study: impact of the certainty gain effect on the energy amounts traded in both the day-ahead and adjustment markets

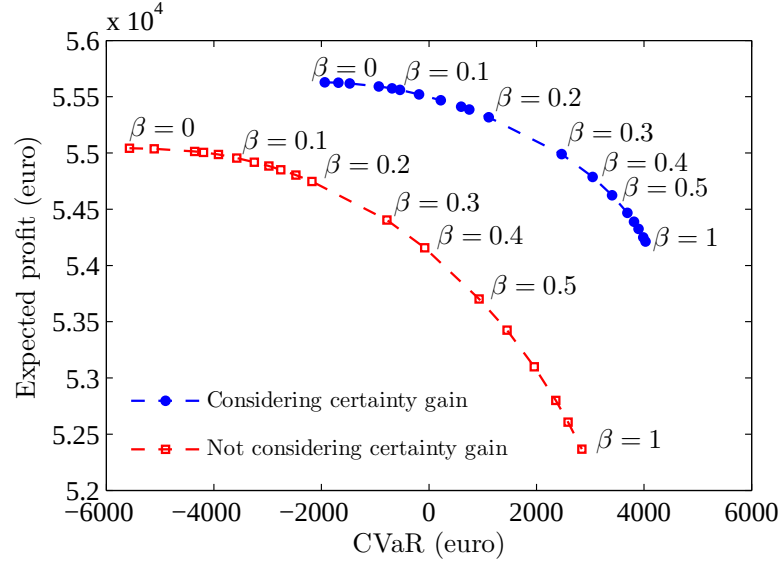


Figure 4.24: Wind producer case study: impact of the certainty gain effect on risk management (efficient frontier)

on the future wind production realization is more accurate.

In terms of profit, the improvements due to the certainty gain effect can be summarized by means of Fig. 4.24, which represents the optimal pairs (CVaR, expected profit) obtained for different values of the risk aversion parameter β . The resulting curve is usually referred to as *efficient frontier* (see Section A.4 in Appendix A). Note that the certainty gain effect produces a significant displacement of the efficient frontier upward and to the right, that is, a displacement in the direction of increasing expected profits and increasing CVaRs. This displacement can be amplified by adding to the market structure additional adjustment markets throughout the scheduling horizon in a sequential manner.

Observe that a significant CVaR gain is attained for a moderate expected profit decrease. For instance, if the weighting factor β is increased from 0 to 1, the ratio $-\frac{\Delta \text{CVaR}}{\Delta \mathcal{E}\{R\}}$ is equal to 4.21 if the certainty gain effect is accounted for in the offering strategy model. However, this ratio drops to 3.14 if such an effect is disregarded. Therefore, adjustment trading floors are efficacious

structural elements to enhance the competitiveness degree of wind producers in an electricity market.

In the analysis carried out until now, the quality of the offering decisions has been tested on the same scenario tree that is used to determine them. Therefore, the results previously discussed constitute an *in-sample* study, which relies on the assumption that the methodology for the scenario tree construction provides a precise representation of the uncertainty involved in the decision-making process. Next, we complement the in-sample analysis with an *out-of-sample* assessment, whose aim is to evaluate the effectiveness of the decisions obtained by solving the proposed stochastic offering models in terms of actual real-world outcomes.

For comparative purposes, we consider four different offering strategies, namely:

Strategy S_1 consists in offering the hourly production forecast at zero price in the day-ahead market and none in the adjustment market.

Strategy S_2 is analogous to strategy S_1 , but trading in the adjustment market the difference of power resulting from the wind production forecast update. This difference is offered to be sold at zero price or to be bought at a sufficiently high price.

Strategy S_3 consists in submitting in the day-ahead market the energy quantities at zero price that are obtained by solving model (4.40)–(4.45).

In strategy S_4 the wind producer offers in the day-ahead market according to model (4.56)–(4.68), which takes into account the possibility of trading in the adjustment market afterward. Once the day-ahead market is cleared and prior to the closure of the adjustment market, the wind producer offers in this latter market using model (4.69)–(4.74).

Note that strategies S_1 and S_2 are based on forecasts while strategies S_3 and S_4 are built upon stochastic programming. For the sake of a fair comparison, forecasts and scenario trees are elaborated using the same time series models. On the other hand, strategies S_1 and S_3 ignore the existence of an adjustment

Table 4.5: Actual profit per month and strategy in M€

Strategy	Month			
	February	April	July	November
S ₁	2.540	2.588	1.643	2.223
S ₂	2.553	2.641	1.651	2.230
S ₃	2.613	2.648	1.650	2.266
S ₄	2.618	2.665	1.644	2.302

market within the market organization, whereas strategies S₂ and S₄ take advantage of this trading floor.

Any of these four strategies make use of the information available up to time period $t - 1$ in order to obtain the decisions to be made at period t . Once actual market prices and wind power production in that period are disclosed, thus becoming historical data to be accounted for in the design of subsequent offering strategies, the actual profit of the wind producer can be computed. Thus, the simulation process is run day by day.

The performance of the offering strategies described above is next appraised in terms of the economic benefit that the wind power producer would have obtained by implementing each of these strategies during four months of 2008, one per season, specifically, February, April, July, and November. Table 4.5 provides the actual profit attained by the wind producer per month and per strategy. Bold font is used to highlight the highest profit value in each column. In light of the results shown in this table, the following comments are in order:

1. The application of strategy S₄ yields the highest total profit in February, April and November. Thus, in general terms, we can state that strategy S₄, according to which the wind producer offers in both the day-ahead and adjustment markets using stochastic models (4.56)–(4.68) and (4.69)–(4.74) with $\beta = 0$, outperforms the rest of strategies. In turn, strategy S₃ provides a higher profit than strategies S₁ and S₂ in these three months. Therefore, strategies built upon a stochastic approach are generally more profitable than those simply based on forecasting.

2. The implementation of strategy S_2 , which makes use of the adjustment market, results in a higher profit than the application of strategy S_1 in all months. Likewise, the actual profit obtained by putting strategy S_4 into practice is higher than that achieved via strategy S_3 in February, April and November. Consequently, we corroborate that adjustment markets allow wind producers to increase their competitiveness in the pool.
3. The results obtained in July are not in keeping with the general conclusions stated in the two previous points. By inspection of the historical data, we find that the features of February, April and November are more similar, while significantly differs from those of July in the following two aspects:
 - (a) The wind speed in July is much lower than in the other months. As a result, it turns out that the wind production average in July is approximately half of that in the other months.
 - (b) The sign of the net system imbalance changes more dramatically in July.

Under these circumstances, the results reveal that strategy S_2 is an appropriate alternative to stochastic offering models. While all strategies rely on wind speed data, strategies S_3 and S_4 make use of more sources of information, e.g., imbalance prices. The usage of multi-source information can naturally enhance the performance of strategies S_3 and S_4 if they count on efficient enough scenario-generation methods. In the case of July, the sign of the system imbalance varies drastically and consequently, the seasonal ARIMA models fail to capture such drastic changes. Thus, the performance of the stochastic programming models worsens as a result of using an inadequate uncertainty modeling. To assess how improvements in the modeling of the imbalance price ratios can increase the efficiency of strategies based on stochastic programming, we produce imbalance price scenarios by introducing small standard normal perturbations over their actual values. Using this improved, although impractical scenario set, the actual profit obtained via strategy S_4 in

July is equal to 1.708 M€, which represents a 3.89% increase. Hence, the uncertainty modeling constitutes a key issue in the design of offering strategies built upon stochastic programming.

To conclude this case study, we point out that the CPU time required to solve the stochastic model (4.56)–(4.68) is lower than four seconds in all the simulations carried out using CPLEX 11.2 [15] under GAMS [14] on a Linux-based server with four processors clocking at 2.6 GHz and 32 GB of RAM.

4.8 Summary

The competitive sale of wind generation in current electricity markets constitutes a significant challenge for wind power producers due to the inherent randomness of the underlying energy resource: the wind. The market strategy of a wind power producer must be profit effective while able to damp as much as possible the profit variability caused by significant wind fluctuation. In addition to uncertainty on wind availability, wind producers must also cope with uncertain market prices as any other market agent.

As a result of the time lapse existing between the day-ahead market and the real-time operation of the power system, wind power producers have significant difficulty to fulfill the production scheduling settled in this market. Consequently, they are forced to correct their energy deviations in a balancing market through which the real-time energy balance between generation and demand is guaranteed.

However, the price formation mechanism in the balancing market is such that any amount of energy produced above the one stipulated in the day-ahead market is paid at a price equal or lower than day-ahead price. Similarly, any amount of energy bought in this market to cover a deficit of generation is charged at a price equal or higher than the day-ahead market price. Therefore, energy trading in the balancing market entails an opportunity cost for those market agents incurring deviations. This cost is usually referred to as *imbalance cost*. Hence, as a wind producer needs to resort to this balancing mechanism, its profitability diminishes.

This chapter focuses on the design of a multi-stage stochastic programming model to build energy offers allowing wind producers to shrewdly compete in the different short-term trading floors available in a fully-fledged electricity market, namely, the day-ahead, adjustment, and balancing markets. The strategy offer obtained from the proposed model is aimed at maximizing the expected profit of the short-term wind energy trading, while controlling profit variability and minimizing the need for balancing energy. Uncertainties affecting decision variables, i.e., wind availability and market prices, are modeled via a set of scenarios, whose cardinality is conveniently reduced to make the resulting linear optimization problem tractable and solvable in a reasonable time using commercially available software. Both the scenario tree building process and the offering model formulation accounts for the knowledge of wind power stochastic behavior gained during the hours between the day-ahead and adjustment market closures. This knowledge gain increases the competitiveness of the wind power producer in the electricity market, a phenomenon which is referred to as *certainty gain effect* in this chapter.

The main technical aspects of the proposed model, as well as the characteristics of the decisions derived from its use, are highlighted and discussed by means of a detailed example and a realistic case study. Additionally, the performance of the stochastic offering model is evaluated via in-sample and out-of-sample studies.

Chapter 5

Market Clearing under Uncertainty

5.1 Introduction

Electric energy systems are exposed to a great variety of uncertainties. Restructuring and liberalization in the electricity sector require the design of operating procedures able to hold a competitive market framework for electricity trading under and in spite of these uncertainties. Moreover, the exceptional features of the commodity *electricity* as regards its transmission, production, and consumption make this task even more challenging. For example, the generation and demand of electricity at a given node of the power system have a global impact on the entire system. As a result, the actions to be executed in order to cope with uncertain conditions without compromising the market consistency call for a comprehensive management able to involve, if required, the participation of every agent in the system.

The reserve constitutes the instrument used to face power system uncertainties, thus enabling energy transactions in electricity markets while guaranteeing a secure utilization of the electrical infrastructure. Roughly speaking, the concept of reserve refers to the ability of a power system to voluntarily alter the production and consumption levels of its generators and loads in response to unpredictable changes in system conditions.

In pursuit of an all-embracing restructuring and liberalization, the reserve, like the energy, is currently managed, scheduled, and traded via electricity markets. In this chapter, we propose a *stochastic security-constrained market-clearing* tool for the simultaneous dispatch of energy and reserve that allows the system operator to solve the tradeoff between economic efficiency and system security. The reserve scheduling that results from the proposed tool translates into an optimal pre-positioning of generating units and loads to efficaciously respond to the realization of uncertainties in accordance with the available resources and their cost.

The magnitude and characteristics of the uncertainties present in power systems are crucial to the planning of reserve. In this chapter, we primarily focus on the uncertainties related to wind power production. For completeness, we also deal with equipment failures but, in this respect, we limit ourselves to highlight the most important features of this source of uncertainty, specifically those concerning its statistical treatment and its detrimental effect on power system reliability. The management of the uncertainty associated with the demand is analogous to that of wind generation and therefore, is not considered here.

5.2 Market Clearing

5.2.1 Overview

Electricity markets usually include a variety of trading floors for market agents to carry out short-term energy transactions according to their market strategy. The set of these sequentially arranged short-term trading floors is referred to as the *pool*.

The pool typically consists of the day-ahead market, the adjustment markets, and the balancing markets. The day-ahead market is generally the most important one in terms of trading volume and as such, is often seen as a reference by the rest of trading structures. In the adjustment markets, market participants can take corrective actions that are aimed to reduce or eliminate the differences between their expected production and the energy dispatch set-

tled in the day-ahead market. Finally, the balancing markets are organized to ensure the power balance between generation and demand on a real-time basis.

In the pool, the Market Operator receives energy offers from producers and energy bids from consumers, and determines, for the 24 hours of the trading day, the power production of every producer, the consumption level of every consumer, and the price at which every producer/consumer is paid/charged for its energy production/consumption, [195]. The aim is to maximize the net social welfare. This process is known as *market clearing* and constitutes the *leitmotiv* of this chapter. The technical literature is rich in references on methods to clear electricity markets, e.g., [19, 23, 39, 40, 84, 86, 87, 185, 195, 198, 200, 205].

5.2.2 Procedures: Simplicity vs Optimality

The market-clearing procedure applied by the Market Operator should produce a feasible dispatch, be transparent and fair, and maximize the net social welfare. The data required by the market-clearing procedure is the offering and bidding information provided by producers and consumers, respectively. The economic offering information submitted by any producer for every hour generally consists of a set of energy blocks and their corresponding prices. Likewise, a generator can complement this simple economic data by declaring a start-up price and by providing some information on its inter-temporal technical limitations, i.e., its minimum up and down times, and ramping capabilities.

Market-clearing procedures currently implemented in electricity markets are significantly different from one another. As an example, the market clearing algorithm used in the market of the Iberian Peninsula, [3], consists of a sequence of single-period auctions, where inter-temporal constraints are enforced ex post using simple heuristic rules. The independent consideration of every hour may result in inefficient solutions. Therefore, in this market, the simplicity and transparency of the clearing algorithm prevails over the optimality of the resulting energy dispatch. In general, this criterion is predominant in European electricity markets. For the sake of optimality and maximum

economic efficiency, a multi-period market-clearing procedure is required in order to take into consideration inter-temporal constraints within a global optimization problem, [23]. In the particular case of the electricity markets established in the East Coast of the US (e.g., [4] and [5]), their functioning is presently based on a sophisticated market-clearing procedure in which network constraints are taken into account through a *dc load flow model*.

5.2.3 Security Constraints

Day-ahead markets are cleared well in advance of the scheduling horizon, e.g., 14-38 hours ahead. As a result, the outcomes of the market-clearing procedures are subject to uncertainty as for their final implementation due to unforeseen disturbances occurring in real-time such as generator and line outages (also referred to as *contingencies*) or unpredictable fluctuations of the wind power production. The ability of a power system to survive a credible set of these disturbances without extensive load disconnections is usually understood as *system security*, [165].

In terms of security, two types of actions can be distinguished. Preventive security actions are designed to put the system in a state such that in the event of a credible disturbance enough resources are available to fast perform the corrective security actions that guarantee the normal operation of the system once the disturbance has taken place.

The reserves are the services traded in the market to physically materialize the preventive and corrective security actions by means of their scheduling and their subsequent deployment, respectively. In practice, scheduling reserves means operating the system at less than its full capacity, while the deployment of reserves usually translates into the redispatch of units previously committed in the day-ahead market, the voluntary curtailment of specific loads, and/or even the quick start-up of extra generating units to palliate unexpected shortages of energy supply. That is, the *deployment of reserve* is the collection of security corrective actions through which the *scheduled reserve* is materialized, i.e., transformed into actual energy production or consumption in the aftermath of a disturbance.

Reserve consists of spinning and non-spinning reserves. Spinning reserve is the portion of reserve comprising the generation synchronized to the system and the load voluntarily interruptible. Non-spinning reserve is the generating reserve not synchronized to the system but capable of serving demand within a given time frame. Therefore, its use entails the start-up of the production unit providing this service.

Spinning reserve can be up- or down-going. A generating unit providing up-/down-spinning reserve must be ready to increase/decrease its production if required. In the case of a consumer, the criterion is the opposite. A load committed to provide up-/down-spinning reserve can be ordered to decrease/increase its consumption if necessary.

Nowadays, there exist mainly two paradigms concerning how reserves should be traded and scheduled in electricity markets, [87]. The first one starts from the premise that energy and reserve are weakly coupled services, and consequently, reserves are scheduled in a series of auctions run once energy schedules have been determined in a separate market, [198,200]. This is the current practice in the wholesale electricity market of the Iberian Peninsula, for example. As the aforementioned premise is generally false, sequential market-clearing procedures may provide solutions too far away from the optimal operating point, or even infeasible in a first approximation. On the contrary, the other paradigm is based on simultaneous market-clearing procedures, which assume that energy and reserves are in fact strongly interrelated and allow avoiding the need for *ad hoc* operator intervention, uneconomical out-of-merit operation, the start-up of extra units, as well as unnecessary load shedding, [185,205]. This approach is usually associated with the US East Coast electricity markets and is very often called *security-constrained market clearing*.

5.2.4 Approaches for Setting Reserve Requirements

A key issue is how to establish the amount of reserves required in a power system. In order to deal with this problem, two different approaches, deterministic or probabilistic, can be applied. The *deterministic* approach, [165], is normally based on rule-of-thumb type criteria such as scheduling enough reserve to cover

the loss of the largest production unit, or to supply a portion of the hourly demand, or even a combination of both. Nevertheless, note that these preventive security measures do not account for the probability of occurrence of the contingencies or wind power fluctuations they are supposed to cover. Therefore, if these probabilities are high enough, the amount of scheduled reserves may be insufficient. Conversely, if the probabilities of occurrence are sufficiently low, reserve requirements may be overestimated. In the *probabilistic* approach, [33, 71, 84, 93, 96], however, reserve needs are set by considering both the probability of occurrence of every possible disturbance and the damage caused to the system by each of them. This normally requires, on the one hand, the evaluation of a huge number of possible disturbances, which may result in computational intractability, and on the other, the acquisition of the statistical failure data necessary to compute outage probabilities; data usually exposed to important inaccuracies. In order to circumvent these drawbacks, we propose:

1. To consider a *hybrid* approach [39, 40, 86] according to which just a reduced set of the most credible contingencies (equipment outages) is accounted for to estimate the probabilities of failures and expected outcomes.
2. To use the scenario reduction techniques described in Chapter 3 of this thesis to trim down the number of scenarios required to appropriately model wind production uncertainty.

A market-clearing procedure combining a probabilistic/hybrid approach to set the reserve needs and the implementation of preventive security actions constitutes a so-called *stochastic security-constrained market-clearing procedure*, which is the subject of study in this chapter.

5.2.5 Solution Algorithm

From a mathematical point of view, numerous techniques have been proposed in the technical literature to solve market-clearing procedures, namely, heuristics, dynamic programming, Lagrangian relaxation, mixed-integer linear pro-

gramming, etc. Among these methodologies, Lagrangian relaxation [55] used to be one of the most employed approaches because of its capability of solving large-scale problems. Its main flaw is that, as a consequence of the non-convex nature of the market-clearing problems we are dealing with, heuristic algorithms are required to find feasible solutions, which may be suboptimal. On the contrary, mixed-integer linear programming [49, 155] guarantees convergence to the optimal solution in a finite number of steps while providing a flexible and accurate modeling framework. Furthermore, efficient mixed-integer algorithms such as the branch-and-cut technique are currently available within commercial solvers for optimization. For all these reasons, mixed-integer linear programming is the solution methodology selected to solve the stochastic security-constrained market-clearing procedure presented in this chapter. Needless to say, the price to pay is the need to formulate a *mixed-integer linear* model.

In this line, the main drawback of this type of market-clearing models with stochastic security is, by far, the computational burden involved. The market-clearing methodology described in this chapter is built on a global optimization framework accounting for every state of the system under every possible realization of the uncertain events (equipment failures and/or wind power). For large-scale systems, this translates into a two-stage stochastic programming problem with a computationally heavy second stage including many variables and constraints. Although, currently, the technical literature offers various techniques to alleviate this burden (some of which have been studied in Chapter 3), they may be insufficient.

Hopefully, the steady progress in computing technologies could well overcome these difficulties in the future. In this sense, it is comforting to know that the complex algorithms used at present to run the large areas covered by some of the ISOs in the US would have been deemed impractical 25 years ago due to their computational requirements. Moreover, in the meantime, the market-clearing model presented in this dissertation can be used by the system operator as a tool to evaluate the reserve needs from a market viewpoint in the short, mid and long term [148]. This analysis would be based on historical records and forecasts of market agents' behavior, equipment failures, and wind

power production.

5.3 Stochastic Security Metrics

In this section, we elaborate on the concept of *stochastic security metric*.

5.3.1 Security Metric Definition

A *security metric* is a measure quantifying the level of security of a power system, usually expressed in terms of the load involuntarily shed as a result of a disturbance (equipment failure or unexpected wind power/demand variation). A *stochastic security metric* should take into account the probability of occurrence of the disturbances capable of threatening the system security.

In the context of a stochastic security-constrained market clearing, stochastic security metrics are used to appraise the risk that load shedding incidents take place following the occurrence of disturbances.

5.3.2 Standard Metrics

In the technical literature, [33, 86], several stochastic metrics have been commonly used, namely, the *expected load not served* (ELNS), the *loss-of-load probability* (LOLP), and the *loss-of-load expectation* (LOLE).

The ELNS measures the average amount of energy not supplied to loads as a result of load shedding events. As its own name indicates, the expected load not served is a weighted average energy value accounting for both the probability of disturbances and the damage that these disturbances cause to the system in terms of lost load.

The LOLP is defined as the probability that disturbances lead to load shedding. As opposed to the ELNS, however, the loss-of-load probability is a dimensionless number that does not provide any information on the *severity* of disturbances, i.e., on the energy not supplied. This lack of a clear physical meaning makes the LOLP a less intuitive metric to work with by system operators.

The LOLE appraises the expected number of hours during which loss-of-load events could happen. As the LOLP, the loss-of-load expectation fails to provide an estimation of the damage done to the system by disturbances.

From a mathematical viewpoint, both the LOLE and the LOLP require the use of binary variables to be incorporated into a mixed-integer linear programming problem, [39, 86]. On the contrary, the ELNS can be expressed linearly, without binary variables, as follows:

$$\text{ELNS}_j = \sum_{\omega=1}^{N_\Omega} \sum_{t=1}^{N_T} \pi_\omega P_{jtw}^{\text{L, shed}} d_t, \quad (5.1)$$

where ELNS_j is the expected load not served to consumer j during the scheduling horizon, π_ω is the probability of occurrence of scenario ω , d_t is the duration of time period t , and $P_{jtw}^{\text{L, shed}}$ is the load shedding imposed on consumer j in period t and scenario w , given by:

$$P_{jtw}^{\text{L, shed}} = \sum_{j:(j,n) \in M_L} P_{jtw}^{\text{L}} - \sum_{i:(i,n) \in M_G(t,\omega)} P_{itw}^{\text{G}} + \sum_{r:(n,r) \in \Lambda(t,\omega)} P_{tw}^{\text{F}}(n, r), \quad \forall n, \forall t, \forall \omega. \quad (5.2)$$

In equation (5.2), P_{itw}^{G} , P_{jtw}^{L} , and $P_{tw}^{\text{F}}(n, r)$ are, respectively, the power output of unit i , the power consumed by load j , and the power flow through line (n, r) , in period t and scenario ω .

Logically, the involuntary load shedding cannot be negative, nor greater than the actual load,

$$0 \leq P_{jtw}^{\text{L, shed}} \leq P_{jtw}^{\text{L}}, \quad \forall j, \forall t, \forall \omega. \quad (5.3)$$

For all the reasons previously indicated, the expected load not served is usually deemed as a superior security assessment index and as such, it is the stochastic security metric used in the subsequent market-clearing formulation.

5.3.3 Security Criteria Based on the ELNS

The ELNS can be used to enforce a stochastic security criterion in a market-clearing procedure in three different fashions, [39, 40]. The first one consists in

including a set of constraints limiting the magnitude of the ELNS either node-by-node, area-by-area, or over the entire system. Likewise, these constraints can be imposed over the whole scheduling horizon, or just over a subset of time periods (e.g., hour-by-hour, peak hours, off-peak hours, etc.). The second way is to add a penalty function, increasing monotonically with ELNS, to the objective function of the market-clearing problem through the so-called *value of lost load* provided by each consumer. The third criterion is just a combination of the two previous ones.

Imposing bounds on the ELNS has three drawbacks, namely:

1. The upper bound limits would have to be specified by a regulatory agency by means of an arduous and non-discriminatory process calling for transparency and equity.
2. In cases where there is no enough reserve or sufficient transmission capacity, the specified upper bounds on ELNS may not be satisfied, leading to an infeasible market-clearing problem.
3. In the absence of an economic penalty of the ELNS, the upper bound would be reached as a way to minimize the social cost of scheduling and deploying reserves albeit there may be sufficient reserve resources and transmission capacity available to avoid load shedding events.

For simplicity and in order to circumvent these shortcomings, in the market-clearing formulation presented next, we just use a stochastic security criterion solely based on an economic penalty of the ELNS within the objective function. Therefore, according to this criterion, the system operator is willing to allow some involuntary load shedding if the disturbances causing loss-of-load events occur with sufficiently low probabilities and the corresponding increase in expected social cost is small.

5.4 Uncertainty Characterization

In this section, uncertainties pertaining to equipment failures and wind power production are characterized in the form of scenarios.

5.4.1 Equipment Failures

Each *failure scenario* f corresponds to a pair (k, τ) expressing the occurrence of a contingency k within a time period τ . The scenario in which no contingency happens during the whole scheduling horizon defines the so-called *pre-contingency state* of the system and, hereinafter, is denoted as scenario $f = 0$.

A contingency is defined as an equipment failure (generator or transmission line outages). It is widely accepted that the time between two consecutive equipment failures, as well as the time to repair a failure, both follow exponential distributions. These distributions are determined, respectively, by the mean time to failure (MTTF) and the mean time to repair (MTTR), which are obtained from historical data, [33, 34].

The MTTR is usually much greater than the 24 hourly periods that typically comprise the scheduling horizon of day-ahead electricity markets. This way, repairs are ignored so that once some equipment fails it is assumed to be unavailable for the rest of the scheduling horizon. For the sake of tractability and taking into account their low probability, we also assume that non-simultaneous contingencies (sequential failures of different pieces of equipment) are not possible.

Probabilities π_f are derived from the probabilities associated with the following two random events:

Random event $A(k, \tau)$: *contingency k happens within the time period τ ,*

$$P[A(k, \tau)] = \int_{\tau-1}^{\tau} \lambda_k e^{-\lambda_k t} dt = e^{-\lambda_k \tau} (e^{\lambda_k} - 1), \quad (5.4)$$

where λ_k is the inverse of the mean time to the occurrence of contingency k .

Random event $B(k)$: *contingency k does not occur during the entire scheduling horizon,*

$$P[B(k)] = 1 - \int_0^T \lambda_k e^{-\lambda_k t} dt = e^{-\lambda_k T}, \quad (5.5)$$

where T is the time length of the scheduling horizon (e.g., 24 hours).

Therefore, under the assumption of statistically independent contingencies, the probability π_0 of the pre-contingency scenario $f = 0$, in which no contin-

gency occurs during the whole scheduling horizon, can be computed as the product of the probabilities $P[B(k)]$ for all k , i.e.,

$$\pi_0 = \prod_{k=1}^K P[B(k)] = \prod_{k=1}^K e^{-\lambda_k T}, \quad (5.6)$$

where K is the cardinal of the set of pre-selected contingencies.

On the other hand, the probability π_f of the scenario f in which contingency k occurs during the time period τ given that the remainder of system components are available for the entire scheduling horizon can be calculated as

$$\pi_f = \pi(k, \tau) = P[A(k, \tau)] \prod_{\substack{y=1 \\ y \neq k}}^K P[B(y)] = e^{-\lambda_k \tau} (e^{\lambda_k} - 1) \prod_{\substack{y=1 \\ y \neq k}}^K e^{-\lambda_y T}. \quad (5.7)$$

It should be noted that probabilities π_0 and π_f do not add up to 1 because sequential contingencies have been disregarded.

5.4.2 Wind Power Production

The power produced by a wind farm q , with $q = 1, 2, \dots, N_W$, in time period t is modeled as a random variable denoted by P_{qt}^W . The collection $\mathbf{P}_q^W$ of dependent random variables P_{qt}^W throughout a scheduling horizon of N_T periods, i.e., $\mathbf{P}_q^W = \{P_{qt}^W, t = 1, 2, \dots, N_T\}$, constitutes a stochastic process. As stated in Chapter 3, in stochastic programming the treatment of a stochastic process within an optimization problem usually requires its discretization into a set of scenarios ϖ describing its plausible realizations.

In the context of this chapter, each wind production scenario ϖ is a matrix of $N_W \times N_T$ wind power values, i.e.,

$$\text{Scenario } \varpi : \begin{pmatrix} P_{11\varpi}^W & \cdots & P_{1N_T\varpi}^W \\ \vdots & \ddots & \vdots \\ P_{N_W1\varpi}^W & \cdots & P_{N_W N_T\varpi}^W \end{pmatrix},$$

where $P_{qt\varpi}^W$ represents the realization of random variable P_{qt}^W in scenario ϖ .

The matrix form is required because the power produced by different wind farms may not be independent. Consequently, a scenario generation method such as the one proposed in Section 3.3.6 should be used to construct the set of wind production scenarios.

Each scenario ϖ has a probability of occurrence π_ϖ such that the summation of the probabilities over all wind power scenarios is equal to 1, i.e., $\sum_{\varpi=1}^{N_\varpi} \pi_\varpi = 1$.

The treatment of the uncertainty associated with wind power in a stochastic programming framework should generally comprise two phases, namely, scenario generation and scenario reduction. The first one (scenario generation) consists in the accurate modeling of the wind power stochastic behavior by generating a sufficient number of scenarios representing the most plausible realizations of wind production throughout the scheduling horizon. The second phase (scenario reduction) is required because the size of this initial scenario set is usually too large, resulting in an optimization model that is intractable. Hence, to achieve tractability, statistical techniques [74, 100, 101, 152] are applied in order to reduce the number of scenarios while retaining the essential features of the original scenario set (see Chapter 3).

5.4.3 Scenario Tree

The set of failure and wind production scenarios, together with the decision timing of the problem, can be arranged in a two-stage scenario tree as illustrated in Fig. 5.1. Each scenario ω in the tree is defined as a pair (f, ϖ) representing the parallel occurrence of the failure event f and the wind power realization ϖ . On the assumption that the wind power production and the happening of equipment failures are statistically independent, the probability π_ω of each scenario ω in the tree is computed as

$$\pi_\omega = \pi(f, \varpi) = \pi_f \times \pi_\varpi, \quad (5.8)$$

where, of necessity, it holds that $\sum_{\omega=1}^{N_\Omega} \pi_\omega = 1$.

The first stage in the tree (root node) represents the electricity market,

where decisions pertaining to the scheduling of energy and reserve are made. The second stage (leaf nodes) constitutes the real-time operation of the power system which revolves around the deployment of reserves in order to accommodate the specific realization (scenario) of the uncertain variables (wind power productions and equipment failures). From an intuitive point of view, the aim of the stochastic programming in the market-clearing procedure is to *locate* the root node at an *equidistant* position with respect to the leaf nodes.

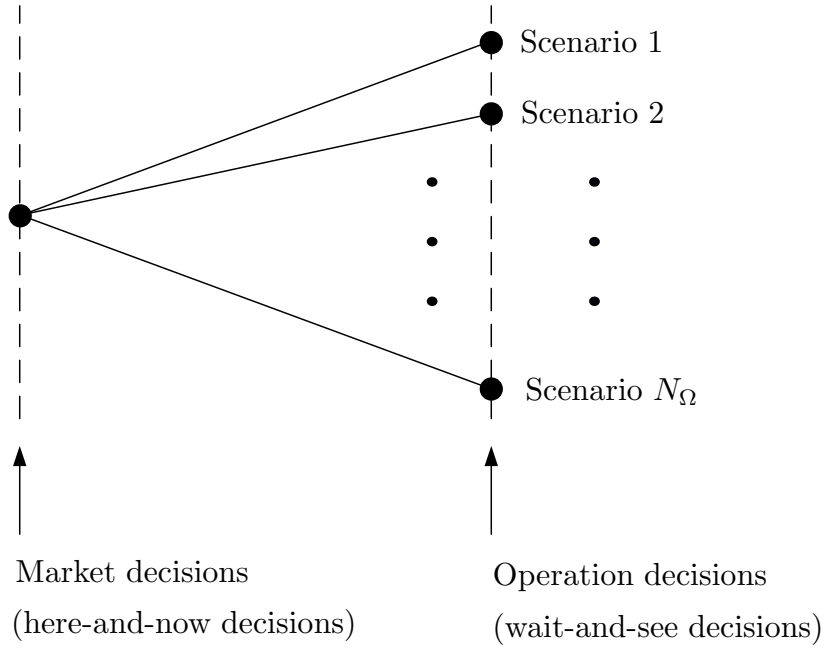


Figure 5.1: Scenario tree example for the stochastic security-constrained market-clearing problem

5.5 Market-Clearing Model

In order to account for the uncertainty that affects the operation of a power system, in particular, wind production and equipment failures, the market clearing is formulated as a two-stage stochastic programming problem, [35], where the first stage represents the electricity market, its constraints and

rules, and the second stage represents the power system, its real-time operation and physical limitations. Thus, the variables pertaining to the first stage correspond to the market decisions (preventive actions), which consider the plausible realizations of the uncertain events and their possible impact on the real-time operation of the power system (corrective actions).

The structuring of the stochastic programming problem into two stages is widely used in electricity markets where decisions must be usually made before the uncertain events materialize (e.g., one day in advance in a day-ahead market).

5.5.1 Assumptions

The following assumptions are considered to formulate the proposed market-clearing procedure with stochastic security.

1. As stated in Section 5.4.1, only single contingencies can occur over the length of the scheduling horizon, where such a contingency may be a simultaneous compounded failure. That is, non-simultaneous, sequential contingencies are disregarded.
2. The proposed market-clearing formulation is intended to determine the reserve services required to economically face every possible, but single, contingency k occurring at any specific time period τ of the scheduling horizon. This means that once a contingency has taken place, the system operator should employ a model analogous to the one presented here so as to reassess reserve needs.
3. Wind generation is assumed to be a regulated activity and thus, wind producers do not compete in the market. Consequently, wind generation is either considered a negative demand or spilled. This is equivalent to assuming that all the wind producers offer their production at zero price. This is the case in which wind power is treated in most energy systems throughout the world.
4. A linear representation of the network is considered through a dc load flow model.

It should be pointed out that these assumptions are made for convenience, simplicity, or the sake of computational tractability.

5.5.2 Variables

Variables that do not depend on any particular scenario realization in each time period are the following.

1. Start-up and shut-down plan of each generating unit (u_{it}) and the associated costs (C_{it}^{SU}).
2. Scheduled power output for each generating unit ($P_{it}^{\text{G,S}}$).
3. Scheduled power consumption for each load ($P_{it}^{\text{L,S}}$).
4. The down/up spinning reserve scheduled for each generating unit ($R_{it}^{\text{D}}, R_{it}^{\text{U}}$).
5. The non-spinning reserve scheduled for each off-line generating unit (R_{it}^{NS}).
6. The down/up reserve scheduled for each load ($R_{jt}^{\text{D}}, R_{jt}^{\text{U}}$).

These variables are *first-stage* variables and constitute *here-and-now* decisions, i.e., decisions pertaining to the market clearing that are made before the realization of any one of the scenarios.

Variables pertaining to each particular scenario in each time period are enumerated below.

1. The changes in the start-up and shut-down plan of each generating unit required for making feasible the deployment of non-spinning reserves ($v_{it\omega}$) and the costs derived from these changes ($C_{it\omega}^{\text{A}}$).
2. The deployment of down/up spinning reserve by each generating unit with scheduled spinning reserve ($r_{it\omega}^{\text{D}}, r_{it\omega}^{\text{U}}$).
3. The deployment of non-spinning reserve by each generating unit with scheduled non-spinning reserve ($r_{it\omega}^{\text{NS}}$).
4. The deployment of down/up reserve by each load with scheduled reserve ($r_{jt\omega}^{\text{D}}, r_{jt\omega}^{\text{U}}$).

5. The involuntarily load shed by each consumer $(P_{jtw}^{L, \text{shed}})$ and the wind power spillage (S_{gtw}) from each wind farm.
6. The network related variables, namely, angle of every node (δ_{ntw}) , flow through each line (P_{tw}^F) , and real-time power productions (P_{itw}^G) and consumptions (P_{jtw}^L) .

These variables, which do depend on the particular scenarios, are related to the actual operation of the power system, are *second-stage* variables and constitute *wait-and-see* decisions.

5.5.3 Structure

Within the proposed two-stage stochastic programming model, four parts can be distinguished. The objective function to be minimized, included in Section 5.5.4, accounts for both the costs pertaining to the electricity market and the costs incurred during the real-time operation of the power system. Three sets of constraints are included: the constraints modeling the market and its rules (in Section 5.5.5), those modeling the real-time operation of the power system once a particular scenario ω is realized (in Section 5.5.6), and finally, the linking constraints (in Section 5.5.7), which bind the market decisions to the actual operation of the power system through the deployment of reserves provided by units and loads.

The resulting model is formulated as a large-scale mixed-integer linear programming problem that can be solved using commercially available software, [14].

5.5.4 Objective function

The objective function below to be minimized is the expected cost (EC):

$$\begin{aligned}
EC &= \sum_{t=1}^{N_T} EC_t = \\
&= \sum_{t=1}^{N_T} \sum_{i=1}^{N_G} C_{it}^{\text{SU}} \\
&+ \sum_{t=1}^{N_T} d_t \left[\sum_{i=1}^{N_G} (C_{it}^{R^U} R_{it}^U + C_{it}^{R^D} R_{it}^D + C_{it}^{R^{\text{NS}}} R_{it}^{\text{NS}}) \right. \\
&\quad \left. + \sum_{j=1}^{N_L} (C_{jt}^{R^U} R_{jt}^U + C_{jt}^{R^D} R_{jt}^D) \right] \\
&+ \sum_{\omega=1}^{N_\Omega} \pi_\omega \left\{ \sum_{t=1}^{N_T} \sum_{i=1}^{N_G} C_{it\omega}^A \right. \\
&\quad \left. + \sum_{t=1}^{N_T} d_t \left[\sum_{i=1}^{N_G} \sum_{m=1}^{N_{Oit}} \lambda_{Git}(m) p_{Git\omega}(m) - \sum_{j=1}^{N_L} \sum_{\ell=1}^{N_{Bjt}} \lambda_{Ljt}(\ell) p_{Ljt\omega}(\ell) \right. \right. \\
&\quad \left. \left. + \sum_{j=1}^{N_L} V_{jt}^{\text{LOL}} P_{jt\omega}^{\text{L, shed}} \right] \right\}, \tag{5.9}
\end{aligned}$$

where EC_t is the expected cost of the system in period t and $C_{it}^{R^U}$, $C_{it}^{R^D}$, and $C_{it}^{R^{\text{NS}}}$ are, respectively, the offer costs of the up-, down-, and non-spinning reserves of unit i in period t . Likewise, $C_{jt}^{R^U}$ and $C_{jt}^{R^D}$ represent, respectively, the offer costs of the up- and down-spinning reserves of load j in period t . Moreover, every unit i submits a series of N_{Oit} selling offers comprising each an energy block $p_{Git}^{\max}(m)$ and a price $\lambda_{Git}(m)$. Analogously, every load j submits a set of N_{Bjt} purchasing bids including each a block of energy $p_{Ljt}^{\max}(\ell)$ and a price $\lambda_{Ljt}(\ell)$.

The objective function above includes six terms, from the second line to the seventh line in equation (5.9), namely:

1. The start-up offer cost of the generating units,
2. the offer cost of contracting up/down spinning and non-spinning reserves from generating units,

3. the offer cost of contracting up/down spinning reserves from loads,
4. the cost resulting from the adjustments of the start-up and shut-down plan of generating units previously established by the market clearing,
5. the energy offer cost of the generating units minus the demand utility,
6. the cost of the load shedding.

The energy offer cost of the generating units is introduced as a second-stage cost since its computation requires the knowledge of the availability of production units throughout the scheduling horizon and therefore, this cost is dependent on the scenario realization.

5.5.5 Electricity Market Constraints

The constraints below model the functioning of the market and consequently, involve first-stage variables, i.e., variables that do not depend on scenario ω . The functioning of the market specifically relies on the equilibrium between generation offers and demand bids. In addition, the decisions made in the electricity market should account for the contribution of wind generation to the energy supply. Needless to say, this contribution is uncertain by the time the market is cleared and therefore, the scheduling of production units and loads should be carried out with the aim of leaving enough *room* in the power network to accommodate the eventual outcome of the wind power production at minimum cost. Mathematically, the *room* left in the power system for the wind generation is modeled through a continuous variable $P_t^{\text{W},\text{S}}$, which is added to the market equilibria, i.e.,

$$\sum_{i=1}^{N_G} P_{it}^{\text{G},\text{S}} + P_t^{\text{W},\text{S}} = \sum_{j=1}^{N_L} P_{jt}^{\text{L},\text{S}}, \quad \forall t. \quad (5.10)$$

Variable $P_t^{\text{W},\text{S}}$ can be interpreted as the amount of wind power *scheduled* in the electricity market for period t and is logically bounded by the wind generation

limits,

$$0 \leq P_t^{W,S} \leq \sum_{q=1}^{N_W} P_q^{W,\max}, \quad \forall t, \quad (5.11)$$

where $P_q^{W,\max}$ is the installed power capacity of the wind farm q .

The participation of producers and consumers in the electricity market is also restricted to their production and consumption limits. This is stated by means of constraints (5.12) and (5.13), respectively.

$$P_i^{G,\min} u_{it} \leq P_{it}^{G,S} \leq P_i^{G,\max} u_{it}, \quad \forall i, \forall t \quad (5.12)$$

$$P_{jt}^{L,\min} \leq P_{jt}^{L,S} \leq P_{jt}^{L,\max}, \quad \forall j, \forall t. \quad (5.13)$$

$P_{jt}^{L,\min}$ and $P_{jt}^{L,\max}$ are parameters submitted as part of the demand-side bids. In the case of inelastic demand, the two limits are equal, that is, $P_{jt}^{L,\min} = P_{jt}^{L,\max} = P_{jt}^{L,S}$.

According to the rationale behind a security-constrained market-clearing procedure, in which energy and reserve are simultaneously dispatched, the scheduling of reserves proceeds as follows: if generating unit i has been committed to supply energy in period t (i.e., $u_{it} = 1$), then it can also provide spinning reserve (upward or downward) for that period as stated through equations (5.14) and (5.15),

$$0 \leq R_{it}^U \leq R_{it}^{U,\max} u_{it}, \quad \forall i, \forall t \quad (5.14)$$

$$0 \leq R_{it}^D \leq R_{it}^{D,\max} u_{it}, \quad \forall i, \forall t. \quad (5.15)$$

Otherwise ($u_{it} = 0$), generating unit i can still contribute to cover the need for non-spinning reserves in the system as expressed by equation (5.16),

$$0 \leq R_{it}^{NS} \leq R_{it}^{NS,\max} (1 - u_{it}), \quad \forall i, \forall t. \quad (5.16)$$

On the other hand, since loads are assumed to be always connected to the grid, they can only provide spinning reserve,

$$0 \leq R_{jt}^U \leq R_{jt}^{U,\max}, \quad \forall j, \forall t \quad (5.17)$$

$$0 \leq R_{jt}^D \leq R_{jt}^{D, \max}, \quad \forall j, \forall t. \quad (5.18)$$

Finally, the scheduling program must also account for the cost associated with the expected start-ups required for its practical implementation. This cost is modeled through the constraints below

$$C_{it}^{\text{SU}} \geq \lambda_{it}^{\text{SU}}(u_{it} - u_{i,t-1}), \quad \forall i, \forall t \quad (5.19)$$

$$C_{it}^{\text{SU}} \geq 0, \quad \forall i, \forall t. \quad (5.20)$$

Specifically, equation (5.20) states that the start-up cost of a generating unit must be either zero or positive. The nature of the optimization problem seeking to minimize the expected cost incurred by the market participants guarantees that variable C_{it}^{SU} , which represents the start-up cost of generating unit i in period t , takes the minimum value that is possible. In particular, this value is equal to λ_{it}^{SU} if $u_{it} = 1$ and $u_{i,t-1} = 0$, i.e., if unit i is started up in period t , and 0 otherwise. Further details on the modeling of the start-up cost of a generating unit are provided in [22, 47, 160].

Note that the rules governing the functioning of the market, setting the balance point and consequently, determining the scheduling program, can be much more complex than those presented above. For instance, the market balance can be enforced per node by incorporating network considerations into the market formulation. However, there is no need for this, and neither advantage nor improvement is expected to result from a market representation with a higher degree of sophistication. This is so because the stochastic programming framework, on which this market-clearing paradigm is built, guarantees the enforcement of all relevant constraints. This means that market decisions are made accounting for the physical limitations imposed by the subsequent real-time operation of the power system, mathematically modeled via the second-stage of the stochastic programming problem. In other words, the scheduling program satisfying the market balance (5.10) envisions all *plausible paths* to be followed by system variables during the real-time operation and thus, constitutes an optimal pre-positioning of generators and loads to respond to transmission capacity problems, operation limits, failure events,

and/or wind power fluctuations, which is the issue of interest in this chapter.

5.5.6 Real-Time Operating Constraints

These constraints model the real-time operation of the power system and therefore, involve second-stage variables, i.e., variables that do depend on scenarios ω .

The actual operation of a power system revolves around the continuous power balance between production and consumption, which is formulated as

$$\begin{aligned} \sum_{i:(i,n) \in M_G(t,\omega)} P_{it\omega}^G - \sum_{j:(j,n) \in M_L} (P_{jt\omega}^L - P_{jt\omega}^{L,\text{shed}}) + \sum_{q:(q,n) \in M_W} (P_{qt\omega}^W - S_{qt\omega}) \\ - \sum_{r:(n,r) \in \Lambda(t,\omega)} P_{tw}^F(n,r) = 0, \forall n, \forall t, \forall \omega, \end{aligned} \quad (5.21)$$

where $P_{tw}^F(n,r)$ is the power flow through line (n,r) from n to r , which, in accordance with a dc load flow model, is given by

$$P_{tw}^F(n,r) = B(n,r)(\delta_{nt\omega} - \delta_{rt\omega}), \quad \forall (n,r) \in \Lambda(t,\omega), \forall t, \forall \omega. \quad (5.22)$$

Note that the set of transmission lines, Λ , and the mapping of the set of generating units into the set of nodes, M_G , are contingent on scenario ω , since contingencies often translate into modifications of unit availability and system topology.

Observe in equations (5.21) that the modeling of the real-time operation of the power system includes the actual power $P_{qt\omega}^W$ injected into the grid from wind farm q . The continuous variable $S_{qt\omega}$ represents the power produced by wind farm q that is *spilled* (and therefore not utilized) in period t and scenario ω . The amount of wind power spilled has to be greater or equal than zero, and smaller or equal than the actual production of wind power, i.e.,

$$0 \leq S_{qt\omega} \leq P_{qt\omega}^W, \quad \forall q, \forall t, \forall \omega. \quad (5.23)$$

Wind power spillage can be due to both economic and technical criteria. As we already know, the uncertain nature of wind generation requires

scheduling reserves to guarantee and preserve system security. This operation entails a cost. Consequently, if the benefit inherent to the cost-free character of wind energy is smaller than the cost associated with the management of its unpredictability, then wind spillage turns out to be profitable. Likewise, the variability of wind generation calls for a flexible operation of the power system. Hence, the physical limitations of the electrical infrastructure impose a cap on the amount of wind power that can be injected into the network and the remaining wind power production has to be spilled in consequence.

Constraints (5.24) and (5.25) below set the generation limits of each unit for each period and scenario. Note that these inequalities are analogous to the first-stage constraints (5.12), except that the ones below are enforced per scenario through the on/off variables $v_{it\omega}$.

$$P_{it\omega}^G \geq P_i^{G, \min} v_{it\omega}, \quad \forall i \in G(t, \omega), \forall t, \forall \omega \quad (5.24)$$

$$P_{it\omega}^G \leq P_i^{G, \max} v_{it\omega}, \quad \forall i \in G(t, \omega), \forall t, \forall \omega, \quad (5.25)$$

where $G(t, \omega)$ is the set of available generating units in period t and scenario ω .

Set of constraints (5.26) expresses the power output as the sum of the accepted energy blocks, whose lower and upper limits are imposed by (5.27). That is, equations (5.26) and (5.27) enable us to approximate the energy offer cost function of generating units by blocks. This formulation of the energy offers implies for every unit that the declared costs of the energy blocks are monotonically increasing. To model non-convex energy offer cost functions, additional binary variables are required, [22].

$$P_{it\omega}^G = \sum_{m=1}^{N_{Oit}} p_{Git\omega}(m), \quad \forall i, \forall t, \forall \omega \quad (5.26)$$

$$0 \leq p_{Git\omega}(m) \leq p_{Git}^{\max}(m), \quad \forall m, \forall i, \forall t, \forall \omega. \quad (5.27)$$

Logically, for every generating unit i that is not available in period t and scenario ω , i.e., $\forall i \notin G(t, \omega)$, it holds that $v_{it\omega} = 0$ and as a result, $P_{it\omega}^G = 0$.

Similarly to equations (5.26) and (5.27), constraints (5.28) and (5.29) serve

us to model the declared utility function of demands by blocks,

$$P_{jt\omega}^L = \sum_{\ell=1}^{N_{Bjt}} p_{Ljt\omega}(\ell), \quad \forall j, \forall t, \forall \omega \quad (5.28)$$

$$0 \leq p_{Ljt\omega}(\ell) \leq p_{Ljt}^{\max}(\ell), \quad \forall \ell, \forall j, \forall t, \forall \omega. \quad (5.29)$$

The line flow limits in each period and scenario are enforced by the group of constraints (5.30), hereinafter referred to as *transmission capacity constraints*.

$$-P^{\text{F}, \max}(n, r) \leq P_{tw}^{\text{F}}(n, r) \leq P^{\text{F}, \max}(n, r), \quad \forall (n, r) \in \Lambda(t, \omega), \quad \forall t, \forall \omega. \quad (5.30)$$

Lastly, note that the amount of involuntary load shedding imposed on each consumer has to be positive and smaller than or equal to its actual demand. This is mathematically expressed as

$$0 \leq P_{jt\omega}^{\text{L, shed}} \leq P_{jt\omega}^{\text{L}}, \quad \forall j, \forall t, \forall \omega. \quad (5.31)$$

According to equations (5.21) and (5.31), load shedding variables can be mathematically seen as fictitious generators, located at load nodes, with a maximum capacity equal to the magnitude of the corresponding load.

Finally, this part of the formulation, which involves the real-time operation of the system (second-stage variables), can also include ramping constraints and minimum up/down time of generating units as stated in [22, 47]. Ramping constraints are imposed on variables $P_{it\omega}^{\text{G}}$, which represent the power outputs of generating units during the actual operation of the power system, by using binary variables $v_{it\omega}$ representing their on/off status. Minimum up/down time constraints are, in turn, imposed through these 0/1 variables. For conciseness, the mathematical formulation of these constraints is moved to Appendix C.

5.5.7 Linking constraints

These constraints relate market decisions to the real-time operation of the power system through the deployment of reserves provided by units and loads. Hence, from a stochastic programming perspective, these coupling constraints

encompass both first- and second-stage variables.

The power outputs of generating units on a real-time basis result from the power productions settled in the electricity market clearing and the different types of reserves deployed to accommodate system failures and wind production uncertainty. This is stated as follows:

$$P_{it\omega}^G = P_{it}^{G,S} + r_{it\omega}^U + r_{it\omega}^{NS} - r_{it\omega}^D, \quad \forall t, \forall \omega, \forall i \in G(t, \omega). \quad (5.32)$$

Needless to say, for every unit i that is not available in period t and scenario ω , i.e., $\forall i \notin G(t, \omega)$, it should be satisfied that $r_{it\omega}^U = r_{it\omega}^{NS} = r_{it\omega}^D = 0$.

An analogous decomposition can be carried out for the power consumed by each load,

$$P_{jt\omega}^L = P_{jt}^{L,S} - r_{jt\omega}^U + r_{jt\omega}^D, \quad \forall j, \forall t, \forall \omega. \quad (5.33)$$

The amount of reserve that a generating unit i is willing to deploy in time period t as a response to a disturbance is limited to the quantity established in the electricity market for each type of reserve. Constraints (5.34)–(5.36) below constitute the mathematical implementation of these limits for up-, down-, and non-spinning reserves, respectively,

$$0 \leq r_{it\omega}^U \leq R_{it}^U, \quad \forall i, \forall t, \forall \omega \quad (5.34)$$

$$0 \leq r_{it\omega}^D \leq R_{it}^D, \quad \forall i, \forall t, \forall \omega \quad (5.35)$$

$$0 \leq r_{it\omega}^{NS} \leq R_{it}^{NS}, \quad \forall i, \forall t, \forall \omega. \quad (5.36)$$

Similar constraints are to be enforced for each load j with scheduled reserve,

$$0 \leq r_{jt\omega}^U \leq R_{jt}^U, \quad \forall j, \forall t, \forall \omega \quad (5.37)$$

$$0 \leq r_{jt\omega}^D \leq R_{jt}^D, \quad \forall j, \forall t, \forall \omega. \quad (5.38)$$

The deployment of reserves during the real-time operation of the power system can entail changes in the start-up and shut-down planning settled in the electricity market. These changes require the adjustment of the start-up costs computed in the market (first-stage), adjustment that is mathematically mod-

eled through the following set of equations:

$$C_{it\omega}^A = C_{it\omega}^{\text{SU}} - C_{it}^{\text{SU}}, \quad \forall i, \forall t, \forall \omega \quad (5.39)$$

$$C_{it\omega}^{\text{SU}} \geq \lambda_{it}^{\text{SU}} (v_{it\omega} - v_{i(t-1)\omega}), \quad \forall i, \forall t, \forall \omega \quad (5.40)$$

$$C_{it\omega}^{\text{SU}} \geq 0, \quad \forall i, \forall t, \forall \omega. \quad (5.41)$$

Note that variable $C_{it\omega}^{\text{SU}}$ accounts for the start-up cost incurred by generating unit i during the real-time operation of the power system in period t and scenario ω . Thus, $C_{it\omega}^A$ is zero if the commitment status of unit i in period t is as scheduled in the market. Its value is positive if unit i has to be started up in period t against what was set in the market and it is negative if a scheduled start-up for unit i in period t is finally cancelled due to the realization of scenario ω .

On the other hand, it should be noted that, in objective function (5.9), the start-up costs are separated into two terms: the first one, C_{it}^{SU} , accounts for the costs resulting from the scheduled start-ups of each generating unit (first-stage costs). The second term, $C_{it\omega}^A$, covers the costs caused by the adjustments required in the unit commitment plan in order to respond to a particular disturbance (second-stage costs). However, from a computational point of view, this splitting of the start-up costs is not necessary, since the ultimate start-up costs incurred by generators, including both scheduling and the subsequent adjustments, can be merged into the single second-stage term, $C_{it\omega}^{\text{SU}}$, through (5.39). This way, constraints (5.19), (5.20) and (5.39), together with variables C_{it}^{SU} and $C_{it\omega}^A$, can be eliminated.

The formulation of the proposed market-clearing procedure ends up with the constraints modeling the non-anticipatory character of the information. Specifically, these constraints take into account that the system operator cannot anticipate a failure event and thus, no corrective action aimed to relieve it can be executed before its occurrence. This is a direct consequence of the discrete nature of equipment contingencies, which manifests itself as a system failure occurring at a certain instant τ in time within the scheduling horizon. Prior to that instant, there is no sign of the impending failure, there is no way to preempt it and therefore, the system operator cannot react against it.

In contrast, wind power constitutes a stochastic process whose uncertainty is revealed on a continuous basis, which compels the system operator to undertake corrective measures intended to alleviate its unpredictable fluctuations from the very beginning of the scheduling horizon. Being so, the aim of the non-anticipativity constraints (5.42) and (5.43) is to guarantee that the corrective actions executed by the system operator have to be the same for identical wind power realizations as long as no equipment failure takes place.

Let us consider the subset $\Upsilon = \{\omega = (f, \varpi), f = 0, \varpi = 1, \dots, N_\varpi\}$ of scenarios ω such that no equipment failure occurs during the entire scheduling horizon. Non-anticipativity constraints are stated as follows:

$$\begin{aligned} P_{jtw}^{\text{L, shed}} &= P_{jtw'}^{\text{L, shed}}, \quad r_{jtw}^{\text{U}} = r_{jtw'}^{\text{U}}, \quad r_{jtw}^{\text{D}} = r_{jtw'}^{\text{D}}, \\ \forall j, \forall t < \tau, \forall \omega, \omega' : \omega &= (0, \varpi), \omega' = (f, \varpi'), f = (k, \tau), \varpi = \varpi' \end{aligned} \quad (5.42)$$

$$\begin{aligned} r_{itw}^{\text{U}} &= r_{itw'}^{\text{U}}, \quad r_{itw}^{\text{D}} = r_{itw'}^{\text{D}}, \quad r_{itw}^{\text{NS}} = r_{itw'}^{\text{NS}}, \\ \forall i, \forall t < \tau, \forall \omega, \omega' : \omega &= (0, \varpi), \omega' = (f, \varpi'), f = (k, \tau), \varpi = \varpi' \end{aligned} \quad (5.43)$$

That is, in equations (5.42) and (5.43), scenario ω belongs to subset Υ , while scenario ω' does not, but both of them share the same wind power realization, $\varpi = \varpi'$. In plain words, the amount and type of reserves that are deployed, as well as the amount of load that is involuntarily curtailed in two scenarios with identical wind power realizations, are forced to be the same up to the occurrence of an equipment failure. Additionally, from equations (5.42) and (5.43), it can be inferred that $v_{itw} = v_{itw'}$ under the same circumstances.

In case that wind power uncertainty is not dealt with in the market-clearing procedure, non-anticipativity constraints should be reformulated as

$$L_{jtw}^{\text{shed}} = 0, \quad \forall j, \forall t < \tau, \forall \omega = (k, \tau) \quad (5.44)$$

$$r_{itw}^{\text{U}} = r_{itw}^{\text{D}} = r_{itw}^{\text{NS}} = 0, \quad \forall i, \forall t < \tau, \forall \omega = (k, \tau) \quad (5.45)$$

$$r_{jtw}^{\text{U}} = r_{jtw}^{\text{D}} = 0, \quad \forall j, \forall t < \tau, \forall \omega = (k, \tau), \quad (5.46)$$

while if equipment failures are disregarded, non-anticipativity constraints are

not required. Note that equations (5.44)–(5.46) ensure that, in the absence of wind generation, load shedding and reserve deployment actions can be solely performed in response to a failure event, but not in advance. As a result, these equations fix the decision variables pertaining to scenario $f = 0$, in which no contingency occurs during the entire scheduling horizon, to the values set in the market.

5.5.8 Formulation

Lastly, the complete formulation of the proposed market-clearing procedure to cope with wind production uncertainty and equipment failures is presented below for clarity.

$$\begin{aligned}
\text{Minimize EC} &= \sum_{t=1}^{N_T} \text{EC}_t = \\
&= \sum_{t=1}^{N_T} \sum_{i=1}^{N_G} C_{it}^{\text{SU}} \\
&+ \sum_{t=1}^{N_T} d_t \left[\sum_{i=1}^{N_G} (C_{it}^{R^U} R_{it}^U + C_{it}^{R^D} R_{it}^D + C_{it}^{R^{\text{NS}}} R_{it}^{\text{NS}}) \right. \\
&\quad \left. + \sum_{j=1}^{N_L} (C_{jt}^{R^U} R_{jt}^U + C_{jt}^{R^D} R_{jt}^D) \right] \\
&+ \sum_{\omega=1}^{N_\Omega} \pi_\omega \left\{ \sum_{t=1}^{N_T} \sum_{i=1}^{N_G} C_{it\omega}^{\text{A}} \right. \\
&\quad + \sum_{t=1}^{N_T} d_t \left[\sum_{i=1}^{N_G} \sum_{m=1}^{N_{Oit}} \lambda_{Git}(m) p_{Git\omega}(m) - \sum_{j=1}^{N_L} \sum_{\ell=1}^{N_{Bjt}} \lambda_{Ljt}(\ell) p_{Ljt\omega}(\ell) \right. \\
&\quad \left. \left. + \sum_{j=1}^{N_L} V_{jt}^{\text{LOL}} P_{jt\omega}^{\text{L, shed}} \right] \right\}, \tag{5.47}
\end{aligned}$$

subject to

$$\sum_{i=1}^{N_G} P_{it}^{\text{G,S}} + P_t^{\text{W,S}} = \sum_{j=1}^{N_L} P_{jt}^{\text{L,S}}, \quad \forall t \tag{5.48}$$

$$P_i^{G, \min} u_{it} \leq P_{it}^{G, S} \leq P_i^{G, \max} u_{it}, \quad \forall i, \forall t \quad (5.49)$$

$$P_{jt}^{L, \min} \leq P_{jt}^{L, S} \leq P_{jt}^{L, \max}, \quad \forall j, \forall t \quad (5.50)$$

$$0 \leq P_t^{W, S} \leq \sum_{q=1}^{N_W} P_q^{W, \max}, \quad \forall t \quad (5.51)$$

$$0 \leq R_{it}^U \leq R_{it}^{U, \max} u_{it}, \quad \forall i, \forall t \quad (5.52)$$

$$0 \leq R_{it}^D \leq R_{it}^{D, \max} u_{it}, \quad \forall i, \forall t \quad (5.53)$$

$$0 \leq R_{it}^{NS} \leq R_{it}^{NS, \max} (1 - u_{it}), \quad \forall i, \forall t \quad (5.54)$$

$$0 \leq R_{jt}^U \leq R_{jt}^{U, \max}, \quad \forall j, \forall t \quad (5.55)$$

$$0 \leq R_{jt}^D \leq R_{jt}^{D, \max}, \quad \forall j, \forall t \quad (5.56)$$

$$C_{it}^{SU} \geq \lambda_{it}^{SU} (u_{it} - u_{i,t-1}), \quad \forall i, \forall t \quad (5.57)$$

$$C_{it}^{SU} \geq 0, \quad \forall i, \forall t \quad (5.58)$$

$$\begin{aligned} \sum_{i:(i,n) \in M_G(t, \omega)} P_{it\omega}^G - \sum_{j:(j,n) \in M_L} (P_{jt\omega}^L - P_{jt\omega}^{L, \text{shed}}) + \sum_{q:(q,n) \in M_W} (P_{qt\omega}^W - S_{qt\omega}) \\ - \sum_{r:(n,r) \in \Lambda(t, \omega)} P_{t\omega}^F(n, r) = 0, \quad \forall n, \forall t, \forall \omega \end{aligned} \quad (5.59)$$

$$P_{t\omega}^F(n, r) = B(n, r)(\delta_{nt\omega} - \delta_{rt\omega}), \quad \forall (n, r) \in \Lambda(t, \omega), \forall t, \forall \omega \quad (5.60)$$

$$P_{it\omega}^G \geq P_i^{G, \min} v_{it\omega}, \quad \forall i \in G(t, \omega), \forall t, \forall \omega \quad (5.61)$$

$$P_{it\omega}^G \leq P_i^{G, \max} v_{it\omega}, \quad \forall i \in G(t, \omega), \forall t, \forall \omega \quad (5.62)$$

$$P_{it\omega}^G = \sum_{m=1}^{N_{Oit}} p_{Git\omega}(m), \quad \forall i, \forall t, \forall \omega \quad (5.63)$$

$$0 \leq p_{Git\omega}(m) \leq p_{Git}^{\max}(m), \quad \forall m, \forall i, \forall t, \forall \omega \quad (5.64)$$

$$P_{jt\omega}^L = \sum_{\ell=1}^{N_{Bjt}} p_{Ljt\omega}(\ell), \quad \forall j, \forall t, \forall \omega \quad (5.65)$$

$$0 \leq p_{Ljt\omega}(\ell) \leq p_{Ljt}^{\max}(\ell), \quad \forall \ell, \forall j, \forall t, \forall \omega \quad (5.66)$$

$$-P^{\text{F}, \max}(n, r) \leq P_{t\omega}^{\text{F}}(n, r) \leq P^{\text{F}, \max}(n, r), \quad \forall (n, r) \in \Lambda(t, \omega), \quad \forall t, \forall \omega \quad (5.67)$$

$$0 \leq P_{jt\omega}^{\text{L}, \text{shed}} \leq P_{jt\omega}^{\text{L}}, \quad \forall j, \forall t, \forall \omega \quad (5.68)$$

$$0 \leq S_{qt\omega} \leq P_{qt\omega}^{\text{W}}, \quad \forall q, \forall t, \forall \omega \quad (5.69)$$

$$P_{it\omega}^{\text{G}} = P_{it}^{\text{G}, \text{S}} + r_{it\omega}^{\text{U}} + r_{it\omega}^{\text{NS}} - r_{it\omega}^{\text{D}}, \quad \forall i \in G(t, \omega), \forall t, \forall \omega \quad (5.70)$$

$$P_{jt\omega}^{\text{L}} = P_{jt}^{\text{L}, \text{S}} - r_{jt\omega}^{\text{U}} + r_{jt\omega}^{\text{D}}, \quad \forall j, \forall t, \forall \omega \quad (5.71)$$

$$0 \leq r_{it\omega}^{\text{U}} \leq R_{it}^{\text{U}}, \quad \forall i, \forall t, \forall \omega \quad (5.72)$$

$$0 \leq r_{it\omega}^{\text{D}} \leq R_{it}^{\text{D}}, \quad \forall i, \forall t, \forall \omega \quad (5.73)$$

$$0 \leq r_{it\omega}^{\text{NS}} \leq R_{it}^{\text{NS}}, \quad \forall i, \forall t, \forall \omega \quad (5.74)$$

$$0 \leq r_{jt\omega}^{\text{U}} \leq R_{jt}^{\text{U}}, \quad \forall j, \forall t, \forall \omega \quad (5.75)$$

$$0 \leq r_{jt\omega}^{\text{D}} \leq R_{jt}^{\text{D}}, \quad \forall j, \forall t, \forall \omega \quad (5.76)$$

$$C_{it\omega}^{\text{A}} = C_{it\omega}^{\text{SU}} - C_{it}^{\text{SU}}, \quad \forall i, \forall t, \forall \omega \quad (5.77)$$

$$C_{it\omega}^{\text{SU}} \geq \lambda_{it}^{\text{SU}}(v_{it\omega} - v_{i,t-1,\omega}), \quad \forall i, \forall t, \forall \omega \quad (5.78)$$

$$C_{it\omega}^{\text{SU}} \geq 0, \quad \forall i, \forall t, \forall \omega \quad (5.79)$$

$$P_{jt\omega}^{\text{L}, \text{shed}} = P_{jt\omega'}^{\text{L}, \text{shed}}; \quad r_{jt\omega}^{\text{U}} = r_{jt\omega'}^{\text{U}}; \quad r_{jt\omega}^{\text{D}} = r_{jt\omega'}^{\text{D}}, \\ \forall j, \forall t < \tau, \forall \omega, \omega' : \omega = (0, \varpi), \omega' = (f, \varpi'), f = (k, \tau), \varpi = \varpi' \quad (5.80)$$

$$r_{it\omega}^{\text{U}} = r_{it\omega'}^{\text{U}}; \quad r_{it\omega}^{\text{D}} = r_{it\omega'}^{\text{D}}; \quad r_{it\omega}^{\text{NS}} = r_{it\omega'}^{\text{NS}},$$

$$\forall i, \forall t < \tau, \forall \omega, \omega' : \omega = (0, \varpi), \omega' = (f, \varpi'), f = (k, \tau), \varpi = \varpi' \quad (5.81)$$

$$v_{it\omega} = 0, \forall t, \forall \omega, \forall i \notin G(t, \omega) \quad (5.82)$$

$$r_{it\omega}^U = r_{it\omega}^{NS} = r_{it\omega}^D = 0, \forall t, \forall \omega, \forall i \notin G(t, \omega) \quad (5.83)$$

$$u_{it}, v_{it\omega} \in \{0, 1\}, \forall i, \forall t, \forall \omega. \quad (5.84)$$

The objective function (5.47) is the expected cost of power system operation, which includes the costs pertaining to the electricity-market stage (start-up planning of generating units and reserve scheduling) and the costs derived from the real-time operation of the power system (*ad hoc* adjustments to the start-up and shut-down planning of generating units, demand supply, reserve deployment, and load shedding actions).

The group of equations (5.48)–(5.58) is related to the electricity-market stage, which basically boils down to the balance between energy offers by producers and energy bids by consumers (5.48), generation limits and demand bounds (5.49)–(5.51), the reserve scheduling constraints (5.52)–(5.56), and the modeling of the costs pertaining to the start-up planning of generating units (5.57)–(5.58).

The set of constraints (5.59)–(5.69) models the real-time operation of the power system, which revolves around the real-time power balance between generation and demand (5.59) and accommodates the power production from wind farms. The power balance equations are complemented with a dc model of the transmission network (5.60), the generations limits of production units (5.61)–(5.62), the decomposition of generator power outputs and demands by blocks (5.63)–(5.66), the transmission capacity constraints (5.67), and the load shedding and wind spillage bounds (5.68)–(5.69).

The set of equations (5.70)–(5.79), referred to as *linking constraints*, couple market decisions with operating decisions made in real time through the deployment of reserves by generating units and loads (5.70)–(5.76), and the changes in the start-up and shut-down planning of units that make such a deployment possible (5.77)–(5.79).

Equations (5.80) and (5.81) enforce the non-anticipatory character of in-

formation in the decision-making process.

Constraints (5.82) and (5.83) are imposed on generating units that are not available as a result of an unexpected failure.

Finally, constraints (5.84) declare variables u_{it} and $v_{it\omega}$ as binary.

5.5.9 Wind Spillage *Cost*

The expected cost defined in (5.9) does not account for other relevant aspects of wind power production, as those linked to the amount of system carbon emissions avoided by wind sources and to the pressure imposed by some market policies promoting wind power integration. In order to fill this gap, we redefine the objective function z of the proposed market-clearing model as the summation of the expected cost (5.9) and the so-called (expected) wind spillage *cost*, i.e.,

$$z = \text{EC} + \sum_{\omega=1}^{N_{\Omega}} \sum_{t=1}^{N_T} \pi_{\omega} V_{qt}^{\text{S}} S_{qt\omega}, \quad (5.85)$$

where V_{qt}^{S} is the cost associated with the spillage of the power generated by wind farm q in period t .

Note that the wind spillage term is not included into the expected cost as it is just an *indirect* cost difficult to determine and whose only purpose is to serve the system operator as a mechanism to alter the energy and reserve dispatch program in favor of wind generation.

5.6 Metrics for the Economic Appraisal of Wind Generation

Concerning financial issues, two conflicting and opposite effects should be taken into account while appraising wind generation. On the one hand, the integration of wind power into an electricity system leads to a reduction of the operation cost as wind generation, with zero fuel cost, usually replaces units with significant fuel costs (thermal units). On the other hand, since wind generation is variable and difficult to predict accurately, it contributes to system

uncertainty and requires the system operator to schedule additional reserve capacity to maintain system security. This increase in the required reserve translates into an increase of the operation cost.

Therefore, these two conflicting effects should be considered while assessing the integration of wind generation and its profitability. For this purpose, three performance metrics are defined below. Specifically, these metrics are used to clarify the two conflicting economic effects of wind generation: the first effect being the fuel cost reduction and the second one, the reserve cost increase due to wind uncertainty. Other metrics and methods defined to assess the wind capacity value (reliability-based approaches or simpler metrics based on wind capacity factor) are found in [2].

5.6.1 Average Benefit (AB)

The *Average Benefit* in \$/MWh is a measure of the decrease in expected cost as a result of injecting an additional MWh of wind power into the network,

$$AB = \frac{\sum_{t=1}^{N_T} \frac{DC_t^0 - EC_t^*}{d_t \sum_{q=1}^{N_W} \hat{P}_{qt}^W}}{N_T} \quad (5.86)$$

where EC_t^* is the objective function optimal value in period t (see equation (5.9)), $\hat{P}_{qt}^W$ represents the power output forecast for wind farm q in period t and DC_t^0 is the *deterministic* cost of the system in period t without wind power generation. Note that DC_t^0 is obtained by solving the proposed model for a deterministic case in which no wind is considered.

5.6.2 Average Uncertainty Cost (AUC)

The *Average Uncertainty Cost* in \$/MWh is a measure of the equivalent deterministic production cost of *fictitious* wind generators producing exactly their forecast power outputs,

$$AUC = \frac{\sum_{t=1}^{N_T} \frac{EC_t^* - DC_t^F}{d_t \sum_{q=1}^{N_W} \hat{P}_{qt}^W}}{N_T} \quad (5.87)$$

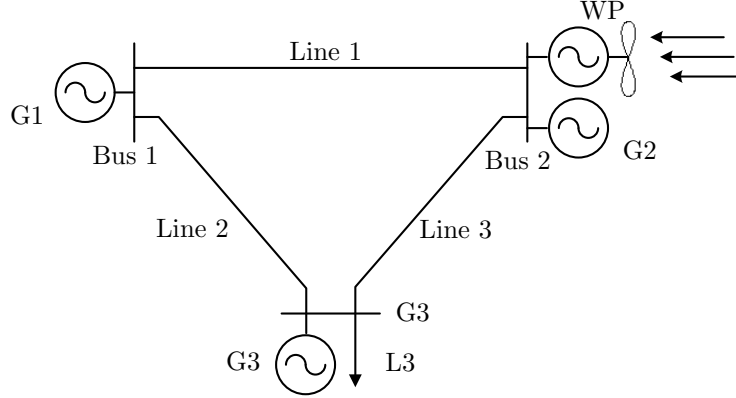


Figure 5.2: Three-bus system

where DC_t^F is the *deterministic* cost in period t assuming that the wind power production is perfectly known and coincides with its forecast. DC_t^F is obtained by solving the proposed model for a deterministic case in which only the wind power forecast scenario is considered.

5.6.3 Net Average Benefit (NAB)

The *Net Average Benefit* in \$/MWh is a measure of the profitability of the wind power generation injected into the system,

$$\text{NAB} = \text{AB} - \text{AUC}. \quad (5.88)$$

5.7 Market-Clearing Example

In this example, the market-clearing procedure formulated in Section 5.5.8 is applied to the three-node system depicted in Fig. 5.2. The data defining the base case for this system have been extracted from [40]. Line reactances are all 0.13 p.u. The capacities of all lines are the same and equal to 55 MW.

Data for the generating units are given in Table 5.1.

For the sake of simplicity, we assume that the energy and reserve offers of the units are the same throughout the scheduling horizon consisting of four

Table 5.1: Generating unit data

Generator i	1	2	3
$P_i^{G, \min}$ (MW)	10	10	10
$P_i^{G, \max}$ (MW)	100	100	50
λ_{it}^{SU} (\$)	100	100	100
λ_{Git} (\$/MWh)	30	40	20
$C_{it}^{R^U}$ (\$/MWh)	5.0	7.0	8.0
$C_{it}^{R^D}$ (\$/MWh)	5.0	7.0	8.0
$C_{it}^{R^{\text{NS}}}$ (\$/MWh)	4.5	5.5	7.0

Table 5.2: Hourly demand at bus 3

Period t	1	2	3	4
$P_{jt}^{\text{L}, \text{S}}$ (MW)	30	80	110	40

hourly periods. Each unit offers a single block of energy at a cost of λ_{Git} and a start-up cost offer of λ_{it}^{SU} . The maximum amounts of reserve services offered by the units are the largest possible, i.e., $P_i^{G, \max} - P_i^{G, \min}$ for up and down-spinning reserves and $P_i^{G, \max}$ for non-spinning reserves. Minimum up- and down-time constraints are not considered in this example and the ramping capabilities of generating units are the largest possible, i.e., $P_i^{G, \max}$ MW/h. Furthermore, the three units are assumed to be off-line at the beginning of the study horizon ($t = 0$).

A load, with the hourly demand profile shown in Table 5.2, is located at bus 3 (see Fig. 5.2).

This load offers to sell up- and down-spinning reserves at \$70/MWh. The load can be partially curtailed up to 10% of the hourly demand in order to provide additional up-spinning reserve. Likewise, it can increase its consumption the same quantity so as to contribute to down-spinning reserve. Beyond these limits, the value of lost load is assumed to be \$1000/MWh.

As depicted in Fig. 5.2, the three-node system also includes a wind plant (WP) at bus 2, whose installed capacity is 60 MW.

Next, we comprehensively analyze the individual and combined effects of equipment failures and wind production on the results yielded by the proposed market-clearing procedure.

5.7.1 Equipment Failures

First, we assume that the power production of the wind farm at node 2 is zero. Likewise, we suppose that:

1. Units 1 and 2, and transmission lines 1 and 3, are 100% reliable, i.e., they never fail.
2. Time between two consecutive failures of unit 3 follows an exponential distribution characterized by a MTTF equal to 500 h.
3. Analogously, time between two consecutive failures of line 2 follows an exponential distribution with a MTTF of 1000 h.

By indexing the failures of unit 3 and line 2 with $k = 1, 2$, respectively, and considering that any one of these single contingencies can occur during any one of the four hours of the scheduling horizon, $\tau = 1, \dots, 4$, and that the joint failure of unit 3 and line 2 is not possible, the following nine scenarios f can be defined:

Scenario $f = 0$, in which no contingency happens during the entire scheduling horizon.

Scenarios $f = 1, \dots, 4$, defined by pairs $(k, \tau) = (1, \tau)$ with $\tau = f$, where the failure of unit 3 takes place in time period τ .

Scenarios $f = 5, \dots, 8$, defined by pairs $(k, \tau) = (2, \tau)$ with $\tau = f - 4$, where the outage of line 2 occurs in time period τ .

In order to compute the probabilities of occurrence associated with the nine failures scenarios above, we need first to calculate the probabilities of random

events $A(k, \tau)$ and $B(k)$ defined in Section 5.4.1:

$$\begin{aligned}
 P[A(1, \tau)] &= \exp(-\lambda_1 \tau) (\exp(\lambda_1) - 1) = \exp\left(-\frac{1}{500} \frac{\text{h}}{\text{h}}\right) \left(\exp\left(\frac{1}{500} \frac{\text{h}}{\text{h}}\right) - 1\right) \\
 &= 1.9980 \times 10^{-3} \\
 P[A(2, \tau)] &= \exp(-\lambda_2 \tau) (\exp(\lambda_2) - 1) = \exp\left(-\frac{1}{1000} \frac{\text{h}}{\text{h}}\right) \left(\exp\left(\frac{1}{1000} \frac{\text{h}}{\text{h}}\right) - 1\right) \\
 &= 0.9995 \times 10^{-3} \\
 P[B(1)] &= \exp(-\lambda_1 T) = \exp\left(-\frac{4}{500} \frac{\text{h}}{\text{h}}\right) = 0.9920 \\
 P[B(2)] &= \exp(-\lambda_2 T) = \exp\left(-\frac{4}{1000} \frac{\text{h}}{\text{h}}\right) = 0.9960.
 \end{aligned}$$

Then, the probability π_0 that none of contingencies k occurs during the scheduling horizon is given by

$$\pi_0 = \prod_{k=1}^K P[B(k)] = P[B(1)]P[B(2)] = 0.9920 \times 0.9960 = 0.9880.$$

Likewise, the probability $\pi_f, f = 1, \dots, 4$ that the failure of unit 3 occurs during the interval $\tau = f$, given that all other system components are available for the entire scheduling horizon, is

$$\begin{aligned}
 \pi_f &= \pi(1, f) = P[A(1, f)] \prod_{\substack{y=1 \\ y \neq 1}}^2 P[B(y)] = P[A(1, f)]P[B(2)] \\
 &= 1.9980 \times 10^{-3} \times 0.9960 = 1.9900 \times 10^{-3}; \quad \forall f = 1, \dots, 4.
 \end{aligned}$$

Finally, the probability $\pi_f, f = 5, \dots, 8$ that only the outage of line 2 in time period $\tau = f - 4$ happens during the whole scheduling horizon is

$$\begin{aligned}
 \pi_f &= \pi(2, f - 4) = P[A(2, f - 4)] \prod_{\substack{y=1 \\ y \neq 2}}^2 P[B(y)] = P[A(2, f - 4)]P[B(1)] \\
 &= 0.9995 \times 10^{-3} \times 0.9920 = 0.9915 \times 10^{-3}; \quad \forall f = 5, \dots, 8.
 \end{aligned}$$

Note that $\sum_{f=0}^9 \pi_f \lesssim 1$, as the missing probability corresponds to the simultaneous and sequential outages of unit 3 and line 2, which are significantly

Table 5.3: Scheduling program. Powers in MW

Period t	P_t^S			R_t^U				R_t^D				R_t^{NS}		
	G1	G2	G3	G1	G2	G3	L3	G1	G2	G3	L3	G1	G2	G3
1	0	0	30	0	0	0	0	0	0	0	0	0	0	0
2	30	0	50	0	0	0	0	0	0	0	0	0	0	0
3	60	0	50	22.5	0	0	0	5	0	0	0	0	0	0
4	0	0	40	0	0	0	0	0	0	0	0	40	0	0

more unlikely and are not considered here. Their sum is indeed approximately equal to 0.999926. A simple criterion to force these probabilities to add up to one consists in just distributing proportionally the missing probability amongst them. By rounding off to five decimal places the new probabilities π'_f become

$$\begin{aligned}\pi'_0 &= \frac{\pi_0}{\sum_{f=1}^9 \pi_f} = 0.98807 \\ \pi'_f &= \frac{\pi_f}{\sum_{f=1}^9 \pi_f} = 1.99015 \times 10^{-3}; \quad \forall f = 1, \dots, 4 \\ \pi'_f &= \frac{\pi_f}{\sum_{f=1}^9 \pi_f} = 0.99157 \times 10^{-3}; \quad \forall f = 5, \dots, 8,\end{aligned}$$

which add up to 1.00000.

Table 5.3 provides the scheduling program resulting from the market-clearing procedure.

The expected social cost is \$7050.7. The total amount of scheduled reserves is 67.5 MW, which represents approximately the 26% of the demand. Table 5.4 provides the amount of load that is involuntarily shed per scenario and period.

Therefore, the expected load not served (ELNS) associated with the proposed scheduling program is

$$\begin{aligned}\text{ELNS} &= (30 + 2 \times 50 + 3 \times 27.5) \times 1.99015 \times 10^{-3} + \\ &\quad + 3 \times 5 \times 0.99157 \times 10^{-3} = 0.4378 \text{ MWh.}\end{aligned}$$

[illegible]

As those loss-of-load incidents respond solely to economic criteria, a simple manner to avoid them is increasing the value of the lost load. As an example, if this value is increased up to \$25 000/MWh, no load shedding occurs during the entire scheduling horizon under any scenario. Table 5.5 details the scheduling program derived from the market-clearing procedure for this new value of lost load.

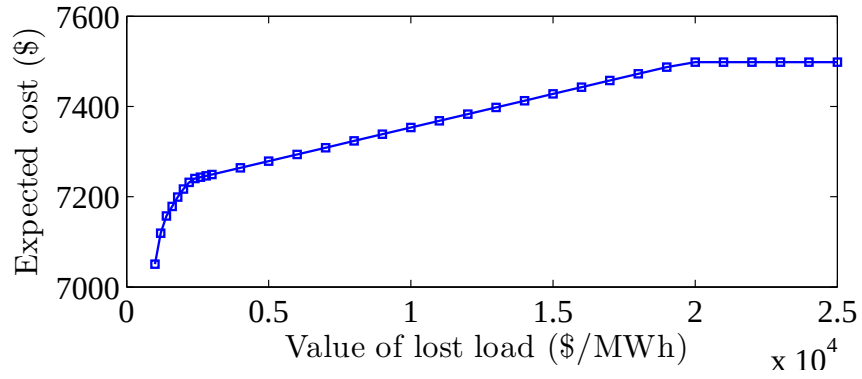
[illegible]

In this case, note that the quantity of scheduled reserves represents 67% of the total demand. Likewise, the expected social cost associated with this new program amounts to \$7498, which translates into an increase of 6.34% with respect to the previous one.

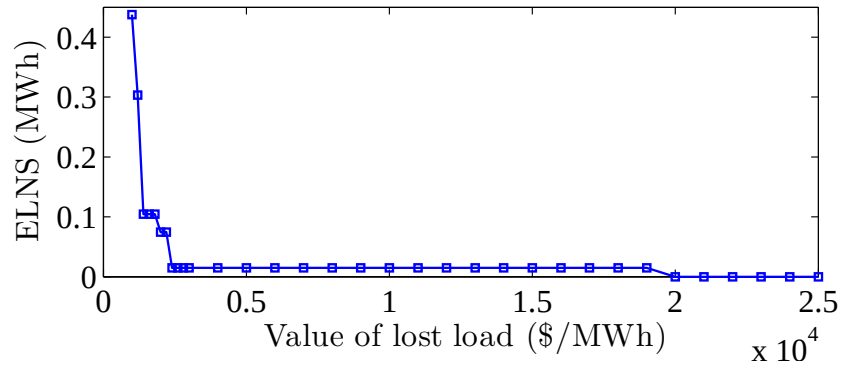
Fig. 5.3 shows the evolution of the expected social cost, the expected load not served (ELNS), and the total amount of reserves scheduled in the market as a function of the value of lost load. It can be inferred from this figure that the changes undergone by these three magnitudes are clearly interrelated. The linear stretch characterizing most part of the evolution of the expected cost corresponds to a range of the value of lost load in which both the ELNS and the total scheduled reserve remain stable. The scheduling programs generated by the market-clearing procedure for a value of lost load within this range are all identical. As a result, the increase rate of the referred linear evolution is precisely proportional to the value of lost load, with the amount of load shed being the proportionality constant.

On the other hand, in light of the results provided in this figure, there is no doubt that the reduction of the frequency and magnitude of load shedding events leads to an increase in the amount of reserve scheduled in the electricity market and consequently, to a rise in the expected cost of system operation. In plain words, prevention implies cost. As an example, observe that the initial stretch of the ELNS evolution is characterized by a sharp drop, which coincides with a brisk increase in expected cost and total scheduled reserve.

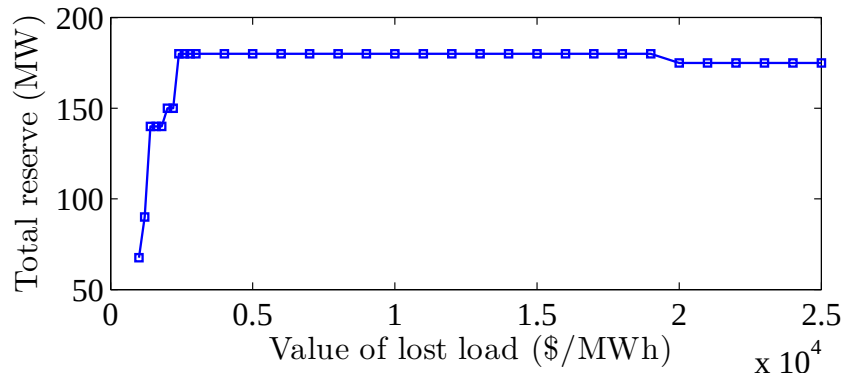
Nonetheless, there is an exception in this general behavior. If we examine in detail Fig. 5.3(c), we can appreciate a small decrease in the total reserve occurring exactly for the value of lost load from which the ELNS becomes zero and the expected cost becomes flat. That is, this step change in the amount of reserve scheduled in the market is intended to eliminate load shedding events once and for all. This exception can be easily understood by means of Tables 5.6 and 5.7. The first one provides a comparison between the scheduling programs obtained for two different values of lost load, \$15 000/MWh and \$25 000/MWh. Note that these values are, respectively, smaller and greater than the value of lost load for which the referred decrease in reserve takes place. The numbers presented in square brackets correspond to those results



(a)



(b)



(c)

Figure 5.3: Impact of the magnitude of the value of lost load on: (a) the expected social cost; (b) the expected amount of load not served (ELNS); (c) the total amount of scheduled reserve

Table 5.6: Reserve schedule for $V^{\text{LOL}} = \$15\,000/\text{MWh}$ [$\$25\,000/\text{MWh}$]. Powers in MW

Period t	P_t^S			R_t^U				R_t^D				R_t^{NS}		
	G1	G2	G3	G1	G2	G3	L3	G1	G2	G3	L3	G1	G2	G3
1	0	0	30	0	0	0	0	0	0	0	0	30	0	0
2	30	0	50	50	0	0	0	0	0	0	0	0	0	0
3	60	0	50	0	0	0	0 [5]	5	0	0	0	0	55 [45]	0
4	0	0	40	0	0	0	0	0	0	0	0	40	0	0

Table 5.7: Involuntary load shedding (MWh) for $V^{\text{LOL}} = \$15\,000/\text{MWh}$ [$\$25\,000/\text{MWh}$]

Period t	Scenario f								
	0	1	2	3	4	5	6	7	8
1	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	5 [0]	5 [0]	5 [0]	0
4	0	0	0	0	0	0	0	0	0

in which the two schedules differ, with the value enclosed corresponding to $V^{\text{LOL}} = \$25\,000/\text{MWh}$. Table 5.7 illustrates the magnitude of load shedding events associated with both scheduling programs. The format used for this second table is analogous to that of Table 5.6.

The only differences that can be detected between both scheduling programs concern period 3, a period characterized by the highest demand level (110 MW). The loss of line 2 (connecting nodes 1 and 3) brings about a substantial reduction of the network transmission capability, which, in particular, poses great difficulties for the system to meet the demand during period 3. Specifically, in the absence of this line, the maximum load that can be supplied at node 3 is 105 MW, 50 of which are generated by unit 3 located at that

node. The remainder 55 MW come from unit 1 through lines 1 and 3, which become both saturated as a consequence. Against this situation, the system operator has two alternatives. The first one basically consists in doing nothing and thereby, if the outage of line 2 finally materializes, those 5 MW missing to satisfy the complete demand are irremediably shed without the consent of the corresponding consumer. This corrective action would entail a very high cost, but the probability of having to put it into practice is also very low. Therefore, this operation strategy turns out to be the most economic option if the value of lost load is sufficiently low (e.g., \$15 000/MWh). In contrast, the system operator can *ask for the consent* of the consumer at node 3 to reduce its consumption in 5 MW in the case that the outage of line 2 comes into effect. Needless to say, this is speaking in a figurative sense. In fact, this *request* is negotiated in the electricity market through the scheduling and contracting of reserves with the load. This market operation also entails a cost. However, unlike the cost associated with the involuntary load shedding, this cost is usually much lower (as the system operator has the permission from the load), but *certain*, that is, it constitutes a cost to incur irrespective of the scenario that eventually materializes. Consequently, this option becomes profitable for values of lost load high enough (e.g., \$25 000/MWh). These two possible situations are depicted in Fig. 5.4.

On the other hand, the system operator *contracts* non-spinning reserve with unit 2 in order to cover the likely loss of the production of unit 3 without the need for resorting to involuntary load shedding actions. For this purpose, 55 MW of non-spinning reserve from unit 2 are scheduled if the value of lost load is equal to \$15 000/MWh. On the contrary, if this value is \$25 000/MWh, just 45 MW of this reserve are required since the system operator can also make use of the additional 5 MW of up-spinning reserve that, in this case, are contracted from the demand to eliminate the loss-of-load events pertaining to the outage of line 2. This issue explains the total reserve reduction in Fig. 5.3(c) and is further clarified in Fig. 5.5.

Lastly, Table 5.8 displays a detailed breakdown of the expected social costs both in absolute terms and in percentages of the total value (number indicated in parentheses). System operation costs include the cost of the energy

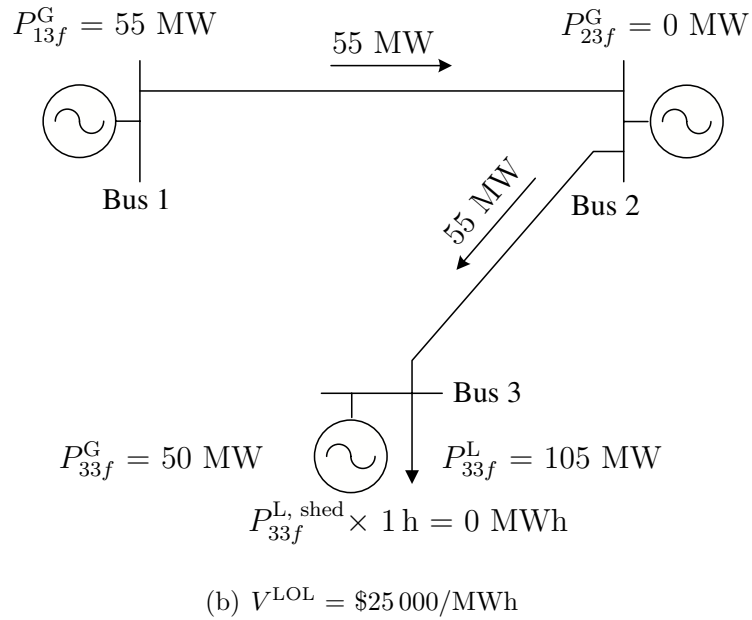
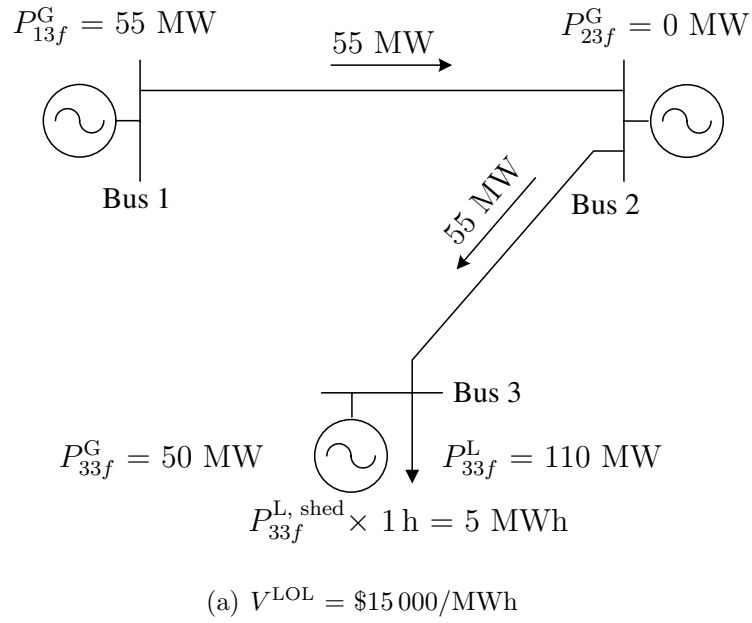


Figure 5.4: Effect of the outage of line 2 during period $t = 3$ in failure scenarios $f = 5, 6$, and 7

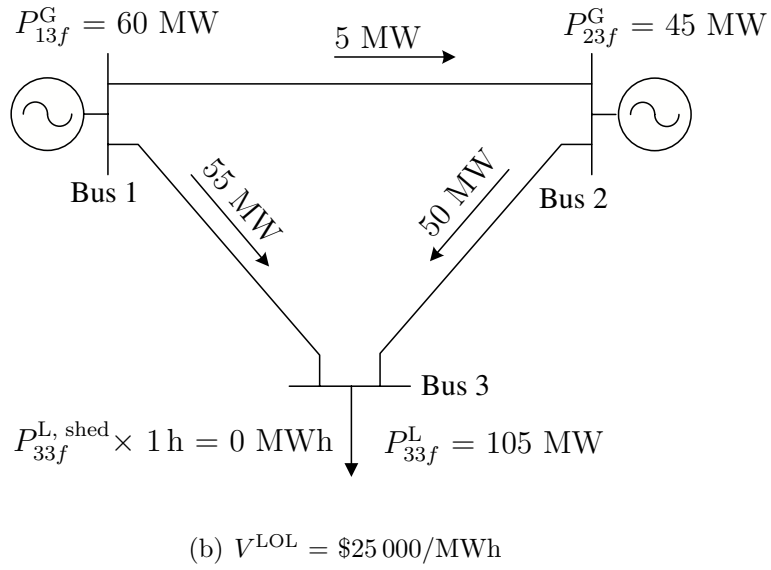
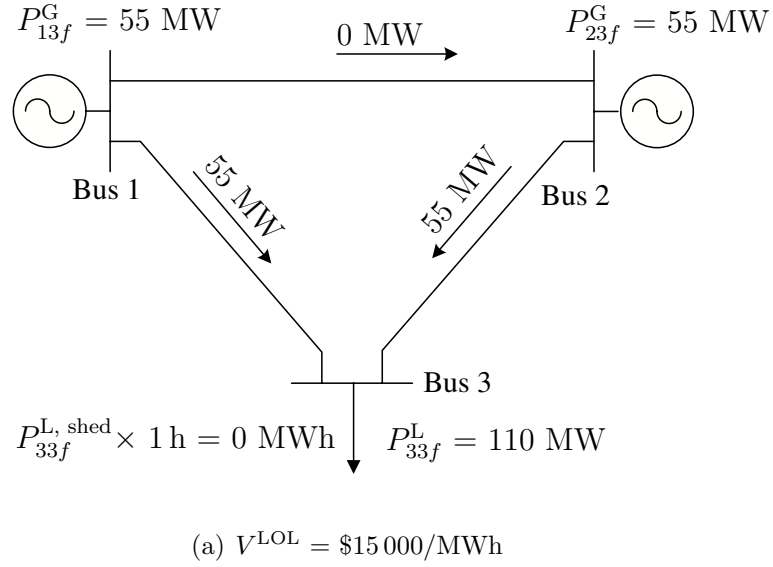


Figure 5.5: Effect of the outage of unit 3 during period $t = 3$ in failure scenarios $f = 1, 2$, and 3

Table 5.8: Breakdown of expected social costs in \$ (%)

V^{LOL} (\$/MWh)	Total	Start- up	System operation	Reserve scheduling	Energy not served
1000	7051 (100)	200 (3)	6096 (86)	318 (5)	438 (6)
15 000	7428 (100)	200 (3)	6112 (82)	893 (12)	223 (3)
25 000	7498 (100)	200 (3)	6110 (81)	1188 (16)	0 (0)

required to materialize the deployment of reserves plus the cost of the energy transactions carried out in the electricity market. It can be observed that the *cost* associated with the scheduling of reserves is an increasing function of the value of lost load, as opposed to the decreasing trend exhibited by the cost of the energy not served. In short, this table highlights the intuitive fact that, in order to eliminate likely load shedding events, thus making the system security prevail over economic objectives, the system operator has to reinforce the preventive actions, i.e., to schedule a greater amount of reserves and/or to schedule more expensive reserves. Whichever the option chosen, this task entails additional costs.

5.7.2 Wind Power Production

In the following, we disregard possible equipment failures and we assume that the power produced by the wind farm located at bus 2 can be statistically modeled by three scenarios: *forecast*, *high*, and *low*, with probabilities 0.6, 0.2, and 0.2, respectively. The power values characterizing these scenarios over the scheduling horizon are listed in Table 5.9. We examine then the influence of several basic aspects of wind generation on the scheduling program derived from the proposed market-clearing model and on its associated expected cost composition. In particular, we appraise the impact of: i) wind unit location; ii) network congestion; iii) wind spillage cost, and iv) wind penetration and uncertainty levels.

To start with, Fig. 5.6 offers a breakdown of the total expected cost that

Table 5.9: Wind power scenarios

Period t	$P_{t\varpi}^W$ (MW)		
	Forecast	High	Low
1	6	9	2
2	20	30	13
3	35	50	25
4	8	12	6

results from solving model (5.47)–(5.84) for three different situations, namely:

- a) The wind power production is perfectly known and coincides with the forecast.
- b) The three-node system is dispatched excluding wind generation.
- c) The power produced by the wind generator is uncertain. Its stochastic behavior is modeled via the three wind power scenarios provided in Table 5.9.

By comparing case a) or c) with case b), it becomes clear that the inclusion of wind energy sources into the generation mix of this small-scale power system leads to a substantial reduction in the running costs, that is, in the costs of energy production. In instances a) and c), the energy supply to the load at node 3 can be partially covered by the wind unit. As a result, a part of the energy that otherwise would be produced by the other three units, with significant fuel costs, is displaced by renewable and free energy. On the other hand, case c) shows an expected cost somewhat greater than that of case a). As can be observed, the difference is mainly due to the cost of the reserves required to accommodate the uncertain variability of wind generation, cost that cannot be disregarded, albeit it is significantly smaller than the cost associated with the energy production.

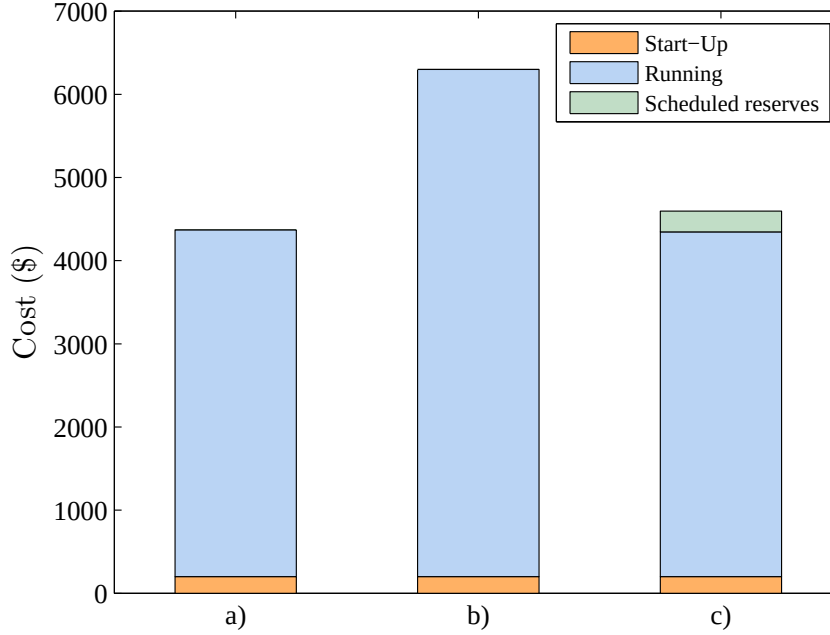


Figure 5.6: Three relevant cases: a) No wind generation uncertainty; b) no wind generation; c) uncertain wind generation

Impact of wind unit location and network congestion

Next, we explore how the location of the wind unit impacts on the expected cost. The magnitude of this impact is dependent on both the wind potential at the specific site and the network conditions. More precisely, the wind site characteristics shape the stochastic behavior of the power produced by the wind unit installed in it, and the capacity of the network determines the capability of this wind unit to evacuate its uncertain and variable power production toward the consumption nodes. Being so, let us assume that the wind characteristics are the same all over the three-node system. Consequently, the three wind power scenarios listed in Table 5.9 serve us to model the wind power production irrespective of the node at which the wind unit is placed. Then, the network condition remains as the only cause responsible for the impact of wind unit location on the expected cost.

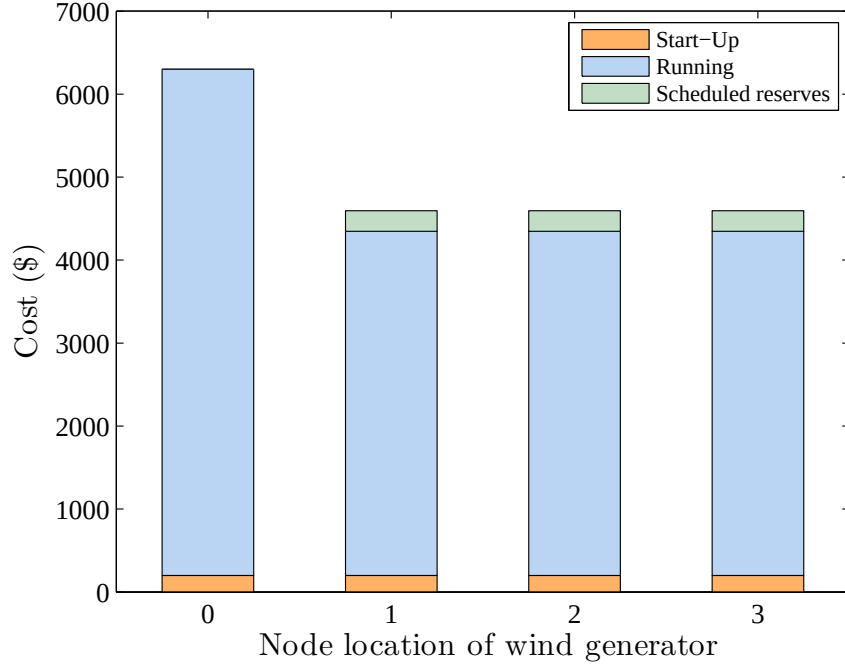


Figure 5.7: Base case ($P^{\text{F}, \max}(2, 3) = 55$ MW). Expected cost breakdown

Fig. 5.7 illustrates the breakdown of the expected cost as a function of the wind unit position within the power system. The node location “0” stands for the case in which no wind generation is considered. Note that the expected cost and its components are identical regardless of the node at which the wind unit is located. The reason for this is that the grid has room enough to accommodate and take advantage of the whole wind power production. According to the network data provided at the beginning of this example, the transmission capacities of all the lines are equal to 55 MW. For the sake of clarity, hereinafter, the three-node system with these capacity values will be referred to as *base case*. We also define the *network-constrained* case, which is equal to the former except for the fact that the capacity of line 3 is reduced to 25 MW ($P^{\text{F}, \max}(2, 3) = 25$ MW) in order to highlight the effect of network bottlenecks on the expected cost. Fig. 5.8 is analogous to Fig. 5.7, but for the network-constrained case. Observe that if the wind unit is located at node 2,

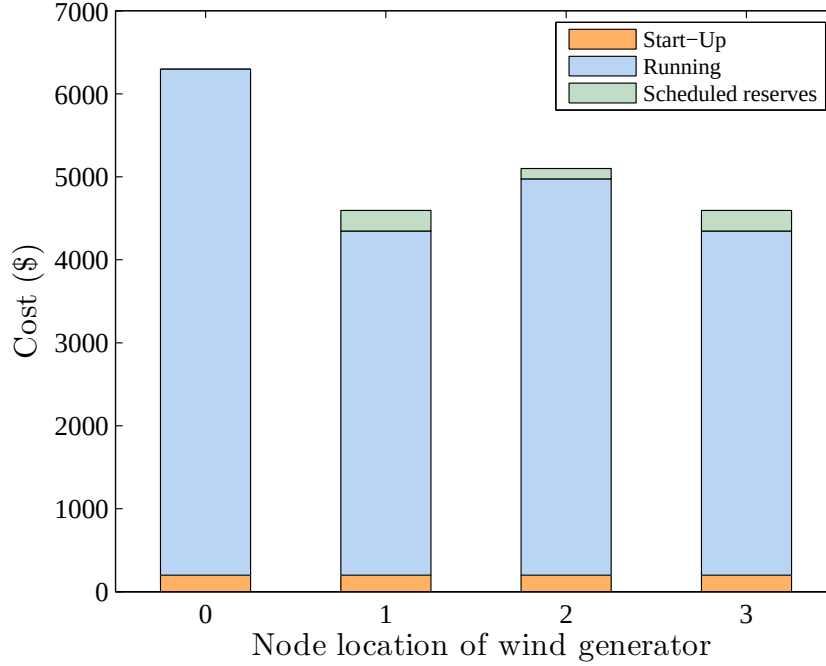


Figure 5.8: Network-constrained case ($P^{\text{F}, \max}(2, 3) = 25$ MW). Expected cost breakdown

the expected cost suffers an important rise, which is the result of a substantial increase in running costs and a small reduction in reserve costs. This rise is justified by a significant upsurge of wind spillage phenomena. If wind is spilled, the power system can be operated under a diminished level of uncertainty and less reserves need to be scheduled in consequence. On the contrary, the share of renewable and free energy in the electricity supply decreases and costly energy resources have to be used instead.

The performance metrics (AB, AUC and NAB) defined in Section 5.6 can be used to summarize the information regarding the impact of network congestion on the profitability of wind generation. Table 5.10 provides the value of these metrics computed for both the base case and the network-constrained one. Additionally, these metrics are also calculated for each possible node location of the wind unit. It should be noted in Table 5.10 that, under no congestion ($P^{\text{F}, \max}(2, 3) = 55$ MW), the three metrics take on the same values

Table 5.10: Impact of network congestion on performance metrics

Performance metrics	$P^{\text{F}, \max}(2, 3) = 55 \text{ MW}$	$P^{\text{F}, \max}(2, 3) = 25 \text{ MW}$	
	WP at buses 1, 2 or 3	WP at buses 1 or 3	WP at bus 2
AB	20.95	20.95	17.34
AUC	4.05	4.05	3.37
NAB	16.90	16.90	13.97

regardless of where the wind unit is located. However, under network congestion ($P^{\text{F}, \max}(2, 3) = 25 \text{ MW}$), the metrics do change depending upon the node at which the wind power unit is placed: if it is located at nodes 1 or 3, the metric values are the same and equal to those obtained under no congestion, but if it is located at node 2 the three metric values are comparatively smaller. In this latter case, the *average benefit* (AB) decreases since the system cannot fully benefit from the wind potential at that node due to the physical restrictions imposed by the network. The *average uncertainty cost* (AUC) also decreases as the network acts as a filter cutting off, and thus reducing, the uncertain variability of wind power production. From a numerical perspective, the AUC decreases because the associated deterministic cost DC_t^{F} , computed from the wind power forecast, is also a congested case and then the deterministic and stochastic problems yield similar results. As the reduction in AB is greater than the decrease of AUC, the *net average benefit* finally diminishes. This means that, under economic criteria, nodes 1 or 3 are more suitable sites than node 2 to locate the wind farm if the capacity of line 3 is limited to 25 MW.

On the other hand, network bottlenecks can have an important effect on the scheduling program provided by the market-clearing model proposed in this chapter. As an illustration of this, in Table 5.11, the scheduling and dispatching programs obtained for both the base and network-constrained cases with the wind unit located at node 2 are compared. The results shown in square brackets for period 3 correspond to the network-constrained case. For

Table 5.11: Comparison of scheduling programs. Powers in MW

Period t	$P_t^{W, S}$	$P_t^{G, S}$			R_t^U				R_t^D				R_t^{NS}		
		G1	G2	G3	G1	G2	G3	L3	G1	G2	G3	L3	G1	G2	G3
1	6	0	0	24	0	0	4	0	0	0	0	0	0	0	0
2	30	0	0	50	0	0	0	0	0	0	0	0	17	0	0
3	35	25	0	50	10	0	0	0	15	0	0	0	0	0	0
	[15]	[45]	[0]	[50]	[0]	[0]	[0]	[0]	[0]	[0]	[0]	[0]	[0]	[0]	[0]
4	8	0	0	32	0	0	2	0	0	0	0	0	0	0	0

conciseness, the results of periods 1, 2 and 4 for this case are omitted because they are identical to the corresponding ones in the base case. Observe that the hourly wind production scheduled in period 2 is equal to the *high* wind power scenario for both the base case and the network-constrained one. This result, which may be surprising a priori, poses the following relevant question: would it not be more sensible to program a wind power production equal to the value forecasted for that period? Intuitively, we may feel tempted to respond *yes* because, first, this is precisely the solution schedule for the rest of time periods (except for period 3 in the network-constrained case, issue that we discuss below), and second, the *forecast* wind power scenario is by far the one with the highest probability (0.6). However, our tireless search for the economic optimum lead us to answer with an emphatic *no* for the following reason: to guarantee supplying the load at bus 3 for any realization of wind power production in period 2, the market operator contracts 17 MW of reserve with unit 1 for that period. Clearing the market with a wind power schedule in period 2 of 30 MW results in unit 1 being out of the energy market in that period and thus allows the market operator to contract those 17 MW of reserve from the non-spinning reserve offer of this unit. In contrast, if we fix the wind power scheduled in period 2 to its forecast value (20 MW), then unit 1 is committed in the energy market. Consequently, the required 17 MW of reserve are obtained from the spinning reserve offer of unit 1, which is more expensive and as a result, a higher expected cost is obtained.

On the other hand, in the network-constrained case, the optimal-market

clearing solution is achieved by scheduling only 15 MW of wind power in period 3. This is the maximum amount of wind power that can be injected at bus 2 without making use of any type of reserve. Consequently, the excess of wind generation is spilled, reserves are not required and unit 1 is committed (with a scheduled power of 45 MW) in order to meet the period load of 110 MW by transferring power through line 2.

Impact of wind spillage cost

So as to gain a greater insight into the economic effects of wind spillage, we consider below a wind spillage cost equal to \$100/MWh in the network-constrained case. This value is arbitrary, which is immaterial for the purposes of this example. Yet, this cost should be in practice sufficiently high to account for other benefits associated with wind power that are not included in the model, e.g., avoided emissions, and to constitute a numerical translation of the policies intended to encourage the integration of wind energy. Ultimately, this cost translates into a dispatching priority to the wind farm regardless of the operation cost. Fig. 5.9 shows the breakdown of the expected cost, i.e., the breakdown of variable EC in objective function (5.85), if the market-clearing problem is solved taking into account such a spillage cost. Once more, if the wind unit is placed at node 2, line 3 becomes congested and the wind potential cannot be completely exploited. However, unlike in a free cost wind spillage situation (see Fig. 5.8), in this case, the system operator tries its best to curb the waste of wind as much as possible through a more flexible operation of the power system based on the scheduling of additional and more expensive reserves (e.g., reserve from the load). This enhanced flexibility enables a greater use of the wind sources and therefore, the running costs keep almost unchanged, or become even lower. Fig. 5.10 clarifies this issue by means of two pie charts illustrating the breakdown in percentages of the expected cost for the two different considered values of wind spillage cost, 0 and \$100/MWh, respectively. We insist that the term *expected cost* refers to variable EC in objective function (5.85). Note that the portion corresponding to the cost of reserves increases from 2% to 18% when the wind spillage becomes costly. This

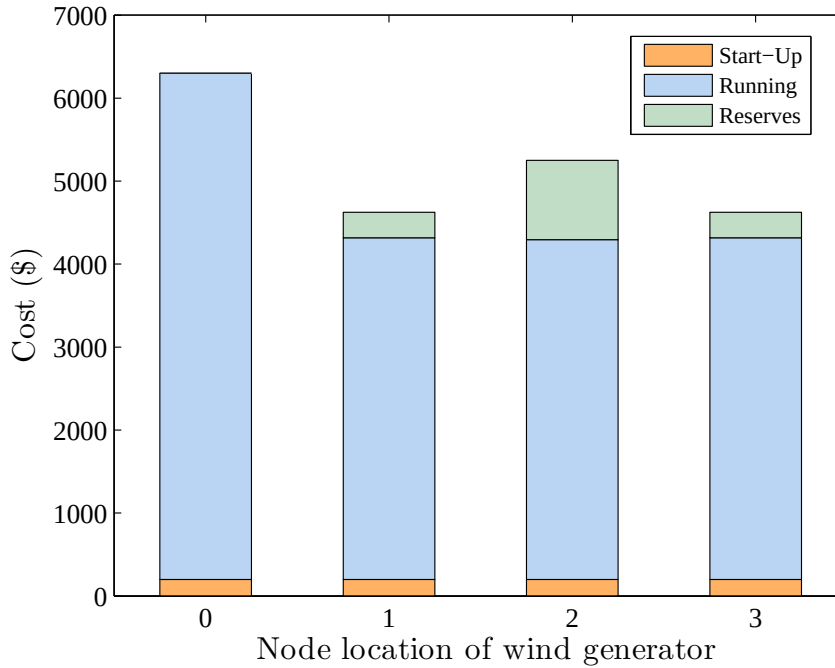


Figure 5.9: Network-constrained case with a cost of wind power spillage equal to \$100/MWh. Expected cost breakdown

increase is aimed at getting the power system ready to *absorb* the largest possible amount of wind energy. Nonetheless, in spite of this additional expense devoted to strength system reserves, some of the wind power production still has to be irremediably spilled due to network congestion. This stresses the fundamental role that the network plays in the integration of wind generation: managing system uncertainties requires not only generating units and loads willing to alter their energy production and consumption if needed, but also an electrical infrastructure robust enough to make such alterations feasible.

Impact of wind penetration and uncertainty levels

Lastly, we analyze the impact of the wind integration level on the three-node system. For this purpose, the base case and the network-constrained one are next solved for a large set of conditions characterized by different wind power

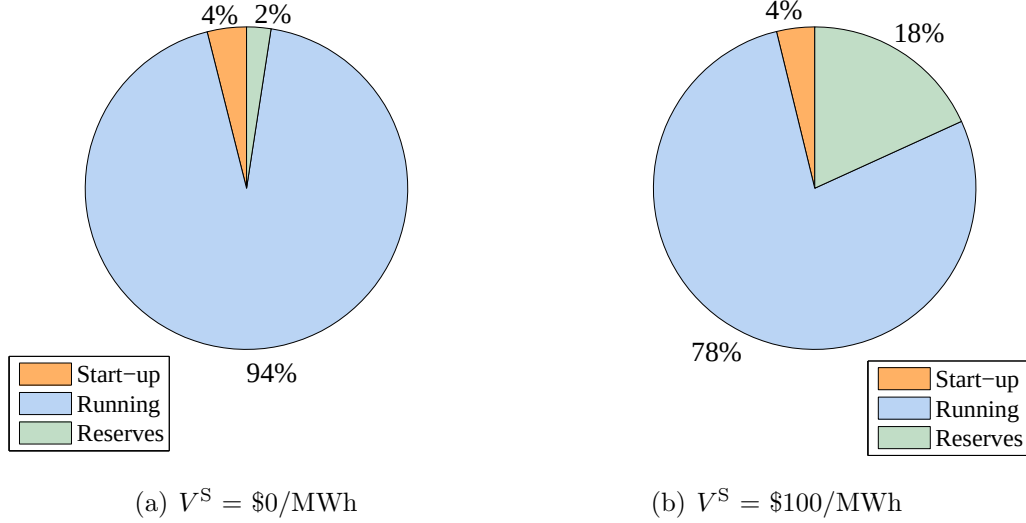


Figure 5.10: Network-constrained case. Wind unit at node 2. Effect of wind spillage cost on the expected cost breakdown

penetration and uncertainty levels. The wind power penetration level is defined by setting the wind power forecast scenario to a percentage of the hourly demand (horizontal axis in Figs. 5.11–5.13), while the two other scenarios (*low* and *high*) are built as given percentages of the wind power forecast, below and above respectively. The uncertainty level is defined as a measure of how far from the wind power forecast scenario the *low* and *high* scenarios are. The uncertainty level considered ranges from 0% to 100% at 10% increments in such a way that $x\%$ uncertainty means that the *high* and *low* scenarios are equal to the wind power forecast multiplied by $(1 + x/100)$ and $(1 - x/100)$, respectively.

The eleven curves depicted in Fig. 5.11 show the evolution of the expected social cost in the base case as the wind power penetration increases, representing each curve a different uncertainty level. A decreasing trend of the expected cost can be observed as the wind power penetration increases. However, it should be noted that this trend is less marked as the uncertainty associated to wind production grows.

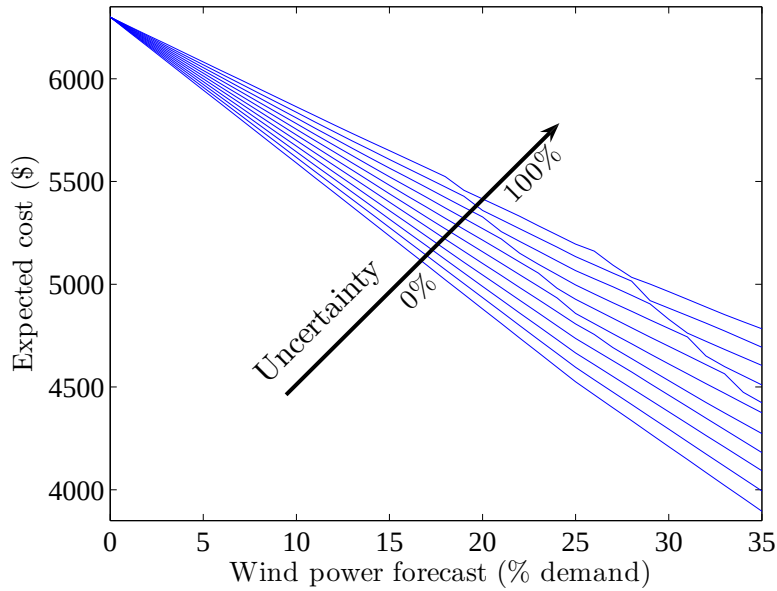


Figure 5.11: Wind unit at node 2. Expected cost as a function of wind penetration and wind uncertainty. Base case

Fig. 5.12 illustrates the evolution of the expected cost for the network-constrained case. The same wind power penetration and uncertainty levels as in Fig. 5.11 are considered. By comparing both figures, the constraining effect of the network is made evident. Qualitatively, it can be observed that the effect of the network is twofold. On the one hand, the slopes of the eleven curves decrease for increasing wind power penetration levels, which translates into a decrease of the reduction rate of the expected cost. The reason for this is that the network imposes a *cap* on the amount of wind power that can be injected into the system. Thus, once the network becomes congested, the wind is spilled and consequently, the system cannot benefit from high wind scenarios. On the other hand, the curves corresponding to low uncertainty levels stay closer to one another in comparison with those of Fig. 5.11. This is due to the fact that for those conditions in which the wind power penetration level is high ($> 20\%$ approximately) and the three wind power scenarios are close enough (uncertainty level below 50% approximately), not only the *high*

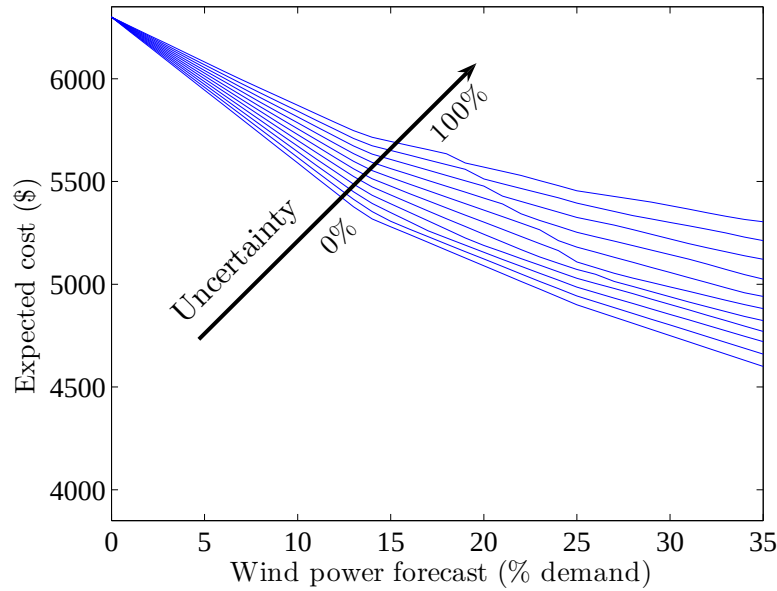


Figure 5.12: Wind unit at node 2. Expected cost as a function of wind penetration and wind uncertainty. Network-constrained case

realization of wind may cause network congestion, but also the lowest ones, and then the differences among the values of the expected cost corresponding to these conditions are reduced.

Finally, in Fig. 5.13, the curves corresponding to a 50% level of uncertainty for the previously analyzed cases are depicted together for comparison. The impact of network congestion on the expected cost is clearly highlighted by comparing curves a) and b).

Curve c) represents a new instance in which the unit ramping capabilities are reduced so that ramping constraints become binding. Observe that the effects caused by binding ramping constraints and network congestion are similar. Inter-temporal limits of generating units (ramping capabilities and minimum up and down times), together with the transmission network capacity, determine how flexible a power system is to accommodate wind power fluctuations. Thus, the more restrictive these limits are, the more wind will be irremediably wasted.

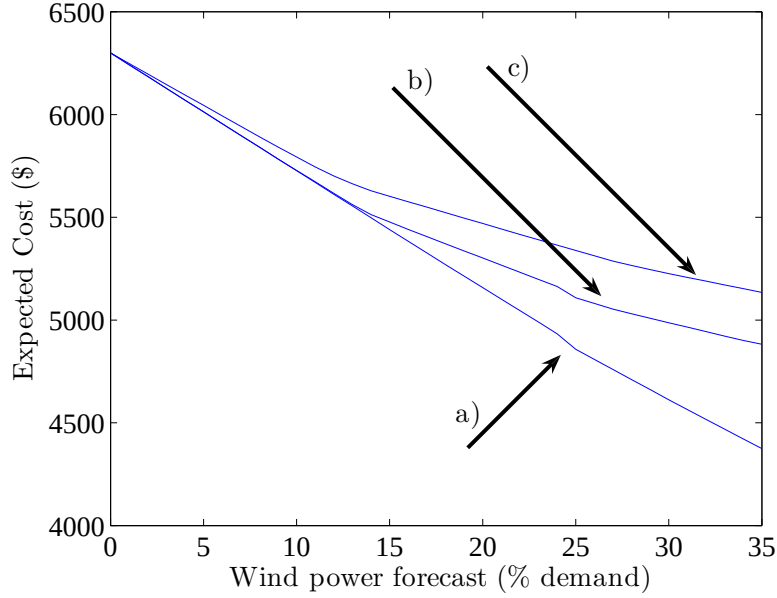


Figure 5.13: a) Base case, b) $P^{F, \max}(2, 3) = 25\text{MW}$, and c) $P^{F, \max}(2, 3) = 25\text{MW}$ & binding ramping constraints

5.7.3 Equipment Failures & Wind Power Production

Next, we examine the combined impact of equipment failures and wind production uncertainty on the energy and reserve dispatch of the three-node system. To this end, we consider the three wind power scenarios listed in Table 5.9 and the nine contingencies enumerated in Section 5.7.1. Thus, each scenario ω in the market-clearing formulation (5.47)–(5.84) is indeed constituted by a pair of a failure event f and a wind production realization ϖ , i.e., $\omega = (f, \varpi)$, with $f = 0, \dots, 8$, $\varpi = 1, \dots, 3$, and $\omega = 1, \dots, N_F \times N_\varpi = 27$, where N_F is the number of failure incidents including the pre-contingency state of the system.

If we consider that the scenario index ω runs as follows,

$$w = f + 1, \quad f = 0, \dots, 8 \text{ and } \varpi = 1 \text{ (forecast wind power scenario)}$$

$$w = f + 10, \quad f = 0, \dots, 8 \text{ and } \varpi = 2 \text{ (high wind power scenario)}$$

$$w = f + 19, \quad f = 0, \dots, 8 \text{ and } \varpi = 3 \text{ (low wind power scenario),}$$

then the scenarios in which no equipment failures occur correspond to $\omega = 1, 10$, and 19 . Being so, the somewhat involved non-anticipativity constraints (5.42) and (5.43) should be formulated as

$$\begin{aligned} P_{t\omega}^{L, \text{shed}} &= P_{t1}^{L, \text{shed}}, \omega = 2, \dots, 9 \\ P_{t\omega}^{L, \text{shed}} &= P_{t10}^{L, \text{shed}}, \omega = 11, \dots, 18 \\ P_{t\omega}^{L, \text{shed}} &= P_{t19}^{L, \text{shed}}, \omega = 20, \dots, 27, \end{aligned}$$

$$\begin{aligned} r_{it\omega}^U &= r_{it1}^U, \forall i, \omega = 2, \dots, 9 \\ r_{it\omega}^U &= r_{it10}^U, \forall i, \omega = 11, \dots, 18 \\ r_{it\omega}^U &= r_{it19}^U, \forall i, \omega = 20, \dots, 27, \end{aligned}$$

$$\begin{aligned} r_{it\omega}^D &= r_{it1}^D, \forall i, \omega = 2, \dots, 9 \\ r_{it\omega}^D &= r_{it10}^D, \forall i, \omega = 11, \dots, 18 \\ r_{it\omega}^D &= r_{it19}^D, \forall i, \omega = 20, \dots, 27, \end{aligned}$$

$$\begin{aligned} r_{it\omega}^{\text{NS}} &= r_{it1}^{\text{NS}}, \forall i, \omega = 2, \dots, 9 \\ r_{it\omega}^{\text{NS}} &= r_{it10}^{\text{NS}}, \forall i, \omega = 11, \dots, 18 \\ r_{it\omega}^{\text{NS}} &= r_{it19}^{\text{NS}}, \forall i, \omega = 20, \dots, 27, \end{aligned}$$

$$\begin{aligned} r_{jt\omega}^U &= r_{jt1}^U, \forall j, \omega = 2, \dots, 9 \\ r_{jt\omega}^U &= r_{jt10}^U, \forall j, \omega = 11, \dots, 18 \\ r_{jt\omega}^U &= r_{jt19}^U, \forall j, \omega = 20, \dots, 27, \end{aligned}$$

$$\begin{aligned} r_{jt\omega}^D &= r_{jt1}^D, \forall j, \omega = 2, \dots, 9 \\ r_{jt\omega}^D &= r_{jt10}^D, \forall j, \omega = 11, \dots, 18 \\ r_{jt\omega}^D &= r_{jt19}^D, \forall j, \omega = 20, \dots, 27, \end{aligned}$$

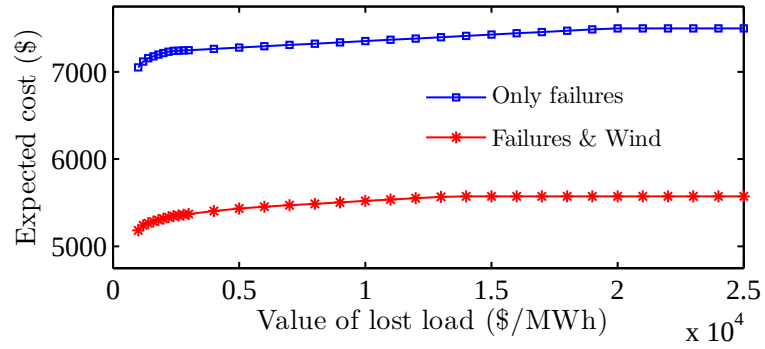
for all $t < \tau$, where τ represents the time period in which the contingency associated with scenario ω takes place.

This family of equations guarantees that no corrective actions other than those aimed at accommodating wind power variability can be executed until the occurrence of a failure event.

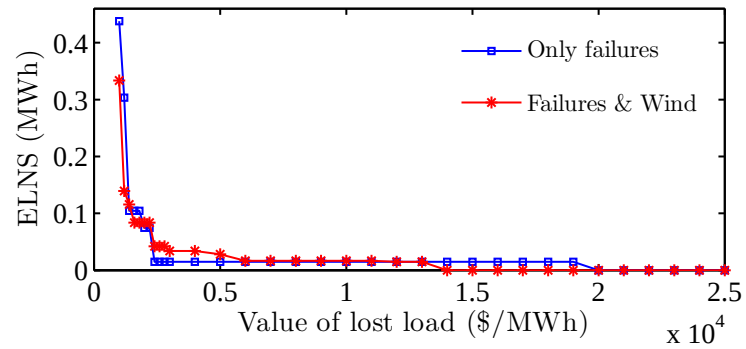
In order to assess the *added value* of wind generation in a power system with components that are susceptible to fail, we contrast the results obtained in this mixed variant of the market-clearing example with those corresponding to the *wind-free* situation previously analyzed in Section 5.7.1. Specifically, Fig. 5.14 offers a comparison of the expected cost of system operation, the expected load not served, and the total amount of reserve scheduled in the market as the value of lost load increases for these two different situations.

The reduction in the total expected cost that is achieved when introducing the wind unit into the three-node system is apparent in Fig. 5.14(a). In numbers, this decrease keeps around 20-25% in all instances. Additionally, integrating wind resources requires the scheduling of a higher amount of reserves, as can be observed in Fig. 5.14(c). This increase in reserve is intended to take full advantage of the cost-free power produced by the wind farm and, needless to say, is conditional on the flexibility and capacity of the power system grid and its users (production units and loads). Further, the reserve that is additionally dispatched can be also deployed in the aftermath of a contingency, and this is the reason why, with wind generation, the system operator leads the expected load not served to zero for values of lost load lower than those required if the wind unit is taken out of the system, as shown in Fig. 5.14(b).

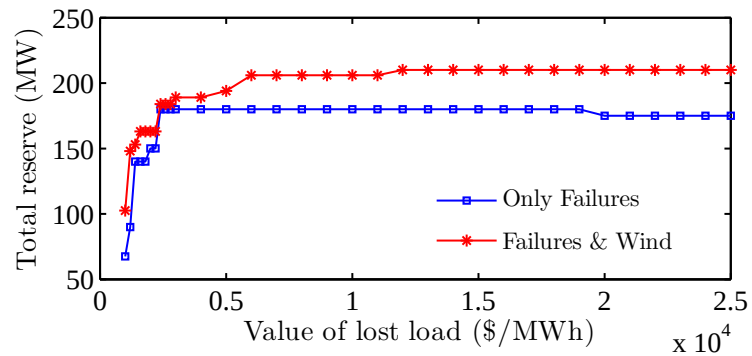
In fact, wind generation can be used to enhance system security in terms of expected load not served, especially if wind farms are placed close to consumption nodes. To illustrate this, we reduce the transmission capacity of line 3 to 35 MW. If so, in the absence of wind generation, both the failure of unit 3 and the outage of line 2 entail load shedding actions regardless of the value of lost load that is considered, as highlighted by the blue line in Fig. 5.15(b). Under these *wind-free* circumstances, load shedding events cannot be completely eliminated because of the existence of network bottlenecks. In contrast, if a wind farm with a power production following the scenarios in



(a)



(b)



(c)

Figure 5.14: Equipment failures and wind production. Impact of the magnitude of the value of lost load on: (a) the expected social cost, (b) the expected amount of load not served (ELNS), and (c) the total amount of scheduled reserve

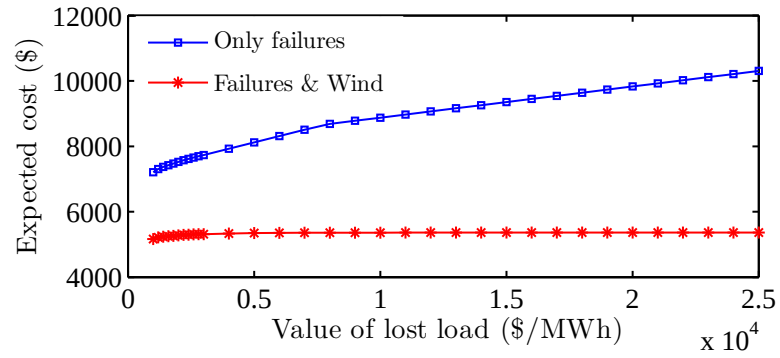
Table 5.9 is located at node 3, the system operator can easily avoid involuntary load curtailments, as evidenced by the red line in Fig. 5.15(b), while significantly reducing again the expected cost of system operation (Fig. 5.15(a)) at the expense of increasing the total amount of scheduled reserve (Fig. 5.15(c)).

Logically, in all fairness, decisions on the siting of a wind farm are generally subject to multiple constraints. Indeed, wind often blows stronger in certain specific regions far away from large consumption areas, but this unfortunate fact does not necessarily imply that wind generation cannot be shrewdly managed to strengthen the reliability and security of power systems. Needless to say, any other generation technology that is additionally connected to node 3 can be also used to reduce the amount of load that is involuntarily shed as a result of failure events, but wind generation may serve for this purpose while simultaneously pushing down the costs of system operation. Moreover, we should point out that the comparison illustrated in Fig. 5.15 has no intention other than showing that wind production, albeit uncertain, does not necessarily undermine the power system security if measured in terms of the expected load not served at least.

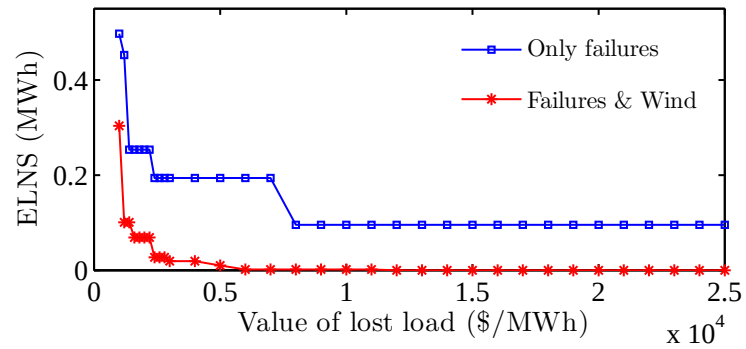
5.8 Market-Clearing Case Studies

In this section, the proposed electricity market-clearing model with stochastic security is further tested over a 24-h scheduling horizon on the 24-bus power system described in Appendix B. This system comprises 17 loads, 12 generating units, and 34 transmission lines.

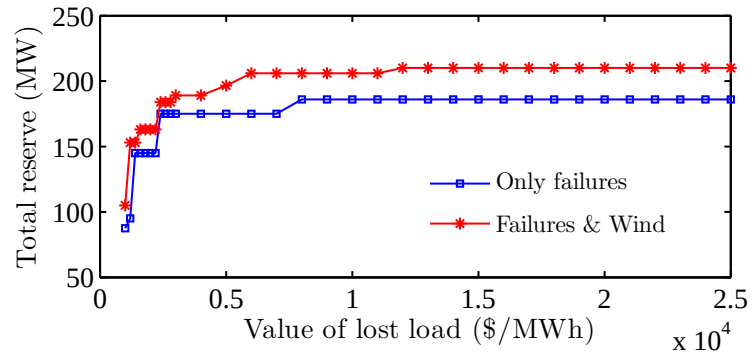
We separately examine the effect of the equipment failures and the wind generation uncertainty on the market-clearing results in two different case studies. This way, the magnitude and nature of the economic impact of these two markedly different sources of uncertainty can be conveniently characterized and compared. Moreover, the proposed market-clearing procedure is computationally very demanding and thus, the joint consideration of failure events and wind power production may easily lead to intractability. Lastly, we insist that the scope of this thesis focuses on the short-term management of wind



(a)



(b)



(c)

Figure 5.15: Equipment failures and wind production. Wind unit at bus 3 and $P^{\text{F}, \max}(2, 3) = 35$ MW. Impact of the magnitude of the value of lost load on: (a) the expected social cost, (b) the expected amount of load not served (ELNS), and (c) the total amount of scheduled reserve

generation and it is in this field where we claim contributions.

5.8.1 Equipment Failure Case Study

This first case study is aimed to appraise the impact of unit failures on both the magnitude and the composition of the expected cost of system operation. To this end, we consider a set of credible contingencies made up of all the single outages of the six generating units located at buses 13, 18 and 21–23. The capacities of these units sum up to 2351 MW, which represents almost 71% of the total capacity installed in the power system (3325 MW). The remaining six units are assumed to be 100% reliable and consequently, they never fail. Likewise, neither sequential, nor compounded failures are taken into account in the analysis. Being so, the number of scenarios to be considered in the market-clearing formulation is 145, which results from the outage of every generating unit susceptible to failure occurring within any time period of the scheduling horizon (6×24), plus the scenario free of contingencies ($6 \times 24 + 1 = 145$).

As stated in Section 5.4.1, the time between to consecutive equipment failures is modeled using an exponential distribution univocally specified by the mean time to failure (MTTF). For simplicity and without loss of generality, in this case study we assume that the six generating units subject to a possible failure have all the same MTTF. Thus, we analyze three cases, namely, A, B, and C, each characterized by a different value of the mean time to failure: 3000, 1000, and 200 hours, respectively, as indicated in Table 5.12. Note that, in relative terms, case A corresponds to a robust power system, while case C yields an electric system with a notably reduced level of reliability. Case B constitutes a situation halfway between cases A and C. The probability π_0 of the pre-contingency scenario, in which no failure event happens during the entire scheduling horizon, is indeed equal to 0.95, 0.87, and 0.55 for cases A, B, and C, respectively.

In order to make the proposed market-clearing problem tractable and solvable in a reasonable time using an ordinary computer, non-spinning reserves are not modeled. The mathematical treatment of such a type of reserve within a mixed-integer programming formulation requires binary variables $v_{it\omega}$ in prob-

Table 5.12: Mean time to failure (MTTF) of generating units

Case	MTTF (h)
A	3000
B	1000
C	200

lem (5.47)–(5.84) to model the commitment status of every generating unit in each period and scenario. Therefore, the number of these variables drastically increases with the number of scenarios considered. Hence, if non-spinning reserves are excluded from the market-clearing problem, the on/off state of every generating unit different from the one failing remains equal to the commitment schedule settled in the electricity market throughout the whole scheduling horizon. Logically, in the aftermath of a failure event, the binary variable representing the on/off status of the affected generating unit becomes necessarily zero. Disruptions caused by unit outages are then handled by means of spinning reserves and/or load shedding. From a programming viewpoint, non-spinning reserves are ruled out of the market-clearing problem by enforcing that $v_{it\omega} = u_{it}$, for all i, t and ω , with the consequent reduction in binary variables. However, in spite of this simplifying measure, the resulting market clearing is still hard to solve and includes 1 471 318 constraints and 1 004 197 variables, out of which 216 are binary. Note that, in terms of notation, the subscript ω here can be replaced by the subscript f used to index failure scenarios given that wind production is not included in this case study.

Table 5.13 shows the breakdown of the expected cost for the three previously referred cases, A, B, and C. The numbers in parentheses represent the proportion in percentages of each expected cost component with respect to the total. A value of lost load equal to \$2000/MWh is considered. By simple inspection of this table, we come to the conclusion that the reliability level of a power system has a relevant impact on how costly its operation is. Just observe the rise experienced by the expected total cost as the system becomes less reliable (from case A to case C). Particularly, note the remarkable increase

Table 5.13: Breakdown of the expected social cost in \$ (%) for $V^{\text{LOL}} = \$2000/\text{MWh}$

Case	Total	Start-up	System operation	Reserve scheduling	Energy not served
A	470 094 (100)	5521 (1)	420 328 (89)	38 675 (8)	5570 (1)
B	479 744 (100)	5503 (1)	423 348 (88)	42 523 (9)	8370 (2)
C	511 423 (100)	5436 (1)	434 734 (85)	42 719 (8)	28 534 (6)

of the cost associated with the expected load not served as the system reliability worsens. In an attempt to curb likely load shedding events, the amount of reserve scheduled in the electricity market increases and so does its associated cost, at least in absolute terms. The cost of system operation (that related to the energy production and consumption) is impacted by the system reliability as well, albeit to a lesser extent relatively speaking. Its rise has to do with the fact that more expensive, but also more flexible generating units (faster and with wider ramp limits) are committed to the energy dispatch program resulting from the market clearing so as to enhance the availability of spinning reserves.

As seen in the example of Section 5.7.1, a straightforward manner to *force* the scheduling of additional reserve consists in increasing the value of lost load. In this respect, Table 5.14 displays both the expected load not served and the amount of programmed reserve for different values of this parameter, specifically 2000, 4000 and \$10 000/MWh. Observe that, in cases A and B, as the value of lost load becomes higher, the megawatts of demand that are expected to be involuntarily shed decrease as a result of planning more reserve. This statement does not hold, however, for case C, in which both the amount of expected load not served and the quantity of scheduled reserve remain unchanged and equal to 14.27 and 9450 MW, respectively. The technical limitations and physical restrictions of the power system and its agents (generating units and loads) impose an upper bound on the amount of reserve that can be ensured. This bound is precisely 9450 MW. Besides, with this quantity of reserve, the

Table 5.14: Expected load not served (ELNS) and amount of scheduled reserve for different values of lost load (V^{LOL})

(a) ELNS (MW)				(b) Total reserve (MW)			
Case	V^{LOL} (\$/MWh)			Case	V^{LOL} (\$/MWh)		
	2000	4000	10 000		2000	4000	10 000
A	2.79	1.53	1.47	A	8913	9360	9450
B	4.19	4.10	4.10	B	9400	9450	9450
C	14.27	14.27	14.27	C	9450	9450	9450

expected load not served throughout the scheduling horizon can be reduced, in case C, up to 14.27 MW at the most.

Finally, the costs of energy and security per megawatt and hour are shown in Table 5.15 for different values of lost load and cases A, B, and C. The cost of energy is defined as the sum of the start-up and system operation costs. In order to express this cost in per megawatt and hour, the above sum is divided by the time length of the scheduling horizon (24 h) and the total capacity installed in the power system (3325 MW). Similarly, the cost of security is computed as the sum of the cost derived from load shedding events and the cost pertaining to the scheduling of reserves. Since the need for reserves and the occurrence of load shedding incidents are directly linked to unit failures, the result of this sum is divided by the time span of the scheduling horizon (24 h) and the aggregated capacity of all the units that are susceptible to a failure (2351 MW). The cost of energy represents the cost stemming from the energy supply, while the cost of security expresses the cost of balancing supply and demand at all buses, in all time periods and for all scenarios. Observe in Table 5.15 that the cost of energy per megawatt and hour is not affected, at least appreciably, by the value of lost load or the reliability level of the power system. On the contrary, the cost of security is considerably influenced by both factors. In this sense, as the value of lost load increases and/or the reliability of system components deteriorates, the cost of security rises. This is a direct consequence of the scheduling of additional reserves and/or the upsurge of load

Table 5.15: Costs of energy and security in \$ per MWh

(a) Energy				(b) Security			
Case	V^{LOL} (\$/MWh)			Case	V^{LOL} (\$/MWh)		
	2000	4000	10 000		2000	4000	10 000
A	5.34	5.34	5.34	A	0.78	0.86	1.02
B	5.37	5.37	5.37	B	0.90	1.05	1.48
C	5.52	5.52	5.52	C	1.26	1.77	3.29

shedding phenomena and their associated cost.

On the other hand, note also that the cost of security keeps well below the cost of energy in all cases. In the following case study, we witness how the marked uncertain nature of wind power production pushes the cost of system security up, much more than unit failures, at the same time that its zero energy production cost leads to a significant reduction in the cost of energy.

Lastly, the CPU time required to solve the market-clearing procedure with equipment failures keep below 40 minutes in all instances using CPLEX 11.2.1 under GAMS 23.0 [14] on a Linux-based server with two processors Quad Core AMD Opteron 2356 clocking at 2.3 GHz and 8 GB of RAM.

5.8.2 Wind Production Case Study

The objective of this second case study is threefold:

1. To provide additional insight into the impact of wind generation on the cost of scheduling reserves by using a more realistic data set. For computational simplicity and convenience, we ignore unit failures.
2. To identify non-spinning reserves and inter-temporal constraints as the main origin of the high computational burden associated with stochastic security-constrained market-clearing models. In this respect, we should point out here that, unless stated otherwise in the text, all the market-clearing problems solved in this case study include non-spinning reserves

and inter-temporal constraints (ramping limits and minimum up and down times of generating units).

3. To list and evaluate some straightforward strategies aimed at alleviating such a computational burden.

As explained in Section 5.4.2, the accurate modeling of the wind power stochastic behavior usually requires generating a sufficient number of scenarios representing the most plausible realizations of wind power throughout the scheduling horizon. For this purpose, we can use one of the scenario generation techniques described in Chapter 3. However, there also exists a simple alternative to these more elaborate methodologies: we can just build the scenario set directly from historical records, that is, we can just employ *historical scenarios*. Needless to say, the use of this alternative is contingent on the availability of a good and extensive data collection.

In this case study, 3018 equiprobable wind speed scenarios are constructed using the wind speed data publicly available in [1] and collected from a location in the state of Kansas (US) by Dr. Gary Johnson. This wind data set comprises average hourly speed measurements between May 1995 and February 2005. We consider a wind farm having 100 commercial wind generators, all of which correspond to the same type of turbine model, model A or model B, with nameplate capacities of 1.5 and 2.5 MW, respectively. Therefore, the total installed wind power capacity of the farm is 150 or 250 MW depending on which of the two turbine models is used.

As stated in Section 3.3.5, to transform those 3018 wind speed scenarios into wind power scenarios, we just aggregate the power output of each individual wind turbine comprising the wind farm, which, in turn, is estimated from its characteristic power curve. Fig. 5.16 depicts the power curves of turbine models A and B, which express wind turbine power output as a function of the inputting wind speed.

The size of the scenario set so obtained (3018 scenarios) is too large, resulting in an optimization model that is intractable (with 290 000 binary variables approximately). Hence, to achieve tractability, statistical techniques, as those described in Chapter 3, are applied in order to reduce the number of scenarios

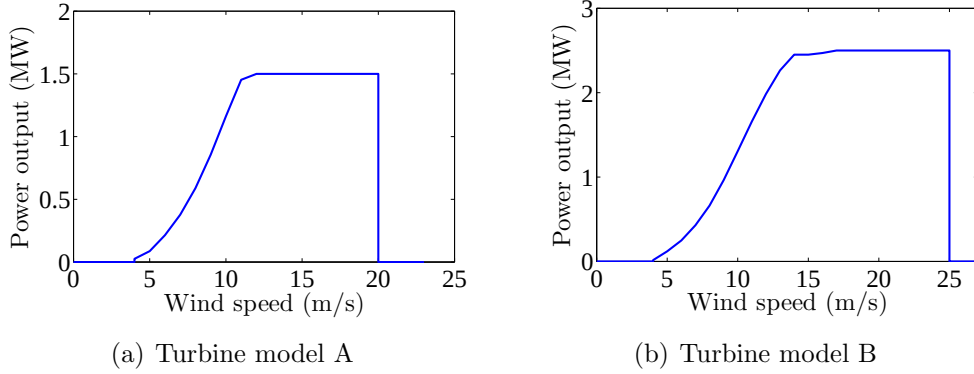


Figure 5.16: Turbine speed-power curves

while retaining the essential features of the original scenario set. The reduced scenario set resulting from this process includes 20 scenarios.

Table 5.16 provides a breakdown of the expected cost of the system into the cost pertaining to the energy supply and the cost of security for two different locations of the wind plant (WP), node 7 and node 20, and two different installed wind power capacities (150 and 250 MW). It also provides the costs of energy and security per installed MW and hour (as defined in the previous case study). Given that the expected amount of load not served is zero for all the variants examined in this case study, the cost of security becomes equivalent to the cost associated with the scheduling of reserves. Wind potential is assumed to be the same at nodes 7 and 20, that is, the same wind power scenarios can be used to model the stochastic behavior of wind generation at both sites. Since the wind generation is assumed to be free, the cost of the energy is obtained by dividing the total cost of the energy by the length of the scheduling horizon (24 h) and the total power capacity of the electric system excluding the wind power capacity (3325 MW). On the contrary, as wind generation is responsible for the need of reserves, the cost of security is obtained by dividing the total cost of the scheduled reserve by the installed wind power capacity in the system (150 or 250 MW) and the time span of the scheduling horizon (24 h). Note that the cost of security is comparatively relevant. In this sense, it should be taken into account that, on the one hand, wind constitutes an important

Table 5.16: Costs of energy and security

WP at bus #	Installed wind capacity (MW)	Total Cost (\$)		Cost (\$) per MWh	
		Energy	Security	Energy	Security
7	150	384 233	13 107	4.81	3.64
7	250	376 701	16 897	4.72	2.82
20	150	384 245	13 093	4.82	3.64
20	250	376 644	16 835	4.72	2.81

source of cheap renewable energy but, on the other hand, reserves are required in order to accommodate its unpredictable variability and thus, to maintain system security and reliability. Therefore, the integration of wind generation into a power system reduces the cost of energy at the expense of increasing the cost of security, i.e., the cost of being able to balance supply and demand at all buses, at all times, and for all conditions.

In this same vein, we can compare the results provided in Table 5.16 with those presented in Table 5.15 pertaining to the effect of unit failures on the costs of energy and security. From the comparison, it can be concluded that wind production makes the maintenance of system security economically more costly than equipment failures. In general, one megawatt of wind generation needs to be backed up with a greater amount of reserves than one megawatt produced by a conventional unit subject to a possible failure. In contrast, a megawatt obtained from the force of wind is free, while a megawatt resulting from the combustion of fossil fuels is not. Therefore, wind generation contributes to reduce the cost of energy.

Regarding computational issues, the mathematical programming problem associated with the proposed market-clearing model is large-scale, as it includes 228 562 constraints, 153 409 continuous variables, and 4536 binary variables. The model is hard to solve and the computational burden is mostly due to the binary variables required to determine the on/off status of generating units. Note that in order to model the deployment of non-spinning reserves, one binary variable, v_{itw} , is required per unit, period and scenario. Hence, it seems

Table 5.17: Valuation of non-spinning reserves (NSR). Expected costs

WP at bus #	Installed wind capacity (MW)	Expected cost (\$)		Cost increase (%)
		With NSR	Without NSR	
7	150	397 340	397 610	0.07
7	250	393 598	394 426	0.21
20	150	397 338	397 607	0.07
20	250	393 479	394 217	0.19

Table 5.18: Valuation of non-spinning reserves (NSR). CPU times

WP at bus #	Installed wind capacity (MW)	CPU time (s)		Time reduction (%)
		With NSR	Without NSR	
7	150	1555	21	98.6
7	250	23 448	21	99.9
20	150	1536	21	98.6
20	250	18 355	22	99.9

reasonable to question whether it is worthwhile to include non-spinning reserves in the model. Tables 5.17 and 5.18 show, respectively, a comparison between the expected costs and CPU times with and without non-spinning reserve. Computation times have been obtained using CPLEX 11.2.1 under GAMS 23.0 [14] on the same machine used in the previous case study.

In all cases, excluding non-spinning reserve implies an increase of the expected cost lower than 0.21% with respect to the solution including non-spinning reserve. However, the decrease in computation time is dramatic (above 98.6%), which is mostly due to the fact that the number of binary variables without non-spinning reserve is 216 vs 4536 with non-spinning reserve.

The inter-temporal constraints, which include the ramping limits, the minimum up and down times, and the start-up cost modeling of generating units,

also make the solution to the proposed market-clearing procedure much more involved. These constraints affect and couple decision variables in different time periods and thus, if excluded from the market-clearing formulation, the reserve and energy dispatch covering the whole scheduling horizon can be performed hour by hour. That is, if inter-temporal constraints are disregarded, the stochastic security-constrained market-clearing formulation presented in this chapter can be decomposed into N_T single-period market-clearing problems, where N_T is the number of time periods spanning the scheduling horizon. Needless to say, such a decomposition drastically reduces the computational burden of the market-clearing process at the expense of increasing the system operating costs. As an example, the calculation time keeps below 30 seconds in all instances listed in Table 5.16, provided that inter-temporal constraints are removed from the market-clearing formulation. In practice, these constraints can be enforced *ex post* by using, for example, some *ad hoc* rules as in the case of the Electricity Market of the Iberian Peninsula [3] or by setting the outcomes of each single-period market clearing as the initial conditions for the subsequent one.

Moreover, all the results presented in Table 5.17 pertain to optimal solutions, i.e., to solutions that are obtained considering a *mip gap* of 0%. The *mip gap* is defined for mixed-integer programming (mip) minimization problems as the difference in percent between the objective function value associated with the best integer solution found and the objective function lower bound. Hence, an easy manner of mitigating the computational demand of the proposed market-clearing model is to work with a *mip gap* greater than zero at the risk of getting a suboptimal solution. Table 5.19 shows the expected cost increase stemming from the consideration of *mip gaps* equal to 1 and 0.5%. Likewise, Table 5.20 includes the corresponding reduction in calculation time. Percentages are given with respect to the market-clearing solution that includes non-spinning reserves in Tables 5.17 and 5.18. Observe that a very remarkable decrease in *cpu time* can be achieved at the expense of a small increase in the expected cost.

These tables also highlight the computational advantages obtained from the application of a *warm-start strategy*. This strategy consists in using the

Table 5.19: Expected cost increase (%) resulting from the application of several strategies to relieve the computational burden

WP at bus #	Installed wind capacity (MW)	Mip gap (%)	
		1	0.5
7	150	0.54	0.19
7	250	0.66	0.13
20	150	0.64	0.20
20	250	0.47	0.09

Table 5.20: Time reduction (%) achieved through the application of several strategies to relieve the computational burden

WP at bus #	Installed wind capacity (MW)	Mip gap (%)		Warm start (%)
		1	0.5	
7	150	81.2	75.2	24.7
7	250	94.6	91.1	81.3
20	150	85.9	76.0	29.2
20	250	90.9	88.4	57.7

market-clearing solution without non-spinning reserve as the initial first integer feasible solution of the mip algorithm to solve the market-clearing model with non-spinning reserves. Logically, this tactic does not entail any increase in the expected cost, as it just affects the search, but not the final solution. In contrast, it brings about a substantial reduction in computational time (see Table 5.20), because the search for the optimal solution is usually shortened. Needless to say, the warm-start strategy can be combined with a positive mip gap.

Finally, we conclude this section presenting several guidelines to further relieve the computational burden of the market-clearing problem with stochastic security. These guidelines have to do with modeling simplifications and are intended to reduce the number of binary variables v_{itw} used to model non-

spinning reserves. These simplifications are the following.

1. To limit the number of generating units that provide non-spinning reserve. This is a realistic assumption since not all the generating units meet the required conditions to offer this ancillary service.
2. To bundle the hourly wind power production in intervals greater than 1 hour. This way of proceeding may be appropriate and necessary in long- and medium-term analysis, in which, as indicated in Section 5.2.5, the proposed model can be used as a planning tool for reserve valuation based on a market-clearing procedure.
3. To reduce the number of scenarios representing the stochastic behavior of wind power. This reduction has been applied in this case study as a mandatory measure to make the problem tractable, but it can also be employed to enhance computational efficiency thanks to the rapid development of accurate and efficacious scenario reduction techniques (see Chapter 3).

Nowadays, all the simplifying measures discussed in this case study can potentially make the implementation of stochastic security procedures feasible in real-world applications until the progress in computing technologies enables us to solve the fully-fledged market-clearing model described in this chapter within the required time limits.

5.9 Summary

The time gap existing between market agreements on electricity transactions and their realization during the real-time operation of the power system leads to uncertainties as for the availability of system resources at the operation time. Reserves are the ancillary services intended to accommodate these uncertainties, enabling the actual implementation of the market outcomes in real time while maintaining the normal functioning of the physical system.

In this chapter, a stochastic security-constrained market-clearing procedure is presented as a promising tool to resolve the tradeoff between economic ef-

efficiency and system security when dealing with equipment failures and wind power fluctuations. According to the rationale behind the proposed procedure, energy and reserve are simultaneously traded within a common optimization framework. The safety level linked to the resulting scheduling programs is appraised through a security measure, the *expected load not served*, that accounts for the probability of occurrence and magnitude of disturbances.

Mathematically, this market-clearing procedure is formulated as a large-scale mixed-integer linear programming problem that is computationally very demanding in general. Consequently, some simplifications may have to be implemented in order to achieve the optimal or an approximate solution in a reasonable time for realistic applications using commercially available software.

The most relevant aspects of the proposed market-clearing formulation are pointed up using several small examples and larger-scale case studies based on a 3-node and a 24-node electric systems, respectively.

Aritmética
(Vida *or* the Muse)

Nada.

.
.
.

Nace de la soledad.

Asoma tras el velo de una media sonrisa.
Se hace a golpe de yugo y tesón,
con grandes dosis de frustración.

Al morir, queda un vacío inmenso que
sólo el advenimiento de una nueva búsqueda,
se antoja capaz de llenar.

El proceso,
discurre sin más sentido que el sentido en sí mismo.
El hallazgo,
lejos de tornarse recompensa,
se convierte en la antesala de una nueva meta.

Es como,
el pan que sacia al hambriento,
el rumor que mueve al curioso,
el valor que ha perdido el pusilánime
o la inspiración que se rinde al poeta.
Consagrado al descubrimiento,
avocado a las sombras trémulas de las vanguardias,
sin juicio, cuándo loco, estúpido o iluminado,
vago inmaterial y perdido por este mundo difuminado.

Copenhague, 2 de septiembre de 2010

A stylized, handwritten signature in black ink, consisting of several loops and a long horizontal stroke at the end.

Chapter 6

Closure

In this closing chapter, the thesis is summarized, conclusions are drawn and proposals for future work are provided.

6.1 Summary and Conclusions

Wind power is shaping up as the safest bet of the international community to curb greenhouse gas emissions from the electricity sector, while guaranteeing the energy supply in the coming years. The integration of large amounts of wind generation into current electric energy systems is setting and is expected to set new challenges for power system operators and planners.

The features of wind power that are to alter the functioning of today's power systems on a technical level are:

1. Wind power production is highly variable. Consequently, flexibility on both the generation and demand sides needs to be enhanced to cope with wind power fluctuations in an efficient manner.
2. Wind generation is contingent on weather conditions and as such, is highly uncertain. Reserves need to be carefully scheduled and allocated to accommodate unpredictable wind power changes.
3. Wind power integration increases the number of dispersed power sources (wind farms) in the network, thus extending the range of possible power-

flow configurations. Although this makes the performance and reliability problems much more challenging, it also provides opportunities for designing grids that are more versatile than traditional networks.

4. Wind farms are usually located in areas where wind resources are abundant. In turn, these locations are often far away from large energy pockets. Therefore, transmission issues prove to be of special relevance in the large-scale integration of wind power. New methods for making the best use of wind generation in existing networks need to be developed, while promoting investment for transmission expansion.

Likewise, wind generation is expected to have a significant impact on the economic front of power systems, namely:

1. Wind power volatility is passed on to electricity prices, thus increasing the financial risk faced by conventional producers, consumers and retailers.
2. The free-cost nature of wind energy pushes down the average value of the electricity price, particularly if wind producers receive extra-payments outside the market (subsidies). It is fair to wonder if electricity markets will provide the proper economic incentives for producers to invest in the flexible generation technologies required to continue integrating larger amounts of wind.
3. The organization of electricity markets should include efficient mechanisms to provide and price balancing energy under competition. In this sense, more emphasis should be placed on reserve and real-time energy markets.

In this dissertation short-term decision-making tools for wind producers and market operators are developed, illustrated and analyzed.

When it comes to participate in electricity markets, a wind producer seeks to maximize the profit obtained from selling its uncertain production. To this end, its market strategy should account for the characteristics of the trading floors available within the market organization. In this thesis, we consider three representative trading arenas for short-term electricity transactions, namely:

The day-ahead market, which is cleared one day in advance of actual energy delivery. The bulk of the energy to be delivered the next day is usually negotiated in this market, which is often seen as a reference in consequence.

The adjustment market, which is cleared some hours after the day-ahead market and thus, closer to actual energy delivery. As a result, wind producers can participate in this market with a reduced level of uncertainty about its final production (certainty gain effect). However, the trading volume in the adjustment market is generally much lower than in the day-ahead market. Consequently, the clearing price of the adjustment market tends to be more volatile than the electricity price in the day-ahead market.

The balancing market, in which the real-time energy deviations incurred by wind producers with respect to the scheduling program settled in the aforementioned markets are covered by dispatchable power sources. By definition, the prices at which a wind power producer can sell its overproduction or purchase its generation shortage in the balancing market are less favorable than the day-ahead market price. Consequently, a wind producer that needs balancing energy to fulfill obligations contracted in previous markets incurs an opportunity cost known as *imbalance cost*.

We propose a three-stage stochastic programming model to build optimal offering curves for the day-ahead market taking into account the availability of an adjustment market and the cost derived from the balancing process. This way, the offering model allows the wind producer to resolve the tradeoff among selling energy in the day-ahead market with lower price volatility, but higher production uncertainty; trading in the adjustment market with higher price volatility, but lower production uncertainty; and selling in the balancing market at less competitive prices on average. The Conditional Value-at-Risk is used to control the risk associated with profit variability. After simple manipulations, the offering model can be formulated as a linear programming problem easy to solve.

Numerical results indicate that:

1. The short-term trading problem of a wind power producer is essentially stochastic and as such should be treated. This is inferred from the positive character of the Value of the Stochastic Solution linked to the solution provided by the risk-neutral variant of the proposed offering models and from the results in terms of actual profit yielded by the out-of-sample assessment of such models.
2. A multi-stage stochastic modeling of the wind producer problem constitutes a natural way to reproduce the sequential order in which decisions are made and uncertain information is revealed. The number of stages in the wind producer problem corresponds to the number of short-term trading arenas comprising the market organization.
3. Risk management in the short term generally makes sense for the wind producer problem because:
 - (a) A high decrease in the risk of profit variability is achieved for a comparatively small reduction in expected profit.
 - (b) No clear financial instruments are currently available for wind producers to hedge against the risk associated with production uncertainty in the long run.
 - (c) Long-term forecasting of wind power production is very difficult.
4. Energy offers for the day-ahead market anticipating the improvement in wind power forecasts as the adjustment market-clearing time approaches (certainty gain effect) enable the wind power producer to increase its expected profit while reducing the level of risk involved.
5. Adjustment markets can be seen as structural elements within electricity markets to increase the competitiveness degree of wind power producers.
6. The advantage of using a stochastic model instead of a deterministic one to devise offering strategies becomes more apparent as the description of the stochastic phenomena (wind production and market prices) becomes increasingly accurate.

Wind power uncertainty affects not only energy trading decisions by wind producers, but also the clearing of the electricity market by the market operator. In this thesis, we design a market-clearing tool for the simultaneous dispatch of energy and reserve that allows accommodating wind power variability and equipment failures in a reliable and economical fashion. Reserve requirements are driven by the tradeoff between the cost of the expected load not served and the energy and reserve offers. The procedure is based on a two-stage stochastic programming model with recourse in which the first-stage represents the functioning of the electricity market, its constraints and rules, and the second stage represents the power system, its real-time operation and physical limitations imposed by the network and its users. Both stages are strongly coupled by the *linking constraints*, which restrict the amount of reserves that are deployed (recourse variables) after power system uncertainties (wind power production and equipment failures) disclose. Mathematically, the market-clearing tool is formulated as a mixed-integer optimization problem, whose level of detail makes it hard to solve. Therefore, modeling simplifications may have to be implemented for real-world applications.

Examples and realistic case studies are used to illustrate and test the proposed tool, concluding that:

1. Market-clearing procedures able to cope with power system uncertainties are imperative in order to resolve the tradeoff between system security and economic efficiency.
2. Network conditions play a fundamental role in the design of the preventive and corrective actions aimed to manage wind variability and possible equipment failures.
3. The value of lost load has a great impact on the dispatch program resulting from the market clearing, especially on the reserve schedule. In fact, the value assigned to the involuntary load shedding can be seen as a parameter useful to strengthen system security by programming more reserves.
4. Comparatively, the cost related to the energy supply is barely affected

by the reliability level of system components or the value of lost load. In contrast, the cost of security, which includes the cost of the reserve scheduling and the cost associated with the expected load not served, is strongly dependent on both factors.

5. Wind generation decreases expected operation costs, but increases the costs pertaining to scheduling of reserves.
6. The reserve cost due to wind power uncertainty is relevant in comparison with the costs related to energy production.
7. Network congestion may seriously hinder the cost reduction achievable by wind generation.
8. The consideration of a wind spillage cost may have a significant impact on scheduled reserves and on generation/demand scheduling.

The decision-making tools developed for wind producers and market operators need to be fed with appropriate scenario sets modeling the uncertainties involved. In this thesis, scenario generation methods based on ARIMA models are used to produce wind power and market price scenarios. In particular, we propose a novel scenario generation technique for quasi-contemporaneous processes that enable us to replicate the most salient features of wind speeds at multiple sites, namely:

1. The marginal distribution of the wind speed at each location by *normalization* of the historical wind speed data.
2. The dynamic behavior of wind speed series over time by appropriate univariate ARMA fits.
3. The spatial interrelations among wind speeds at the multiple sites. To this end, we make use of an orthogonal transformation that allows reproducing the cross-correlograms of the historical residuals that result from the univariate ARMA fits.

The ability of the proposed technique to capture these characteristics is highlighted through realistic case studies.

Lastly, efficient techniques are used to reduce the cardinality of the scenario sets provided by the ARIMA-based scenario generation methods while preserving as intact as possible the statistical information embedded in them.

6.2 Recommendations for Future Work

Based on the elaborations in this thesis, continuing research in various fields is conceivable and to some extent also required for a successful practical application of the presented methods.

1. The performance of the proposed decision-making tools are contingent on the uncertainty modeling, area in which there is significant room for improvement. Specifically:
 - (a) The procedure for generating wind power scenarios can be refined by incorporating information from numerical weather predictions. Even though local winds exhibit a strong autoregressive behavior in the short run, thus motivating the use of ARMA models to describe their dynamic, physical models are often required to track major changes in wind speed regime for horizons greater than a couple of hours [146].
 - (b) Likewise, the proper transformation of wind speed into wind power, which in this thesis is simplified by using the manufacturer's power curve of the installed wind turbine model, is another issue that lends itself to further investigation.
 - (c) As to market prices, the modeling of energy prices in the balancing or real-time market may be much improved by incorporating explanatory variables such as demand forecasts or the generation capacity excess or shortfall [224]. These exogenous variables could provide valuable information on the sign of the net system imbalance.

2. The offering model developed in this thesis for trading wind energy in short-term electricity markets is built on the assumption that wind producers have no capability to alter market prices. This hypothesis holds for the day-ahead market, but may be unrealistic for both the adjustment and balancing markets, where the energy volume negotiated is comparatively much smaller. Therefore, it would be interesting to model the wind producer as a price maker in these two markets within its short-term trading problem like other authors have already done for thermal producers [180].
3. Nowadays, there is no clear incentive for wind producers to engage in long-term firm energy contracts due to its production uncertainty. A line of research in this respect consists in analyzing the profitability of composing portfolios based on renewable sources with complementary production availability [206]. The offering model proposed in this dissertation could be extended then to include the possibility of trading in markets for long-term energy transactions (e.g., futures market) via these portfolios.
4. Most of the improvements that can be implemented in the market-clearing tool proposed in this thesis have to do with simplifications or algorithms intended to speed up its solution. In this vein, if the non-anticipatory character of the energy and reserve scheduling (here-and-now decisions) is relaxed, the resulting problem can be easily solved as it decomposes by scenario. Therefore, decomposition techniques [55, 197, 199] may be particularly suited to tackle the proposed market-clearing problem.
5. The market-clearing tool presented in this dissertation needs to be complemented with a fair pricing mechanism for energy and reserve that allows producers to recover generation costs from consumers while guaranteeing revenue reconciliation. How to price commodities in electricity markets is a field of active research. In the context of stochastic security under wind power uncertainty, reserve pricing is a topic of special

importance.

6. The proposed market-clearing tool can be used to perform some interesting analyses such as:
 - (a) How demand flexibility can contribute to make the maximum use of wind generation.
 - (b) How wind power uncertainty impacts the operation and costs of conventional generation.

To conclude, we would like to point out that, apart from the topics for future research outlined in this section, which are directly derived from the problems dealt with in this thesis, wind integration is expected to pose unique challenges in the planning and operation of today's electric energy systems. This just means tons of entertainment for the current and next generation of researchers in power systems.

Appendix A

Introduction to Stochastic Programming

In the main body of this dissertation, stochastic programming is referred to several times as the framework upon which the proposed mathematical models are built. This appendix introduces the basic concepts required to formulate and evaluate decision-making problems using stochastic programming.

A.1 Why stochastic programming?

Without loss of generality, let us consider the following linear optimization problem:

$$\underset{x}{\text{Maximize}} \quad c^\top x \tag{A.1}$$

$$\text{subject to} \quad Ax \leq b \tag{A.2}$$

$$x \geq 0, \tag{A.3}$$

where x is the decision variable vector, and matrix A and vectors b and c are input data. If parameters A , b and c are perfectly determined, solution algorithm for linear optimization problems (e.g., the well-known simplex method) can be used to find the *best* value of the decision variable vector x .

Let us suppose, however, that some of these input data are contingent on

the realization λ_ω of a certain random parameter λ . Mathematically, we can write $A_\omega = A(\lambda_\omega)$, $b_\omega = b(\lambda_\omega)$, and $c_\omega = c(\lambda_\omega)$.

If the decision-making process is such that the decision variable vector x needs to be determined before the realization λ_ω of random parameter λ , problem (A.1)–(A.3) becomes a *stochastic linear program*. Prior to undertaking the solution of a problem like this, we need to be aware of the implications of uncertain data in the following three aspects:

Feasibility. In principle, constraint (A.2) should be satisfied for any possible realization of random parameter λ . Therefore, the issue of how to guarantee the feasibility of decision vector x becomes considerably more involved in stochastic optimization.

Optimality. As cost vector c is assumed to be uncertain, objective function (A.1) is not a real-valued function anymore, but a family of random variables $f(x, \omega) = c_\omega^\top x$. That is, for each feasible decision vector x , $f(x, \omega)$ can be viewed as the value of random variable $f(x, \cdot)$ at the argument ω . Therefore, optimizing in a *stochastic* sense translates into finding the decision vector x that yields the *best* random variable $f(x, \omega)$. For this purpose, we need to specify a criterion to discriminate the goodness of each member of the family. It is a common practice to rank random variables $f(x, \omega)$ according to their expectations and pick the biggest (in a maximization problem). Needless to say, other modes of ranking are possible, as we will see later on in Section A.4, where we introduce the notion of *risk*.

Solution algorithm. Optimization problem (A.1)–(A.3) needs to be recast so that solution algorithms for linear programming problems (e.g., the simplex method) can be used to obtain the optimal value of decision vector x in the terms previously discussed. If, for example, random variable λ affecting problem (A.1)–(A.3) is continuous, a discrete approximation of this variable in the form of a set Ω of plausible outcomes or scenarios is required in most of the cases to make such a problem solvable. Each outcome or scenario $\omega \in \Omega$ has associated a probability of occurrence π_ω such that $\sum_{\omega \in \Omega} \pi_\omega = 1$.

Stochastic programming provides us with the concepts and tools required to deal with the implications of having uncertain data in an optimization problem for decision making. In the next section, we focus on the group of stochastic programming problems with recourse, which exhibit a particular structure and encompass the mathematical models described in this dissertation.

A.2 Stochastic Programming Problems with Recourse

Recourse problems are defined as stochastic programs in which some decisions, referred to as *recourse actions*, can be made after uncertainty is revealed [35]. Recourse problems are classified by their number of *stages*. Each stage represents a point in time where decisions are made. The number of stages of a recourse problem is dependent on how decisions are sequentially made in relation to how the uncertain input information is disclosed over time. In this line, all decisions that are made based on identical information on uncertain parameters are grouped in the same stage [90].

The simplest recourse problem is the two-stage stochastic programming problem, in which decisions are divided into two groups, namely:

- Decisions that have to be made before the realization of uncertain parameters. These decisions are known as *first-stage* or *here-and-now* decisions and do not depend on each possible realization of the random parameters.
- Decisions that are made after the actual values of uncertain parameters are disclosed. These decisions are called *second-stage*, *wait-and-see*, or *recourse* decisions and are dependent on each plausible value of the random parameters. The term *recourse* refers to the fact that second-stage decisions enable the decision maker to adapt to the actual outcomes of the random events.

Schematically, the sequence of decisions and events in a two-stage stochastic programming problem can be expressed as follows:

1. First-stage decisions, x , are made.

2. The actual outcome λ_ω of random parameter λ is realized, and
3. second-stage decisions, $y(x, \omega)$, are made accordingly.

Note that we intentionally write $y(x, \omega)$ to stress that wait-and-see decisions differ as functions of the outcome of the random event and of the first-stage decisions.

If this decision sequence is repeated more than once, then we have a multi-stage stochastic programming problem (with a number of stages greater than two). An example of a multi-stage decision-making process with s stages is the following:

1. Decisions x^1 are made.
2. Uncertain parameter λ^1 is realized as $\lambda_{\omega^1}^1$.
3. Decisions $x^2(x^1, \omega^1)$ are made.
4. Uncertain parameter λ^2 is realized as $\lambda_{\omega^2}^2$.
5. Decisions $x^3(x^1, \omega^1, x^2, \omega^2)$ are made.
- ...
- 2s-2. Uncertain parameter λ^{s-1} is realized as $\lambda_{\omega^{s-1}}^{s-1}$.
- 2s-1. Decisions $x^s(x^1, \omega^1, \dots, x^{s-1}, \omega^{s-1})$ are made.

In all stochastic programming problems, and especially in multi-stage problems, it is necessary to enforce the *non-anticipatory* character of information [35]. That is, if the outcomes of the random events are identical up to stage k , then the values taken by decision variables must be identical up to that stage as well.

The *non-anticipativity* of decisions is implicitly accounted for in the previous decision sequence. Specifically, observe that decisions x^1 are independent of the future realization of uncertain parameters $\{\lambda^1, \dots, \lambda^{s-1}\}$. In the next stage, however, decisions x^2 are contingent on the outcome of random parameter λ^1 , but they are unique for all random parameters that are realized in the

future, namely, $\{\lambda^2, \dots, \lambda^{s-1}\}$. Analogous considerations hold for the rest of stages.

Mathematically, a two-stage stochastic linear programming problem can be formulated as

$$\begin{aligned} & \underset{x}{\text{Maximize}} \quad c^\top x + \mathcal{E}_\omega\{Q(x, \omega)\} \\ & \text{subject to} \quad Ax \leq b \\ & \quad \quad \quad x \geq 0 \end{aligned} \tag{A.4}$$

being

$$\begin{aligned} Q(x, \omega) = & \left\{ \underset{y_\omega}{\text{Maximize}} \quad q_\omega^\top y_\omega \right. \\ & \text{subject to} \quad W_\omega y_\omega \leq h_\omega - T_\omega x \\ & \quad \quad \quad \left. y_\omega \geq 0 \right\}, \quad \forall \omega \in \Omega, \end{aligned} \tag{A.5}$$

where x and y_ω are the first- and second-stage decision variable vector, respectively, c , q_ω , b , h_ω , A , T_ω , and W_ω are known vectors and matrices of appropriate size, and $\mathcal{E}_\omega$ is the expectation operator over the outcome index ω . Note that, in problem (A.4)–(A.5), the mathematical expectation is used as the criterion for ranking the objective function random variables $\{f(x, \omega) = c^\top x + Q(x, \omega), Ax \leq 0, x \geq 0\}$. Moreover, observe that the uncertainty involved in problem (A.4)–(A.5) is assumed to be properly represented by means of a finite set Ω of scenarios, realizations, or outcomes ω .

Subproblem (A.5), wherein decisions y_ω are made after uncertainty is disclosed, is named *recourse problem*.

Under rather general assumptions [35], the two-stage stochastic programming problem (A.4)–(A.5) can be equivalently expressed as

$$\begin{aligned} & \underset{x, y_\omega}{\text{Maximize}} \quad c^\top x + \sum_{\omega \in \Omega} \pi_\omega q_\omega^\top y_\omega \\ & \text{subject to} \quad Ax \leq b \\ & \quad \quad \quad W_\omega y_\omega \leq h_\omega - T_\omega x, \quad \forall \omega \in \Omega \end{aligned} \tag{A.6}$$

$$\begin{aligned} x &\geq 0 \\ y_\omega &\geq 0, \forall \omega \in \Omega, \end{aligned}$$

where π_ω is the probability of occurrence of scenario ω .

Formulation (A.6) is known as the *deterministic equivalent problem* of the stochastic programming problem (A.4)–(A.5) and can be directly processed by solution algorithms for linear programs.

Lastly, both the stochastic problem (A.4)–(A.5) and its associated deterministic equivalent problem (A.6) can be easily extended to include a number of stages greater than two.

A.3 Quality Metrics

When dealing with decision-making problems subject to uncertain input data, it is always a good practice to answer the following two questions:

- Why to use a stochastic approach to tackle the decision-making problem instead of a deterministic one? Deterministic problems are fabricated out of stochastic models by replacing the uncertain parameters by their expected or forecast values. Therefore, deterministic problems are simpler and easier to solve than stochastic ones.
- How much do we gain by improving the forecasting techniques to identify the most likely scenarios?

Two metrics are usually computed to respond to these questions, namely, the Value of Stochastic Solution (VSS) and the Expected Value of Perfect Information (EVPI), respectively. These measures, which are widely used in stochastic programming [35], are explained below.

A.3.1 The Value of Stochastic Solution

The Value of Stochastic Solution quantifies the economic advantage of using a stochastic programming approach over a deterministic one to cope with

decision-making problems under uncertainty. The VSS for a two-stage stochastic programming problem is calculated in four steps:

Step 1: First, stochastic programming problem (A.6) is solved, yielding an optimal objective function value denoted by z^{s*} . Superscript s stands for “stochastic”.

Step 2: Next, the deterministic problem that results from replacing the uncertain parameters in (A.6) by their expected values is solved. To be more precise, this problem can be formulated as follows:

$$\begin{aligned}
 & \underset{x^d, \hat{y}}{\text{Maximize}} && c^\top x^d + \hat{q}^\top \hat{y} \\
 & \text{subject to} && Ax^d \leq b \\
 & && \hat{W}\hat{y} \leq \hat{h} - \hat{T}x^d \\
 & && x^d \geq 0 \\
 & && \hat{y} \geq 0,
 \end{aligned} \tag{A.7}$$

where $\hat{q} = q(\mathcal{E}_\omega \{\lambda_\omega\})$, $\hat{W} = W(\mathcal{E}_\omega \{\lambda_\omega\})$, $\hat{h} = h(\mathcal{E}_\omega \{\lambda_\omega\})$, and $\hat{T} = T(\mathcal{E}_\omega \{\lambda_\omega\})$. Let us represent the optimal decision vector of problem (A.7) as x^{d*} , where superscript d stands for “deterministic”.

Step 3: The original stochastic programming problem (A.6) is then solved fixing the value of the first-stage variable vector (x) to that obtained in the previous step (x^{d*}). The resulting problem can be formulated as

$$\begin{aligned}
 & \underset{y_\omega}{\text{Maximize}} && z^d = c^\top x^{d*} + \sum_{\omega \in \Omega} \pi_\omega q_\omega^\top y_\omega \\
 & \text{subject to} && W_\omega y_\omega \leq h_\omega - T_\omega x^{d*}, \forall \omega \in \Omega \\
 & && y_\omega \geq 0, \forall \omega \in \Omega,
 \end{aligned} \tag{A.8}$$

which provides an optimal objective function value denoted by z^{d*} . Note that problem (A.8) decomposes by scenario and as such, is generally easy to solve.

Step 4: Finally, the Value of the Stochastic Solution is computed as

$$\text{VSS} = z^{\text{s}*} - z^{\text{d}*}. \quad (\text{A.9})$$

In reference [78], some criteria to properly extend the application of the VSS to multi-stage stochastic programming problems are provided.

A.3.2 The Expected Value of Perfect Information

The Expected Value of Perfect Information is an index that quantifies how much a decision maker is *willing to pay* for obtaining perfect information about the future.

The EVPI is computed as follows:

Step 1: Stochastic programming problem (A.6) is first solved, yielding an optimal objective function value equal to $z^{\text{s}*}$.

Step 2: Second, the problem that results from (A.6) if the non-anticipativity of decisions is relaxed, i.e., if all variables are defined as scenario-dependent, is solved. This problem, usually referred to as the *wait-and-see* problem, can be stated as

$$\begin{aligned} \underset{x_\omega, y_\omega}{\text{Maximize}} \quad & z^{\text{ws}} = \sum_{\omega \in \Omega} \pi_\omega (c^\top x_\omega + q_\omega^\top y_\omega) \\ \text{subject to} \quad & Ax_\omega \leq b \\ & W_\omega y_\omega \leq h_\omega - T_\omega x_\omega, \quad \forall \omega \in \Omega \\ & x_\omega \geq 0 \\ & y_\omega \geq 0, \quad \forall \omega \in \Omega. \end{aligned} \quad (\text{A.10})$$

It is said that decisions in problem (A.10) are made with perfect information. This problem can be decomposed by scenario, which makes it easier to solve.

Step 3: Let us denote by $z^{\text{ws}*}$ the optimal objective function value of problem

(A.10), where “ws” stands for “wait-and-see”. The EVPI is given by

$$\text{EVPI} = z^{\text{ws}*} - z^{\text{s}*}. \quad (\text{A.11})$$

When assessing the performance of a given stochastic programming model, aside from the calculation of the quality metrics described in this section, the model should be validated, if possible, using real-world data (out-of-sample analysis).

A.4 Risk Management

Defining

$$f(x, \omega) = c^\top x + \max_{y_\omega} \{q_\omega^\top y_\omega : W_\omega y_\omega = h_\omega - T_\omega x, y_\omega \geq 0\}, \quad (\text{A.12})$$

problem (A.6) can be equivalently expressed in compact formulation as

$$\text{Maximize}_x \quad \mathcal{E}_\omega \{f(x, \omega)\} \quad (\text{A.13})$$

subject to

$$x \in X, \quad (\text{A.14})$$

where X represents the feasible region of the decision variable vector x .

The objective of problem (A.13)–(A.14) is to maximize the expected value of function $f(x, \omega)$, which can correspond, for instance, to the profit achieved by a wind power producer in a given trading horizon. Notwithstanding the fact that ranking a random variable by its expected value is advantageous in many aspects, its main drawback is that the remaining characteristics of the distribution associated with the random variable are disregarded, particularly those providing information about how *dispersed* its possible outcomes are. For example, a random variable representing a profit distribution with an expected value that is acceptable to the decision maker could also exhibit a non-negligible probability of experiencing negative profits (losses).

The notion of *risk* stems from the possibility of obtaining a decision vector

x leading to a profit distribution with undesirable properties, e.g., a high probability of low profit. In order to control the variability of profit $f(x, \omega)$, a *risk measure* is included in the formulation of the decision-making problem. This risk measure is a function $r_\omega \{f(x, \omega)\}$ that assigns to a given random variable representing profit $\{f(x, \omega), \forall \omega \in \Omega\}$ a real number characterizing the risk associated with that profit. The risk function $r_\omega \{f(x, \omega)\}$ can be incorporated into problem (A.13)–(A.14) as follows:

$$\text{Maximize}_x \quad \mathcal{E}_\omega \{f(x, \omega)\} - \beta r_\omega \{f(x, \omega)\} \quad (\text{A.15})$$

subject to

$$x \in X, \quad (\text{A.16})$$

where parameter $\beta \in [0, \infty)$ is a weighting factor used to resolve the tradeoff between expected profit and risk aversion. If $\beta = 0$, the risk term in the objective function (A.15) vanishes and the resulting problem boils down to (A.13)–(A.14), which is known as the *risk-neutral problem*. As β increases, the expected profit becomes less important in relation to the risk term and consequently, the decision maker becomes more risk averse.

The optimal solution obtained from problem (A.15)–(A.16) can be expressed as a function of parameter β , and thus represented in the so-called *efficient frontier* [139]. The efficient frontier is a curve made up of pairs (expected profit, risk) in such a way that it is impossible to find a set of decision variables yielding greater expected profit and less risk simultaneously. This way, a solution with greater expected profit than that of an efficient point can only be obtained at the cost of experiencing a higher risk, and viceversa. Fig. A.1 illustrates an example of efficient frontier for problem (A.15)–(A.16). Note that small values of β provide solutions with high expected profit and also high risk. In contrast, large values of β yield solutions with smaller expected profit and smaller risk. Efficient frontiers are therefore useful for decision makers to face the tradeoff between expected profit and risk. Finally, we underline that efficient frontiers are composed of a finite set of efficient points (thus, they are not continuous) and the line resulting from joining efficient points is not necessarily either convex or concave.

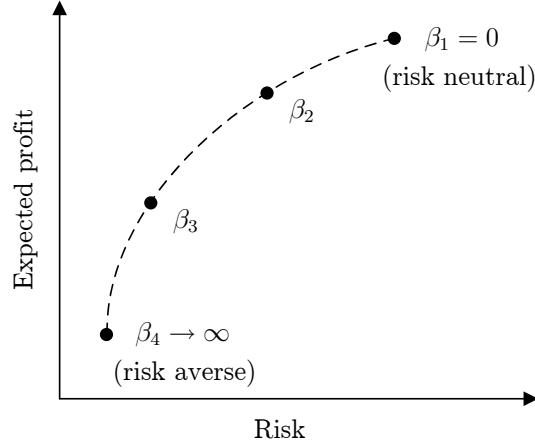


Figure A.1: Example of efficient frontier

Several risk measures have been analyzed in the technical literature, namely, the variance, the shortfall probability, the expected shortfall, the Value-at-Risk (VaR), and the Conditional Value-at-Risk (CVaR) [139, 174, 175, 210]. Among all of them, the CVaR is deemed superior because:

1. It can be included in problem (A.13)–(A.14) using a linear formulation.
2. It is able to quantify “fat tails” in probability distributions and,
3. it is a coherent risk measure, i.e., it exhibits good properties such as translation invariance, subadditivity, positive homogeneity, and monotonicity [25, 188].

For a given $\alpha \in (0, 1)$, the Conditional Value-at-Risk, CVaR_α , is defined as the expected value of the profit smaller than the $(1 - \alpha)$ -quantile of the profit distribution. This quantile is also called the Value-at-Risk at confidence level α , VaR_α , and can be mathematically expressed as

$$\text{VaR}_\alpha(x) = \max \left\{ \zeta : P(\omega | f(x, \omega) < \zeta) \leq 1 - \alpha \right\}, \quad \forall \alpha \in (0, 1), \quad (\text{A.17})$$

where P stands for “probability”.

In turn, the $\text{CVaR}_\alpha(x)$ for a discrete distribution is defined as, [189, 190]

$$\text{CVaR}_\alpha(x) = \max \left\{ \zeta - \frac{1}{1-\alpha} \mathcal{E}_\omega \left\{ \max\{\zeta - f(x, \omega), 0\} \right\} \right\}, \quad \forall \alpha \in (0, 1). \quad (\text{A.18})$$

If all profit scenarios are equiprobable, $\text{CVaR}_\alpha(x)$ is computed as the expected profit of the $(1 - \alpha) \times 100\%$ worst scenarios.

The CVaR can be embedded into problem (A.13)–(A.14) as follows:

$$\underset{x}{\text{Maximize}} \quad \mathcal{E}_\omega \{f(x, \omega)\} + \beta \text{CVaR}_\alpha(x) \quad (\text{A.19})$$

subject to

$$\begin{aligned} \text{CVaR}_\alpha(x) &= \zeta - \frac{1}{1-\alpha} \sum_{\omega \in \Omega} \pi_\omega \eta_\omega \\ \zeta - f(x, \omega) &\leq \eta_\omega, \quad \forall \omega \in \Omega \\ \eta_\omega &\geq 0, \quad \forall \omega \in \Omega \\ x &\in X. \end{aligned} \quad (\text{A.20})$$

Lastly, Fig. A.2 illustrates the concept of the Conditional Value-at-Risk on both the probability density function (pdf) and the cumulative distribution function of a certain continuous random variable representing profit.

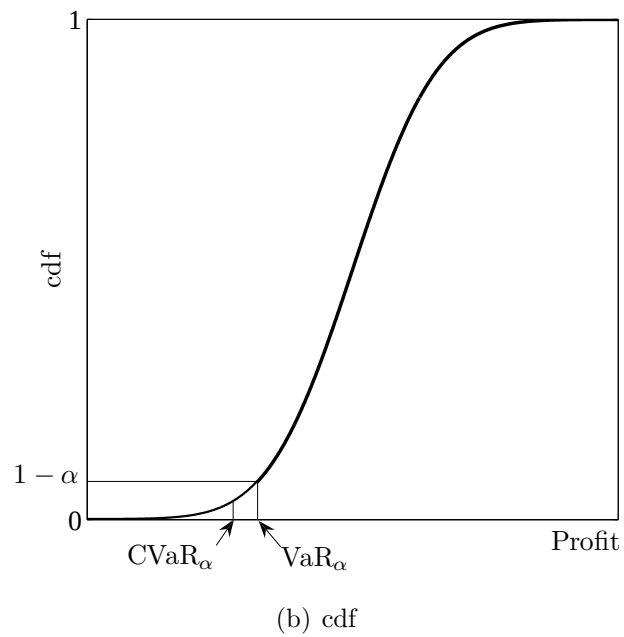
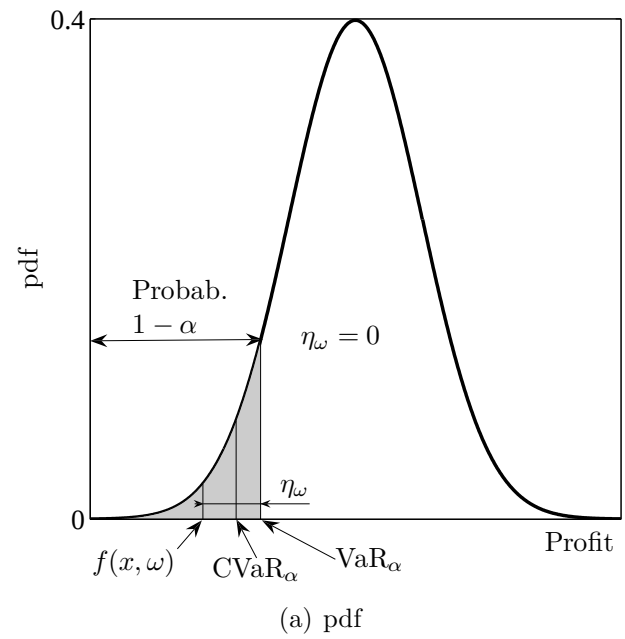


Figure A.2: Illustration of the concept of the Conditional Value-at-Risk

Appendix B

24-Node System Data

This appendix is devoted to describe the characteristics of the 24-node system that we use to examine the market-clearing tool proposed in Chapter 5. Unless stated otherwise in the main body of this dissertation, the characteristics listed next apply integrally.

B.1 Network data

The 24-node system considered in this book is based on the single-area version of the IEEE Reliability Test System–1996 [85]. The system comprises 34 transmission lines as illustrated in Fig. B.1. Line resistances are null and thus, active power losses are disregarded. The values of reactance and capacity of transmission lines are listed in Table B.1. Line reactances are given in per unit on a 100-MVA base.

B.2 Generator data

Technical data for the generators are given in Table B.2. The power system includes 12 generating units. Unit 10 is a hydro generator whereas the rest of them are all thermal. In particular, units 8 and 9 are nuclear power plants. The nuclear and hydro generators are assumed to be must-run units. Table B.3 indicates the node location of each unit.

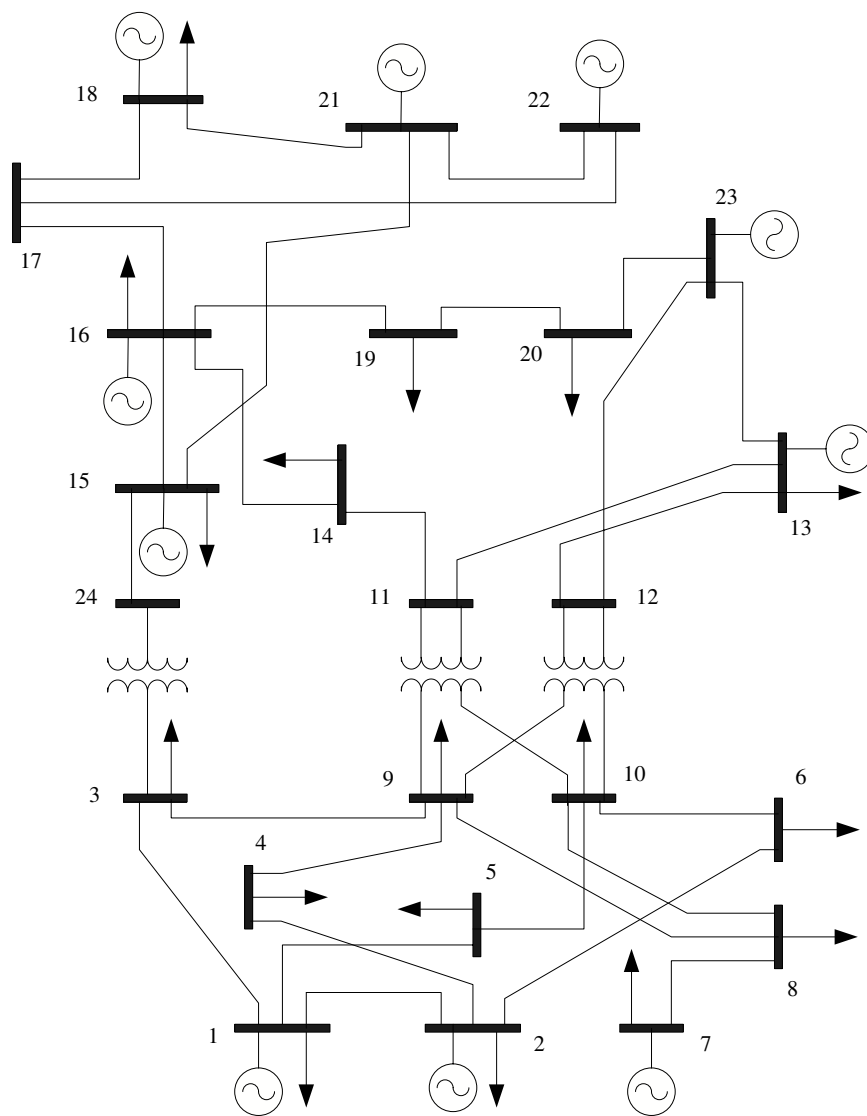


Figure B.1: 24-node system

Table B.1: 24-node system: reactance and capacity of transmission lines

From node	To node	Reactance (p.u.)	Capacity (MVA)
1	2	0.0146	175
1	3	0.2253	175
1	5	0.0907	175
2	4	0.1356	175
2	6	0.2050	175
3	9	0.1271	175
3	24	0.0840	400
4	9	0.1110	175
5	10	0.0940	175
6	10	0.0642	175
7	8	0.0652	175
8	9	0.1762	175
8	10	0.1762	175
9	11	0.0840	400
9	12	0.0840	400
10	11	0.0840	400
10	12	0.0840	400
11	13	0.0488	500
11	14	0.0426	500
12	13	0.0488	500
12	23	0.0985	500
13	23	0.0884	500
14	16	0.0594	500
15	16	0.0172	500
15	21	0.0249	1000
15	24	0.0529	500
16	17	0.0263	500
16	19	0.0234	500
17	18	0.0143	500
17	22	0.1069	500
18	21	0.0132	1000
19	20	0.0203	1000
20	23	0.0112	1000
21	22	0.0692	500

Table B.2: Technical data of generating units

Unit #	1	2	3	4	5	6	7	8	9	10	11	12
$P^{G,\min}$ (MW)	30.4	30.4	75	206.85	12	54.25	54.25	100	100	300	108.5	140
$P^{G,\max}$ (MW)	152	152	300	591	60	155	155	400	400	300	310	350
RD (MW/h)	152	152	300	540	60	155	155	400	400	300	310	240
RU (MW/h)	152	152	300	540	60	155	155	400	400	300	310	240
SD (MW/h)	152	152	300	540	60	155	155	400	400	300	310	240
SU (MW/h)	152	152	300	540	60	155	155	400	400	300	310	240
D^T (h)	4	4	8	10	2	8	8	1	1	0	8	48
U^T (h)	8	8	8	12	4	8	8	1	1	0	8	24
S^0 (h)	0	0	2	1	1	2	0	0	0	0	0	0
V^0 (h)	22	22	0	0	0	0	10	769	16	24	10	300
u^0 (h)	1	1	0	0	0	0	1	1	1	1	1	1

$P^{G,\min}$: minimum power output; $P^{G,\max}$: maximum power output; RD: ramp-down limit; RU: ramp-up limit; SD: shut-down ramp limit; SU: start-up ramp limit; D^T : minimum down time; U^T : minimum up time; S^0 : time periods the generating unit has been offline at the beginning of the scheduling horizon; V^0 : time periods the generating unit has been online at the beginning of the scheduling horizon; u^0 : initial commitment status (1 online, 0 otherwise)

Table B.3: Node location of generating units

Unit #	1	2	3	4	5	6	7	8	9	10	11	12
Node location	1	2	7	13	15	15	16	18	21	22	23	23

Table B.4: Size (MWh) of the energy blocks offered by generating units

Unit #	Block #			
	1	2	3	4
1	30.40	45.60	45.60	30.40
2	30.40	45.60	45.60	30.40
3	75.00	75.00	90.00	60.00
4	206.85	147.75	118.20	118.20
5	12.00	18.00	18.00	12.00
6	54.25	38.75	31.00	31.00
7	54.25	38.75	31.00	31.00
8	100.00	100.00	120.00	80.00
9	100.00	100.00	120.00	80.00
10	300.00	0.00	0.00	0.00
11	108.50	77.50	62.00	62.00
12	140.00	87.50	52.50	70.00

For simplicity, the energy and reserve offers of the units are assumed to be the same throughout the scheduling horizon consisting of 24 hourly periods. Tables B.4 and B.5 provide, respectively, the blocks of energy submitted by each generating unit and their corresponding offer costs. In Table B.6 the start-up cost *declared* by each unit is listed. Each generator offers the maximum possible amount of up and down-spinning reserve ($P^{G,\max} - P^{G,\min}$), both at a cost equal to 25% of its highest offer cost of energy production. However, only generators located at buses 7, 15 and 16 offer non-spinning reserve. We consider that these units constitute the set of generators with the required features to provide this reserve service. Then, each of these generating units offers the maximum possible amount of non-spinning reserve ($P^{\max}$) at a cost equal to 20% of its highest offer cost of energy production.

Table B.5: Offer costs (\$/MWh) submitted by generating units

Unit #	Block #			
	1	2	3	4
1	11.46	11.96	13.89	15.97
2	11.46	11.96	13.89	15.97
3	18.60	20.03	21.67	22.72
4	19.20	20.32	21.22	22.13
5	23.41	23.78	26.84	30.40
6	9.92	10.25	10.68	11.26
7	9.92	10.25	10.68	11.26
8	5.31	5.38	5.53	5.66
9	5.31	5.38	5.53	5.66
10	0.00	0.00	0.00	0.00
11	9.92	10.25	10.68	11.26
12	10.08	10.66	11.09	11.72

Table B.6: Start-up costs of generating units

Unit #	Start-up cost (\$)	Unit #	Start-up cost (\$)
1	1430.4	7	312.0
2	1430.4	8	0.0
3	1725.0	9	0.0
4	3056.7	10	0.0
5	437.0	11	624.0
6	312.0	12	2298.0

Table B.7: Load profile

Hour	System demand (MW)	Hour	System demand (MW)
12-1 am	1775.835	noon-1 pm	2517.975
1-2	1669.815	1-2	2517.975
2-3	1590.300	2-3	2464.965
3-4	1563.795	3-4	2464.965
4-5	1563.795	4-5	2623.995
5-6	1590.300	5-6	2650.500
6-7	1961.370	6-7	2650.500
7-8	2279.430	7-8	2544.480
8-9	2517.975	8-9	2411.955
9-10	2544.480	9-10	2199.915
10-11	2544.480	10-11	1934.865
11-noon	2517.975	11-12	1669.815

B.3 Demand data

The load profile for the considered 24-hour scheduling horizon is provided in Table B.7. The 24-node system represented in Fig. B.1 includes 17 loads. Both the nodal location of these loads and their contribution in percentage to the total system demand are listed in Table B.8. All these loads offer identical rates of \$50/MWh for up and down-spinning reserves. Loads can be curtailed from their normal levels up to 2% of their scheduled consumption for all hours to provide up-spinning reserves. Likewise, the demands can increase the same quantity to contribute to down-spinning reserve. Beyond these limits, the value of lost load is assumed to be \$2000/MWh.

Table B.8: Node location and distribution of the total system demand

Load #	Node #	% of system load
1	1	3.8
2	2	3.4
3	3	6.3
4	4	2.6
5	5	2.5
6	6	4.8
7	7	4.4
8	8	6.0
9	9	6.1
10	10	6.8
11	13	9.3
12	14	6.8
13	15	11.1
14	16	3.5
15	18	11.7
16	19	6.4
17	20	4.5

Appendix C

Inter-Temporal Constraints of Generators

In this appendix, the inter-temporal constraints of generators, (ramping limits and minimum up and down times) are formulated.

C.1 Ramping Limits

Consider the following additional notation:

RD_i	Ramp-down limit of unit i .
RU_i	Ramp-up limit of unit i .
SD_i	Shut-down ramp limit of unit i .
SU_i	Start-up ramp limit of unit i .
u_i^0	Initial commitment status of unit i (1 online, 0 otherwise).

Following [47], the ramp-up and start-up ramp rates of each generating unit i that is available in time period t under scenario ω are enforced by equations (C.1) below. Analogously, the shut-down ramp and ramp-down limits are imposed through equations (C.2) and (C.3).

$$\begin{aligned}
P_{it\omega}^G &\leq P_{i(t-1)\omega}^G + \text{RU}_i v_{i(t-1)\omega} \\
&\quad + \text{SU}_i (v_{it\omega} - v_{i(t-1)\omega}) \\
&\quad + P_i^{G, \max} (1 - v_{it\omega}), \quad \forall i \in G(t, \omega), \forall t, \forall \omega \quad (\text{C.1})
\end{aligned}$$

$$\begin{aligned}
P_{it\omega}^G &\leq P_i^{G, \max} v_{i(t+1)\omega} \\
&\quad + \text{SD}_i (v_{it\omega} - v_{i(t+1)\omega}), \\
&\quad \forall i \in G(t, \omega), \forall t = 1, 2, \dots, N_T - 1, \forall \omega \quad (\text{C.2})
\end{aligned}$$

$$\begin{aligned}
P_{i(t-1)\omega}^G - P_{it\omega}^G &\leq \text{RD}_i v_{it\omega} \\
&\quad + \text{SD}_i (v_{i(t-1)\omega} - v_{it\omega}) \\
&\quad + P_i^{G, \max} (1 - v_{i(t-1)\omega}), \quad \forall i \in G(t, \omega), \forall \omega, \forall t. \quad (\text{C.3})
\end{aligned}$$

Note that terms $P_i^{G, \max} (1 - v_{it\omega})$, $P_i^{G, \max} v_{i(t+1)\omega}$ and $P_i^{G, \max} (1 - v_{i(t-1)\omega})$ in equations (C.1), (C.2), and (C.3), respectively, are added to avoid constraint conflicts.

C.2 Minimum Up- and Down-Time Constraints

The following parameters are required to formulate the generator minimum up- and down-time limitations:

D_i^T	Minimum down time of unit i .
U_i^T	Minimum up time of unit i .
S_i^0	Time periods unit i has been off-line at the beginning of the scheduling horizon.
V_i^0	Time periods unit i has been online at the beginning of the scheduling horizon.
u_i^0	Initial commitment status of unit i (1 online, 0 otherwise).

The minimum up-time constraints are written as follows:

$$\sum_{t=1}^{T_i^{\text{ON}}} (1 - v_{it\omega}) = 0, \quad \forall i, \quad \forall \omega \quad (\text{C.4})$$

$$\sum_{t_1=t}^{t+U_i^{\text{T}}-1} v_{it_1\omega} \geq U_i^{\text{T}} (v_{it\omega} - v_{i(t-1)\omega})$$

$$\forall i, \forall t = T_i^{\text{ON}} + 1, \dots, N_{\text{T}} - U_i^{\text{T}} + 1, \quad \forall \omega \quad (\text{C.5})$$

$$\sum_{t_1=t}^{N_{\text{T}}} [v_{it_1\omega} - (v_{it\omega} - v_{i(t-1)\omega})] \geq 0$$

$$\forall i, \forall t = N_{\text{T}} - U_i^{\text{T}} + 2, \dots, N_{\text{T}}, \quad \forall \omega, \quad (\text{C.6})$$

where T_i^{ON} is the number of initial periods during which unit i must be online. T_i^{ON} is mathematically expressed as

$$T_i^{\text{ON}} = \text{Min}\{N_{\text{T}}, (U_i^{\text{T}} - V_i^0) u_i^0\}. \quad (\text{C.7})$$

Equations (C.4) constrain the initial status of the units according to T_i^{ON} . Equations (C.5) are enforced throughout the subsequent periods to comply with the minimum up-time constraint during all the possible sets of consecutive periods of size U_i^{T} . Lastly, constraints (C.6) model the final $U_i^{\text{T}} - 1$ periods in which if unit i is started up, it remains online until the end of the time horizon.

Minimum down-time constraints are obtained from the minimum up-time limitations (C.4)–(C.6) if $v_{it\omega}$, U_i^{T} , and V_i^0 are replaced by $1 - v_{it\omega}$, D_i^{T} , and S_i^0 , respectively. That is,

$$\sum_{t=1}^{T_i^{\text{OFF}}} v_{it\omega} = 0, \quad \forall i, \quad \forall \omega \quad (\text{C.8})$$

$$\sum_{t_1=t}^{t+D_i^{\text{T}}-1} (1 - v_{it_1\omega}) \geq D_i^{\text{T}} (v_{i(t-1)\omega} - v_{it\omega})$$

$$\forall i, \forall t = T_i^{\text{OFF}} + 1, \dots, N_T - D_i^T + 1, \quad \forall \omega \quad (\text{C.9})$$

$$\sum_{t_1=t}^{N_T} [1 - v_{it_1\omega} - (v_{i(t-1)\omega} - v_{it\omega})] \geq 0$$

$$\forall i, \forall t = N_T - D_i^T + 2, \dots, N_T, \quad \forall \omega, \quad (\text{C.10})$$

where T_i^{OFF} is the number of initial periods during which unit i must be off-line. T_i^{OFF} is mathematically expressed as

$$T_i^{\text{OFF}} = \text{Min}\{N_T, (D_i^T - S_i^0) (1 - u_i^0)\}. \quad (\text{C.11})$$

If scenario ω involves the outage of generating unit i in time period τ , parameter N_T in equations (C.4)–(C.11) should be replaced by the number of periods until failure, which coincides numerically with τ .

Bibliography

- [1] Electrical and Computer Engineering Department of Kansas State University, Dr. Gary Johnson's Renewable Energy and Tesla Coil Page. Available at <http://eece.ksu.edu/~gjohnson/>, 2010.
- [2] IEA WIND Task 25. Design and Operation of Power Systems with Large Amounts of Wind Power. Available at <http://www.ieawind.org>, 2010.
- [3] Market Operator of the Electricity Market of the Iberian Peninsula, OMEL. Available at <http://www.omel.es>, 2010.
- [4] New York Electricity Market, NY ISO. Available at <http://www.nyiso.com/>, 2010.
- [5] Pennsylvania–New Jersey–Maryland Electricity Market, PJM ISO. Available at <http://www.pjm.com/>, 2010.
- [6] Red Eléctrica de España, S.A. Sistema de Información del Operador del Sistema, e-sios. Available at <http://www.esios.ree.es>, 2010.
- [7] Red Eléctrica de España, S.A. Transmission System Operator of the Spanish power system. Available at <http://www.ree.es>, 2010.
- [8] SAS, Statistical Analysis Systems. SAS Institute. Available at <http://www.sas.com>, 2010.

- [9] SCA Statistical System. Scientific Computing Associates.
Available at <http://www.scausa.com>, 2010.
- [10] SmartGrids: European Technology Platform for the Electricity Networks of the Future.
Available at <http://www.smartgrids.eu/>, 2010.
- [11] SPSS, Statistical Package for the Social Sciences. SPSS Inc.
Available at <http://www.spss.com>, 2010.
- [12] Stochastic Programming Community Home Page.
Available at <http://stoprog.org/>, 2010.
- [13] The Electric Reliability Council of Texas (ERCOT) market.
Available at <http://www.ercot.com>, 2010.
- [14] The GAMS Development Corporation Website.
Available at <http://www.gams.com/>, 2010.
- [15] The ILOG CPLEX Website.
Available at <http://www.ilog.com/products/cplex/>, 2010.
- [16] Wind Power Integration in Liberalised Electricity Markets (WILMAR) project.
Available at <http://www.wilmar.risoe.dk.>, 2010.
- [17] T. Ackermann, J. Rivier Abbad, I. M. Dudurych, I. Erlich, H. Holttinen, J. Runge Kristoffersen, and Ejnar Sørensen. European Balancing Act. *IEEE Power & Energy Magazine*, 5(6):91–103, Nov.-Dec. 2007.
- [18] T. Ackermann (Ed). *Wind Power in Power Systems*. John Wiley & Sons, Chichester, England, 2005.
- [19] J. Aghaei, H. A. Shayanfar, and N. Amjady. Joint market clearing in a stochastic framework considering power system security. *Applied Energy*, 86(9):1675–1682, Sep. 2009.

- [20] J. L. Angarita, J. Usaola, and J. Martínez-Crespo. Combined hydro-wind generation bids in a pool-based electricity market. *Energy Policy*, 79(7):1038–1046, Jul. 2009.
- [21] J. M. Angarita and J. G. Usaola. Combining hydro-generation and wind energy biddings and operation on electricity spot markets. *Electric Power Systems Research*, 77(5–6):393–400, Apr. 2007.
- [22] J. M. Arroyo and A. J. Conejo. Optimal response of a thermal unit to an electricity spot market. *IEEE Transactions on Power Systems*, 15(3):1098–1104, Aug. 2000.
- [23] J. M. Arroyo and A. J. Conejo. Multiperiod auction for a pool-based electricity market. *IEEE Transactions on Power Systems*, 17(4):1225–1231, Nov. 2002.
- [24] J. M. Arroyo and F. D. Galiana. Energy and reserve pricing in security and network-constrained electricity markets. *IEEE Transactions on Power Systems*, 20(2):634–643, May 2005.
- [25] P. Artzner, F. Delbaen, J. M. Eber, and D. Heath. Coherent measures of risk. *Mathematical Finance*, 9(3):203–228, Jul. 1999.
- [26] Danish Wind Industry Association. Wind Turbine Power Calculator. Available at <http://www.windpower.org/>, 2010.
- [27] A. Baillo, M. Ventosa, M. Rivier, and A. Ramos. Optimal offering strategies for generation companies operating in electricity spot markets. *IEEE Transactions on Power Systems*, 19(2):745–753, May 2004.
- [28] R. Barth, H. Brand, P. Meibom, and C. Weber. A stochastic unit-commitment model for the evaluation of the impacts of integration of large amounts of intermittent wind power. In *Probabilistic Methods Applied to Power Systems (PMAPS)*, pages 26–33, Stockholm, Sweden, 2006.

- [29] R. J. Barthelmie, F. Murray, and S. C. Pryor. The economic benefit of short-term forecasting for wind energy in the UK electricity market. *Energy Policy*, 36(5):1687–1696, May 2008.
- [30] G. N. Bathurst and G. Strbac. Value of combining energy storage and wind in short-term energy and balancing markets. *Electric Power Systems Research*, 67(1):1–8, Oct. 2003.
- [31] G. N. Bathurst, J. Weatherill, and G. Strbac. Trading wind generation in short term energy markets. *IEEE Transactions on Power Systems*, 17(3):782–789, Aug. 2002.
- [32] L. E. Benitez, P. C. Benitez, and G. C. van Kooten. The economics of wind power with energy storage. *Energy Economics*, 30(4):1973–1989, Jul. 2008.
- [33] R. Billinton and R. N. Allan. *Reliability Evaluation of Power Systems*. Springer, New York, 1996.
- [34] R. Billinton and R. Karki. Application of Monte Carlo simulation to generating system well-being analysis. *IEEE Transactions on Power Systems*, 14(3):1172–1177, Aug. 1999.
- [35] J. R. Birge and F. Louveaux. *Introduction to stochastic programming*. Springer, New York, 1997.
- [36] S. Borenstein. The long-run efficiency of real-time electricity pricing. *Energy Journal*, 26(3):93–116, 2005.
- [37] S. Borenstein and S. P. Holland. On the efficiency of competitive electricity markets with time-invariant retail prices. *RAND Journal of Economics*, 36(3):469–493, Autumn 2005.
- [38] F. Bouffard and F. D Galiana. Stochastic security for operations planning with significant wind power generation. *IEEE Transactions on Power Systems*, 23(2):306–316, May 2008.

- [39] F. Bouffard, F. D. Galiana, and A. J. Conejo. Market-clearing with stochastic security—Part I: formulation. *IEEE Transactions on Power Systems*, 20(4):1818–1826, Nov. 2005.
- [40] F. Bouffard, F. D. Galiana, and A. J. Conejo. Market-clearing with stochastic security—Part II: case studies. *IEEE Transactions on Power Systems*, 20(4):1827–1835, Nov. 2005.
- [41] G. E. P. Box, G. M. Jenkins, and G. C. Reinsel. *Time Series Analysis: Forecasting and Control*. John Wiley & Sons, New Jersey, USA, 2008.
- [42] T. H Bradley and A. A. Frank. Design, demonstrations and sustainability impact assessments for plug-in hybrid electric vehicles. *Renewable & sustainable energy reviews*, 13(1):104–117, Jan. 2009.
- [43] P. J. Brockwell and R. A. Davis. *Time Series: Theory and Methods*. Springer, New York, 2009.
- [44] M. Brower. Development of Eastern Regional Wind Resource and Wind Plant Output Datasets.
Available at <http://www.nrel.gov/wind/integrationdatasets/pdfs/eastern>, 2010.
- [45] D. J. Burke and M. J. O'Malley. Maximizing firm wind connection to security constrained transmission networks. *IEEE Transactions on Power Systems*, 25(2):749–759, May 2010.
- [46] M. Carrión. *Medium-Term Decision Making for Consumers, Retailers, and Producers in Electricity Markets via Stochastic Programming*. PhD thesis, Universidad de Castilla-La Mancha, Ciudad Real, España.
Available at http://www.uclm.es/area/gsee/tesis/Tesis_MCarrion.pdf, 2008.
- [47] M. Carrión and J. M. Arroyo. A computationally efficient mixed-integer linear formulation for the thermal unit commitment problem. *IEEE Transactions on Power Systems*, 21(3):1371–1378, Aug. 2006.

- [48] M. Carrión, A. B. Philpott, A. J. Conejo, and J. M. Arroyo. A stochastic programming approach to electric energy procurement for large consumers. *IEEE Transactions on Power Systems*, 22(2):744–754, May 2007.
- [49] E. Castillo, A. J. Conejo, P. Pedregal, R. García-Bertrand, and N. Alguacil. *Building and Solving Mathematical Programming Models in Engineering and Science*. John Wiley and Sons, New York, 2002.
- [50] E. D. Castronuovo and J. A. P. Lopes. Optimal operation and hydro storage sizing of a wind-hydro power plant. *International Journal of Electrical Power & Energy Systems*, 26(10):771–778, Dec. 2004.
- [51] E. D. Castronuovo and J. A. Peças Lopes. On the optimization of the daily operation of a wind-hydro power plant. *IEEE Transactions on Power Systems*, 19(3):1599–1606, Aug. 2004.
- [52] A. Ceña. The impact of wind energy on the electricity price and on the balancing power costs: the Spanish case. In *the European Wind Energy Conference (EWEC)*, Marseille, France, Mar. 2009.
- [53] A. N. Celik. Assessing the suitability of wind speed distribution functions based on wind power density. *Renewable Energy*, 28(10):1563–1574, Aug. 2003.
- [54] C. Chatfield. *The Analysis of Time Series: An Introduction*. Chapman & Hall, 2003.
- [55] A. J. Conejo, E. Castillo, R. Mínguez, and R. García-Bertrand. *Decomposition Techniques in Mathematical Programming*. Springer, New York, 2006.
- [56] A. J. Conejo, J. Contreras, R. Espínola, and M. A. Plazas. Forecasting electricity prices for a day-ahead pool-based electric energy market. *International Journal of Forecasting*, 21(3):435–462, Jul.-Sep. 2005.

- [57] A. J. Conejo, F. J. Nogales, M. Carrión, and J. M. Morales. Electricity pool prices: long-term uncertainty characterization for futures-market trading and risk management. *Journal of the Operational Research Society*, 61(2):235–245, Feb. 2010.
- [58] J. Contreras, R. Espínola, F. J. Nogales, and A. J. Conejo. ARIMA models to predict next-day electricity prices. *IEEE Transactions on Power Systems*, 18(3):1014–1020, Aug. 2003.
- [59] A. Costa, A. Crespo, J. Navarro, G. Lizcano, H. Madsen, and E. Feitosa. A review on the young history of the wind power short-term prediction. *Renewable & Sustainable Energy Reviews*, 12(6):1725–1744, Aug. 2008.
- [60] L. M. Costa, F. Bourry, and G. Kariniotakis. Stochastic optimization techniques for the optimal combination of wind power generation and energy storage in a market environment. In *the European Wind Energy Conference (EWEC)*, Brussels, Belgium, Apr. 2008.
- [61] Global Wind Energy Council. Global Wind 2009 Report. Available at <http://www.gwec.net>, 2010.
- [62] G. Dany. Power reserve in interconnected systems with high wind power production. In *Proceedings IEEE Power Tech*, Porto, Portugal, Sept. 2001.
- [63] Red Eléctrica de España. Requisitos de respuesta frente a huecos de tensión de las instalaciones eólicas. Available at http://www.ree.es/operacion/pdf/po/P0_resol_12.3_Respuesta_huecos_eolica.pdf, 2006.
- [64] Red Eléctrica de España. Requisitos técnicos de las instalaciones eólicas, fotovoltaicas y todas aquellas instalaciones de producción cuya tecnología no emplee un generador síncrono conectado directamente a la red. P.O.12.2 (*draft*). Available at <http://www.aeeolica.es/userfiles/file/procedimiento-verificacion/BORRADOR-DE-LA-SEPARATA-DEL-P.O.12.2.PDF>, 2010.

- [65] E. Delarue and W. D'haeseleer. Adaptive mixed-integer programming unit commitment strategy for determining the value of forecasting. *Applied Energy*, 85(4):171–181, Apr. 2008.
- [66] E. A. DeMeo, W. Grant, M. R. Milligan, and M. J. Schuerger. Wind plant integration. *IEEE Power & Energy Magazine*, 3(6):38–46, Nov.-Dec. 2005.
- [67] E. A. DeMeo, G. A. Jordan, C. Kalich, J. King, M. R. Milligan, C. Murrey, B. Oakleak, and M. J. Schuerger. Accommodating wind's natural behavior. *IEEE Power & Energy Magazine*, 5(6):68–77, Nov.-Dec. 2007.
- [68] P. Denholm, E. Ela, B. Kirby, and M. Milligan. The Role of Energy Storage with Renewable Electricity Generation. National Renewable Energy Laboratory (NREL)
Available at <http://www.nrel.gov/docs/fy10osti/47187.pdf>, Jan. 2010.
- [69] E. Denny and M. O'Malley. Wind generation, power system operation, and emissions reduction. *IEEE Transactions on Power Systems*, 21(1):341–347, Feb. 2006.
- [70] E. Denny and M. O'Malley. Quantifying the total net benefits of grid integrated wind. *IEEE Transactions on Power Systems*, 20(2):605–615, May 2007.
- [71] T. S. Dillon, K. W. Edwin, and R. J. Taud. Integer programming approach to the problem of optimal unit commitment with probabilistic reserve determination. *IEEE Transactions on Power Apparatus and Systems*, PAS-97(6):2154–2166, Nov.-Dec. 1978.
- [72] R. Doherty and M. O'Malley. A new approach to quantify reserve demand in systems with significant installed wind capacity. *IEEE Transactions on Power Systems*, 20(2):587–595, May 2005.

- [73] J. Dupačová, G. Consigli, and S. W. Wallace. Scenarios for multistage stochastic programs. *Annals of Operations Research*, 100(1-4):25–53, Dec. 2000.
- [74] J. Dupačová, N. Gröwe-Kuska, and W. Römisch. Scenario reduction in stochastic programming: An approach using probability metrics. *Mathematical Programming*, 95(3):493–511, Mar. 2003.
- [75] A. Eichhorn and W. Römisch. Polyhedral risk measures in stochastic programming. *SIAM Journal on Optimization*, 16(1):69–95, 2005.
- [76] C. K. Ekman and S. H. Jensen. Prospects for large scale electricity storage in Denmark. *Energy Conversion and Management*, 51(6):1140–1147, Jun. 2010.
- [77] T. H. M. El-Fouly, H. H. Zeineldin, E. F. El-Saadany, and M. M. A. Salama. Impact of wind generation control strategies, penetration level and installation location on electricity market prices. *IET Renewable Power Generation*, 2(3):162–169, Sep. 2008.
- [78] L. F. Escudero, A. Garín, M. Merino, and G. Pérez. The value of the stochastic solution in multistage problems. *TOP*, 15(1):48–64, Jul. 2007.
- [79] European Union (EU). Climate change: Commission welcomes final adoption of Europe’s climate and energy package. Europa press releases RAPID.
Available at <http://europa.eu/rapid/pressReleasesAction.do?reference=IP/08/1998>, Dec. 2008.
- [80] European Wind Energy Association (EWEA). Making 180 GW a reality by 2020. Position Paper.
Available at <http://www.ewea.org/index.php?id=182>, Oct. 2007.
- [81] European Wind Energy Association (EWEA). Pure Power: Wind energy targets for 2020 and 2030. Report.
Available at <http://www.ewea.org/index.php?id=178>, Dec. 2009.

- [82] A. Fabbri, T. G. S. Román, J. R. Abbad, and V. H. M. Quezada. Assessment of the cost associated with wind generation prediction errors in a liberalized electricity market. *IEEE Transactions on Power Systems*, 20(3):1440–1446, Aug. 2005.
- [83] D. Feretic, Z. Tomsic, and N. Cavlina. Feasibility analysis of wind-energy utilization in Croatia. *Energy*, 24(3):239–246, Mar. 1999.
- [84] M. Flynn, W. P. Sheridan, J. D. Dillon, and M. J. O'Malley. Reliability and reserve in competitive electricity market scheduling. *IEEE Transactions on Power Systems*, 16(1):78–87, Feb. 2001.
- [85] Reliability Test System Task Force. The IEEE Reliability Test System—1996. *IEEE Transactions on Power Systems*, 14(3):1010–1020, Aug. 1999.
- [86] F. D. Galiana and F. Bouffard. An electricity market with a probabilistic spinning reserve criterion. *IEEE Transactions on Power Systems*, 19(1):300–307, Feb. 2004.
- [87] F. D. Galiana, F. Bouffard, J. M. Arroyo, and J. F. Restrepo. Scheduling and pricing of coupled energy and primary, secondary, and tertiary reserves. *Proceedings of the IEEE*, 93(11):1970–1983, Nov. 2005.
- [88] A. Garcia, J. L. Torres, E. Prieto, and A. de Francisco. Fitting wind speed distributions: a case study. *Solar Energy*, 62(2):139–144, Feb. 1998.
- [89] J. García-González, R. Moraga Ruiz de la Muela, L. Matres Santos, and A. Mateo González. Stochastic joint optimization of wind generation and pumped-storage units in an electricity market. *IEEE Transactions on Power Systems*, 23(2):460–468, May 2008.
- [90] H. I. Gassmann and A. Prékopa. On stages and consistency checks in stochastic programming. *Operations Research Letters*, 33(2):171–175, Mar. 2005.

- [91] G. Giebel, R. Brownsword, and G. Kariniotakis. The state of the art in short-term prediction of wind power: A literature overview. Available at <http://anemos.cma.fr> (deliverable D.1.1. of ANEMOS Project, 2003).
- [92] H. A. Gil and G. Joos. Generalized estimation of average displaced emissions by wind generation. *IEEE Transactions on Power Systems*, 22(3):1035–1043, Aug. 2007.
- [93] H. B. Gooi, D. P. Mendes, K. R. W. Bell, and D. S. Kirschen. Optimal scheduling of spinning reserve. *IEEE Transactions on Power Systems*, 14(4):1485–1492, Nov. 1999.
- [94] The Brattle Group. Quantifying Demand Response Benefits in PJM. Cambridge, MA. Available at <http://sites.energetics.com/MADRI/pdfs/brattlegroupreport.pdf>, Jan. 2007.
- [95] N. Gröwe-Kuska, H. Heitsch, and W. Römisch. Scenario reduction and scenario tree construction for power management problems. *Proceedings IEEE PowerTech*, Bologna, Italy, Jun. 2003.
- [96] J. D. Guy. Security constrained unit commitment. *IEEE Transactions on Power Apparatus and Systems*, PAS-90(3):1385–1390, May-Jun. 1971.
- [97] A. C Harvey. *Time Series Models*. John Wiley & Sons Inc, New York, USA, 1981.
- [98] L. Hecker, Z. Zhou, D. Osborn, and J. Lawhorn. Value based transmission planning process for joint coordinated system plan. In *Power & Energy Society General Meeting, 2009. PES'09. IEEE*, pages 2416–2421, Calgary, Canada, July 2009.
- [99] K. Hedman and G. Sheblé. Comparing hedging methods for wind power: using pumped storage hydro units vs options purchasing. In *International Conference on Probabilistic Methods Applied to Power Systems PMAPS*, pages 61–66, Stockholm, Sweden, Jun. 2006.

- [100] H. Heitsch and W. Römisch. Scenario reduction algorithms in stochastic programming. *Computational Optimization and Applications*, 24(2-3):187–206, Feb.-Mar. 2003.
- [101] H. Heitsch and W. Römisch. A note on scenario reduction for two-stage stochastic programs. *Operations Research Letters*, 35:731–738, Nov. 2007.
- [102] J. P. Hennessey. Some aspects of wind power statistics. *Journal of Applied Meteorology*, 16(2):119–128, 1977.
- [103] J. L. Higle and S. Sen. Stochastic decomposition – an algorithm for 2-stage linear-programs with recourse. *Mathematics for Operations Research*, 16(3):650–669, Aug. 1991.
- [104] J. L. Higle and S. Sen. *Stochastic Decomposition. A Statistical Method for Large Scale Stochastic Linear Programming*. Kluwer, Norwell, 1996.
- [105] E. Hirst. Integrating wind output with bulk power operations and wholesale electricity markets. *Wind Energy*, 5(1):19–36, May 2002.
- [106] S. P. Holland and E. T Mansur. The short-run effects of time-varying prices in competitive electricity markets. *Energy Journal*, 27(4):127–155, 2006.
- [107] H. Holttinen. *Hourly Wind Power Variations and their Impact on the Nordic Power System Operation*. PhD thesis, Helsinki University of Technology, Helsinki, Finland.
Available at http://www.regie-energie.qc.ca/audiences/3526-04/MemoiresParticip3526/Memoire_CCVK_15_hh.lic-thesis.finalpub.pdf, 2003.
- [108] H. Holttinen. Impact of hourly wind power variations on the system operation in the Nordic countries. *Wind Energy*, 8(2):197–218, Apr.-Jun. 2005.

- [109] H. Holttinen and S. Tuhkanen. The effect of wind power on CO₂ abatement in the Nordic countries. *Energy Policy*, 32(14):1639–1652, Sep. 2004.
- [110] Y.-Y. Hong and C.-Y. Hsiao. Locational marginal price forecasting in deregulated electricity markets using artificial intelligence. *IEEE Proceedings–Generation, Transmission and Distribution*, 149(5):621–626, Sep. 2002.
- [111] H. Hotelling. New light on the correlation coefficient and its transforms. *Journal of the Royal Statistical Society Series B-Statistical Methodology*, 15(2):193–232, 1953.
- [112] K. Høyland, M. Kaut, and S. W. Wallace. A heuristic for moment-matching scenario generation. *Computational Optimization and Applications*, 24(2-3):169–185, Feb. 2003.
- [113] K. Høyland and S. W. Wallace. Generating scenario trees for multistage decision problems. *Management Science*, 47(2):295–307, Feb. 2001.
- [114] H. Ibrahim, A. Ilinca, and J. Perron. Energy storage systems – characteristics and comparisons. *Renewable & Sustainable Energy Reviews*, 12(2):1221–1250, Jun. 2008.
- [115] M. Ilic, F. D. Galiana, and L. Fink. *Power System Restructuring: Engineering and Economics*. Springer, Massachusetts, 2010.
- [116] T. Jonsson, P. Pinson, and H. Madsen. On the market impact of wind energy forecasts. *Energy Economics*, 32(2):313–320, Mar. 2010.
- [117] P. Kall and S. W. Wallace. *Stochastic Programming*. John Wiley & Sons, Ltd., Chichester, West Sussex, UK, 1994.
- [118] E. S. Karapidakis and N. D. Hatziargyriou. Online preventive dynamic security of isolated power systems using decision trees. *IEEE Transactions on Power Systems*, 17(2):297–304, May 2002.

- [119] M. Kaut and S. W. Wallace. Evaluation of scenario-generation methods for stochastic programming. *SPEPS*, Paper 14-2003.
Available at <http://edoc.hu-berlin.de/browsing/speps/>.
- [120] R. J. Kaye, F. F. Wu, and P. Varaiya. Pricing for system security. *IEEE Transactions on Power Systems*, 10(2):575–583, May 1995.
- [121] W. Kempton and J. Tomic. Vehicle-to-grid power implementation: From stabilizing the grid to supporting large-scale renewable energy. *Journal of Power Sources*, 144(1):280–294, Jun. 2005.
- [122] D. S. Kirschen and G. Strbac. *Fundamentals of Power System Economics*. John Wiley & Sons, Ltd., Chichester, West Sussex, UK, 2004.
- [123] J. Kiviluoma and P. Meibom. Influence of wind power, plug-in electric vehicles, and heat storages on power system investments. *Energy*, 35(3):1244–1255, Mar. 2010.
- [124] M. Korpaas, A. T. Holen, and R. Hildrum. Operation and sizing of energy storage for wind power plants in a market system. *International Journal of Electrical Power & Energy Systems*, 25(8):599–606, Oct. 2003.
- [125] P. Kundur. *Power System Stability and Control*. McGraw-Hill Professional, New York, 1994.
- [126] National Renewable Energy Laboratory. Eastern Wind Dataset.
Available at <ftp://ftp2.nrel.gov/pub/ewits/TimeSeries/LandBased/2006/>, 2010.
- [127] D. P. Lalas, H. Tselepidaki, and G. Theoharatos. An analysis of wind power potential in Greece. *Solar Energy*, 30(6):497–505, 1983.
- [128] L. Landberg, G. Giebel, H. A. Nielsen, T. Nielsen, and H. Madsen. Short-term prediction - An overview. *Wind Energy*, 6(3):273–280, Jul.-Sep. 2003.

- [129] M. Lei, L. Shiyan, J. Chuanwen, L. Hongling, and Z. Yan. A review on the forecasting of wind speed and generated power. *Renewable & Sustainable Energy Reviews*, 13(4):915–920, May 2009.
- [130] T. Li and M. Shahidehpour. Risk-constrained generation asset arbitrage in power systems. *IEEE Transactions on Power Systems*, 22(3):1130–1139, Aug. 2007.
- [131] L. Liu and G. P. Hudak. *Forecasting and Time Series Analysis Using the SCA Statistical System*. Scientific Computing Associates, 1994.
- [132] P.-L. Liu and A. Der Kiureghian. Multivariate distribution models with prescribed marginals and covariances. *Probabilistic Engineering Mechanics*, 1(2):105–112, 1986.
- [133] P. J. Luickx, E. D. Delarue, and W. D. D’haeseleer. Considerations on the backup of wind power: Operational backup. *Applied Energy*, 85(9):787–799, Sep. 2008.
- [134] I. Y. F. Lun and J. C. Lam. A study of Weibull parameters using long-term wind observations. *Renewable Energy*, 20(2):145–153, Jun. 2000.
- [135] H. Lund. Large-scale integration of wind power into different energy systems. *Energy*, 30(13):2402–2412, Oct. 2005.
- [136] H. Lund and W. Kempton. Integration of renewable energy into the transport and electricity sectors through V2G. *Energy Policy*, 36(9):3578–3587, Sep. 2008.
- [137] Y.V. Makarov, C. Loutan, J. Ma, and P. de Mello. Operational Impacts of Wind Generation on California Power Systems. *IEEE Transactions on Power Systems*, 24(2):1039–1050, May 2009.
- [138] P. Mandal, T. Senjyu, N. Urasaki, T. Funabashi, and A. K. Srivastava. A novel approach to forecast electricity price for PJM using neural network and similar days method. *IEEE Transactions on Power Systems*, 22(4):2058–2065, Nov. 2007.

- [139] H. M. Markowitz. *Portfolio Selection: Efficient Diversification of Investments*. Wiley, New York, 1959.
- [140] A. Mas-Collel, M. D. Whinston, and J. R. Green. *Microeconomic Theory*. Oxford University Press, New York, 1995.
- [141] J. Matevosyan and L. Söder. Minimization of imbalance cost trading wind power on the short-term power market. *IEEE Transactions on Power Systems*, 21(3):1396–1404, Aug. 2006.
- [142] M. A. Matos and R. Bessa. Operating reserve adequacy evaluation using uncertainties of wind power forecast. In *Proceedings IEEE Power Tech*, pages 1704–1711, Bucharest, Romania, Jun.-Jul. 2009.
- [143] A. C. Miller and T. R. Rice. Discrete approximations of probability distributions. *Management Science*, 29(3):352–362, Mar. 1983.
- [144] M. S. Miranda and R. W. Dunn. Spatially correlated wind speed modelling for generation adequacy studies in the UK. In *Proceedings of the IEEE Power Engineering Society General Meeting*, pages 4783–4788, Tampa, Florida, Jun. 2007.
- [145] R. Moesgaard and P. E. Morthorst. The Impact of Wind Power on Electricity Prices in Denmark. In *the European Wind Energy Conference (EWEC)*, Brussels, Belgium, Apr. 2008.
- [146] C. Monteiro, R. Bessa, V. Miranda, A. Botterud, J. Wang, and G. Conzelmann. Wind Power Forecasting: State-of-the-Art 2009. Argonne National Laboratory. Available at <http://www.dis.anl.gov/pubs/65613.pdf>, 2009.
- [147] J. M. Morales, L. Baringo, A. J. Conejo, and R. Mínguez. Probabilistic power flow with correlated wind sources. *IET Generation, Transmission & Distribution*, 4(5):641–651, 2010.
- [148] J. M. Morales, A. J. Conejo, and J. Pérez-Ruiz. Economic valuation of reserves in power systems with high penetration of wind power. *IEEE Transactions on Power Systems*, 24(2):900–910, May 2009.

- [149] J. M. Morales, A. J. Conejo, and J. Pérez-Ruiz. Short-term trading for a wind power producer. *IEEE Transactions on Power Systems*, 25(1):554–564, Feb. 2010.
- [150] J. M. Morales, A. J. Conejo, and J. Pérez-Ruiz. Simulating the impact of wind production on locational marginal prices. *IEEE Transactions on Power Systems*, in press, 2010.
- [151] J. M. Morales, R. Mínguez, and A. J. Conejo. A methodology to generate statistically dependent wind speed scenarios. *Applied Energy*, 87(3):843–855, Mar. 2010.
- [152] J. M. Morales, S. Pineda, A. J. Conejo, and M. Carrión. Scenario reduction for futures market trading in electricity markets. *IEEE Transactions on Power Systems*, 24(2):878–888, May 2009.
- [153] P. E. Morthorst. Wind power and the conditions at a liberalized power market. *Wind Energy*, 6(3):297–308, Jul.-Sep. 2003.
- [154] F. Müsgens and K. Neuhoff. Modelling dynamic constraints in electricity markets and the costs of uncertain wind output. *Cambridge Working Papers in Economics*, Faculty of Economics, University of Cambridge Available at <http://www.dspace.cam.ac.uk/bitstream/1810/131648/1/eprg0514.pdf>, Mar. 2006.
- [155] G. L. Nemhauser and L. A. Wolsey. *Integer and Combinatorial Optimization*. Wiley-Interscience, New York, 1999.
- [156] North American Electric Reliability Corporation (NERC). Accommodating High Levels of Variable Generation. Special Report. Available at http://www.nerc.com/news_pr.php?npr=283, Apr. 2009.
- [157] H. Nfaoui, J. Buret, and A. A. M. Sayig. Wind characteristics and wind energy potential in Morocco. *Solar Energy*, 63(1):51–60, Jul. 1998.
- [158] F. J. Nogales and A. J. Conejo. Electricity price forecasting through transfer function models. *Journal of the Operational Research Society*, 57:350–356, 2006.

- [159] F. J. Nogales, J. Contreras, A. J. Conejo, and R. Espínola. Forecasting next-day electricity prices by time series models. *IEEE Transactions on Power Systems*, 17(2):342–348, May 2002.
- [160] M. P. Nowak and W. Römisch. Stochastic Lagrangian relaxation applied to power scheduling in a hydro-thermal system under uncertainty. *Annals of Operations Research*, 100:251–272, 2000.
- [161] National Renewable Energy Laboratory (NREL). Wind Energy Curtailment. Case Studies.
Available at <http://www.nrel.gov/docs/fy10osti/46716.pdf>, Oct. 2009.
- [162] National Renewable Energy Laboratory (NREL). Dynamic Maps, Gis Data, & Analysis Tools. Wind Maps.
Available at <http://www.nrel.gov/gis/wind.html>, 2010.
- [163] US Department of Energy (DOE). 20% Wind Energy by 2030: Increasing Wind Energy’s Contribution to US Electricity Supply. DOE Office of Energy Efficiency and Renewable Energy Report, Jul. 2008.
Available at http://www1.eere.energy.gov/windandhydro/wind_2030.html.
- [164] M. Olsson and L. Söder. Modeling real-time balancing power market prices using combined SARIMA and Markov processes. *IEEE Transactions on Power Systems*, 23(2):443–450, May 2008.
- [165] IEEE/CIGRE Joint Task Force on Stability Terms and Definitions. Definition and classification of power system stability. *IEEE Transactions on Power Systems*, 19(2):1387–1401, May 2004.
- [166] M. A. Ortega-Vazquez and D. S. Kirschen. Optimizing the spinning reserve requirements using a cost/benefit analysis. *IEEE Transactions on Power Systems*, 22(1):24–33, Feb. 2007.
- [167] M. A. Ortega-Vazquez and D. S. Kirschen. Estimating the spinning reserve requirements in systems with significant wind power generation

- penetration. *IEEE Transactions on Power Systems*, 24(1):114–124, Feb. 2009.
- [168] P. A. Østergaard. Ancillary services and the integration of substantial quantities of wind power. *Applied Energy*, 83(5):451–463, May 2006.
- [169] H. T. Pao. Forecasting electricity market pricing using artificial neural networks. *Energy Conversion and Management*, 48(3):907–912, Mar. 2007.
- [170] K. A. Papadogiannis and N. D. Hatziargyriou. Optimal allocation of primary reserve services in energy markets. *IEEE Transactions on Power Systems*, 19(1):652–659, Feb. 2004.
- [171] G. Papaefthymiou and D. Kurowicka. Using copulas for modeling stochastic dependence in power system uncertainty analysis. *IEEE Transactions on Power Systems*, 24(1):40–49, Feb. 2009.
- [172] B. Parsons, M. Milligan, B. Zavadil, D. Brooks, B. Kirby, K. Dragoon, and J. Caldwell. Grid impacts of wind power: A summary of recent studies in the United States. *Wind Energy*, 7(2):87–108, Apr.-Jun. 2004.
- [173] D. Peña, G. C. Tiao, and R. S. Tsay (Eds). *A Course in Time Series Analysis*. John Wiley, New York, 2001.
- [174] G. Pflug. Some remarks on the value-at-risk and the conditional value-at-risk. In S. Uryasev, editor, *Probabilistic Constrained Optimization: Methodology and Applications*. Kluwer Academic Publishers, Norwell, MA, 2000.
- [175] G. C. Pflug and W. Römisch. *Modeling, Measuring and Managing Risk*. World Scientific Publishing Company, Hackensack, 2007.
- [176] W. F. Pickard, A. Q. Shen, and N. J. Hansing. Parking the power: Strategies and physical limitations for bulk energy storage in supply-demand matching on a grid whose input power is provided by intermittent sources. *Renewable & sustainable energy reviews*, 13(8):1934–1945, Oct. 2009.

- [177] S. Pineda and A. J. Conejo. Scenario reduction for risk-averse electricity trading. *IET Generation, Transmission & Distribution*, in press 2010.
- [178] P. Pinson, C. Chevallier, and G. N. Kariniotakis. Trading wind generation from short-term probabilistic forecasts of wind power. *IEEE Transactions on Power Systems*, 22(3):1148–1156, Aug. 2007.
- [179] P. Pinson, H. Madsen, H. A. Nielsen, G. Papaefthymiou, and B. Klöckl. From probabilistic forecasts to statistical scenarios of short-term wind power production. *Wind Energy*, 12(1):51–62, Jan. 2009.
- [180] M. A. Plazas, A. J. Conejo, and F. J. Prieto. Multimarket optimal bidding for a power producer. *IEEE Transactions on Power Systems*, 20(4):2041–2050, Nov. 2005.
- [181] A. Prékopa. *Stochastic Programming*. Kluwer Academic Publishers, Dordrecht, 1995.
- [182] S. T. Rachev. *Probability Metrics and the Stability of Stochastic Models*. Wiley, Chichester, 1991.
- [183] P. Ramírez and J. A. Carta. Influence of the data sampling interval in the estimation of the parameters of the Weibull wind speed probability density distribution: a case study. *Energy Conversion and Management*, 46(15-16):2419–2438, Sep. 2005.
- [184] B. Ramsay and A. J. Wang. A neural network based estimator for electricity spot-pricing with particular reference to weekend and public holidays. *Neurocomputing*, 23(1-3):47–57, Dec. 1998.
- [185] N. S. Rau. Optimal dispatch of a system based on offers and bids—a mixed integer lp formulation. *IEEE Transactions on Power Systems*, 14(1):274–279, Feb. 1999.
- [186] Renewable Energy Research Laboratory. Center for Energy Efficiency & Renewable Energy. University of Massachusetts at Amherst. RERL Wind Data.

- Available at http://www.ceere.org/rer1/publications/resource_data/index.html, 2010.
- [187] Regional Greenhouse Gas Initiative (RGGI). Overview of RGGI CO₂ Budget Trading Program. Available at http://www.rggi.org/docs/program_summary_10_07.pdf, Oct. 2007.
- [188] R. T. Rockafellar. *Coherent Approaches to Risk in Optimization under Uncertainty*. Tutorials in Operations Research. OR tools and applications: glimpses of future technologies, INFORMS, 2007.
- [189] R. T. Rockafellar and S. Uryasev. Optimization of conditional value-at-risk. *Journal of Risk*, 2(3):21–41, 2000.
- [190] R. T. Rockafellar and S. Uryasev. Conditional value-at-risk for general loss distributions. *Journal of Banking & Finance*, 26(7):1443–1471, Jul. 2002.
- [191] C. P. Rodríguez and G. J. Anders. Energy price forecasting in the Ontario competitive power system market. *IEEE Transactions on Power Systems*, 19(1):366–374, Feb. 2004.
- [192] A. Ruszczyński and A. Shapiro (Eds). *Handbooks of Operations Research and Management Science. Stochastic Programming, vol. 10*. Elsevier, Amsterdam, 2003.
- [193] M. J. Schervish. P values: What they are and what they are not. *American Statistician*, 50(3):203–206, Aug. 1996.
- [194] R. A. Schlueter, G. L. Park, M. Lotfalian, H. Shayanfar, and J. Dorsey. Modification of power system operation for significant wind generation penetration. *IEEE Transactions on Power Apparatus and Systems*, PAS-102(1):153–161, 1983.
- [195] F. C. Schweppe, M. C. Caramanis, R. D. Tabors, and R. E. Bohn. *Spot Pricing of Electricity*. Springer, Norwell, Massachusetts.

- [196] J. V. Seguro and T. W. Lambert. Modern estimation of the parameters of the Weibull wind speed distribution for wind energy analysis. *Journal of the Wind Engineering and Industrial Aerodynamics*, 85(1):75–84, Mar. 2000.
- [197] S. Sen and H. D. Sherali. Decomposition with branch-and-cut approaches for two-stage stochastic mixed-integer programming. *Mathematical Programming*, 106(2):203–223, Jun. 2006.
- [198] M. Shahidehpour, H. Yamin, and Z. Li. *Market Operations in Electric Power Systems: Forecasting, Scheduling, and Risk Management*. IEEE-Wiley, New York, 2002.
- [199] H. D. Sherali and B. M. P. Fraticelli. A modification of Benders’ decomposition algorithm for discrete subproblems: An approach for stochastic programs with integer recourse. *Journal of Global Optimization*, 22(1-4):319–342, Jan. 2002.
- [200] H. Singh and A. Papalexopoulos. Competitive procurement of ancillary services by an independent system operator. *IEEE Transactions on Power Systems*, 14(2):498–504, May 1999.
- [201] R. Sioshansi. Evaluating the impacts of real-time pricing on the cost and value of wind generation. *IEEE Transactions on Power Systems*, 25(2):741–748, May 2010.
- [202] R. Sioshansi and W. Short. Evaluating the impacts of real-time pricing on the usage of wind generation. *IEEE Transactions on Power Systems*, 24(2):516–524, May 2009.
- [203] K. Skytte. The regulating power market on the Nordic power exchange Nord Pool: an econometric analysis. *Energy Economics*, 21(4):295–308, Aug. 1999.
- [204] L. Söder. Reserve margin planning in a wind-hydro thermal power system. *IEEE Transactions on Power Systems*, 8(2):564–571, May 1993.

- [205] S. Stoft. *Power System Economics: Designing Markets for Electricity*. IEEE-Wiley, New York, 2002.
- [206] A. Street, L. A. Barroso, B. Flach, M. V. Pereira, and S. Granville. Risk constrained portfolio selection of renewable sources in hydrothermal electricity markets. *IEEE Transactions on Power Systems*, 24(3):1136–1144, Aug. 2009.
- [207] P. Sullivan, W. Short, and N. Blair. Modeling the Benefits of Storage Technologies to Wind Power. National Renewable Energy Laboratory (NREL)
Available at <http://www.nrel.gov/docs/fy08osti/43510.pdf>, Jun. 2008.
- [208] B. R. Szkuta, L. A. Sanabria, and T. S. Dillon. Electricity price short-term forecasting using artificial neural networks. *IEEE Transactions on Power Systems*, 14(3):851–857, Aug. 1999.
- [209] J. Tastu, P. Pinson, E. Kotwa, H. Madsen, and H. A. Nielsen. Spatio-temporal analysis and modeling of short-term wind power forecast errors. *Wind Energy*, published online. DOI: 10.1002/we.401, 2010.
- [210] S. Tiedemann. *Risk Measures with Preselected Tolerance Levels in Two-Stage Stochastic Mixed-Integer Programming*. Cuvillier Verlag, Göttingen, Germany, 2007.
- [211] A. Tuohy, P. Meibom, E. Denny, and M. OMalley. Unit commitment for systems with significant wind penetration. *IEEE Transactions on Power Systems*, 24(2), May 2009.
- [212] B. C. Ummels, M. Gibescu, E. Pelgrum, W. L. Kling, and A. J. Brand. Impacts of wind power on thermal generation unit commitment and dispatch. *IEEE Transaction on Energy Conversion*, 22(1):44–51, Mar. 2007.
- [213] J. Usaola. Probabilistic load flow in systems with wind generation. *IET Generation, Transmission & Distribution*, 3(12):1031–1041, Dec. 2009.

- [214] J. Usaola. Probabilistic load flow with correlated wind power injections. *Electric Power Systems Research*, 80(5):528–536, May 2010.
- [215] V. Vittal. The impact of renewable resources on the performance and reliability of the electricity grid. *The Bridge*, 40(1):5–12, Mar. 2010.
- [216] J. Wang, M. Shahidehpour, and Z. Li. Security-constrained unit commitment with volatile wind power generation. *IEEE Transactions on Power Systems*, 23(3):1319–1327, Aug. 2008.
- [217] W. Wangdee and R. Billinton. Considering load-carrying capability and wind speed correlation of WECS in generation adequacy assessment. *IEEE Transactions on Energy Conversion*, 21(3):734–741, Sep. 2006.
- [218] R. Wiser and G. Barbose. Renewables Portfolio Standards in the United States: A Status Report with Data Through 2007. Lawrence Berkeley National Laboratory, Report LBNL 154-E, Apr. 2008.
Available at <http://eetd.lbl.gov/ea/ems/reports/lbnl-154e-revised.pdf>.
- [219] S. Wong and J. D. Fuller. Pricing energy and reserves using stochastic optimization in an alternative electricity market. *IEEE Transactions on Power Systems*, 22(2):631–638, May 2007.
- [220] T. Wu, M. Rothleder, Z. Alaywan, and A. D. Papalexopoulos. Pricing energy and ancillary services in integrated market systems by an optimal power flow. *IEEE Transactions on Power Systems*, 19(1):339–347, Feb. 2004.
- [221] Y.-K. Wu and J.-S. Hong. A literature review of wind forecasting technology in the world. In *Proceedings IEEE Power Tech*, pages 504–509, Lausanne, Switzerland, Jul. 2007.
- [222] K. Xie and R. Billinton. Considering wind speed correlation of WECS in reliability evaluation using the time-shifting technique. *Electric Power Systems Research*, 79(4):687–693, Apr. 2009.

- [223] M. A. B. Zammit, D. J. Hill, and R. J. Kaye. Designing ancillary services markets for power system security. *IEEE Transactions on Power Systems*, 15(2):675–680, May 2000.
- [224] H. Zareipour, C. A. Cañizares, K. Bhattacharya, and J. Thomson. Application of public-domain market information to forecast Ontario’s wholesale electricity prices. *IEEE Transactions on Power Systems*, 21(4):1707–1717, Nov. 2006.
- [225] L. Zhang, P. B. Luh, and K. Kasiviswanathan. Energy clearing price prediction and confidence interval estimation with cascaded neural networks. *IEEE Transactions on Power Systems*, 18(1):99–105, Feb. 2003.

Index

- 24-node system data, 285
 - demand data, 291
 - generator data, 285
 - network data, 285
- Market clearing, 181, 182
 - AB, 213
 - assumptions, 195
 - AUC, 213
 - case studies, 242
 - economic metrics, 212
 - equipment failures, 191
 - example, 214
 - formulation, 208
 - inter-temporal constraints, 293
 - linking constraints, 204
 - market constraints, 199
 - minimum up and down times, 294
 - model, 194
 - NAB, 214
 - objective function, 197
 - ramping limits, 293
 - real-time operating constraints, 202
 - reserve requirements, 185
 - scenario tree, 193
 - security constraints, 184
 - security criteria, 189
 - security metrics, 188
 - solution algorithm, 186
 - structure, 197
 - summary, 255
 - uncertainty characterization, 190
 - variables, 196
 - wind power production, 192
 - wind spillage cost, 212
- Stochastic programming, 271
 - EVPI, 278
 - quality metrics, 276
 - recourse problems, 273
 - risk management, 279
 - VSS, 276
- Uncertainty characterization, 53, 131, 190
 - adjustment prices, 131
 - ARMA models, 59
 - contemporaneous processes, 71
 - cross-correlations, 71
 - day-ahead prices, 131
 - dependencies, 71
 - equipment failures, 191
 - forward selection, 100

- imbalance prices, 131
 - market price case study, 79
 - non-contemporaneous processes, 71
 - normality, 67
 - probability distance, 98
 - quasi-contemporaneous processes, 71
 - scenario generation, 56
 - scenario reduction, 97
 - scenario reduction examples, 105, 108
 - scenario tree, 56, 137, 193
 - stationarity, 62
 - stochastic process, 54
 - stochastic processes, 71
 - summary, 113
 - wind power production, 133, 192
 - wind speed case studies, 82
 - wind speed conversion, 68
 - model, 139
 - offering curves, 143
 - revenue, 126
 - risk modeling, 144
 - scenario tree, 137
 - summary, 178
 - uncertainty characterization, 131
 - wind power production, 133
- Wind integration, 11
- Wind power trading, 115
- adjustment market, 145
 - adjustment prices, 131
 - case study, 163
 - certainty gain effect, 129
 - day-ahead prices, 131
 - decision framework, 116
 - electricity price formation, 119
 - example, 149
 - formulation, 146
 - imbalance cost, 126
 - imbalance prices, 121, 131